VOLUME I: FOUNDATIONS

The Quantum Theory of Fields

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ABSTRACT: Weinberg's QFT, typeset in nicer notation. Typesetting done entirely by coding agents, namely GPT 5.6.

Contents

Preface to Volume I

Why another book on quantum field theory? Today the student of quantum field theory can choose from among a score of excellent books, several of them quite up-to-date. Another book will be worth while only if it offers something new in content or perspective.

As to content, although this book contains a good amount of new material, I suppose the most distinctive thing about it is its generality; I have tried throughout to discuss matters in a context that is as general as possible. This is in part because quantum field theory has found applications far removed from the scene of its old successes, quantum electrodynamics, but even more because I think that this generality will help to keep the important points from being submerged in the technicalities of specific theories. Of course, specific examples are frequently used to illustrate general points, examples that are chosen from contemporary particle physics or nuclear physics as well as from quantum electrodynamics.

It is, however, the perspective of this book, rather than its content, that provided my chief motivation in writing it. I aim to present quantum field theory in a manner that will give the reader the clearest possible idea of *why* this theory takes the form it does, and why in this form it does such a good job of describing the real world.

The traditional approach, since the first papers of Heisenberg and Pauli on general quantum field theory, has been to take the existence of fields for granted, relying for justification on our experience with electromagnetism, and 'quantize' them — that is, apply to various simple field theories the rules of canonical quantization or path integration. Some of this traditional approach will be found here in the historical introduction presented in Chapter 1. This is certainly a way of getting rapidly into the subject, but it seems to me that it leaves the reflective reader with too many unanswered questions. Why should we believe in the rules of canonical quantization or path integration? Why should we adopt the simple field equations and Lagrangians that are found in the literature? For that matter, why have fields at all? It does not seem satisfactory to me to appeal to experience; after all, our purpose in theoretical physics is not just to describe the world as we find it, but to explain — in terms of a few fundamental principles — why the world is the way it is.

The point of view of this book is that quantum field theory is the way it is because (aside from theories like string theory that have an infinite number of particle types) it is the only way to reconcile the principles of quantum mechanics (including the cluster decomposition property) with those of special relativity. This is a point of view I have held for many years, but it is also one that has become newly appropriate. We have learned in recent years to think of our successful quantum field theories, including quantum electrodynamics, as 'effective field theories,' low-energy approximations to a deeper theory that may not even be a field theory, but something different like a string theory. On this basis, the reason that quantum field theories describe physics at accessible energies is that *any* relativistic quantum theory will look at sufficiently low energy like a quantum field theory. It is therefore important to understand the rationale for quantum field theory in terms of the principles of relativity and quantum mechanics. Also, we think differently now about some of the problems of quantum field theories, such as non-renormalizability and 'triviality,' that used to bother us when we thought of these theories as truly fundamental, and the discussions here will reflect these changes. This is intended to be a book on quantum field theory for the era of effective field theories.

The most immediate and certain consequences of relativity and quantum mechanics are the properties of particle states, so here particles come first — they are introduced in Chapter 2 as ingredients in the representation of the inhomogeneous Lorentz group in the Hilbert space of quantum mechanics. Chapter 3 provides a framework for addressing the fundamental dynamical question: given a state that in the distant past looks like a certain collection of free particles, what will it look like in the future? Knowing the generator of time-translations, the Hamiltonian, we can answer this question through the perturbative expansion for the array of transition amplitudes known as the S-matrix. In Chapter 4 the principle of cluster decomposition is invoked to describe how the generator of time-translations, the Hamiltonian, is to be constructed from creation and annihilation operators. Then in Chapter 5 we return to Lorentz invariance, and show that it requires these creation and annihilation operators to be grouped together in causal quantum fields. As a spin-off, we deduce the CPT theorem and the connection between spin and statistics. The formalism is used in Chapter 6 to derive the Feynman rules for calculating the S-matrix.

It is not until Chapter 7 that we come to Lagrangians and the canonical formalism. The rationale here for introducing them is not that they have proved useful elsewhere in physics (never a very satisfying explanation) but rather that this formalism makes it easy to choose interaction Hamiltonians for which the S-matrix satisfies various assumed symmetries. In particular, the Lorentz invariance of the Lagrangian density ensures the existence of a set of ten operators that satisfy the algebra of the Poincaré group and, as we show in Chapter 3, this is the key condition that we need to prove the Lorentz invariance of the S-matrix. Quantum electrodynamics finally appears in Chapter 8. Path integration is introduced in Chapter 9, and used to justify some of the hand-waving in Chapter 8 regarding the Feynman rules for quantum electrodynamics. This is a somewhat later introduction of path integrals than is fashionable these days, but it seems to me that although path integration is by far the best way of rapidly deriving Feynman rules from a given Lagrangian, it rather obscures the quantum mechanical reasons underlying these calculations.

Volume I concludes with a series of chapters, 10–14, that provide an introduction to the calculation of radiative corrections, involving loop graphs, in general field theories. Here too the arrangement is a bit unusual; we start with a chapter on non-perturbative methods, in part because the results we obtain help us to understand the necessity for field and mass renormalization, without regard to whether the theory contains infinities or not. Chapter 11 presents the classic one-loop calculations of quantum chromodynamics, both as an opportunity to explain useful calculational techniques (Feynman parameters, Wick rotation, dimensional and Pauli–Villars regularization), and also as a concrete example of renormalization in action. The experience gained in Chapter 11 is extended to all orders and general theories in Chapter 12, which also describes the modern view of non-renormalizability that is appropriate to effective field theories. Chapter 13 is a digression on the special problems raised by massless particles of low energy or parallel momenta. The Dirac equation for an electron in an external electromagnetic field, which historically appeared almost at the very start of relativistic quantum mechanics, is not seen here until Chapter 14, on bound state problems, because this equation should not be viewed (as Dirac did) as a relativistic version of the Schrödinger equation, but rather as an approximation to a true relativistic quantum theory, the quantum field theory of photons and electrons. This chapter ends with a treatment of the Lamb shift, bringing the confrontation of theory and experiment up to date.

The reader may feel that some of the topics treated here, especially in Chapter 3, could more properly have been left to textbooks on nuclear or elementary particle physics. So they might, but in my experience these topics are usually either not covered or covered poorly, using specific dynamical models rather than the general principles of symmetry and quantum mechanics. I have met string theorists who have never heard of the relation between time-reversal invariance and final-state phase shifts, and nuclear theorists who do not understand why resonances are governed by the Breit–Wigner formula. So in the early chapters I have tried to err on the side of inclusion rather than exclusion.

Volume II will deal with the advances that have revived quantum field theory in recent years: non-Abelian gauge theories, the renormalization group, broken symmetries, anomalies, instantons, and so on.

I have tried to give citations both to the classic papers in the quantum theory of fields and to useful references on topics that are mentioned but not presented in detail in this book. I did not always know who was responsible for material presented here, and the mere absence of a citation should not be taken as a claim that the material presented here is original. But some of it is. I hope that I have improved on the original literature or standard textbook treatments in several places, as for instance in the proof that symmetry operators are either unitary or antiunitary; the discussion of superselection rules; the analysis of particle degeneracy associated with unconventional representations of inversions; the use of the cluster decomposition principle; the derivation of the reduction formula; the derivation of the external field approximation; and even the calculation of the Lamb shift.

I have also supplied problems for each chapter except the first. Some of these problems aim simply at providing exercise in the use of techniques described in the chapter; others are intended to suggest extensions of the results of the chapter to a wider class of theories.

In teaching quantum field theory, I have found that each of the two volumes of this book provides enough material for a one-year course for graduate students. I intended that this book should be accessible to students who are familiar with non-relativistic quantum mechanics and classical electrodynamics. I assume a basic knowledge of complex analysis and matrix algebra, but topics in group theory and topology are explained where they are introduced.

This is not a book for the student who wants immediately to begin calculating Feynman graphs in the standard model of weak, electromagnetic, and strong interactions. Nor is this a book for those who seek a higher level of mathematical rigor. Indeed, there are parts of this book whose lack of rigor will bring tears to the eyes of the mathematically inclined reader. Rather, I hope it will suit the physicists and physics students who want to understand *why* quantum field theory is the way it is, so that they will be ready for whatever new developments in physics may take us beyond our present understandings.

* * *

Much of the material in this book I learned from my interactions over the years with numerous other physicists, far too many to name here. But I must acknowledge my special intellectual debt to Sidney Coleman, and to my colleagues at the University of Texas: Arno Bohm, Luis Boya, Phil Candelas, Bryce DeWitt, Cecile DeWitt-Morette, Jacques Distler, Willy Fischler, Josh Feinberg, Joaquim Gomis, Vadim Kaplunovsky, Joe Polchinski, and Paul Shapiro. I owe thanks for help in the preparation of the historical introduction to Gerry Holton, Arthur Miller, and Sam Schweber. Thanks are also due to Alyce Wilson, who prepared the illustrations and typed the LaTeX input files until I learned how to do it, and to Terry Riley for finding countless books and articles. I am grateful to Maureen Storey and Alison Woollatt of Cambridge University Press for helping to ready this book for publication, and especially to my editor, Rufus Neal, for his friendly good advice.

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Notation

Typesetter's note: see `NOTATION.md` for the specific notation upgrades I implemented.

Latin indices $i, j, k, \ldots$ generally run over the three spatial coordinates $1, 2, 3$. Greek indices $\mu, \nu, \ldots$ run over spacetime coordinates in the order $0, 1, 2, 3$, with x^0 the time coordinate. Repeated indices are summed unless otherwise indicated.

The spacetime metric is mostly plus,

$$\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1),$$

so an on-shell four-momentum satisfies $p^2 = -m^2$, and spacelike separations satisfy $(x - x')^2 \geq 0$. The d'Alembertian is

$$\Box = \partial_\mu \partial^\mu = \nabla^2 - \partial_t^2.$$

The Levi–Civita tensor is totally antisymmetric, with $\epsilon^{0123} = +1$.

Spatial three-vectors are bold, and a hat denotes a unit vector: $\widehat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$. A dot denotes a time derivative.

Dirac matrices obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \qquad \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3.$$

With this metric convention, $\beta = i\gamma^0$ and the Dirac adjoint is $\bar{u} = u^\dagger \beta$. The step function $\theta(s)$ equals 1 for $s > 0$ and 0 for $s < 0$.

Complex conjugation, transpose, and Hermitian conjugation are written A^*, A^T, and $A^\dagger$, respectively. The identity is $\mathbf{1}$. The abbreviations +H.c. and +c.c. indicate addition of the Hermitian conjugate and complex conjugate of the preceding terms.

States use Dirac notation: $|p, \sigma\rangle$ for one-particle states, $|\Omega\rangle$ for the vacuum, and $\langle\phi|\psi\rangle$ for inner products. Asymptotic states are labeled $|\alpha\rangle_{\mathrm{in}}$ and $|\beta\rangle_{\mathrm{out}}$. Positive- and negative-frequency field pieces are written $\psi^{(+)}$ and $\psi^{(-)}$, avoiding the original overloading of $+$ and $-$ for both field frequencies and asymptotic states.

The interaction Hamiltonian is denoted

$$H_I(t) = \int d^3x \, \mathcal{H}_I(x),$$

in place of the original bare V notation. Other operators are generally left unhatted when the context is unambiguous.

Except in Chapter 1, $\hbar = c = 1$. Throughout, $-e$ is the rationalized charge of the electron, so $\alpha = e^2/(4\pi) \simeq 1/137$. Parentheses at the end of quoted numerical data give the uncertainty in the final digits.

1 Historical Introduction

Our immersion in the present state of physics makes it hard for us to understand the difficulties of physicists even a few years ago, or to profit from their experience. At the same time, a knowledge of our history is a mixed blessing—it can stand in the way of the logical reconstruction of physical theory that seems to be continually necessary.

I have tried in this book to present the quantum theory of fields in a logical manner, emphasizing the deductive trail that ascends from the physical principles of special relativity and quantum mechanics. This approach necessarily draws me away from the order in which the subject in fact developed. To take one example, it is historically correct that quantum field theory grew in part out of a study of relativistic wave equations, including the Maxwell, Klein–Gordon, and Dirac equations. For this reason it is natural that courses and treatises on quantum field theory introduce these wave equations early, and give them great weight. Nevertheless, it has long seemed to me that a much better starting point is Wigner's definition of particles as representations of the inhomogeneous Lorentz group, even though this work was not published until 1939 and did not have a great impact for many years after. In this book we start with particles and get to the wave equations later.

This is not to say that particles are necessarily more fundamental than fields. For many years after 1950 it was generally assumed that the laws of nature take the form of a quantum theory of fields. I start with particles in this book, not because they are more fundamental, but because what we know about particles is more certain, more directly derivable from the principles of quantum mechanics and relativity. If it turned out that some physical system could not be described by a quantum field theory, it would be a sensation; if it turned out that the system did not obey the rules of quantum mechanics and relativity, it would be a cataclysm.

In fact, lately there has been a reaction against looking at quantum field theory as fundamental. The underlying theory might not be a theory of fields or particles, but perhaps of something quite different, like strings. From this point of view, quantum electrodynamics and the other quantum field theories of which we are so proud are mere 'effective field theories,' low-energy approximations to a more fundamental theory. The reason that our field theories work so well is not that they are fundamental truths, but that any relativistic quantum theory will look like a field theory when applied to particles at sufficiently low energy. On this basis, if we want to know why quantum field theories are the way they are, we have to start with particles.

But we do not want to pay the price of altogether forgetting our past. This chapter will therefore present the history of quantum field theory from earliest times to 1949, when it finally assumed its modern form. In the remainder of the book I will try to keep history from intruding on physics.

One problem that I found in writing this chapter is that the history of quantum field theory is from the beginning inextricably entangled with the history of quantum mechanics itself. Thus, the reader who is familiar with the history of quantum mechanics may find some material here that he or she already knows, especially in the first section, where I discuss the early attempts to put together quantum mechanics with special relativity. In this case I can only suggest that the reader should skip on to the less familiar parts.

On the other hand, readers who have no prior familiarity with quantum field theory may find parts of this chapter too brief to be altogether clear. I urge such readers not to worry. This chapter is not intended as a self-contained introduction to quantum field theory, and is not needed as a basis for the rest of the book. Some readers may even prefer to start with the next chapter, and come back to the history later. However, for many readers the history of quantum field theory should serve as a good introduction to quantum field theory itself.

I should add that this chapter is not intended as an original work of historical scholarship. I have based it on books and articles by real historians, plus some historical reminiscences and original physics articles that I have read. Most of these are listed in the bibliography given at the end of this chapter, and in the list of references. The reader who wants to go more deeply into historical matters is urged to consult these listed works.

A word about notation. In order to keep some of the flavor of past times, in this chapter I will show explicit factors of $\hbar$ and c (and even h), but in order to facilitate comparison with modern physics literature, I will use the more modern rationalized electrostatic units for charge, so that the fine structure constant $\alpha \simeq 1/137$ is $e^2/4\pi\hbar c$. In subsequent chapters I will mostly use the 'natural' system of units, simply setting $\hbar = c = 1$.

1.1 Relativistic Wave Mechanics

Wave mechanics started out as relativistic wave mechanics. Indeed, as we shall see, the founders of wave mechanics, Louis de Broglie and Erwin Schrödinger, took a good deal of their inspiration from special relativity. It was only later that it became generally clear that relativistic wave mechanics, in the sense of a relativistic quantum theory of a fixed number of particles, is an impossibility. Thus, despite its many successes, relativistic wave mechanics was ultimately to give way to quantum field theory. Nevertheless, relativistic wave mechanics survived as an important element in the formal apparatus of quantum field theory, and it posed a challenge to field theory, to reproduce its successes.

The possibility that material particles can like photons be described in terms of waves was first suggested [1] in 1923 by Louis de Broglie. Apart from the analogy with radiation, the chief clue was Lorentz invariance: if particles are described by a wave whose phase at position $\mathbf{x}$ and time t is of the form $2\pi(\boldsymbol{\kappa}\cdot\mathbf{x}-\nu t)$, and if this phase is to be Lorentz invariant, then the vector $\boldsymbol{\kappa}$ and the frequency ν must transform like $\mathbf{x}$ and t, and hence like $\mathbf{p}$ and E. In order for this to be possible $\boldsymbol{\kappa}$ and ν must have the same velocity dependence as $\mathbf{p}$ and E, and therefore must be proportional to them, with the same constant of proportionality. For photons, one had the Einstein relation $E = h\nu$, so it was natural to assume that, for material particles,

$$\boldsymbol{\kappa} = \mathbf{p}/h, \qquad \nu = E/h, \tag{1.1.1}$$

just as for photons. The group velocity $\partial\nu/\partial\boldsymbol{\kappa}$ of the wave then turns out to equal the particle velocity, so wave packets just keep up with the particle they represent.

By assuming that any closed orbit contains an integral number of particle wavelengths $\lambda = 1/|\boldsymbol{\kappa}|$, de Broglie was able to derive the old quantization conditions of Niels Bohr and Arnold Sommerfeld, which though quite mysterious had worked well in accounting for atomic spectra. Also, both de Broglie and Walter Elsasser [2] suggested that de Broglie's wave theory could be tested by looking for interference effects in the scattering of electrons from crystals; such effects were established a few years later by Clinton Joseph Davisson and Lester H. Germer [3]. However, it was still unclear how the de Broglie relations (1.1.1) should be modified for non-free particles, as for instance for an electron in a general Coulomb field.

Wave mechanics was by-passed in the next step in the history of quantum mechanics, the development of matrix mechanics [4] by Werner Heisenberg, Max Born, Pascual Jordan and Wolfgang Pauli in the years 1925–1926. At least part of the inspiration for matrix mechanics was the insistence that the theory should involve only observables, such as the energy levels, or emission and absorption rates. Heisenberg's 1925 paper opens with the manifesto: 'The present paper seeks to establish a basis for theoretical quantum mechanics founded exclusively upon relationships between quantities that in principle are observable.' This sort of positivism was to reemerge at various times in the history of quantum field theory, as for instance in the introduction of the S-matrix by John Wheeler and Heisenberg (see Chapter 3) and in the revival of dispersion theory in the 1950s (see Chapter 10), though modern quantum field theory is very far from this ideal. It would take us too far from our subject to describe matrix mechanics in any detail here.

As everyone knows, wave mechanics was revived by Erwin Schrödinger. In his 1926 series of papers [5], the familiar non-relativistic wave equation is suggested first, and then used to rederive the results of matrix mechanics. Only later, in the sixth section of the fourth paper, is a relativistic wave equation offered. According to Dirac [6], the history is actually quite different: Schrödinger first derived the relativistic equation, then became discouraged because it gave the wrong fine structure for hydrogen, and then some months later realized that the non-relativistic approximation to his relativistic equation was of value even if the relativistic equation itself was incorrect! By the time that Schrödinger came to publish his relativistic wave equation, it had already been independently rediscovered by Oskar Klein [7] and Walter Gordon [8], and for this reason it is usually called the 'Klein–Gordon equation.'

Schrödinger's relativistic wave equation was derived by noting first that, for a 'Lorentz electron' of mass m and charge e in an external vector potential $\mathbf{A}$ and Coulomb potential ϕ, the Hamiltonian H and momentum $\mathbf{p}$ are related by[1]

$$0 = (H + e\phi)^2 - c^2(\mathbf{p} + e\mathbf{A}/c)^2 - m^2c^4. \tag{1.1.2}$$

For a free particle described by a plane wave $\exp\{2\pi i(\boldsymbol{\kappa}\cdot\mathbf{x} - \nu t)\}$, the de Broglie relations (1.1.1) can be obtained by the identifications

$$\mathbf{p} = h\boldsymbol{\kappa} \longrightarrow -i\hbar\boldsymbol{\nabla}, \qquad E = h\nu \longrightarrow i\hbar\frac{\partial}{\partial t}, \tag{1.1.3}$$

[1] This is Lorentz invariant, because the quantities $\mathbf{A}$ and ϕ have the same Lorentz transformation property as $c\mathbf{p}$ and E. Schrödinger actually wrote H and $\mathbf{p}$ in terms of partial derivatives of an action function, but this makes no difference to our present discussion.

where $\hbar$ is the convenient symbol (introduced later by Dirac) for $h/2\pi$. By an admittedly formal analogy, Schrödinger guessed that an electron in the external fields $\mathbf{A}, \phi$ would be described by a wave function $\psi(\mathbf{x}, t)$ satisfying the equation obtained by making the same replacements in (1.1.2):

$$0 = \left[\left(i\hbar \frac{\partial}{\partial t} + e\phi \right)^2 - c^2 \left(-i\hbar \boldsymbol{\nabla} + \frac{e\mathbf{A}}{c} \right)^2 - m^2 c^4 \right] \psi(\mathbf{x}, t). \tag{1.1.4}$$

In particular, for the stationary states of hydrogen we have $\mathbf{A} = 0$ and $\phi = e/4\pi r$, and ψ has the time-dependence $\exp(-iEt/\hbar)$, so (1.1.4) becomes

$$0 = \left[\left(E + \frac{e^2}{4\pi r} \right)^2 - c^2 \hbar^2 \nabla^2 - m^2 c^4 \right] \psi(\mathbf{x}). \tag{1.1.5}$$

Solutions satisfying reasonable boundary conditions can be found for the energy values [9]

$$E = mc^2 \left[1 - \frac{\alpha^2}{2n^2} - \frac{\alpha^4}{2n^4} \left(\frac{n}{\ell + \frac{1}{2}} - \frac{3}{4} \right) + \cdots \right], \tag{1.1.6}$$

where $\alpha = e^2/4\pi\hbar c$ is the 'fine structure constant,' roughly $1/137$; n is a positive-definite integer, and ℓ, the orbital angular momentum in units of $\hbar$, is an integer with $0 \leq \ell \leq n-1$. The α^2 term gave good agreement with the gross features of the hydrogen spectrum (the Lyman, Balmer, etc. series) and, according to Dirac [6], it was this agreement that led Schrödinger eventually to develop his non-relativistic wave equation. On the other hand, the α^4 term gave a fine structure in disagreement with existing accurate measurements of Friedrich Paschen [10].

It is instructive here to compare Schrödinger's result with that of Arnold Sommerfeld [10c], obtained using the rules of the old quantum theory:

$$E = mc^2 \left[1 - \frac{\alpha^2}{2n^2} - \frac{\alpha^4}{2n^4} \left(\frac{n}{k} - \frac{3}{4} \right) + \cdots \right]. \tag{1.1.7}$$

where m is the electron mass. Here k is an integer between 1 and n, which in Sommerfeld's theory is given in terms of the orbital angular momentum $\ell\hbar$ as $k = \ell + 1$. This gave a fine structure splitting in agreement with experiment: for instance, for $n = 2$ Eq. (1.1.7) gives two levels ($k = 1$ and $k = 2$), split by the observed amount $\alpha^4 mc^2/32$, or 4.53×10^{-5} eV. In contrast, Schrödinger's result (1.1.6) gives an $n = 2$ fine structure splitting $\alpha^4 mc^2/12$, considerably larger than observed.

Schrödinger correctly recognized that the source of this discrepancy was his neglect of the spin of the electron. The splitting of atomic energy levels by non-inverse-square electric fields in alkali atoms and by weak external magnetic fields (the so-called anomalous Zeeman effect) had revealed a multiplicity of states larger than could be accounted for by the Bohr–Sommerfeld theory; this led George Uhlenbeck and Samuel Goudsmit [11] in 1925 to suggest that the electron has an intrinsic angular momentum $\hbar/2$. Also, the magnitude of the Zeeman splitting [12] allowed them to estimate further that the electron has a magnetic moment

$$\mu = \frac{e\hbar}{2mc}. \tag{1.1.8}$$

It was clear that the electron's spin would be coupled to its orbital angular momentum, so that Schrödinger's relativistic equation should not be expected to give the correct fine structure splitting.

Indeed, by 1927 several authors [13] had been able to show that the spin–orbit coupling was able to account for the discrepancy between Schrödinger's result (1.1.6) and experiment. There are really two effects here: one is a direct coupling between the magnetic moment (1.1.8) and the magnetic field felt by the electron as it moves through the electrostatic field of the atom; the other is the relativistic 'Thomas precession' caused (even in the absence of a magnetic moment) by the circular motion of the spinning electron [14]. Together, these two effects were found to lift the level with total angular momentum $j = \ell + \frac{1}{2}$ to the energy (1.1.7) given by Sommerfeld for $k = \ell + 1 = j + \frac{1}{2}$, while the level with $j = \ell - \frac{1}{2}$ was lowered to the value given by Sommerfeld for $k = \ell = j + \frac{1}{2}$. Thus the energy was found to depend only on n and j, but not separately on ℓ:

$$E = mc^2 \left[1 - \frac{\alpha^2}{2n^2} - \frac{\alpha^4}{2n^4} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) + \cdots \right]. \tag{1.1.9}$$

By accident Sommerfeld's theory had given the correct magnitude of the splitting in hydrogen ($j + \frac{1}{2}$ like k runs over integer values from 1 to n) though it was wrong as to the assignment of orbital angular momentum values ℓ to these various levels. In addition, the multiplicity of the fine structure levels in hydrogen was now predicted to be 2 for $j = \frac{1}{2}$ and $2(2j + 1)$ for $j > \frac{1}{2}$ (corresponding to ℓ values $j \pm \frac{1}{2}$), in agreement with experiment.

Despite these successes, there still was not a thorough relativistic theory which incorporated the electron's spin from the beginning. Such a theory was discovered in 1928 by Paul Dirac. However, he did not set out simply to make a relativistic theory of the spinning electron; instead, he approached the problem by posing a question that would today seem very strange. At the beginning of his 1928 paper [15], he asks 'why Nature should have chosen this particular model for the electron, instead of being satisfied with the point charge.' To us today, this question is like asking why bacteria have only one cell: having spin $\hbar/2$ is just one of the properties that define a particle as an electron, rather than one of the many other types of particles with various spins that are known today. However, in 1928 it was possible to believe that all matter consisted of electrons, and perhaps something similar with positive charge in the atomic nucleus. Thus, in the spirit of the times in which it was asked, Dirac's question can be restated: 'Why do the fundamental constituents of matter have to have spin $\hbar/2$?'

For Dirac, the key to this question was the requirement that probabilities must be positive. It was known [16] that the probability density for the non-relativistic Schrödinger equation is $|\psi|^2$, and that this satisfies a continuity equation of the form

$$\frac{\partial}{\partial t}|\psi|^2 - \frac{i\hbar}{2m} \boldsymbol{\nabla} \cdot (\psi^* \boldsymbol{\nabla} \psi - \psi \boldsymbol{\nabla} \psi^*) = 0$$

so the space-integral of $|\psi|^2$ is time-independent. On the other hand, the only probability density ρ and current $\mathbf{J}$, which can be formed from solutions of the relativistic Schrödinger equation and which satisfy a conservation law,

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{J} = 0, \tag{1.1.10}$$

are of the form

$$\rho = N \,\mathrm{Im}\, \psi^* \left(\frac{\partial}{\partial t} - \frac{ie\phi}{\hbar} \right) \psi, \tag{1.1.11}$$

$$\mathbf{J} = Nc^2 \,\mathrm{Im}\, \psi^* \left(\boldsymbol{\nabla} + \frac{ie\mathbf{A}}{\hbar c} \right) \psi, \tag{1.1.12}$$

with N an arbitrary constant. It is not possible to identify ρ as the probability density, because (with or without an external potential ϕ) ρ does not have definite sign. To quote Dirac's reminiscences [17] about this problem

> I remember once when I was in Copenhagen, that Bohr asked me what I was working on and I told him I was trying to get a satisfactory relativistic theory of the electron, and Bohr said 'But Klein and Gordon have already done that!' That answer first rather disturbed me. Bohr seemed quite satisfied by Klein's solution, but I was not because of the negative probabilities that it led to. I just kept on with it, worrying about getting a theory which would have only positive probabilities.

According to George Gamow [18], Dirac found the answer to this problem on an evening in 1928 while staring into a fireplace at St John's College, Cambridge. He realized that the reason that the Klein–Gordon (or relativistic Schrödinger) equation can give negative probabilities is that the ρ in the conservation equation (1.1.10) involves a time-derivative of the wave function. This in turn happens because the wave function satisfies a differential equation of second order in the time. The problem therefore was to replace this wave equation with another one of first order in time derivatives, like the non-relativistic Schrödinger equation.

Suppose the electron wave function is a multi-component quantity $\psi_n(\mathbf{x})$, which satisfies a wave equation of the form,

$$i\hbar \frac{\partial \psi}{\partial t} = \mathcal{H}\psi, \tag{1.1.13}$$

where $\mathcal{H}$ is some matrix function of space derivatives. In order to have a chance at a Lorentz-invariant theory, we must suppose that because the equation is linear in time-derivatives, it is also linear in space-derivatives, so that $\mathcal{H}$ takes the form:

$$\mathcal{H} = -i\hbar c \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + \beta mc^2, \tag{1.1.14}$$

where α_1, α_2, α_3, and β are constant matrices. From (1.1.13) we can derive the second-order equation

$$\begin{aligned} -\hbar^2 \frac{\partial^2 \psi}{\partial t^2} = \mathcal{H}^2 \psi = & -\hbar^2 c^2 \alpha_i \alpha_j \frac{\partial^2 \psi}{\partial x_i \partial x_j} \\ & - i\hbar mc^3 (\alpha_i \beta + \beta \alpha_i) \frac{\partial \psi}{\partial x_i} + m^2 c^4 \beta^2 \psi. \end{aligned}$$

(The summation convention is in force here; i and j run over the values 1, 2, 3, or x, y, z.) But this must agree with the free-field form of the relativistic Schrödinger equation (1.1.4), which just expresses the relativistic relation between momentum and energy. Therefore, the matrices $\boldsymbol{\alpha}$ and β must satisfy the relations

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 2\delta_{ij}\mathbf{1}, \tag{1.1.15}$$

$$\alpha_i\beta + \beta\alpha_i = 0, \tag{1.1.16}$$

$$\beta^2 = \mathbf{1}, \tag{1.1.17}$$

where δ_{ij} is the Kronecker delta (unity for $i = j$; zero for $i \neq j$) and $\mathbf{1}$ is the unit matrix. Dirac found a set of 4×4 matrices which satisfy these relations

$$\begin{gathered}\alpha_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \qquad \alpha_2 = \begin{bmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{bmatrix}, \\ \alpha_3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}, \qquad \beta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.\end{gathered} \tag{1.1.18}$$

To show that this formalism is Lorentz-invariant, Dirac multiplied Eq. (1.1.13) on the left with β, so that it could be put in the form

$$\left[\hbar c \gamma^\mu \frac{\partial}{\partial x^\mu} + mc^2\right]\psi = 0, \tag{1.1.19}$$

where

$$\gamma^i \equiv -i\beta\alpha_i, \qquad \gamma^0 \equiv -i\beta. \tag{1.1.20}$$

(The Greek indices μ, ν, etc. will now run over the values $0, 1, 2, 3$, with $x^0 = ct$. Dirac used $x_4 = ict$, and correspondingly $\gamma_4 = \beta$.) The matrices γ^μ satisfy the anticommutation relations

$$\frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) = \eta^{\mu\nu} \equiv \begin{cases} +1, & \mu = \nu = 1, 2, 3, \\ -1, & \mu = \nu = 0, \\ 0, & \mu \neq \nu. \end{cases} \tag{1.1.21}$$

Dirac noted that these anticommutation relations are Lorentz-invariant, in the sense that they are also satisfied by the matrices $\Lambda^\mu{}_\nu\gamma^\nu$, where Λ is any Lorentz transformation. He concluded from this that $\Lambda^\mu{}_\nu\gamma^\nu$ must be related to γ^μ by a similarity transformation:

$$\Lambda^\mu{}_\nu\gamma^\nu = S^{-1}(\Lambda)\gamma^\mu S(\Lambda).$$

It follows that the wave equation is invariant if, under a Lorentz transformation $x^\mu \to \Lambda^\mu{}_\nu x^\nu$, the wave function undergoes the matrix transformation $\psi \to S(\Lambda)\psi$. (These matters are discussed more fully, from a rather different point of view, in Chapter 5.)

To study the behavior of electrons in an arbitrary external electromagnetic field, Dirac followed the 'usual procedure' of making the replacements

$$i\hbar\frac{\partial}{\partial t} \longrightarrow i\hbar\frac{\partial}{\partial t} + e\phi, \qquad -i\hbar\boldsymbol{\nabla} \longrightarrow -i\hbar\boldsymbol{\nabla} + \frac{e}{c}\mathbf{A} \tag{1.1.22}$$

as in Eq. (1.1.4). The wave equation (1.1.13) then takes the form

$$\left(i\hbar\frac{\partial}{\partial t} + e\phi\right)\psi = (-i\hbar c\boldsymbol{\nabla} + e\mathbf{A})\cdot\boldsymbol{\alpha}\psi + mc^2\beta\psi. \tag{1.1.23}$$

Dirac used this equation to show that in a central field, the conservation of angular momentum takes the form

$$[\mathcal{H}, -i\hbar\mathbf{r}\times\boldsymbol{\nabla} + \hbar\boldsymbol{\sigma}/2] = 0, \tag{1.1.24}$$

where $\mathcal{H}$ is the matrix differential operator (1.1.14) and $\boldsymbol{\sigma}$ is the 4×4 version of the spin matrix introduced earlier by Pauli [19]

$$\boldsymbol{\sigma} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}\boldsymbol{\alpha}. \tag{1.1.25}$$

Since each component of $\boldsymbol{\sigma}$ has eigenvalues ± 1, the presence of the extra term in (1.1.24) shows that the electron has intrinsic angular momentum $\hbar/2$.

Dirac also iterated Eq. (1.1.23), obtaining a second-order equation, which turned out to have just the same form as the Klein–Gordon equation (1.1.4) except for the presence on the right-hand-side of two additional terms

$$[-e\hbar c\boldsymbol{\sigma}\cdot\mathbf{B} - ie\hbar c\boldsymbol{\alpha}\cdot\mathbf{E}]\psi. \tag{1.1.26}$$

For a slowly moving electron, the first term dominates, and represents a magnetic moment in agreement with the value (1.1.8) found by Goudsmit and Uhlenbeck [11]. As Dirac recognized, this magnetic moment, together with the relativistic nature of the theory, guaranteed that this theory would give a fine structure splitting in agreement (to order $\alpha^4 mc^2$) with that found by Heisenberg, Jordan, and Charles G. Darwin [13]. A little later, an 'exact' formula for the hydrogen energy levels in Dirac's theory was derived by Darwin [20] and Gordon [21]

$$E = mc^2\left(1 + \frac{\alpha^2}{\left\{n - j - \frac{1}{2} + \left[(j+\frac{1}{2})^2 - \alpha^2\right]^{1/2}\right\}^2}\right)^{-1/2}. \tag{1.1.27}$$

The first three terms of a power series expansion in α^2 agree with the approximate result (1.1.9).

This theory achieved Dirac's primary aim: a relativistic formalism with positive probabilities. From (1.1.13) we can derive a continuity equation

$$\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot\mathbf{J} = 0 \tag{1.1.28}$$

with

$$\rho = |\psi|^2, \qquad \mathbf{J} = c\psi^\dagger \boldsymbol{\alpha}\psi, \tag{1.1.29}$$

so that the positive quantity $|\psi|^2$ can be interpreted as a probability density, with constant total probability $\int |\psi|^2 d^3x$. However, there was another difficulty which Dirac was not immediately able to resolve.

For a given momentum $\mathbf{p}$, the wave equation (1.1.13) has four solutions of the plane wave form

$$\psi \propto \exp\left[\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{x} - Et)\right]. \tag{1.1.30}$$

Two solutions with $E = +\sqrt{\mathbf{p}^2c^2 + m^2c^4}$ correspond to the two spin states of an electron with $J_z = \pm\hbar/2$. The other two solutions have $E = -\sqrt{\mathbf{p}^2c^2 + m^2c^4}$, and no obvious physical interpretation. As Dirac pointed out, this problem arises also for the relativistic Schrödinger equation: for each $\mathbf{p}$, there are two solutions of the form (1.1.30), one with positive E and one with negative E.

Of course, even in classical physics, the relativistic relation $E^2 = \mathbf{p}^2c^2 + m^2c^4$ has two solutions, $E = \pm\sqrt{\mathbf{p}^2c^2 + m^2c^4}$. However, in classical physics we can simply assume that the only physical particles are those with positive E. Since the positive solutions have $E > mc^2$ and the negative ones have $E < -mc^2$, there is a finite gap between them, and no continuous process can take a particle from positive to negative energy.

The problem of negative energies is much more troublesome in relativistic quantum mechanics. As Dirac pointed out in his 1928 paper [15], the interaction of electrons with radiation can produce transitions in which a positive-energy electron falls into a negative-energy state, with the energy carried off by two or more photons. Why then is matter stable?

In 1930 Dirac offered a remarkable solution [22]. Dirac's proposal was based on the exclusion principle, so a few words about the history of this principle are in order here.

The periodic table of the elements and the systematics of X-ray spectroscopy had together by 1924 revealed a pattern in the population of atomic energy levels by electrons [23]. The maximum number N_n of electrons in a shell characterized by principal quantum number n is given by twice the number of orbital states with that n

$$N_n = 2\sum_{\ell=0}^{n-1}(2\ell+1) = 2n^2 = 2, 8, 18, \ldots. \tag{1.1.31}$$

Wolfgang Pauli [24] in 1925 suggested that this pattern could be understood if N_n is the total number of possible states in the nth shell, and if in addition there is some mysterious 'exclusion principle' which forbids more than one electron from occupying the same state. He explained the puzzling factor 2 in (1.1.31) as due to a 'peculiar, classically non-describable duplexity' of the electron states, and as we have seen this was understood a little later as due to the spin of the electron [11]. The exclusion principle answered a question that had remained obscure in the old atomic theory of Bohr and Sommerfeld: why do not all the electrons in heavy atoms fall down into the shell of lowest energy? Subsequently Pauli's exclusion principle was formalized by a number of authors [25] as the requirement that the wave function of a multi-electron system is antisymmetric in the coordinates, orbital and spin, of all the electrons. This principle was incorporated into statistical mechanics by Enrico Fermi [26] and Dirac [27], and for this reason particles obeying the exclusion principle are generally called 'fermions,' just as particles like photons for which the wave function is symmetric and which obey the statistics of Bose and Einstein are called 'bosons.' The exclusion principle has played a fundamental role in the theory of metals, white dwarf and neutron stars, etc., as well as in chemistry and atomic physics, but a discussion of these matters would take us too far afield here.

Dirac's proposal was that the positive energy electrons cannot fall down into negative energy states because 'all the states of negative energy are occupied except perhaps a few of small velocity.' The few vacant states, or 'holes,' in the sea of negative energy electrons behave like particles with opposite quantum numbers: positive energy and positive charge. The only particle with positive charge that was known at that time was the proton, and as Dirac later recalled [27a], 'the whole climate of opinion at that time was against new particles' so Dirac identified his holes as protons; in fact, the title of his 1930 article [22] was 'A Theory of Electrons and Protons.'

The hole theory faced a number of immediate difficulties. One obvious problem was raised by the infinite charge density of the ubiquitous negative-energy electrons: where is their electric field? Dirac proposed to reinterpret the charge density appearing in Maxwell's equations as 'the departure from the normal state of electrification of the world.' Another problem has to do with the huge dissimilarity between the observed masses and interactions of the electrons and protons. Dirac hoped that Coulomb interactions between electrons would somehow account for these differences but Hermann Weyl [28] showed that the hole theory was in fact entirely symmetric between negative and positive charge. Finally, Dirac [22] predicted the existence of an electron–proton annihilation process in which a positive-energy electron meets a hole in the sea of negative-energy electrons and falls down into the unoccupied level, emitting a pair of gamma ray photons. By itself this would not have created difficulties for the hole theory; it was even hoped by some that this would provide an explanation, then lacking, of the energy source of the stars. However, it was soon pointed out [29] by Julius Robert Oppenheimer and Igor Tamm that electron–proton annihilation in atoms would take place at much too fast a rate to be consistent with the observed stability of ordinary matter. For these reasons, by 1931 Dirac had changed his mind, and decided that the holes would have to appear not as protons but as a new sort of positively charged particle, of the same mass as the electron [29a].

The second and third of these problems were eliminated by the discovery of the positron by Carl D. Anderson [30], who apparently did not know of this prediction by Dirac. On August 2, 1932, a peculiar cosmic ray track was observed in a Wilson cloud chamber subjected to a 15 kG magnetic field. The track was observed to curve in a direction that would be expected for a positively charged particle, and yet its range was at least ten times greater than the expected range of a proton! Both the range and the specific ionization of the track were consistent with the hypothesis that this was a new particle which differs from the electron only in the sign of its charge, as would be expected for one of Dirac's holes. (This discovery had been made earlier by P.M.S. Blackett, but not immediately published by him. Anderson quotes press reports of evidence for light positive particles in cosmic ray tracks, obtained by Blackett and Giuseppe Occhialini.) Thus it appeared that Dirac was wrong only in his original identification of the hole with the proton.

The discovery of the more-or-less predicted positron, together with the earlier successes of the Dirac equation in accounting for the magnetic moment of the electron and the fine structure of hydrogen, gave Dirac's theory a prestige that it has held for over six decades. However, although there seems little doubt that Dirac's theory will survive in some form in any future physical theory, there are serious reasons for being dissatisfied with its original rationale:

(i) Dirac's analysis of the problem of negative probabilities in Schrödinger's relativistic wave equation would seem to rule out the existence of any particle of zero spin. Yet even in the 1920s particles of zero spin were known—for instance, the hydrogen atom in its ground state, and the helium nucleus. Of course, it could be argued that hydrogen atoms and alpha particles are not elementary, and therefore do not need to be described by a relativistic wave equation, but it was not (and still is not) clear how the idea of elementarity is incorporated in the formalism of relativistic quantum mechanics. Today we know of a large number of spin zero particles—π mesons, K mesons, and so on—that are no less elementary than the proton and neutron. We also know of spin one particles—the $W^{\pm}$ and Z^0—which seem as elementary as the electron or any other particle. Further, apart from effects of the strong interactions, we would today calculate the fine structure of 'mesonic atoms,' consisting of a spinless negative π or K meson bound to an atomic nucleus, from the stationary solutions of the relativistic Klein–Gordon–Schrödinger equation! Thus, it is difficult to agree that there is anything fundamentally wrong with the relativistic equation for zero spin that *forced* the development of the Dirac equation—the problem simply is that the electron happens to have spin $\hbar/2$, not zero.

(ii) As far as we now know, for *every* kind of particle there is an 'antiparticle' with the same mass and opposite charge. (Some purely neutral particles, such as the photon, are their own antiparticles.) But how can we interpret the antiparticles of charged *bosons*, such as the $\pi^{\pm}$ mesons or $W^{\pm}$ particles, as holes in a sea of negative energy states? For particles quantized according to the rules of Bose–Einstein statistics, there is no exclusion principle, and hence nothing to keep positive-energy particles from falling down into the negative-energy states, occupied or not. And if the hole theory does not work for bosonic antiparticles, why should we believe it for fermions? I asked Dirac in 1972 how he then felt about this point; he told me that he did not regard bosons like the pion or $W^{\pm}$ as 'important.' In a lecture [27a] a few years later, Dirac referred to the fact that for bosons 'we no longer have the picture of a vacuum with negative energy states filled up', and remarked that in this case 'the whole theory becomes more complicated.' The next section will show how the development of quantum field theory made the interpretation of antiparticles as holes unnecessary, even though unfortunately it lingers on in many textbooks. To quote Julian Schwinger [30a], 'The picture of an infinite sea of negative energy electrons is now best regarded as a historical curiosity, and forgotten.'

(iii) One of the great successes of the Dirac theory was its correct prediction of the magnetic moment of the electron. This was particularly striking, as the magnetic moment (1.1.8) is twice as large as would be expected for the orbital motion of a charged point particle with angular momentum $\hbar/2$; this factor of 2 had remained mysterious until Dirac's theory. However, there is really nothing in Dirac's line of argument that leads unequivocally to this particular value for the magnetic moment. At the point where we brought electric and magnetic fields into the wave equation (1.1.23), we could just as well have added a 'Pauli term' [31]

$$\kappa c\beta[\gamma^\mu, \gamma^\nu]\psi F_{\mu\nu} \tag{1.1.32}$$

with arbitrary coefficient κ. (Here $F_{\mu\nu}$ is the usual electromagnetic field strength tensor, with $F^{12} = B_3$, $F^{01} = E_1$, etc.) This term could be obtained by first adding a term to the free-field equations proportional to $[\gamma^\mu, \gamma^\nu](\partial^2/\partial x^\mu \partial x^\nu)\psi$, which of course equals zero, and then making the substitutions (1.1.22) as before. A more modern approach would be simply to remark that the term (1.1.32) is consistent with all accepted invariance principles, including Lorentz invariance and gauge invariance, and so there is no reason why such a term should not be included in the field equations. (See Section 12.3.) This term would give an additional contribution proportional to κ to the magnetic moment of the electron, so apart from the possible demand for a purely formal simplicity, there was no reason to expect any particular value for the magnetic moment of the electron in Dirac's theory.

As we shall see in this book, these problems were all eventually to be solved (or at least clarified) through the development of quantum field theory.

1.2 The Birth of Quantum Field Theory

The photon is the only particle that was known as a field before it was detected as a particle. Thus it is natural that the formalism of quantum field theory should have been developed in the first instance in connection with radiation and only later applied to other particles and fields.

In 1926, in one of the central papers on matrix mechanics, Born, Heisenberg, and Jordan [32] applied their new methods to the free radiation field. For simplicity, they ignored the polarization of electromagnetic waves and worked in one space dimension, with coordinate x running from 0 to L; the radiation field $u(x,t)$ if constrained to vanish at these endpoints thus has the same behavior as the displacement of a string with ends fixed at $x = 0$ and $x = L$. By analogy with either the case of a string or the full electromagnetic field, the Hamiltonian was taken to have the form

$$H = \frac{1}{2}\int_0^L \left\{ \left(\frac{\partial u}{\partial t}\right)^2 + c^2 \left(\frac{\partial u}{\partial x}\right)^2 \right\} dx. \tag{1.2.1}$$

In order to reduce this expression to a sum of squares, the field u was expressed as a sum of Fourier components with $u = 0$ at both $x = 0$ and $x = L$:

$$u(x,t) = \sum_{k=1}^{\infty} q_k(t) \sin\left(\frac{\omega_k x}{c}\right), \tag{1.2.2}$$

$$\omega_k = k\pi c/L, \tag{1.2.3}$$

so that

$$H = \frac{L}{4} \sum_{k=1}^{\infty} \left\{ \dot{q}_k^2(t) + \omega_k^2 q_k^2(t) \right\}. \tag{1.2.4}$$

Thus the string or field behaves like sum of independent harmonic oscillators with angular frequencies ω_k, as had been anticipated 20 years earlier by Paul Ehrenfest [32a].

In particular, the 'momentum' $p_k(t)$ canonically conjugate to $q_k(t)$ is determined, as in particle mechanics, by the condition that if H is expressed as a function of the ps and qs, then

$$\dot{q}_k(t) = \frac{\partial}{\partial p_k(t)} H\big(p(t), q(t)\big).$$

This yields a 'momentum'

$$p_k(t) = \frac{L}{2} \dot{q}_k(t), \tag{1.2.5}$$

so the canonical commutation relations may be written

$$[\dot{q}_k(t), q_j(t)] = \frac{2}{L}[p_k(t), q_j(t)] = \frac{-2i\hbar}{L}\delta_{kj}, \tag{1.2.6}$$

$$[q_k(t), q_j(t)] = 0. \tag{1.2.7}$$

Also, the time-dependence of $q_k(t)$ is governed by the Hamiltonian equation of motion

$$\ddot{q}_k(t) = \frac{2}{L}\dot{p}_k(t) = -\frac{2}{L}\frac{\partial H}{\partial q_k(t)} = -\omega_k^2 q_k(t). \tag{1.2.8}$$

The form of the matrices defined by Eqs. (1.2.6)–(1.2.8) was already known to Born, Heisenberg, and Jordan through previous work on the harmonic oscillator. The q-matrix is given by

$$q_k(t) = \sqrt{\frac{\hbar}{L\omega_k}} \Big[a_k \exp(-i\omega_k t) + a_k^\dagger \exp(+i\omega_k t) \Big] \tag{1.2.9}$$

with a_k a time-independent matrix and $a_k^\dagger$ its Hermitian adjoint, satisfying the commutation relations

$$[a_k, a_j^\dagger] = \delta_{kj}, \tag{1.2.10}$$

$$[a_k, a_j] = 0. \tag{1.2.11}$$

The rows and columns of these matrices are labelled with a set of positive integers $n_1, n_2, \ldots$, one for each normal mode. The matrix elements are

$$(a_k)_{n'_1, n'_2, \ldots; n_1, n_2, \ldots} = \sqrt{n_k}\, \delta_{n'_k, n_k - 1} \prod_{j \neq k} \delta_{n'_j, n_j}, \tag{1.2.12}$$

$$(a_k^\dagger)_{n'_1, n'_2, \ldots; n_1, n_2, \ldots} = \sqrt{n_k + 1}\, \delta_{n'_k, n_k + 1} \prod_{j \neq k} \delta_{n'_j, n_j}. \tag{1.2.13}$$

For a single normal mode, these matrices may be written explicitly as

$$a = \begin{bmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & 0 & 0 & \sqrt{3} & \cdots \\ 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \qquad a^\dagger = \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots \\ \sqrt{1} & 0 & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

It is straightforward to check that (1.2.12) and (1.2.13) do satisfy the commutation relations (1.2.10) and (1.2.11).

The physical interpretation of a column vector with integer components $n_1, n_2, \ldots$ is that it represents a state with n_k quanta in each normal mode k. The matrix a_k or $a_k^\dagger$ acting on such a column vector will respectively lower or raise n_k by one unit, leaving all n_ℓ with $\ell \neq k$ unchanged; they may therefore be interpreted as operators which annihilate or create one quantum in the kth normal mode. In particular, the vector with all n_k equal to zero represents the vacuum; it is annihilated by any a_k.

This interpretation is further borne out by inspection of the Hamiltonian. Using (1.2.9) and (1.2.10) in (1.2.4) gives

$$H = \sum_k \hbar\omega_k \left(a_k^\dagger a_k + \frac{1}{2} \right). \tag{1.2.14}$$

The Hamiltonian is then diagonal in the n-representation

$$(H)_{n_1', n_2', \ldots; n_1, n_2, \ldots} = \sum_k \hbar\omega_k \left(n_k + \frac{1}{2} \right) \prod_j \delta_{n_j', n_j}. \tag{1.2.15}$$

We see that the energy of the state is just the sum of energies $\hbar\omega_k$ for each quantum present in the state, plus an infinite zero-point energy $E_0 = \frac{1}{2}\sum_k \hbar\omega_k$. Applied to the radiation field, this formalism justified the Bose method of counting radiation states according to the numbers n_k of quanta in each normal mode.

Born, Heisenberg, and Jordan used this formalism to derive an expression for the r.m.s. energy fluctuations in black-body radiation. (For this purpose they actually only used the commutation relations (1.2.6)–(1.2.7).) However, this approach was soon applied to a more urgent problem, the calculation of the rates for spontaneous emission of radiation.

In order to appreciate the difficulties here, it is necessary to go back in time a bit. In one of the first papers on matrix mechanics, Born and Jordan [33] had assumed in effect that an atom, in dropping from a state β to a lower state α, would emit radiation just like a classical charged oscillator with displacement

$$\mathbf{r}(t) = \mathbf{r}_{\beta\alpha} \exp(-2\pi i \nu t) + \mathbf{r}_{\beta\alpha}^* \exp(2\pi i \nu t), \tag{1.2.16}$$

where

$$h\nu = E_\beta - E_\alpha \tag{1.2.17}$$

and $\mathbf{r}_{\beta\alpha}$ is the β, α element of the matrix associated with the electron position. The energy E of such an oscillator is

$$E = \frac{1}{2} m \left\{ \dot{\mathbf{r}}^2 + (2\pi\nu)^2 \mathbf{r}^2 \right\} = 8\pi^2 m \nu^2 |\mathbf{r}_{\beta\alpha}|^2. \tag{1.2.18}$$

A straightforward classical calculation then gives the radiated power, and dividing by the energy $h\nu$ per photon gives the rate of photon emission

$$A(\beta \to \alpha) = \frac{16\pi^3 e^2 \nu^3}{3hc^3} |\mathbf{r}_{\beta\alpha}|^2. \tag{1.2.19}$$

However, it was not at all clear why the formulas for emission of radiation by a classical dipole should be taken over in this manner in dealing with spontaneous emission.

A little later a more convincing though even less direct derivation was given by Dirac [34]. By considering the behavior of quantized atomic states in an oscillating classical electromagnetic field with energy density per frequency interval u at frequency (1.2.17), he was able to derive formulas for the rates $uB(\alpha \to \beta)$ and $uB(\beta \to \alpha)$ for absorption or induced emission:

$$B(\alpha \to \beta) = B(\beta \to \alpha) \simeq \frac{2\pi^2 e^2}{3h^2} |\mathbf{r}_{\beta\alpha}|^2. \tag{1.2.20}$$

(Note that the expression on the right is symmetric between states α and β, because $\mathbf{r}_{\alpha\beta}$ is just $\mathbf{r}^*_{\beta\alpha}$.) Einstein [34a] had already shown in 1917 that the possibility of thermal equilibrium between atoms and black-body radiation imposes a relation between the rate $A(\beta \to \alpha)$ of spontaneous emission and the rates uB for induced emission or absorption:

$$A(\beta \to \alpha) = \left(\frac{8\pi h \nu^3}{c^3} \right) B(\beta \to \alpha). \tag{1.2.21}$$

Using (1.2.20) in this relation immediately yields the Born–Jordan result (1.2.19) for the rate of spontaneous emission. Nevertheless, it still seemed unsatisfactory that thermodynamic arguments should be needed to derive formulas for processes involving a single atom.

Finally, in 1927 Dirac [35] was able to give a thoroughly quantum mechanical treatment of spontaneous emission. The vector potential $\mathbf{A}(\mathbf{x}, t)$ was expanded in normal modes, as in Eq. (1.2.2), and the coefficients were shown to satisfy commutation relations like (1.2.6). In consequence, each state of the free radiation field was specified by a set of integers n_k, one for each normal mode, and the matrix elements of the electromagnetic interaction $e\dot{\mathbf{r}} \cdot \mathbf{A}$ took the form of a sum over normal modes, with matrix coefficients proportional to the matrices a_k and $a_k^\dagger$ defined in Eqs. (1.2.10)–(1.2.13). The crucial result here is the factor $\sqrt{n_k + 1}$ in Eq. (1.2.13); the probability for a transition in which the number of photons in a normal mode k rises from n_k to $n_k + 1$ is proportional to the square of this factor, or $n_k + 1$. But in a radiation field with n_k photons in a normal mode k, the energy density u per frequency interval is

$$u(\nu_k) = \left(\frac{8\pi \nu_k^2}{c^3} \right) n_k \times h\nu_k,$$

so the rate for emission of radiation in normal mode k is proportional to

$$n_k + 1 = \frac{c^3 u(\nu_k)}{8\pi h \nu_k^3} + 1.$$

The first term is interpreted as the contribution of induced emission, and the second term as the contribution of spontaneous emission. Hence, without any appeal to thermodynamics, Dirac could conclude that the ratio of the rates uB for induced emission and A for spontaneous emission is given by the Einstein relation, Eq. (1.2.21). Using his earlier result (1.2.20) for B, Dirac was thus able to rederive the Born–Jordan formula [33] (1.2.19) for spontaneous emission rate A. A little later, similar methods were used by Dirac to give a quantum mechanical treatment of the scattering of radiation and the lifetime of excited atomic states [36], and by Victor Weisskopf and Eugene Wigner to make a detailed study of spectral line shapes [36a]. Dirac in his work was separating the electromagnetic potential into a radiation field $\mathbf{A}$ and a static Coulomb potential A^0, in a manner which did not preserve the manifest Lorentz and gauge invariance of classical electrodynamics. These matters were put on a firmer foundation a little later by Enrico Fermi [36b]. Many physicists in the 1930s learned their quantum electrodynamics from Fermi's 1932 review.

The use of canonical commutation relations for q and p or a and $a^\dagger$ also raised a question as to the Lorentz invariance of the quantized theory. Jordan and Pauli [37] in 1928 were able to show that the commutators of fields at different spacetime points were in fact Lorentz-invariant. (These commutators are calculated in Chapter 5.) Somewhat later, Bohr and Leon Rosenfeld [38] used a number of ingenious thought experiments to show that these commutation relations express limitations on our ability to measure fields at spacetime points separated by time-like intervals.

It was not long after the successful quantization of the electromagnetic field that these techniques were applied to other fields. At first this was regarded as a 'second quantization'; the fields to be quantized were the wave functions used in one-particle quantum mechanics, such as the Dirac wave function of the electron. The first step in this direction seems to have been taken in 1927 by Jordan [39]. In 1928 an essential element was supplied by Jordan and Wigner [40]. They recognized that the Pauli exclusion principle prevents the occupation number n_k of electrons in any normal mode k (counting spin as well as position variables) from taking any values other than 0 or 1. The electron field therefore cannot be expanded as a superposition of operators satisfying the commutation relations (1.2.10), (1.2.11), because these relations require n_k to take all integer values from 0 to ∞. Instead, they proposed that the electron field should be expanded in a sum of operators $a_k, a_k^\dagger$ satisfying the *anticommutation* relations

$$a_k a_j^\dagger + a_j^\dagger a_k = \delta_{jk}, \tag{1.2.22}$$

$$a_k a_j + a_j a_k = 0. \tag{1.2.23}$$

The relations can be satisfied by matrices labelled by a set of integers $n_1, n_2, \ldots$, one for each normal mode, each integer taking just the values zero and one:

$$(a_k)_{n_1', n_2', \ldots; n_1, n_2, \ldots} = \begin{cases} 1 & n_k' = 0,\ n_k = 1,\ n_j' = n_j \text{ for } j \neq k, \\ 0 & \text{otherwise,} \end{cases} \tag{1.2.24}$$

$$(a_k^\dagger)_{n_1', n_2', \ldots; n_1, n_2, \ldots} = \begin{cases} 1 & n_k' = 1,\ n_k = 0,\ n_j' = n_j \text{ for } j \neq k, \\ 0 & \text{otherwise.} \end{cases} \tag{1.2.25}$$

For instance, for a single normal mode we have just two rows and two columns, corresponding to the values unity and zero of n' and n; the a and $a^\dagger$ matrices take the form

$$a = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad a^\dagger = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The reader may check that (1.2.24) and (1.2.25) do satisfy the anticommutation relations (1.2.22) and (1.2.23).

The interpretation of a column vector characterized by integers $n_1, n_2, \dots$ is that it represents a state with n_k quanta in each normal mode k, just as for bosons. The difference is, of course, that since each n_k takes only the values 0 and 1, there can be at most one quantum in each normal mode, as required by the Pauli exclusion principle. Again, a_k destroys a quantum in normal mode k if there is one there already, and otherwise gives zero; also, $a_k^\dagger$ creates a quantum in normal mode k unless there is one there already, in which case it gives zero. Much later it was shown by Fierz and Pauli [40a] that the choice between commutation and anticommutation relations is dictated solely by the particle's spin: commutators must be used for particles with integer spin like the photon, and anticommutators for particles with half-integer spin like the electron. (This will be shown in a different way in Chapter 5.)

The theory of general quantum fields was first laid out in 1929, in a pair of comprehensive articles by Heisenberg and Pauli [41]. The starting point of their work was the application of the canonical formalism to the fields themselves, rather than to the coefficients of the normal modes appearing in the fields. Heisenberg and Pauli took the Lagrangian L as the space-integral of a local function of fields and spacetime derivatives of fields; the field equations were then determined from the principle that the action $\int L\, dt$ should be stationary when the fields are varied; and the commutation relations were determined from the assumption that the variational derivative of the Lagrangian with respect to any field's time-derivative behaves like a 'momentum' conjugate to that field (except that commutation relations become anticommutation relations for fermion fields). They also went on to apply this general formalism to the electromagnetic and Dirac fields, and explored the various invariance and conservation laws, including the conservation of charge, momentum, and energy, and Lorentz and gauge invariance.

The Heisenberg–Pauli formalism is essentially the same as that described in our Chapter 7, and so for the present we can limit ourselves to a single example which will turn out to be useful later in this section. For a free complex scalar field $\phi(x)$ the Lagrangian is taken as

$$L = \int d^3x \left[\dot{\phi}^\dagger \dot{\phi} - c^2 (\nabla\phi)^\dagger \cdot (\nabla\phi) - \left(\frac{mc^2}{\hbar}\right)^2 \phi^\dagger \phi \right]. \tag{1.2.26}$$

If we subject $\phi(x)$ to an infinitesimal variation $\delta\phi(x)$, the Lagrangian is changed by the amount

$$\delta L = \int d^3x \Big[\dot\phi^\dagger \delta\dot\phi + \dot\phi\, \delta\dot\phi^\dagger - c^2 \boldsymbol{\nabla}\phi^\dagger \cdot \boldsymbol{\nabla}\delta\phi - c^2 \boldsymbol{\nabla}\phi \cdot \boldsymbol{\nabla}\delta\phi^\dagger$$
$$- \left(\frac{mc^2}{\hbar}\right)^2 \phi^\dagger \delta\phi - \left(\frac{mc^2}{\hbar}\right)^2 \phi\, \delta\phi^\dagger \Big]. \tag{1.2.27}$$

It is assumed in using the principle of stationary action that the variation in the fields should vanish on the boundaries of the spacetime region of integration. Thus, in computing the change in the action $\int L\, dt$, we can immediately integrate by parts, and write

$$\delta \int L\, dt = c^2 \int d^4x \left[\delta\phi^\dagger \left(\Box - \left(\frac{mc}{\hbar}\right)^2 \right) \phi + \delta\phi \left(\Box - \left(\frac{mc}{\hbar}\right)^2 \right) \phi^\dagger \right].$$

But this must vanish for any $\delta\phi$ and $\delta\phi^\dagger$, so ϕ must satisfy the familiar relativistic wave equation

$$\left[\Box - \left(\frac{mc}{\hbar}\right)^2 \right] \phi = 0 \tag{1.2.28}$$

and its adjoint. The 'momenta' canonically conjugate to the fields ϕ and $\phi^\dagger$ are given by the variational derivatives of L with respect to $\dot\phi$ and $\dot\phi^\dagger$, which we can read off from (1.2.27) as

$$\pi \equiv \frac{\delta L}{\delta\dot\phi} = \dot\phi^\dagger, \tag{1.2.29}$$

$$\pi^\dagger \equiv \frac{\delta L}{\delta\dot\phi^\dagger} = \dot\phi. \tag{1.2.30}$$

These field variables satisfy the usual canonical commutation relations, with a delta function in place of a Kronecker delta

$$[\pi(\mathbf{x},t), \phi(\mathbf{y},t)] = [\pi^\dagger(\mathbf{x},t), \phi^\dagger(\mathbf{y},t)] = -i\hbar\delta^3(\mathbf{x}-\mathbf{y}), \tag{1.2.31}$$

$$[\pi(\mathbf{x},t), \phi^\dagger(\mathbf{y},t)] = [\pi^\dagger(\mathbf{x},t), \phi(\mathbf{y},t)] = 0, \tag{1.2.32}$$

$$[\pi(\mathbf{x},t), \pi(\mathbf{y},t)] = [\pi^\dagger(\mathbf{x},t), \pi^\dagger(\mathbf{y},t)] = [\pi(\mathbf{x},t), \pi^\dagger(\mathbf{y},t)] = 0, \tag{1.2.33}$$

$$[\phi(\mathbf{x},t), \phi(\mathbf{y},t)] = [\phi^\dagger(\mathbf{x},t), \phi^\dagger(\mathbf{y},t)] = [\phi(\mathbf{x},t), \phi^\dagger(\mathbf{y},t)] = 0. \tag{1.2.34}$$

The Hamiltonian here is given (just as in particle mechanics) by the 'sum' of all canonical momenta times the time-derivatives of the corresponding fields, minus the Lagrangian:

$$H = \int d^3x \left[\pi\dot\phi + \pi^\dagger\dot\phi^\dagger \right] - L \tag{1.2.35}$$

or, using (1.2.26), (1.2.29), and (1.2.30):

$$H = \int d^3x \left[\pi^\dagger\pi + c^2 (\boldsymbol{\nabla}\phi)^\dagger \cdot (\boldsymbol{\nabla}\phi) + \left(\frac{m^2c^4}{\hbar^2}\right) \phi^\dagger\phi \right]. \tag{1.2.36}$$

After the papers by Heisenberg and Pauli one element was still missing before quantum field theory could reach its final pre-war form: a solution to the problem of the negative-energy states. We saw in the last section that in 1930, at just about the time of the

Heisenberg–Pauli papers, Dirac had proposed that the negative-energy states of the electron were all filled, but with only the holes in the negative-energy sea observable, rather than the negative-energy electrons themselves. After Dirac's idea was seemingly confirmed by the discovery of the positron in 1932, his 'hole theory' was used to calculate a number of processes to the lowest order of perturbation theory, including electron–positron pair production and scattering.

At the same time, a great deal of work was put into the development of a formalism whose Lorentz invariance would be explicit. The most influential effort was the 'many-time' formalism of Dirac, Vladimir Fock, and Boris Podolsky [42], in which the state vector was represented by a wave function depending on the spacetime and spin coordinates of all electrons, positive-energy and negative-energy. In this formalism, the total number of electrons of either positive or negative energy is conserved; for instance, production of an electron–positron pair is described as the excitation of a negative-energy electron to a positive-energy state, and the annihilation of an electron and positron is described as the corresponding deexcitation. This many-time formalism had the advantage of manifest Lorentz invariance, but it had a number of disadvantages; In particular, there was a profound difference between the treatment of the photon, described in terms of a quantized electromagnetic field, and that of the electron and positron. Not all physicists felt this to be a disadvantage; the electron field unlike the electromagnetic field did not have a classical limit, so there were doubts about its physical significance. Also, Dirac [42a] conceived of fields as the means by which we observe particles, so that he did not expect particles and fields to be described in the same terms. Though I do not know whether it bothered anyone at the time, there was a more practical disadvantage of the many-time formalism: it would have been difficult to use it to describe a process like nuclear beta decay, in which an electron and antineutrino are created without an accompanying positron or neutrino. The successful calculation by Fermi [43] of the electron energy distribution in beta decay deserves to be counted as one of the early triumphs of quantum field theory.

The essential idea that was needed to demonstrate the equivalence of the Dirac hole theory with a quantum field theory of the electron was provided by Fock [43a] and by Wendell Furry and Oppenheimer [44] in 1933–4. To appreciate this idea from a more modern standpoint, suppose we try to construct an electron field in analogy with the electromagnetic field or the Born–Heisenberg–Jordan field (1.2.2). Since electrons carry a charge, we would not like to mix annihilation and creation operators, so we might try to write the field as

$$\psi(x) = \sum_k u_k(\mathbf{x}) e^{-i\omega_k t} a_k, \tag{1.2.37}$$

where $u_k(\mathbf{x})e^{-i\omega_k t}$ are a complete set of orthonormal plane-wave solutions of the Dirac equation (1.1.13) (with k now labelling the three-momentum, spin, and sign of the energy):

$$\mathcal{H} u_k = \hbar\omega_k u_k, \tag{1.2.38}$$

$$\mathcal{H} \equiv -i\hbar c \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + \beta m c^2, \tag{1.2.39}$$

$$\int u_k^\dagger u_\ell \, d^3x = \delta_{k\ell}, \tag{1.2.40}$$

and a_k are the corresponding annihilation operators, satisfying the Jordan–Wigner anticommutation relations (1.2.22)–(1.2.23). According to the ideas of 'second quantization' or the canonical quantization procedure of Heisenberg and Pauli [41], the Hamiltonian is formed by calculating the 'expectation value' of $\mathcal{H}$ with a 'wave function' replaced by the quantized field (1.2.37)

$$H = \int d^3x\, \psi^\dagger \mathcal{H} \psi = \sum_k \hbar\omega_k a_k^\dagger a_k. \tag{1.2.41}$$

The trouble is, of course, that this is not a positive operator—half the ω_k are negative while the operators $a_k^\dagger a_k$ take only the positive eigenvalues 1 and 0. (See Eqs. (1.2.24) and (1.2.25).) In order to cure this disease, Furry and Oppenheimer picked up Dirac's idea [42] that the positron is the absence of a negative-energy electron; the anticommutation relations are symmetric between creation and annihilation operators, so they defined the positron creation and annihilation operators as the corresponding annihilation and creation operators for negative-energy electrons

$$b_k^\dagger \equiv a_k, \qquad b_k \equiv a_k^\dagger \qquad (\text{for } \omega_k < 0) \tag{1.2.42}$$

where the label k on b denotes a positive-energy positron mode with momenta and spin opposite to those of the electron mode k. The Dirac field (1.2.37) may then be written

$$\psi(x) = \sum_k^{(+)} a_k u_k(x) + \sum_k^{(-)} b_k^\dagger u_k(x), \tag{1.2.43}$$

where $(+)$ and $(-)$ indicate sums over normal modes k with $\omega_k > 0$ and $\omega_k < 0$, respectively, and $u_k(x) \equiv u_k(\mathbf{x})e^{-i\omega_k t}$. Similarly, using the anticommutation relations for the bs, we can rewrite the energy operator (1.2.41) as

$$H = \sum_k^{(+)} \hbar\omega_k a_k^\dagger a_k + \sum_k^{(-)} \hbar|\omega_k| b_k^\dagger b_k + E_0, \tag{1.2.44}$$

where E_0 is the infinite c-number

$$E_0 = -\sum_k^{(-)} \hbar|\omega_k|. \tag{1.2.45}$$

In order for this redefinition to be more than a mere formality, it is necessary also to specify that the physical vacuum is a state $|\Omega\rangle$ containing no positive-energy electrons or positrons:

$$a_k|\Omega\rangle = 0 \qquad (\omega_k > 0), \tag{1.2.46}$$
$$b_k|\Omega\rangle = 0 \qquad (\omega_k < 0). \tag{1.2.47}$$

Hence (1.2.44) gives the energy of the vacuum as just E_0. If we measure all energies relative to the vacuum energy E_0, then the physical energy operator is $H - E_0$; and Eq. (1.2.44) shows that this is a positive operator.

The problem of negative-energy states for a charged spin zero particle was also resolved in 1934, by Pauli and Weisskopf [45], in a paper written in part to challenge Dirac's picture of filled negative-energy states. Here the creation and annihilation operators satisfy commutation rather than anticommutation relations, so it is not possible to interchange the roles of these operators freely, as was the case for fermions. Instead we must return to the Heisenberg–Pauli canonical formalism [41] to decide which coefficients of the various normal modes are creation or annihilation operators.

Pauli and Weisskopf expanded the free charged scalar field in plane waves in a cube of spatial volume $V = L^3$:

$$\phi(\mathbf{x}, t) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} q(\mathbf{k}, t) e^{i\mathbf{k}\cdot\mathbf{x}} \tag{1.2.48}$$

with the wave numbers restricted by the periodicity condition, that the quantities $k_j L/2\pi$ for $j = 1, 2, 3$ should be a set of three positive or negative integers. Similarly the canonically conjugate variable (1.2.29) was expanded as

$$\pi(\mathbf{x}, t) \equiv \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} p(\mathbf{k}, t) e^{-i\mathbf{k}\cdot\mathbf{x}}. \tag{1.2.49}$$

The minus sign is put into the exponent here so that (1.2.29) now becomes:

$$p(\mathbf{k}, t) = \dot{q}^\dagger(\mathbf{k}, t). \tag{1.2.50}$$

The Fourier inversion formula gives

$$q(\mathbf{k}, t) = \frac{1}{\sqrt{V}} \int d^3x\, \phi(\mathbf{x}, t) e^{-i\mathbf{k}\cdot\mathbf{x}}, \tag{1.2.51}$$

$$p(\mathbf{k}, t) = \frac{1}{\sqrt{V}} \int d^3x\, \pi(\mathbf{x}, t) e^{+i\mathbf{k}\cdot\mathbf{x}}, \tag{1.2.52}$$

and therefore the canonical commutation relations (1.2.31)–(1.2.34) yield for the qs and ps:

$$[p(\mathbf{k}, t), q(\mathbf{l}, t)] = \frac{-i\hbar}{V} \int d^3x\, e^{i\mathbf{k}\cdot\mathbf{x}} e^{-i\mathbf{l}\cdot\mathbf{x}} = -i\hbar\delta_{\mathbf{kl}} \tag{1.2.53}$$

$$\begin{aligned}[p(\mathbf{k}, t), q^\dagger(\mathbf{l}, t)] &= [p(\mathbf{k}, t), p(\mathbf{l}, t)] = [p(\mathbf{k}, t), p^\dagger(\mathbf{l}, t)] \\ &= [q(\mathbf{k}, t), q(\mathbf{l}, t)] = [q(\mathbf{k}, t), q^\dagger(\mathbf{l}, t)] = 0\end{aligned} \tag{1.2.54}$$

together with other relations that may be derived from these by taking their Hermitian adjoints. By inserting (1.2.48) and (1.2.49) in the formula (1.2.36) for the Hamiltonian, we can also write this operator in terms of ps and qs:

$$H = \sum_{\mathbf{k}} \left[p^\dagger(\mathbf{k}, t) p(\mathbf{k}, t) + \omega_k^2 q^\dagger(\mathbf{k}, t) q(\mathbf{k}, t) \right], \tag{1.2.55}$$

where

$$\omega_k^2 = c^2\mathbf{k}^2 + \left(\frac{mc^2}{\hbar}\right)^2. \tag{1.2.56}$$

The time-derivatives of the ps are then given by the Hamiltonian equation

$$\dot{p}(\mathbf{k},t) = -\frac{\partial H}{\partial q(\mathbf{k},t)} = -\omega_k^2 q^\dagger(\mathbf{k},t) \tag{1.2.57}$$

(and its adjoint), a result which in the light of Eq. (1.2.50) is just equivalent to the Klein–Gordon–Schrödinger wave equation (1.2.28).

We see that, just as in the case of the 1926 model of Born, Heisenberg, and Jordan [4], the free field behaves like an infinite number of coupled harmonic oscillators. Pauli and Weisskopf could construct p and q operators which satisfy the commutation relations (1.2.53)–(1.2.54) and the 'equations of motion' (1.2.50) and (1.2.57), by introducing annihilation and creation operators $a, b, a^\dagger, b^\dagger$ of *two* different kinds, corresponding to particles and antiparticles:

$$q(\mathbf{k},t) = i\sqrt{\frac{\hbar}{2\omega_k}}\left[a(\mathbf{k})\exp(-i\omega_k t) - b^\dagger(\mathbf{k})\exp(i\omega_k t)\right], \tag{1.2.58}$$

$$p(\mathbf{k},t) = \sqrt{\frac{\hbar\omega_k}{2}}\left[b(\mathbf{k})\exp(-i\omega_k t) + a^\dagger(\mathbf{k})\exp(+i\omega_k t)\right] \tag{1.2.59}$$

where

$$[a(\mathbf{k}), a^\dagger(\mathbf{l})] = [b(\mathbf{k}), b^\dagger(\mathbf{l})] = \delta_{\mathbf{kl}}, \tag{1.2.60}$$

$$[a(\mathbf{k}), a(\mathbf{l})] = [b(\mathbf{k}), b(\mathbf{l})] = 0, \tag{1.2.61}$$

$$\begin{aligned} [a(\mathbf{k}), b(\mathbf{l})] &= [a(\mathbf{k}), b^\dagger(\mathbf{l})] = [a^\dagger(\mathbf{k}), b(\mathbf{l})] \\ &= [a^\dagger(\mathbf{k}), b^\dagger(\mathbf{l})] = 0. \end{aligned} \tag{1.2.62}$$

It is straightforward to check that these operators do satisfy the desired relations (1.2.53), (1.2.54), (1.2.50), and (1.2.57). The field (1.2.48) may be written

$$\begin{aligned} \phi(\mathbf{x},t) = \frac{i}{\sqrt{V}}\sum_{\mathbf{k}}\sqrt{\frac{\hbar}{2\omega_k}}\Big[&a(\mathbf{k})\exp(i\mathbf{k}\cdot\mathbf{x} - i\omega_k t) \\ &- b^\dagger(-\mathbf{k})\exp(-i\mathbf{k}\cdot\mathbf{x} + i\omega_k t)\Big] \end{aligned} \tag{1.2.63}$$

and the Hamiltonian (1.2.55) takes the form

$$H = \sum_{\mathbf{k}} \frac{1}{2}\hbar\omega_k \left[b^\dagger(\mathbf{k})b(\mathbf{k}) + b(\mathbf{k})b^\dagger(\mathbf{k}) + a^\dagger(\mathbf{k})a(\mathbf{k}) + a(\mathbf{k})a^\dagger(\mathbf{k})\right]$$

or, using (1.2.60)–(1.2.62)

$$H = \sum_{\mathbf{k}} \hbar\omega_k \left[b^\dagger(\mathbf{k})b(\mathbf{k}) + a^\dagger(\mathbf{k})a(\mathbf{k})\right] + E_0, \tag{1.2.64}$$

where E_0 is the infinite c-number

$$E_0 = \sum_{\mathbf{k}} \hbar\omega_k. \tag{1.2.65}$$

The existence of two different kinds of operators a and b, which appear in precisely the same way in the Hamiltonian, shows that this is a theory with two kinds of particles with the same mass. As emphasized by Pauli and Weisskopf, these two varieties can be identified as particles and the corresponding antiparticles, and if charged have opposite charges. Thus, as we stressed above, bosons of spin zero as well as fermions of spin 1/2 can have distinct antiparticles, which for bosons cannot be identified as holes in a sea of negative energy particles.

We now can tell whether a and b or $a^\dagger$ and $b^\dagger$ are the annihilation operators by taking the expectation values of commutation relations in the vacuum state $|\Omega\rangle$. For instance, if $a_k^\dagger$ were an annihilation operator it would give zero when applied to the vacuum state, so the vacuum expectation value of (1.2.60) would give

$$-\big\|a(\mathbf{k})|\Omega\rangle\big\|^2 = \langle\Omega|[a(\mathbf{k}), a^\dagger(\mathbf{k})]|\Omega\rangle = +1 \tag{1.2.66}$$

in conflict with the requirement that the left-hand side must be negative-definite. In this way we can conclude that it is a_k and b_k that are the annihilation operators, and therefore

$$a(\mathbf{k})|\Omega\rangle = b(\mathbf{k})|\Omega\rangle = 0. \tag{1.2.67}$$

This is consistent with all commutation relations. Thus, the canonical formalism forces the coefficient of the $e^{+i\omega_k t}$ in the field (1.2.58) to be a creation operator, as it also is in the Furry–Oppenheimer formalism [44] for spin 1/2.

Equations (1.2.64) and (1.2.67) now tell us that E_0 is the energy of the vacuum state. If we measure all energies relative to E_0, then the physical energy operator is $H - E_0$, and (1.2.64) shows that this again is positive.

What about the problem that served Dirac as a starting point, the problem of negative probabilities? As Dirac had recognized, the only probability density ρ, which can be formed from solutions of the Klein–Gordon–Schrödinger free scalar wave equation (1.2.28), and which satisfies a conservation law of the form (1.1.10), must be proportional to the quantity

$$\rho = 2\,\mathrm{Im}\left[\phi^\dagger \frac{\partial\phi}{\partial t}\right] \tag{1.2.68}$$

and therefore is not necessarily a positive quantity. Similarly, in the 'second-quantized' theory, where ϕ is given by Eq. (1.2.63), ρ is not a positive operator. Since $\phi^\dagger(x)$ does not commute with $\phi(x)$ here, we can write (1.2.68) in various forms, which differ by infinite c-numbers; it proves convenient to write it as

$$\rho = \frac{i}{\hbar}\left[\frac{\partial\phi}{\partial t}\phi^\dagger - \frac{\partial\phi^\dagger}{\partial t}\phi\right]. \tag{1.2.69}$$

The space-integral of this operator is then easily calculated to be

$$N \equiv \int \rho\, d^3x = \sum_{\mathbf{k}} \Big(a^\dagger(\mathbf{k})a(\mathbf{k}) - b^\dagger(\mathbf{k})b(\mathbf{k})\Big) \tag{1.2.70}$$

and clearly has eigenvalues of either sign.

However, in a sense this problem appears in quantum field theory for spin 1/2 as well as spin zero. The density operator $\psi^\dagger\psi$ of Dirac is indeed a positive operator, but in order to construct a physical density we ought to subtract the contribution of the filled electron states. In particular, using the plane-wave decomposition (1.2.43), we may write the total number operator as

$$N \equiv \int d^3x\, \psi^\dagger\psi = \sum_{\mathbf{k}}^{(+)} a^\dagger(\mathbf{k})a(\mathbf{k}) + \sum_{\mathbf{k}}^{(-)} b(\mathbf{k})b^\dagger(\mathbf{k}).$$

The anticommutation relations for the bs allow us to rewrite this as

$$N - N_0 = \sum_{\mathbf{k}}^{(+)} a_k^\dagger a_k - \sum_{\mathbf{k}}^{(-)} b_k^\dagger b_k, \tag{1.2.71}$$

where N_0 is the infinite constant

$$N_0 = \sum_{\mathbf{k}}^{(-)} 1. \tag{1.2.72}$$

According to Eqs. (1.2.46) and (1.2.47), N_0 is the number of particles in the vacuum, so Furry and Oppenheimer reasoned that the physical number operator is $N - N_0$, and this now has both negative and positive eigenvalues, just as for spin zero.

The solution to this problem provided by quantum field theory is that neither the ψ of Furry and Oppenheimer nor the ϕ of Pauli and Weisskopf are probability amplitudes, which would have to define conserved positive probability densities. Instead, the physical Hilbert space is spanned by states defined as containing definite numbers of particles and/or antiparticles in each mode. If $|\Phi_n\rangle$ are a complete orthonormal set of such states, then a measurement of particle numbers in an arbitrary state $|\Psi\rangle$ will yield a probability for finding the system in state $|\Phi_n\rangle$, given by

$$P_n = \left|\langle\Phi_n|\Psi\rangle\right|^2, \tag{1.2.73}$$

where $\langle\Phi_n|\Psi\rangle$ is the usual Hilbert space scalar product. Hence, no question as to the possibility of negative probabilities will arise for any spin. The wave fields ϕ, ψ, etc., are not probability amplitudes at all, but *operators* which create or destroy particles in the various normal modes. It would be a good thing if the misleading expression 'second quantization' were permanently retired.

In particular, the operators N and $N - N_0$ of Eqs. (1.2.70) and (1.2.71) are not to be interpreted as total probabilities, but as number operators: specifically, the number of particles *minus* the number of antiparticles. For charged particles, the conservation of charge forces the charge operators to be proportional to these number operators, so the minus signs in (1.2.70) and (1.2.71) allow us immediately to conclude that particles and antiparticles have opposite charge. In this field-theoretic formalism, interactions contribute terms to the Hamiltonian which are of third, fourth, or higher order in field variables, and the rates of various processes are given by using these interaction operators in a time-dependent perturbation theory. The conceptual framework described in the above brief remarks will serve as the basis for much of the work in this book.

Despite its apparent advantages, quantum field theory did not immediately supplant hole theory; rather, the two points of view coexisted for a while, and various combinations of field-theoretic and hole-theoretic ideas were used in calculations of physical reaction rates. This period saw a number of calculations of cross sections to lowest order in powers of e^2 for various processes, such as $e^- + \gamma \to e^- + \gamma$ in 1929 by Klein and Nishina [46]; $e^+ + e^- \to 2\gamma$ in 1930 by Dirac [47]; $e^- + e^- \to e^- + e^-$ in 1932 by Møller [48]; $e^- + Z \to e^- + \gamma + Z$ and $\gamma + Z \to e^+ + e^- + Z$ (where Z denotes the Coulomb field of a heavy atom) in 1934 by Bethe and Heitler [49]; and $e^+ + e^- \to e^+ + e^-$ in 1936 by Bhabha [50]. (Rules for the calculation of such processes are given in Chapter 8, and worked out in detail there for the case of electron–photon scattering.) These lowest-order calculations gave finite results, in reasonable agreement with the experimental data.

Nevertheless, a general feeling of dissatisfaction with quantum field theory (whether or not in the form of hole theory) persisted throughout the 1930s. One of the reasons for this was the apparent failure of quantum electrodynamics to account for the penetrating power of the charged particles in cosmic ray showers, noted in 1936 by Oppenheimer and Franklin Carlson [50a]. Another cause of dissatisfaction that turned out to be related to the first was the steady discovery of new kinds of particles and interactions. We have already mentioned the electron, photon, positron, neutrino, and, of course, the nucleus of hydrogen, the proton. Throughout the 1920s it was generally believed that heavier nuclei are composed of protons and electrons, but it was hard to see how a light particle like the electron could be confined in the nucleus. Another severe difficulty with this picture was pointed out in 1931 by Ehrenfest and Oppenheimer [51]: the nucleus of ordinary nitrogen, N^{14}, in order to have atomic number 7 and atomic weight 14, would have to be composed of 14 protons and 7 electrons, and would therefore have to be a fermion, in conflict with the result of molecular spectroscopy [52] that N^{14} is a boson. This problem (and others) were solved in 1932 with the discovery of the neutron [53], and by Heisenberg's subsequent suggestion [54] that nuclei are composed of protons and neutrons, not protons and electrons. It was clear that a strong non-electromagnetic force of short range would have to operate between neutrons and protons to hold nuclei together.

After the success of the Fermi theory of beta decay, several authors [54a] speculated that nuclear forces might be explained in this theory as due to the exchange of electrons and neutrinos. A few years later, in 1935, Hideki Yukawa proposed a quite different quantum field theory of the nuclear force [55]. In an essentially classical calculation, he found that the interaction of a scalar field with nucleons (protons or neutrons) would produce a nucleon–nucleon potential, with a dependence on the nucleon separation r given by

$$V(r) \propto \frac{1}{r}\exp(-\lambda r) \tag{1.2.74}$$

instead of the $1/r$ Coulomb potential produced by electric fields. The quantity λ was introduced as a parameter in Yukawa's scalar field equation, and when this equation was quantized, Yukawa found that it described particles of mass $\hbar\lambda/c$. The observed range of the strong interactions within nuclei led Yukawa to estimate that $\hbar\lambda/c$ is of the order of 200 electron masses. In 1937 such 'mesons' were discovered in cloud chamber experiments [56] by Seth Neddermeyer and Anderson and by Jabez Curry Street and Edward Carl Stevenson, and it was generally believed that these were the hypothesized particles of Yukawa.

The discovery of mesons revealed that the charged particles in cosmic ray showers are not all electrons, and thus cleared up the problem with these showers that had bothered Oppenheimer and Carlson. At the same time, however, it created new difficulties. Lothar Nordheim [56a] pointed out in 1939 that the same strong interactions by which the mesons are copiously produced at high altitudes (and which are required in Yukawa's theory) should have led to the mesons' absorption in the atmosphere, a result contradicted by their copious appearance at lower altitudes. In 1947 it was shown in an experiment by Marcello Conversi, Ettore Pancini, and Oreste Piccioni [57] that the mesons which predominate in cosmic rays at low altitude actually interact weakly with nucleons, and therefore could not be identified with Yukawa's particle. This puzzle was cleared up by a theoretical suggestion [58], and its subsequent experimental confirmation [59] by Cesare Lattes, Occhialini, and Cecil Powell—there are two kinds of mesons with slightly different masses: the heavier (now called the π meson or pion) has strong interactions and plays the role in nuclear force envisaged by Yukawa; the lighter (now called the μ meson, or muon) has only weak and electromagnetic interactions, and predominates in cosmic rays at sea level, being produced by the decay of π mesons. In the same year, 1947, entirely new kinds of particles (now known as K mesons and hyperons) were found in cosmic rays by George Rochester and Clifford Butler [60]. From 1947 until the present particles have continued to be discovered in a bewildering variety, but to pursue this story would take us outside the bounds of our present survey. These discoveries showed clearly that any conceptual framework which was limited to photons, electrons, and positrons would be far too narrow to be taken seriously as a fundamental theory. But an even more important obstacle was presented by a purely theoretical problem—the problem of infinities.

1.3 The Problem of Infinities

Quantum field theory deals with fields $\psi(x)$ that destroy and create particles at a spacetime point x. Earlier experience with classical electron theory provided a warning that a point electron will have infinite electromagnetic self-mass; this mass is $e^2/6\pi ac^2$ for a surface distribution of charge with radius a, and therefore blows up for $a \to 0$. Disappointingly this problem appeared with even greater severity in the early days of quantum field theory, and although greatly ameliorated by subsequent improvements in the theory, it remains with us to the present day.

The problem of infinities in quantum field theory was apparently first noted in the 1929–30 papers of Heisenberg and Pauli [41]. Soon after, the presence of infinities was confirmed in calculations of the electromagnetic self-energy of a bound electron by Oppenheimer [61], and of a free electron by Ivar Waller [62]. They used ordinary second-order perturbation theory, with an intermediate state consisting of an electron and a photon: for instance, the shift of the energy E_n of an electron in the nth energy level of hydrogen is given by

$$\Delta E_n = \sum_{m,\lambda} \int d^3k \, \frac{|\langle m; \mathbf{k}, \lambda | H' | n \rangle|^2}{E_n - E_m - |\mathbf{k}|c} \,. \tag{1.3.1}$$

where the sums and integral are over all intermediate electron states m, photon helicities λ, and photon momenta $\mathbf{k}$, and H' is the term in the Hamiltonian representing the interaction of radiation and electrons. This calculation gave a self-energy that is formally infinite; further, if this infinity is removed by discarding all intermediate states with photon wave numbers greater than $1/a$, then the self-energy behaves like $1/a^2$ as $a \to 0$. Infinities of this sort are often called ultraviolet divergences, because they arise from intermediate states containing particles of very short wavelength.

These calculations treated the electron according to the rules of the original Dirac theory, without filled negative-electron states. A few years later Weisskopf repeated the calculation of the electron self-mass in the new hole theory, with all negative-energy states full. In this case another term appears in second-order perturbation theory, which in a non-hole-theory language can be described as arising from processes in which the electron in its final state first appears out of the vacuum together with a photon and a positron which then annihilate along with the initial electron. Initially Weisskopf found a $1/a^2$ dependence on the photon wave-number cutoff $1/a$. The same calculation was being carried out (at the suggestion of Bohr) at that time by Carlson and Furry. After seeing Weisskopf's results, Furry realized that while Weisskopf had included an electrostatic term that he and Carlson had neglected, Weisskopf had made a new mistake in the calculation of the magnetic self-energy. After hearing from Furry and correcting his own error, Weisskopf found that the $1/a^2$ terms in the total mass shift cancelled! However, despite this cancellation, an infinity remained: with a wave-number cutoff $1/a$, the self-mass was found to be [63]

$$m_{\text{em}} = \frac{3\alpha}{2\pi} m \ln\left(\frac{\hbar}{mca}\right) . \tag{1.3.2}$$

The weakening of the cut-off dependence, to $\ln a$ as compared with the classical $1/a$ or the early quantum $1/a^2$, was mildly encouraging at the time and turned out to be of great importance later, in the development of renormalization theory.

An infinity of quite a different kind was encountered in 1933, apparently first by Dirac [64]. He considered the effect of an external static nearly uniform charge density $\varepsilon(\mathbf{x})$ on the vacuum, i.e., on the negative-energy electrons in the filled energy levels of hole theory. The Coulomb interaction between $\varepsilon(\mathbf{x})$ and the charge density of the negative-energy electrons produces a "vacuum polarization," with induced charge density

$$\delta\varepsilon = A\varepsilon + B\left(\frac{\hbar}{mc}\right)^2 \nabla^2\varepsilon + \cdots . \tag{1.3.3}$$

The constant B is finite, and of order α. On the other hand, A is logarithmically divergent, of order $\alpha \ln a$, where $1/a$ is the wave-number cutoff.

Infinities also seemed to occur in a related problem, the scattering of light by light. Hans Euler, Bernard Kockel, and Heisenberg [65] showed in 1935–6 that these infinities could be eliminated by using a more-or-less arbitrary prescription suggested earlier by Dirac [66] and Heisenberg [67]. They calculated an effective Lagrangian density for the non-linear electrodynamic effects produced by virtual electron–positron pairs:

$$\mathcal{L} = \frac{1}{2}\left(\mathbf{E}^2 - \mathbf{B}^2\right) + \frac{e^4\hbar}{360\pi^2 m_e^4 c^7}\left[\left(\mathbf{E}^2 - \mathbf{B}^2\right)^2 + 7\left(\mathbf{E}\cdot\mathbf{B}\right)^2\right] + \cdots , \tag{1.3.4}$$

valid for frequencies $\nu \ll m_e c^2/\hbar$. Soon after, Nicholas Kemmer and Weisskopf [68] presented arguments that in this case the infinities are spurious, and that Eq. (1.3.4) can be derived without any subtraction prescription.

One bright spot in the struggle with infinities was the successful treatment of *infrared* divergences, those that arise from the low-energy rather than the high-energy part of the range of integration. In 1937 it was shown by Felix Bloch and Arne Nordsieck [68a] that these infinities cancel provided one includes processes in which arbitrary numbers of low-energy photons are produced. This will be discussed in modern terms in Chapter 13.

Yet another infinity turned up in a calculation by Sidney Michael Dancoff [69] in 1939 of the radiative corrections to the scattering of electrons by the static Coulomb field of an atom. The calculation contained a mistake (one of the terms was omitted), but this was not realized until later [69a].

Throughout the 1930s, these various infinities were seen not merely as failures of specific calculations. Rather, they seemed to indicate a gap in the understanding of relativistic quantum field theory on the most fundamental level, an opinion reinforced by the problems with cosmic rays mentioned in the previous section.

One of the symptoms of this uneasy pessimism was the continued exploration throughout the 1930s and 1940s of alternative formalisms. As Julian Schwinger [69b] later recalled, "The preoccupation of the majority of involved physicists was not with analyzing and carefully applying the known relativistic theory of coupled electron and electromagnetic fields but with changing it." Thus in 1938 Heisenberg [70] proposed the existence of a fundamental length L, analogous to the fundamental action h and fundamental velocity c. Field theory was supposed to work only for distances larger than L, so that all divergent integrals would effectively be cut off at distances L, or momenta $\hbar/L$. Several specific proposals [70a] were made for giving field theory a non-local structure. Some theorists began to suspect that the formalism of state-vectors and quantum fields should be replaced by one based solely on observable quantities, such as the S-matrix introduced by John Archibald Wheeler [71] in 1937 and Heisenberg [72] in 1943, whose elements are the amplitudes for various scattering processes. As we shall see, the concept of the S-matrix has now become a vital part of modern quantum field theory, and for some theorists a pure S-matrix theory became an ideal, especially as a possible solution to the problems of the strong interactions [73]. In yet another direction, Wheeler and Richard Feynman [74] in 1945 attempted to eliminate the electromagnetic field, deriving electromagnetic interactions in terms of an interaction at a distance. They were able to show that a pure retarded (or pure advanced) potential could be obtained by taking into account the interaction not only between source and test charges, but also between these charges and all the other charges in the universe. Perhaps the most radical modification of quantum mechanics suggested during this period was the introduction by Dirac [75] of states of negative probability, as a means of cancelling infinities in sums over states. This idea, of an "indefinite metric" in Hilbert space, has also flourished in quantum field theory, though not in the form originally suggested.

A more conservative idea for dealing with the infinities was also in the air during the 1930s. Perhaps these infinities could all be absorbed into a redefinition, a "renormalization" of the parameters of the theory. For instance, it was already known that in any Lorentz-invariant classical theory the electromagnetic self-energy and self-momentum of an electron *must* take the form of corrections to the mass of the electron; hence the infinities in these quantities can be cancelled by a negative infinity in the "bare" non-electromagnetic mass of the electron, leaving a finite measurable "renormalized" mass. Also, Eq. (1.3.3) shows that the vacuum polarization changes the charge of the electron, from $e = \int d^3x\, \varepsilon$, to

$$e_{\rm TOTAL} = \int d^3x \, (\varepsilon + \delta\varepsilon) = (1 + A)e \, . \tag{1.3.5}$$

Vacuum polarization gives finite results in lowest order if observables like scattering cross-sections are expressed in terms of $e_{\rm TOTAL}$ rather than e. The question was, whether all infinities in quantum field theory could be dealt with in this way. In 1936 Weisskopf [76] suggested that this is the case, and verified that known infinities could be eliminated by renormalization of physical parameters in a variety of sample calculations. However, it was impossible with the calculational techniques then available to show that infinities could always be eliminated in this way, and Dancoff's calculation [69] seemed to show that they could not.

Another effect of the appearance of infinities was a tendency to believe that any effect which turned out to be infinite in quantum field theory was actually not there at all. In particular, the 1928 Dirac theory had predicted complete degeneracy of the $2s_{1/2}$–$2p_{1/2}$ levels of hydrogen to all orders in α; any attempt at a quantum electromagnetic calculation of the splitting of these two levels ran into the problem of the infinite self-energy of a bound electron; therefore the existence of such a splitting was generally not taken seriously. Later Bethe [80] recalled that "This shift comes out infinite in all existing theories, and has therefore always been ignored." This attitude persisted even in the late 1930s, when spectroscopic experiments [77] began to indicate the presence of a $2s_{1/2}$–$2p_{1/2}$ splitting of order 1000 MHz. One notable exception was Edwin Albrecht Uehling [78], who realized that the vacuum polarization effect mentioned earlier would produce a $2s_{1/2}$–$2p_{1/2}$ splitting; unfortunately, as we shall see in Chapter 14, this contribution to the splitting is much smaller than 1000 MHz, and of the wrong sign.

The gloom surrounding quantum field theory began to lift soon after World War II. On June 1–4, 1947, the Conference on the Foundations of Quantum Mechanics at Shelter Island, NY brought theoretical physicists who had been working on the problems of quantum field theory through the 1930s together with a younger generation of theorists who had started scientific work during the war, and — of crucial importance — a few experimental physicists. The discussion leaders were Hans Kramers, Oppenheimer, and Weisskopf. One of the experimentalists (or rather theorist turned experimentalist), Willis Lamb, described a decisive measurement [79] of the $2s_{1/2}$–$2p_{1/2}$ shift in hydrogen. A beam of hydrogen atoms from an oven, many in $2s$ and $2p$ states, was aimed at a detector sensitive only to atoms in excited states. The atoms in $2p$ states can decay very rapidly to the $1s$ ground state by one-photon (Lyman α) emission, while the $2s$ states decay only very slowly by two-photon emission, so in effect the detector was measuring the number of atoms in the metastable $2s$ state. The beam was passed through a magnetic field, which added a known Zeeman splitting to any $2s_{1/2}$–$2p_{1/2}$ splitting naturally present. The beam was also exposed to a microwave-frequency electromagnetic field, with a fixed frequency $\nu \sim 10$ GHz. At a certain magnetic field strength the detector signal was observed to be quenched, indicating that the microwave field was producing resonant transitions from the metastable $2s$ state to the $2p$ state and thence by a rapid Lyman α emission to the ground state. The total (Zeeman plus intrinsic) $2s$–$2p$ splitting at this value of the magnetic field strength would have to be just $h\nu$, from which the intrinsic splitting could be inferred. A preliminary value of 1000 MHz was announced, in agreement with the earlier spectroscopic measurements [77]. The impact of this discovery can be summarized in a saying that was current in Copenhagen when I was a graduate student there in 1954: "Just because something is infinite does not mean it is zero!"

The discovery of the Lamb shift aroused intense interest among the theorists at Shelter Island, many of whom had already been working on improved formalisms for calculation in quantum electrodynamics. Kramers described his work on mass renormalization in the classical electrodynamics of an extended electron [79a], which showed that the difficulties associated with the divergence of the self-energy in the limit of zero radius do not appear explicitly if the theory is reexpressed so that the mass parameter in the formalism is identified with the experimental electron mass. Schwinger and Weisskopf (who had already heard rumors of Lamb's result, and discussed the matter on the trip to Shelter Island) suggested that since the inclusion of intermediate states involving positrons was known to reduce the divergence in energy level shifts from $1/a^2$ to $\ln a$, perhaps the *differences* of the shifts in atomic energy levels might turn out to be finite when these intermediate states were taken into account. (In fact, in 1946, before he learned of Lamb's experiment, Weisskopf had already assigned this problem to a graduate student, Bruce French.) Almost immediately after the conference, during a train ride to Schenectady, Hans Bethe [80] carried out a nonrelativistic calculation, still without including the effects of intermediate states containing positrons, but using a simple cutoff at virtual photon momenta of order $m_e c^2$ to eliminate infinities. He obtained the encouraging approximate value of 1040 MHz. Fully relativistic calculations using the renormalization idea to eliminate infinities were soon thereafter carried out by a number of other authors [81], with excellent agreement with experiment.

Another exciting experimental result was reported at Shelter Island by Isidor I. Rabi. Measurements in his laboratory of the hyperfine structure of hydrogen and deuterium had suggested [82] that the magnetic moment of the electron is larger than the Dirac value $e\hbar/2mc$ by a factor of about 1.0013, and subsequent measurements of the gyromagnetic ratios in sodium and gallium had given a precise value [83]

$$\mu = \frac{e\hbar}{2mc}[1.00118 \pm 0.00003].$$

Learning of these results, Gregory Breit suggested [83a] that they arose from an order α radiative correction to the electron magnetic moment. At Shelter Island, both Breit and Schwinger described their efforts to calculate this correction. Shortly after the conference Schwinger completed a successful calculation of the anomalous magnetic moment of the electron [84]

$$\mu = \frac{e\hbar}{2mc}\left[1 + \frac{\alpha}{2\pi}\right] = \frac{e\hbar}{2mc}[1.001162]$$

in excellent agreement with observation. This, together with Bethe's calculation of the Lamb shift, at last convinced physicists of the reality of radiative corrections.

The mathematical methods used in this period presented a bewildering variety of concepts and formalisms. One approach developed by Schwinger [85] was based on operator methods and the action principle, and was presented by him at a conference at Pocono Manor in 1948, the successor to the Shelter Island Conference. Another Lorentz-invariant operator formalism had been developed earlier by Sin-Itiro Tomonaga [86] and his co-workers in Japan, but their work was not at first known in the West. Tomonaga had grappled with infinities in Yukawa's meson theory in the 1930s. In 1947 he and his group were still out of the loop of scientific communication; they learned about Lamb's experiment from an article in *Newsweek*.

An apparently quite different approach was invented by Feynman [87], and described briefly by him at the Pocono Conference. Instead of introducing quantum field operators, Feynman represented the S-matrix as a functional integral of $\exp(iW)$, where W is the action integral for a set of Dirac particles interacting with a *classical* electromagnetic field, integrated over all Dirac particle trajectories satisfying certain initial and final conditions for $t \to \pm\infty$. One result of great practical importance that came out of Feynman's work was a set of graphical rules for calculating S-matrix elements to any desired order of perturbation theory. Unlike the old perturbation theory of the 1920s and 1930s, these Feynman rules automatically lumped together particle creation and antiparticle annihilation processes, and thereby gave results that were Lorentz-invariant at every stage. We have already seen in Weisskopf's early calculation [63] of the electron self-energy, that it is only in such calculations, including particles and antiparticles on the same footing, that the nature of the infinities becomes transparent.

Finally, in a pair of papers in 1949, Freeman Dyson [88] showed that the operator formalisms of Schwinger and Tomonaga would yield the same graphical rules that had been found by Feynman. Dyson also carried out an analysis of the infinities in general Feynman diagrams, and outlined a proof that these are always precisely the sort which could be removed by renormalization. One of the most striking results that could be inferred from Dyson's analysis was a criterion for deciding which quantum field theories are "renormalizable," in the sense that all infinities can be absorbed into a redefinition of a *finite* number of coupling constants and masses. In particular, an interaction like the Pauli term (1.1.32), which would have changed the predicted magnetic moment of the electron, would spoil the renormalizability of quantum electrodynamics. With the publication of Dyson's papers, there was at last a general and systematic formalism that physicists could easily learn to use, and that would provide a common language for the subsequent applications of quantum field theory to the problems of physics.

I cannot leave the infinities without taking up a puzzling aspect of this story. Oppenheimer [61] in 1930 had already noticed that most of the ultraviolet divergence in the self-energy of a bound electron cancels when one takes the difference between the shifts of two atomic energy levels, and Weisskopf [63] in 1934 had found that most of the divergence in the self-energy of a free electron cancels when one includes intermediate states containing positrons. It would have been natural even in 1934 to guess that including positron intermediate states *and* subtracting the energy shifts of pairs of atomic states would eliminate the ultraviolet divergence in their relative energy shift.[2] There was even experimental evidence [77] for a $2s_{1/2}$–$2p_{1/2}$ energy difference of order 1000 MHz. So why did no one before 1947 attempt a numerical estimate of this energy difference?

Strictly speaking, there was one such attempt [88a] in 1939, but it focused on the wrong part of the problem, the charge radius of the *proton*, which has only a tiny effect on hydrogen energy levels. The calculation gave a result in rough agreement with the early experiments [77]. This was a mistake, as shown in 1939 by Lamb [88b].

A fully relativistic calculation of the Lamb shift including positrons in intermediate states could have been attempted during the 1930s, using the old non-relativistic perturbation theory. As long as one keeps all terms up to a given order, old-fashioned non-relativistic perturbation theory gives the same results as the manifestly relativistic formalisms of Feynman, Schwinger, and Tomonaga. In fact, after Bethe's work, the first precise calculations [81] of the Lamb shift in the USA by French and Weisskopf and Norman Kroll and Lamb were done in just this way, though Tomonaga's group [81] in Japan was already using covariant methods to solve this and other problems.

The one missing element was confidence in renormalization as a means of dealing with infinities. As we have seen, renormalization was widely discussed in the late 1930s. But it had become accepted wisdom in the 1930s, and a point of view especially urged by Oppenheimer [89], that quantum electrodynamics could not be taken seriously at energies of more than about 100 MeV, and that the solution to its problems could be found only in really adventurous new ideas.

Several things happened at Shelter Island to change this expectation. One was news that the problems concerning cosmic rays discussed in the previous section were beginning to be resolved; Robert Marshak presented the hypothesis [58] that there were two types of "meson" with similar masses; the muons that had actually been observed, and the pions responsible for nuclear forces. More important was the fact that now there were reliable experimental values for the Lamb shift and the anomalous magnetic moment that forced physicists to think carefully about radiative corrections. Probably equally important was the fact that the conference brought together theorists who had in their own individual ways been thinking about renormalization as a solution to the problem of infinities. When the revolution came in the late 1940s, it was made by physicists who though mostly young were playing a conservative role, turning away from the search by their predecessors for a radical solution.

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2 Relativistic Quantum Mechanics

The point of view of this book is that quantum field theory is the way it is because (with certain qualifications) this is the only way to reconcile quantum mechanics with special relativity. Therefore our first task is to study how symmetries like Lorentz invariance appear in a quantum setting.

2.1 Quantum Mechanics

First, some good news: quantum field theory is based on the same quantum mechanics that was invented by Schrödinger, Heisenberg, Pauli, Born, and others in 1925–26, and has been used ever since in atomic, molecular, nuclear, and condensed matter physics. The reader is assumed to be already familiar with quantum mechanics; this section provides only the briefest of summaries of quantum mechanics, in the generalized version of Dirac [1].
(i) Physical states are represented by rays in Hilbert space. A Hilbert space is a kind of complex vector space; that is, if $|\Phi\rangle$ and $|\Psi\rangle$ are vectors in the space (often called 'state-vectors') then so is $\xi|\Phi\rangle + \eta|\Psi\rangle$, for arbitrary complex numbers ξ, η. It has a norm[3]: for any pair of vectors there is a complex number $\langle\Phi|\Psi\rangle$, such that

$$\langle\Phi|\Psi\rangle = \langle\Psi|\Phi\rangle^*, \tag{2.1.1}$$

$$\big\langle\Phi \,\big|\, \xi_1\Psi_1 + \xi_2\Psi_2\big\rangle = \xi_1\langle\Phi|\Psi_1\rangle + \xi_2\langle\Phi|\Psi_2\rangle, \tag{2.1.2}$$

$$\big\langle\eta_1\Phi_1 + \eta_2\Phi_2 \,\big|\, \Psi\big\rangle = \eta_1^*\langle\Phi_1|\Psi\rangle + \eta_2^*\langle\Phi_2|\Psi\rangle. \tag{2.1.3}$$

The norm $\langle\Psi|\Psi\rangle$ also satisfies a positivity condition: $\langle\Psi|\Psi\rangle \geq 0$, and vanishes if and only if $|\Psi\rangle = 0$. (There are also certain technical assumptions that allow us to take limits of vectors within Hilbert space.) A ray is a set of normalized vectors (i.e., $\langle\Psi|\Psi\rangle = 1$) with $|\Psi\rangle$ and $|\Psi'\rangle$ belonging to the same ray if $|\Psi'\rangle = \xi|\Psi\rangle$, where ξ is an arbitrary complex number with $|\xi| = 1$.

(ii) Observables are represented by Hermitian operators. These are mappings $|\Psi\rangle \to A|\Psi\rangle$ of Hilbert space into itself, linear in the sense that

$$A\big(\xi|\Psi\rangle + \eta|\Phi\rangle\big) = \xi A|\Psi\rangle + \eta A|\Phi\rangle \tag{2.1.4}$$

and satisfying the reality condition $A^\dagger = A$, where for any linear operator A the adjoint $A^\dagger$ is defined by

$$\langle\Phi|A^\dagger|\Psi\rangle \equiv \langle A\Phi|\Psi\rangle = \big(\langle\Psi|A|\Phi\rangle\big)^*. \tag{2.1.5}$$

(There are also technical assumptions about the continuity of $A|\Psi\rangle$ as a function of $|\Psi\rangle$.) A state represented by a ray $\mathcal{R}$ has a definite value α for the observable represented by an operator A if vectors $|\Psi\rangle$ belonging to this ray are eigenvectors of A with eigenvalue α:

$$A|\Psi\rangle = \alpha|\Psi\rangle \qquad \text{for } |\Psi\rangle \text{ in } \mathcal{R}. \tag{2.1.6}$$

An elementary theorem tells us that for A Hermitian, α is real, and eigenvectors with different αs are orthogonal.

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[3] We shall often use the Dirac bra-ket notation: instead of $\langle\Psi_1|\Psi_2\rangle$, we may write $\langle 1|2\rangle$.

(iii) If a system is in a state represented by a ray $\mathcal{R}$, and an experiment is done to test whether it is in any one of the different states represented by mutually orthogonal rays $\mathcal{R}_1, \mathcal{R}_2, \ldots$ (for instance, by measuring one or more observables) then the probability of finding it in the state represented by $\mathcal{R}_n$ is

$$P(\mathcal{R} \to \mathcal{R}_n) = \big|\langle\Psi|\Psi_n\rangle\big|^2, \tag{2.1.7}$$

where $|\Psi\rangle$ and $|\Psi_n\rangle$ are any vectors belonging to rays $\mathcal{R}$ and $\mathcal{R}_n$, respectively. (A pair of rays is said to be orthogonal if the state-vectors from the two rays have vanishing scalar products.) Another elementary theorem gives a total probability unity:

$$\sum_n P(\mathcal{R} \to \mathcal{R}_n) = 1 \tag{2.1.8}$$

if the state-vectors $|\Psi_n\rangle$ form a complete set.

2.2 Symmetries

A symmetry transformation is a change in our point of view that does not change the results of possible experiments. If an observer O sees a system in a state represented by a ray $\mathcal{R}$ or $\mathcal{R}_1$ or $\mathcal{R}_2, \ldots$, then an equivalent observer O' who looks at the *same* system will observe it in a different state, represented by a ray $\mathcal{R}'$ or $\mathcal{R}'_1$ or $\mathcal{R}'_2, \ldots$, respectively, but the two observers must find the same probabilities

$$P(\mathcal{R} \to \mathcal{R}_n) = P(\mathcal{R}' \to \mathcal{R}'_n). \tag{2.2.1}$$

(This is only a necessary condition for a ray transformation to be a symmetry; further conditions are discussed in the next chapter.) An important theorem proved by Wigner [2] in the early 1930s tells us that for any such transformation $\mathcal{R} \to \mathcal{R}'$ of rays we may define an operator U on Hilbert space, such that if $|\Psi\rangle$ is in ray $\mathcal{R}$ then $U|\Psi\rangle$ is in the ray $\mathcal{R}'$, with U either *unitary* and *linear*

$$\langle U\Phi|U\Psi\rangle = \langle\Phi|\Psi\rangle, \tag{2.2.2}$$

$$U\big(\xi|\Phi\rangle + \eta|\Psi\rangle\big) = \xi U|\Phi\rangle + \eta U|\Psi\rangle \tag{2.2.3}$$

or else *antiunitary* and *antilinear*

$$\langle U\Phi|U\Psi\rangle = \langle\Phi|\Psi\rangle^*, \tag{2.2.4}$$

$$U\big(\xi|\Phi\rangle + \eta|\Psi\rangle\big) = \xi^* U|\Phi\rangle + \eta^* U|\Psi\rangle. \tag{2.2.5}$$

Wigner's proof omits some steps. A more complete proof is given at the end of this chapter in Appendix A.

As already mentioned, the adjoint of a linear operator L is defined by

$$\langle\Phi|L^\dagger|\Psi\rangle \equiv \langle L\Phi|\Psi\rangle. \tag{2.2.6}$$

This condition cannot be satisfied for an antilinear operator, because in this case the right-hand side of Eq. (2.2.6) would be linear in $|\Phi\rangle$, while the left-hand side is antilinear in $|\Phi\rangle$. Instead, the adjoint of an antilinear operator A is defined by

$$\langle\Phi|A^\dagger|\Psi\rangle \equiv \langle A\Phi|\Psi\rangle^* = \langle\Psi|A|\Phi\rangle. \tag{2.2.7}$$

With this definition, the conditions for unitarity or antiunitarity both take the form

$$U^\dagger = U^{-1}. \tag{2.2.8}$$

There is always a trivial symmetry transformation $\mathcal{R} \to \mathcal{R}$, represented by the identity operator $U = \mathbf{1}$. This operator is, of course, unitary and linear. Continuity then demands that any symmetry (like a rotation or translation or Lorentz transformation) that can be made trivial by a continuous change of some parameters (like angles or distances or velocities) must be represented by a linear unitary operator U rather than one that is antilinear and antiunitary. (Symmetries represented by antiunitary antilinear operators are less prominent in physics; they all involve a reversal in the direction of time's flow. See Section 2.6.)

In particular, a symmetry transformation that is infinitesimally close to being trivial can be represented by a linear unitary operator that is infinitesimally close to the identity:

$$U = \mathbf{1} + i\epsilon t \tag{2.2.9}$$

with ϵ a real infinitesimal. For this to be unitary and linear, t must be Hermitian and linear, so it is a candidate for an observable. Indeed, most (and perhaps all) of the observables of physics, such as angular momentum or momentum, arise in this way from symmetry transformations.

The set of symmetry transformations has certain properties that define it as a *group*. If T_1 is a transformation that takes rays $\mathcal{R}_n$ into $\mathcal{R}'_n$, and T_2 is another transformation that takes $\mathcal{R}'_n$ into $\mathcal{R}''_n$, then the result of performing both transformations is another symmetry transformation, which we write T_2T_1, that takes $\mathcal{R}_n$ into $\mathcal{R}''_n$. Also, a symmetry transformation T which takes rays $\mathcal{R}_n$ into $\mathcal{R}'_n$ has an inverse, written T^{-1}, which takes $\mathcal{R}'_n$ into $\mathcal{R}_n$, and there is an identity transformation, $T = \mathbf{1}$, which leaves rays unchanged.

The unitary or antiunitary operators $U(T)$ corresponding to these symmetry transformations have properties that mirror this group structure, but with a complication due to the fact that, unlike the symmetry transformations themselves, the operators $U(T)$ act on vectors in the Hilbert space, rather than on rays. If T_1 takes $\mathcal{R}_n$ into $\mathcal{R}'_n$, then acting on a vector $|\Psi_n\rangle$ in the ray $\mathcal{R}_n$, $U(T_1)$ must yield a vector $U(T_1)|\Psi_n\rangle$ in the ray $\mathcal{R}'_n$, and if T_2 takes this ray into $\mathcal{R}''_n$, then acting on $U(T_1)|\Psi_n\rangle$ it must yield a vector $U(T_2)U(T_1)|\Psi_n\rangle$ in the ray $\mathcal{R}''_n$. But $U(T_2T_1)|\Psi_n\rangle$ is also in this ray, so these vectors can differ only by a phase $\phi_n(T_2, T_1)$

$$U(T_2)U(T_1)|\Psi_n\rangle = e^{i\phi_n(T_2,T_1)}U(T_2T_1)|\Psi_n\rangle. \tag{2.2.10}$$

Furthermore, with one significant exception, the linearity (or antilinearity) of $U(T)$ tells us that these phases are independent of the state $|\Psi_n\rangle$. Here is the proof. Consider any two different vectors $|\Psi_A\rangle, |\Psi_B\rangle$, which are not proportional to each other. Then, applying Eq. (2.2.10) to the state $|\Psi_{AB}\rangle = |\Psi_A\rangle + |\Psi_B\rangle$, we have

$$\begin{aligned} e^{i\phi_{AB}}U(T_2T_1)\big(|\Psi_A\rangle + |\Psi_B\rangle\big) &= U(T_2)U(T_1)\big(|\Psi_A\rangle + |\Psi_B\rangle\big) \\ &= U(T_2)U(T_1)|\Psi_A\rangle + U(T_2)U(T_1)|\Psi_B\rangle \\ &= e^{i\phi_A}U(T_2T_1)|\Psi_A\rangle + e^{i\phi_B}U(T_2T_1)|\Psi_B\rangle. \end{aligned} \tag{2.2.11}$$

Any unitary or antiunitary operator has an inverse (its adjoint) which is also unitary or antiunitary. Multiplying (2.2.11) on the left with $U^{-1}(T_2T_1)$, we have then

$$e^{\pm i\phi_{AB}}\big(|\Psi_A\rangle + |\Psi_B\rangle\big) = e^{\pm i\phi_A}|\Psi_A\rangle + e^{\pm i\phi_B}|\Psi_B\rangle, \tag{2.2.12}$$

the upper and lower signs referring to $U(T_2T_1)$ unitary or antiunitary, respectively. Since $|\Psi_A\rangle$ and $|\Psi_B\rangle$ are linearly independent, this is only possible if

$$e^{i\phi_{AB}} = e^{i\phi_A} = e^{i\phi_B}. \tag{2.2.13}$$

So as promised, the phase in Eq. (2.2.10) is independent of the state-vector $|\Psi_n\rangle$, and therefore this can be written as an operator relation

$$U(T_2)U(T_1) = e^{i\phi(T_2,T_1)}U(T_2T_1). \tag{2.2.14}$$

For $\phi = 0$, this would say that the $U(T)$ furnish a representation of the group of symmetry transformations. For general phases $\phi(T_2, T_1)$, we have what is called a projective representation, or a representation 'up to a phase'. The structure of the Lie group cannot by itself tell us whether physical state-vectors furnish an ordinary or a projective representation, but as we shall see, it can tell us whether the group has any intrinsically projective representations at all.

The exception to the argument that led to Eq. (2.2.14) is that it may not be possible to prepare the system in a state represented by $|\Psi_A\rangle + |\Psi_B\rangle$. For instance, it is widely believed to be impossible to prepare a system in a superposition of two states whose total angular momenta are integers and half-integers, respectively. In such cases, we say that there is a 'superselection rule' between different classes of states [3], and the phases $\phi(T_2, T_1)$ may depend on which of these classes of states the operators $U(T_2)U(T_1)$ and $U(T_2T_1)$ act upon. We will have more to say about these phases and projective representations in Section 2.7. As we shall see there, any symmetry group with projective representations can always be enlarged (without otherwise changing its physical implications) in such a way that its representations can all be defined as non-projective, with $\phi = 0$. Until Section 2.7, we will just assume that this has been done, and take $\phi = 0$ in Eq. (2.2.14).

There is a kind of group, known as a *connected Lie group*, of special importance in physics. These are groups of transformations $T(\theta)$ that are described by a finite set of real continuous parameters, say θ^a, with each element of the group connected to the identity by a path within the group. The group multiplication law then takes the form

$$T(\bar\theta)T(\theta) = T\big(f(\bar\theta,\theta)\big) \tag{2.2.15}$$

with $f^a(\bar\theta,\theta)$ a function of the $\bar\theta$s and θs. Taking $\theta^a = 0$ as the coordinates of the identity, we must have

$$f^a(\theta,0) = f^a(0,\theta) = \theta^a. \tag{2.2.16}$$

As already mentioned, the transformations of such continuous groups must be represented on the physical Hilbert space by unitary (rather than antiunitary) operators $U(T(\theta))$. For a Lie group, these operators can be represented in at least a finite neighborhood of the identity by a power series

$$U\big(T(\theta)\big) = \mathbf{1} + i\theta^a t_a + \frac{1}{2}\theta^b\theta^c t_{bc} + \cdots\,, \tag{2.2.17}$$

where t_a, $t_{bc} = t_{cb}$, etc. are Hermitian operators independent of the θs. Suppose that the $U(T(\theta))$ form an ordinary (i.e., not projective) representation of this group of transformations, i.e.,

$$U\big(T(\bar\theta)\big)U\big(T(\theta)\big) = U\big(T(f(\bar\theta,\theta))\big). \tag{2.2.18}$$

Let us see what this condition looks like when expanded in powers of θ^a and $\bar\theta^a$. According to Eq. (2.2.16), the expansion of $f^a(\bar\theta,\theta)$ to second order must take the form

$$f^a(\bar\theta,\theta) = \theta^a + \bar\theta^a + f^a{}_{bc}\bar\theta^b\theta^c + \cdots \tag{2.2.19}$$

with real coefficients $f^a{}_{bc}$. (The presence of any terms of order θ^2 or $\bar\theta^2$ would violate Eq. (2.2.16).) Then Eq. (2.2.18) reads

$$\begin{aligned}\Big[\mathbf{1} + i\bar\theta^a t_a + \frac{1}{2}\bar\theta^b\bar\theta^c t_{bc} + \cdots\Big]&\Big[\mathbf{1} + i\theta^a t_a + \frac{1}{2}\theta^b\theta^c t_{bc} + \cdots\Big] \\ = \mathbf{1} + i\big(&\theta^a + \bar\theta^a + f^a{}_{bc}\bar\theta^b\theta^c + \cdots\big)t_a \\ &+ \frac{1}{2}\big(\theta^b + \bar\theta^b + \cdots\big)\big(\theta^c + \bar\theta^c + \cdots\big)t_{bc} + \cdots\,.\end{aligned} \tag{2.2.20}$$

The terms of order 1, θ, $\bar\theta$, θ^2, and $\bar\theta^2$ automatically match on both sides of Eq. (2.2.20), but from the $\bar\theta\theta$ terms we obtain a non-trivial condition

$$t_{bc} = -t_b t_c - i f^a{}_{bc} t_a. \tag{2.2.21}$$

This shows that if we are given the structure of the group, i.e., the function $f(\theta,\bar\theta)$, and hence its quadratic coefficient $f^a{}_{bc}$, we can calculate the second-order terms in $U(T(\theta))$ from the generators t_a appearing in the first-order terms. However, there is a consistency condition: the operator t_{bc} must be *symmetric* in b and c (because it is the second derivative of $U(T(\theta))$ with respect to θ^b and θ^c) so Eq. (2.2.21) requires that

$$[t_b, t_c] = iC^a{}_{bc} t_a, \tag{2.2.22}$$

where $C^a{}_{bc}$ are a set of real constants known as *structure constants*

$$C^a{}_{bc} \equiv -f^a{}_{bc} + f^a{}_{cb}. \tag{2.2.23}$$

Such a set of commutation relations is known as a *Lie algebra.* In Section 2.7 we will prove in effect that the commutation relation (2.2.22) is the single condition needed to ensure that this process can be continued: the complete power series for $U(T(\theta))$ may be calculated from an infinite sequence of relations like Eq. (2.2.21), provided we know the first-order terms, the generators t_a. This does not necessarily mean that the operators $U(T(\theta))$ are uniquely determined for all θ^a if we know the t_a, but it does mean that the $U(T(\theta))$ are uniquely determined in at least a finite neighborhood of the coordinates $\theta^a = 0$ of the identity, in such a way that Eq. (2.2.15) is satisfied if θ, $\bar{\theta}$, and $f(\theta, \bar{\theta})$ are in this neighborhood. The extension to all θ^a is discussed in Section 2.7.

There is a special case of some importance, that we will encounter again and again. Suppose that the function $f(\theta, \bar{\theta})$ (perhaps just for some subset of the coordinates θ^a) is simply additive

$$f^a(\theta, \bar{\theta}) = \theta^a + \bar{\theta}^a. \tag{2.2.24}$$

This is the case for instance for translations in spacetime, or for rotations about any one fixed axis (though *not* for both together). Then the coefficients ${f^a}_{bc}$ in Eq. (2.2.19) vanish, and so do the structure constants (2.2.23). The generators then all commute

$$[t_b, t_c] = 0. \tag{2.2.25}$$

Such a group is called *Abelian.* In this case, it is easy to calculate $U(T(\theta))$ for all θ^a. From Eqs. (2.2.18) and (2.2.24), we have for any integer N

$$U\big(T(\theta)\big) = \left[U\left(T\left(\frac{\theta}{N}\right)\right)\right]^N .$$

Letting $N \to \infty$, and keeping only the first-order term in $U(T(\theta/N))$, we have then

$$U\big(T(\theta)\big) = \lim_{N\to\infty}\left[\mathbf{1} + \frac{i}{N}\theta^a t_a\right]^N$$

and hence

$$U\big(T(\theta)\big) = \exp(i t_a \theta^a). \tag{2.2.26}$$

2.3 Quantum Lorentz Transformations

Einstein's principle of relativity states the equivalence of certain 'inertial' frames of reference. It is distinguished from the Galilean principle of relativity, obeyed by Newtonian mechanics, by the transformation connecting coordinate systems in different inertial frames. If x^μ are the coordinates in one inertial frame (with $x^0 = t$ a time coordinate and x^1, x^2, x^3 Cartesian space coordinates, the speed of light being set equal to unity) then in any other inertial frame, the coordinates x'^μ must satisfy

$$\eta_{\mu\nu}\, dx'^\mu dx'^\nu = \eta_{\mu\nu}\, dx^\mu dx^\nu \tag{2.3.1}$$

or equivalently

$$\eta_{\mu\nu}\frac{\partial x'^\mu}{\partial x^\rho}\frac{\partial x'^\nu}{\partial x^\sigma} = \eta_{\rho\sigma}. \tag{2.3.2}$$

Here $\eta_{\mu\nu}$ is the diagonal matrix, with elements

$$\eta_{00} = -1, \qquad \eta_{11} = \eta_{22} = \eta_{33} = +1, \tag{2.3.3}$$

and the summation convention is in force: we sum over any index like μ and ν in Eq. (2.3.2), which appears twice in the same term, once upstairs and once downstairs. These transformations have the special property that the speed of light is the same (in our units, equal to unity) in all inertial frames;[4] a light wave travelling at unit speed satisfies $|d\mathbf{x}/dt| = 1$, or in other words $\eta_{\mu\nu}dx^\mu dx^\nu = d\mathbf{x}^2 - dt^2 = 0$, from which it follows that also $\eta_{\mu\nu}dx'^\mu dx'^\nu = 0$, and hence $|d\mathbf{x}'/dt'| = 1$.

Any coordinate transformation $x^\mu \to x'^\mu$ that satisfies Eq. (2.3.2) is *linear* [3a]

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu \tag{2.3.4}$$

with a^μ arbitrary constants, and $\Lambda^\mu{}_\nu$ a constant matrix satisfying the conditions

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}. \tag{2.3.5}$$

For some purposes, it is useful to write the Lorentz transformation condition in a different way. The matrix $\eta_{\mu\nu}$ has an inverse, written $\eta^{\mu\nu}$, which happens to have the same components: it is diagonal, with $\eta^{00} = -1$, $\eta^{11} = \eta^{22} = \eta^{33} = +1$. Multiplying Eq. (2.3.5) with $\eta^{\sigma\tau}\Lambda^\kappa{}_\tau$ and inserting parentheses judiciously, we have

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\big(\Lambda^\nu{}_\sigma\Lambda^\kappa{}_\tau\eta^{\sigma\tau}\big) = \Lambda^\kappa{}_\rho = \eta_{\mu\nu}\eta^{\nu\kappa}\Lambda^\mu{}_\rho.$$

Multiplying with the inverse of the matrix $\eta_{\mu\nu}\Lambda^\mu{}_\rho$ then gives

$$\Lambda^\nu{}_\sigma\Lambda^\kappa{}_\tau\eta^{\sigma\tau} = \eta^{\nu\kappa}. \tag{2.3.6}$$

These transformations form a group. If we first perform a Lorentz transformation (2.3.4), and then a second Lorentz transformation $x'^\mu \to x''^\mu$, with

$$x''^\mu = \bar\Lambda^\mu{}_\rho x'^\rho + \bar a^\mu = \bar\Lambda^\mu{}_\rho\big(\Lambda^\rho{}_\nu x^\nu + a^\rho\big) + \bar a^\mu$$

then the effect is the same as the Lorentz transformation $x^\mu \to x''^\mu$, with

$$x''^\mu = \big(\bar\Lambda^\mu{}_\rho\Lambda^\rho{}_\nu\big)x^\nu + \big(\bar\Lambda^\mu{}_\rho a^\rho + \bar a^\mu\big). \tag{2.3.7}$$

(Note that if $\Lambda^\mu{}_\nu$ and $\bar\Lambda^\mu{}_\nu$ both satisfy Eq. (2.3.5), then so does $\bar\Lambda^\mu{}_\rho\Lambda^\rho{}_\nu$, so this is a Lorentz transformation. The bar is used here just to distinguish one Lorentz transformation from the other.) The transformations $T(\Lambda, a)$ induced on physical states therefore satisfy the composition rule

$$T(\bar\Lambda, \bar a)T(\Lambda, a) = T(\bar\Lambda\Lambda, \bar\Lambda a + \bar a). \tag{2.3.8}$$

Taking the determinant of Eq. (2.3.5) gives

$$(\mathrm{Det}\,\Lambda)^2 = 1 \tag{2.3.9}$$

so $\Lambda^\mu{}_\nu$ has an inverse, $(\Lambda^{-1})^\nu{}_\mu$, which we see from Eq. (2.3.5) takes the form

$$(\Lambda^{-1})^\nu{}_\mu = \Lambda_\mu{}^\nu \equiv \eta_{\mu\rho}\eta^{\nu\sigma}\Lambda^\rho{}_\sigma. \tag{2.3.10}$$

[4] There is a larger class of coordinate transformations, known as *conformal transformations*, for which $\eta_{\mu\nu}dx'^\mu dx'^\nu$ is proportional though generally not equal to $\eta_{\mu\nu}dx^\mu dx^\nu$, and which therefore also leave the speed of light invariant. Conformal invariance in two dimensions has proved enormously important in string theory and statistical mechanics, but the physical relevance of these conformal transformations in four spacetime dimensions is not yet clear.

The inverse of the transformation $T(\Lambda, a)$ is seen from Eq. (2.3.8) to be $T(\Lambda^{-1}, -\Lambda^{-1}a)$, and, of course, the identity transformation is $T(\mathbf{1}, 0)$.

In accordance with the discussion in the previous section, the transformations $T(\Lambda, a)$ induce a unitary linear transformation on vectors in the physical Hilbert space

$$|\Psi\rangle \longrightarrow U(\Lambda, a)|\Psi\rangle.$$

The operators U satisfy a composition rule

$$U(\bar{\Lambda}, \bar{a})U(\Lambda, a) = U(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a}). \tag{2.3.11}$$

(As already mentioned, to avoid the appearance of a phase factor on the right-hand side of Eq. (2.3.11), it is, in general, necessary to enlarge the Lorentz group. The appropriate enlargement is described in Section 2.7.)

The whole group of transformations $T(\Lambda, a)$ is properly known as the *inhomogeneous Lorentz group*, or *Poincaré group*. It has a number of important subgroups. First, those transformations with $a^\mu = 0$ obviously form a subgroup, with

$$T(\bar{\Lambda}, 0)T(\Lambda, 0) = T(\bar{\Lambda}\Lambda, 0), \tag{2.3.12}$$

known as the *homogeneous Lorentz group*. Also, we note from Eq. (2.3.9) that either $\mathrm{Det}\,\Lambda = +1$ or $\mathrm{Det}\,\Lambda = -1$; those transformations with $\mathrm{Det}\,\Lambda = +1$ obviously form a subgroup of either the homogeneous or the inhomogeneous Lorentz groups. Further, from the 00-components of Eqs. (2.3.5) and (2.3.6), we have

$$(\Lambda^0{}_0)^2 = 1 + \Lambda^i{}_0\Lambda^i{}_0 = 1 + \Lambda^0{}_i\Lambda^0{}_i, \tag{2.3.13}$$

with i summed over the values 1, 2, and 3. We see that either $\Lambda^0{}_0 \geq +1$ or $\Lambda^0{}_0 \leq -1$. Those transformations with $\Lambda^0{}_0 \geq +1$ form a subgroup. Note that if $\Lambda^\mu{}_\nu$ and $\bar{\Lambda}^\mu{}_\nu$ are two such Λs, then

$$(\bar{\Lambda}\Lambda)^0{}_0 = \bar{\Lambda}^0{}_0\Lambda^0{}_0 + \bar{\Lambda}^0{}_1\Lambda^1{}_0 + \bar{\Lambda}^0{}_2\Lambda^2{}_0 + \bar{\Lambda}^0{}_3\Lambda^3{}_0.$$

But Eq. (2.3.13) shows that the three-vector $(\Lambda^1{}_0, \Lambda^2{}_0, \Lambda^3{}_0)$ has length $\sqrt{(\Lambda^0{}_0)^2 - 1}$, and similarly the three-vector
$(\bar{\Lambda}^0{}_1, \bar{\Lambda}^0{}_2, \bar{\Lambda}^0{}_3)$ has length $\sqrt{(\bar{\Lambda}^0{}_0)^2 - 1}$, so the scalar product of these two three-vectors is bounded by

$$\left|\bar{\Lambda}^0{}_1\Lambda^1{}_0 + \bar{\Lambda}^0{}_2\Lambda^2{}_0 + \bar{\Lambda}^0{}_3\Lambda^3{}_0\right| \leq \sqrt{(\Lambda^0{}_0)^2 - 1}\sqrt{(\bar{\Lambda}^0{}_0)^2 - 1}, \tag{2.3.14}$$

and so

$$(\bar{\Lambda}\Lambda)^0{}_0 \geq \bar{\Lambda}^0{}_0\Lambda^0{}_0 - \sqrt{(\Lambda^0{}_0)^2 - 1}\sqrt{(\bar{\Lambda}^0{}_0)^2 - 1} \geq 1.$$

The subgroup of Lorentz transformations with $\mathrm{Det}\,\Lambda = +1$ and $\Lambda^0{}_0 \geq +1$ is known as the *proper orthochronous Lorentz group*. Since it is not possible by a continuous change of parameters to jump from $\mathrm{Det}\,\Lambda = +1$ to $\mathrm{Det}\,\Lambda = -1$, or from $\Lambda^0{}_0 \geq +1$ to $\Lambda^0{}_0 \leq -1$, any Lorentz transformation that can be obtained from the identity by a continuous change of parameters must have $\mathrm{Det}\,\Lambda$ and $\Lambda^0{}_0$ of the same sign as for the identity, and hence must belong to the proper orthochronous Lorentz group.

Any Lorentz transformation is either proper and orthochronous, or may be written as the product of an element of the proper orthochronous Lorentz group with one of the discrete transformations $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$, where $\mathcal{P}$ is the space inversion, whose non-zero elements are

$$\mathcal{P}^0{}_0 = 1, \qquad \mathcal{P}^1{}_1 = \mathcal{P}^2{}_2 = \mathcal{P}^3{}_3 = -1, \tag{2.3.15}$$

and $\mathcal{T}$ is the time-reversal matrix, whose non-zero elements are

$$\mathcal{T}^0{}_0 = -1, \qquad \mathcal{T}^1{}_1 = \mathcal{T}^2{}_2 = \mathcal{T}^3{}_3 = 1. \tag{2.3.16}$$

Thus the study of the whole Lorentz group reduces to the study of its proper orthochronous subgroup, plus space inversion and time-reversal. We will consider space inversion and time-reversal separately in Section 2.6. Until then, we will deal only with the homogeneous or inhomogeneous proper orthochronous Lorentz group.

2.4 The Poincaré Algebra

As we saw in Section 2.2, much of the information about any Lie symmetry group is contained in properties of the group elements near the identity. For the inhomogeneous Lorentz group, the identity is the transformation $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu$, $a^\mu = 0$, so we want to study those transformations with

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu\,, \qquad a^\mu = \epsilon^\mu\,, \tag{2.4.1}$$

both $\omega^\mu{}_\nu$ and ϵ^μ being taken infinitesimal. The Lorentz condition (2.3.5)

reads here

$$\begin{aligned}\eta_{\rho\sigma} &= \eta_{\mu\nu}(\delta^\mu_\rho + \omega^\mu{}_\rho)(\delta^\nu_\sigma + \omega^\nu{}_\sigma)\\ &= \eta_{\rho\sigma} + \omega_{\sigma\rho} + \omega_{\rho\sigma} + O(\omega^2)\,.\end{aligned}$$

We are here using the convention, to be used throughout this book, that indices may be lowered or raised by contraction with $\eta_{\mu\nu}$ or $\eta^{\mu\nu}$

$$\omega_{\sigma\rho} = \eta_{\mu\sigma}\omega^\mu{}_\rho, \qquad \omega^\mu{}_\rho = \eta^{\mu\sigma}\omega_{\sigma\rho}\,.$$

Keeping only the terms of first order in ω in the Lorentz condition (2.3.5), we see that this condition now reduces to the antisymmetry of $\omega_{\mu\nu}$

$$\omega_{\mu\nu} = -\omega_{\nu\mu}\,. \tag{2.4.2}$$

An antisymmetric second-rank tensor in four dimensions has $(4 \times 3)/2 = 6$ independent components, so including the four components of ϵ^μ, an inhomogeneous Lorentz transformation is described by $6 + 4 = 10$ parameters.

Since $U(\mathbf{1}, 0)$ carries any ray into itself, it must be proportional to the unit operator,[5] and by a choice of phase may be made equal to it. For an infinitesimal Lorentz transformation (2.4.1), $U(\mathbf{1} + \omega, \epsilon)$ must then equal $\mathbf{1}$ plus terms linear in $\omega_{\rho\sigma}$ and ϵ_ρ. We write

[5]In the absence of superselection rules, the possibility that the proportionality factor may depend on the state on which $U(\mathbf{1}, 0)$ acts can be ruled out by the same reasoning that we used in Section 2.2 to rule out the possibility that the phases in projective representations of symmetry groups may depend on the states on which the symmetries act. Where superselection rules apply, it may be necessary to redefine $U(\mathbf{1}, 0)$ by phase factors that depend on the sector on which it acts.

this as

$$U(\mathbf{1}+\omega,\epsilon)=\mathbf{1}+\frac{i}{2}\omega_{\rho\sigma}J^{\rho\sigma}-i\epsilon_\rho P^\rho+\cdots\,. \tag{2.4.3}$$

Here $J^{\rho\sigma}$ and P^ρ are ω- and ϵ-independent operators, and the dots denote terms of higher order in ω and/or ϵ. In order for $U(\mathbf{1}+\omega,\epsilon)$ to be unitary, the operators $J^{\rho\sigma}$ and P^ρ must be Hermitian

$$J^{\rho\sigma\dagger}=J^{\rho\sigma}\,,\qquad P^{\rho\dagger}=P^\rho\,. \tag{2.4.4}$$

Since $\omega_{\rho\sigma}$ is antisymmetric, we can take its coefficient $J^{\rho\sigma}$ to be antisymmetric also

$$J^{\rho\sigma}=-J^{\sigma\rho}\,. \tag{2.4.5}$$

As we shall see, P^1, P^2, and P^3 are the components of the momentum operators, J^{23}, J^{31}, and J^{12} are the components of the angular momentum vector, and P^0 is the energy operator, or Hamiltonian.[6]

We now examine the Lorentz transformation properties of $J^{\rho\sigma}$ and P^ρ. We consider the product

$$U(\Lambda,a)\,U(\mathbf{1}+\omega,\epsilon)\,U^{-1}(\Lambda,a)\,,$$

where $\Lambda^\mu{}_\nu$ and a^μ are here the parameters of a new transformation, unrelated to ω and ϵ. According to Eq. (2.3.11), the product $U(\Lambda^{-1},-\Lambda^{-1}a)U(\Lambda,a)$ equals $U(\mathbf{1},0)$, so $U(\Lambda^{-1},-\Lambda^{-1}a)$ is the inverse of $U(\Lambda,a)$. It follows then from (2.3.11) that

$$\begin{aligned}&U(\Lambda,a)U(\mathbf{1}+\omega,\epsilon)U^{-1}(\Lambda,a)\\&\qquad=U\big(\Lambda(\mathbf{1}+\omega)\Lambda^{-1},\Lambda\epsilon-\Lambda\omega\Lambda^{-1}a\big)\,.\end{aligned} \tag{2.4.6}$$

To first order in ω and ϵ, we have then

$$\begin{aligned}&U(\Lambda,a)\left[\frac{1}{2}\omega_{\rho\sigma}J^{\rho\sigma}-\epsilon_\rho P^\rho\right]U^{-1}(\Lambda,a)\\&\qquad=\frac{1}{2}(\Lambda\omega\Lambda^{-1})_{\mu\nu}J^{\mu\nu}-(\Lambda\epsilon-\Lambda\omega\Lambda^{-1}a)_\mu P^\mu\,.\end{aligned} \tag{2.4.7}$$

Equating coefficients of $\omega_{\rho\sigma}$ and ϵ_ρ on both sides of this equation (and using (2.3.10)), we find

$$U(\Lambda,a)J^{\rho\sigma}U^{-1}(\Lambda,a)=\Lambda_\mu{}^\rho\Lambda_\nu{}^\sigma\left(J^{\mu\nu}-a^\mu P^\nu+a^\nu P^\mu\right), \tag{2.4.8}$$

$$U(\Lambda,a)P^\rho U^{-1}(\Lambda,a)=\Lambda_\mu{}^\rho P^\mu\,. \tag{2.4.9}$$

For homogeneous Lorentz transformations (with $a^\mu=0$), these transformation rules simply say that $J^{\mu\nu}$ is a tensor and P^μ is a vector. For pure translations (with $\Lambda^\mu{}_\nu=\delta^\mu{}_\nu$), they tell us that P^ρ is translation-invariant, but $J^{\rho\sigma}$ is not. In particular, the change of the space-space components of $J^{\rho\sigma}$ under a spatial translation is just the usual change of the angular momentum under a change of the origin relative to which the angular momentum is calculated.

[6] We will see that this identification of the angular-momentum generators is forced on us by the commutation relations of the $J^{\mu\nu}$. On the other hand, the commutation relations do not allow us to distinguish between P^μ and $-P^\mu$, so the sign for the $\epsilon_\rho P^\rho$ term in (2.4.3) is a matter of convention. The consistency of the choice in (2.4.3) with the usual definition of the Hamiltonian P^0 is shown in Section 3.1.

Next, let's apply rules (2.4.8), (2.4.9) to a transformation that is itself infinitesimal, i.e., $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega'^\mu{}_\nu$ and $a^\mu = \epsilon'^\mu$, with infinitesimals $\omega'^\mu{}_\nu$ and ϵ'^μ unrelated to the previous ω and ϵ. Using Eq. (2.4.3), and keeping only terms of first order in $\omega'^\mu{}_\nu$ and ϵ'^μ, Eqs. (2.4.8) and (2.4.9) now become

$$i\left[\frac{1}{2}\omega'_{\mu\nu}J^{\mu\nu} - \epsilon'_\mu P^\mu, J^{\rho\sigma}\right] = \omega'^\rho{}_\mu J^{\mu\sigma} + \omega'^\sigma{}_\nu J^{\rho\nu} - \epsilon'^\rho P^\sigma + \epsilon'^\sigma P^\rho\,, \tag{2.4.10}$$

$$i\left[\frac{1}{2}\omega'_{\mu\nu}J^{\mu\nu} - \epsilon'_\mu P^\mu, P^\rho\right] = \omega'^\rho{}_\mu P^\mu\,. \tag{2.4.11}$$

Equating coefficients of $\omega'_{\mu\nu}$ and ϵ'_μ on both sides of these equations, we find the commutation rules

$$i[J^{\mu\nu}, J^{\rho\sigma}] = \eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\nu\sigma}J^{\mu\rho} + \eta^{\mu\sigma}J^{\nu\rho}\,, \tag{2.4.12}$$

$$i[P^\mu, J^{\rho\sigma}] = \eta^{\mu\rho}P^\sigma - \eta^{\mu\sigma}P^\rho\,, \tag{2.4.13}$$

$$[P^\mu, P^\rho] = 0\,. \tag{2.4.14}$$

This is the Lie algebra of the Poincaré group.

In quantum mechanics a special role is played by those operators that are *conserved*, i.e., that commute with the energy operator $H = P^0$. Inspection of Eqs. (2.4.13) and (2.4.14) shows that these are the momentum three-vector

$$\mathbf{P} = \{P^1, P^2, P^3\} \tag{2.4.15}$$

and the angular-momentum three-vector

$$\mathbf{J} = \{J^{23}, J^{31}, J^{12}\} \tag{2.4.16}$$

and, of course, the energy P^0 itself. The remaining generators form what is called the 'boost' three-vector

$$\mathbf{K} = \{J^{10}, J^{20}, J^{30}\}\,. \tag{2.4.17}$$

These are *not* conserved, which is why we do not use the eigenvalues of $\mathbf{K}$ to label physical states. In a three-dimensional notation, the commutation relations (2.4.12), (2.4.13), (2.4.14) may be written

$$[J_i, J_j] = i\epsilon_{ijk}J_k\,, \tag{2.4.18}$$

$$[J_i, K_j] = i\epsilon_{ijk}K_k\,, \tag{2.4.19}$$

$$[K_i, K_j] = -i\epsilon_{ijk}J_k\,, \tag{2.4.20}$$

$$[J_i, P_j] = i\epsilon_{ijk}P_k\,, \tag{2.4.21}$$

$$[K_i, P_j] = iH\delta_{ij}\,, \tag{2.4.22}$$

$$[J_i, H] = [P_i, H] = [H, H] = 0\,, \tag{2.4.23}$$

$$[K_i, H] = iP_i\,, \tag{2.4.24}$$

where i, j, k, etc. run over the values 1, 2, and 3, and ϵ_{ijk} is the totally antisymmetric quantity with $\epsilon_{123} = +1$. The commutation relation (2.4.18) will be recognized as that of the angular-momentum operator.

The pure translations $T(\mathbf{1}, a)$ form a subgroup of the inhomogeneous Lorentz group with a group multiplication rule given by (2.3.7) as

$$T(\mathbf{1}, \bar{a})T(\mathbf{1}, a) = T(\mathbf{1}, \bar{a} + a)\,. \tag{2.4.25}$$

This is additive in the same sense as (2.2.24), so by using (2.4.3) and repeating the same arguments that led to (2.2.26), we find that finite translations are represented on the physical Hilbert space by

$$U(\mathbf{1}, a) = \exp(-iP^\mu a_\mu)\,. \tag{2.4.26}$$

In exactly the same way, we can show that a rotation $R_{\boldsymbol{\theta}}$ by an angle $|\boldsymbol{\theta}|$ around the direction of $\boldsymbol{\theta}$ is represented on the physical Hilbert space by

$$U(R_{\boldsymbol{\theta}}, 0) = \exp(i\mathbf{J} \cdot \boldsymbol{\theta})\,. \tag{2.4.27}$$

It is interesting to compare the Poincaré algebra with the Lie algebra of the symmetry group of Newtonian mechanics, the Galilean group. We could derive this algebra by starting with the transformation rules of the Galilean group and then following the same procedure that was used here to derive the Poincaré algebra. However, since we already have Eqs. (2.4.18)–(2.4.24), it is easier to obtain the Galilean algebra as a low-velocity limit of the Poincaré algebra, by what is known as an *Inönü–Wigner contraction* [4,5]. For a system of particles of typical mass m and typical velocity v, the momentum and the angular-momentum operators are expected to be of order $J \sim 1$, $P \sim mv$. On the other hand, the energy operator is $H = M + W$ with a total mass M and non-mass energy W (kinetic plus potential) of order $M \sim m$, $W \sim mv^2$. Inspection of Eqs. (2.4.18)–(2.4.24) shows that these commutation relations have a limit for $v \ll 1$ of the form

$$\begin{gathered}
[J_i, J_j] = i\epsilon_{ijk}J_k\,, \qquad [J_i, K_j] = i\epsilon_{ijk}K_k\,, \qquad [K_i, K_j] = 0\,,\\
[J_i, P_j] = i\epsilon_{ijk}P_k\,, \qquad [K_i, P_j] = iM\delta_{ij}\,,\\
[J_i, W] = [P_i, W] = 0\,, \qquad [K_i, W] = iP_i\,,\\
[J_i, M] = [P_i, M] = [K_i, M] = [W, M] = 0\,,
\end{gathered}$$

with K of order $1/v$. Note that the product of a translation $\mathbf{x} \to \mathbf{x} + \mathbf{a}$ and a 'boost' $\mathbf{x} \to \mathbf{x} + \mathbf{v}t$ should be the transformation $\mathbf{x} \to \mathbf{x} + \mathbf{v}t + \mathbf{a}$, but this is not true for the action of these operators on Hilbert space:

$$\exp(i\mathbf{K} \cdot \mathbf{v}) \exp(-i\mathbf{P} \cdot \mathbf{a}) = \exp(iM\mathbf{a} \cdot \mathbf{v}/2) \exp(i(\mathbf{K} \cdot \mathbf{v} - \mathbf{P} \cdot \mathbf{a}))\,.$$

The appearance of the phase factor $\exp(iM\mathbf{a} \cdot \mathbf{v}/2)$ shows that this is a projective representation, with a superselection rule forbidding the superposition of states of different mass. In this respect, the mathematics of the Poincaré group is simpler than that of the Galilean group. However, there is nothing to prevent us from formally enlarging the Galilean group, by adding one more generator to its Lie algebra, which commutes with all the other generators, and whose eigenvalues are the masses of the various states. In this case physical states provide an ordinary rather than a projective representation of the expanded symmetry group. The difference appears to be a mere matter of notation, except that with this reinterpretation of the Galilean group there is no need for a mass superselection rule.

2.5 One-Particle States

We now consider the classification of one-particle states according to their transformation under the inhomogeneous Lorentz group.

The components of the energy-momentum four-vector all commute with each other, so it is natural to express physical state-vectors in terms of eigenvectors of the four-momentum. Introducing a label σ to denote all other degrees of freedom, we thus consider state-vectors $|p,\sigma\rangle$ with

$$P^\mu |p,\sigma\rangle = p^\mu |p,\sigma\rangle \,. \tag{2.5.1}$$

For general states, consisting for instance of several unbound particles, the label σ would have to be allowed to include continuous as well as discrete labels. We take as part of the definition of a *one-particle* state, that the label σ is purely discrete, and will limit ourselves here to that case. (However, a specific bound state of two or more particles, such as the lowest state of the hydrogen atom, is to be considered as a one-particle state. It is not an *elementary* particle, but the distinction between composite and elementary particles is of no relevance here.)

Eqs. (2.5.1) and (2.4.26) tell us how the states $|p,\sigma\rangle$ transform under translations:

$$U(\mathbf{1},a)|p,\sigma\rangle = e^{-ip\cdot a}|p,\sigma\rangle \,.$$

We must now consider how these states transform under homogeneous Lorentz transformations.

Using (2.4.9), we see that the effect of operating on $|p,\sigma\rangle$ with a quantum homogeneous Lorentz transformation $U(\Lambda,0) \equiv U(\Lambda)$ is to produce an eigenvector of the four-momentum with eigenvalue Λp

$$\begin{aligned} P^\mu U(\Lambda)|p,\sigma\rangle &= U(\Lambda)\left[U^{-1}(\Lambda)P^\mu U(\Lambda)\right]|p,\sigma\rangle \\ &= U(\Lambda)(\Lambda^{-1\mu}{}_\rho P^\rho)|p,\sigma\rangle \\ &= \Lambda^\mu{}_\rho p^\rho U(\Lambda)|p,\sigma\rangle \,. \end{aligned} \tag{2.5.2}$$

Hence $U(\Lambda)|p,\sigma\rangle$ must be a linear combination of the state-vectors $|\Lambda p,\sigma'\rangle$:

$$U(\Lambda)|p,\sigma\rangle = \sum_{\sigma'} C_{\sigma'\sigma}(\Lambda,p)|\Lambda p,\sigma'\rangle \,. \tag{2.5.3}$$

In general, it may be possible by using suitable linear combinations of the $|p,\sigma\rangle$ to choose the σ labels in such a way that the matrix $C_{\sigma'\sigma}(\Lambda,p)$ is block-diagonal; in other words, so that the $|p,\sigma\rangle$ with σ within any one block by themselves furnish a representation of the inhomogeneous Lorentz group. It is natural to identify the states of a specific particle type with the components of a representation of the inhomogeneous Lorentz group which is irreducible, in the sense that it cannot be further decomposed in this way.[7] Our task now

[7] Of course, different particle species may correspond to representations that are isomorphic, i.e., that have matrices $C_{\sigma'\sigma}(\Lambda,p)$ that are either identical, or identical up to a similarity transformation. In some cases it may be convenient to define particle types as irreducible representations of larger groups that contain the inhomogeneous proper orthochronous Lorentz group as a subgroup; for instance, as we shall see, for massless particles whose interactions respect the symmetry of space inversion it is customary to treat all the components of an irreducible representation of the inhomogeneous Lorentz group including space inversion as a single particle type.

is to work out the structure of the coefficients $C_{\sigma'\sigma}(\Lambda, p)$ in irreducible representations of the inhomogeneous Lorentz group.

For this purpose, note that the only functions of p^μ that are left invariant by all proper orthochronous Lorentz transformations $\Lambda^\mu{}_\nu$, are the invariant square $p^2 \equiv \eta_{\mu\nu}p^\mu p^\nu$, and for $p^2 \leq 0$, also the sign of p^0. Hence, for each value of p^2, and (for $p^2 \leq 0$) each sign of p^0, we can choose a 'standard' four-momentum, say k^μ, and express any p^μ of this class as

$$p^\mu = L^\mu{}_\nu(p)k^\nu \,, \tag{2.5.4}$$

where $L^\mu{}_\nu$ is some standard Lorentz transformation that depends on p^μ, and also implicitly on our choice of the standard k^μ. We can then *define* the states $|p, \sigma\rangle$ of momentum p by

$$|p, \sigma\rangle \equiv N(p)U(L(p))|k, \sigma\rangle \,, \tag{2.5.5}$$

where $N(p)$ is a numerical normalization factor, to be chosen later. Up to this point, we have said nothing about how the σ labels are related for different momenta; Eq. (2.5.5) now fills that gap.

Operating on (2.5.5) with an arbitrary homogeneous Lorentz transformation $U(\Lambda)$, we now find

$$\begin{aligned} U(\Lambda)|p, \sigma\rangle &= N(p)U(\Lambda L(p))|k, \sigma\rangle \\ &= N(p)U(L(\Lambda p))U(L^{-1}(\Lambda p)\Lambda L(p))|k, \sigma\rangle \,. \end{aligned} \tag{2.5.6}$$

The point of this last step is that the Lorentz transformation $L^{-1}(\Lambda p)\Lambda L(p)$ takes k to $L(p)k = p$, and then to Λp, and then back to k, so it belongs to the subgroup of the homogeneous Lorentz group consisting of Lorentz transformations $W^\mu{}_\nu$ that leave k^μ invariant:

$$W^\mu{}_\nu k^\nu = k^\mu \,. \tag{2.5.7}$$

This subgroup is called the *little group* [5]. For any W satisfying Eq. (2.5.7), we have

$$U(W)|k, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}(W)|k, \sigma'\rangle \,. \tag{2.5.8}$$

The coefficients $D(W)$ furnish a representation of the little group; that is, for any elements $\bar{W}, W$ we have

$$\begin{aligned} \sum_{\sigma'} D_{\sigma'\sigma}(\bar{W}W)|k, \sigma'\rangle &= U(\bar{W}W)|k, \sigma\rangle = U(\bar{W})U(W)|k, \sigma\rangle \\ &= U(\bar{W})\sum_{\sigma''} D_{\sigma''\sigma}(W)|k, \sigma''\rangle \\ &= \sum_{\sigma'\sigma''} D_{\sigma'\sigma''}(\bar{W})D_{\sigma''\sigma}(W)|k, \sigma'\rangle \end{aligned}$$

and so

$$D_{\sigma'\sigma}(\bar{W}W) = \sum_{\sigma''} D_{\sigma'\sigma''}(\bar{W})D_{\sigma''\sigma}(W) \,. \tag{2.5.9}$$

In particular, we may apply Eq. (2.5.8) to the little-group transformation

$$W(\Lambda, p) \equiv L^{-1}(\Lambda p)\Lambda L(p) \tag{2.5.10}$$

and then Eq. (2.5.6) takes the form

$$U(\Lambda)|p,\sigma\rangle = N(p)\sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda,p))U(L(\Lambda p))|k,\sigma'\rangle\,,$$

or, recalling the definition (2.5.5):

$$U(\Lambda)|p,\sigma\rangle = \left(\frac{N(p)}{N(\Lambda p)}\right)\sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda,p))|\Lambda p,\sigma'\rangle\,. \tag{2.5.11}$$

Apart from the question of normalization, the problem of determining the coefficients $C_{\sigma'\sigma}$ in the transformation rule (2.5.3) has been reduced to the problem of finding the representations of the little group. This approach, of deriving representations of a group like the inhomogeneous Lorentz group from the representations of a little group, is called the method of induced representations [6].

Table 2.1 gives a convenient choice of the standard momentum k^μ and the corresponding little group for the various classes of four-momenta.

Of these six classes of four-momenta, only (a), (c), and (f) have any known interpretations in terms of physical states. Not much needs to be said here about case (f)—$p^\mu = 0$; it describes the vacuum, which is simply left invariant by $U(\Lambda)$. In what follows we will consider only cases (a) and (c), which cover particles of mass $M > 0$ and mass zero, respectively.

This is a good place to pause, and say something about the normalization of these states. By the usual orthonormalization procedure of quantum mechanics, we may choose the states with standard momentum k^μ to be orthonormal, in the sense that

$$\langle k',\sigma'|k,\sigma\rangle = \delta^3(\mathbf{k}'-\mathbf{k})\delta_{\sigma'\sigma}\,. \tag{2.5.12}$$

(The delta function appears here because $|k,\sigma\rangle$ and $|k',\sigma'\rangle$ are eigenstates of a Hermitian operator with eigenvalues $\mathbf{k}$ and $\mathbf{k}'$, respectively.) This has the immediate consequence that the representation of the little group in Eqs. (2.5.8) and (2.5.11) must be unitary[8]

$$D^\dagger(W) = D^{-1}(W)\,. \tag{2.5.13}$$

[8]The little groups $SO(2,1)$ and $SO(3,1)$ for $p^2 > 0$ and $p^\mu = 0$ have no non-trivial finite-dimensional unitary representations, so if there were any states with a given momentum p^μ with $p^2 > 0$ or $p^\mu = 0$ that transform non-trivially under the little group, there would have to be an infinite number of them.

Table 2.1. Standard momenta and the corresponding little group for various classes of four-momenta. Here κ is an arbitrary positive energy, say 1 eV. The little groups are mostly pretty obvious: $SO(3)$ is the ordinary rotation group in three dimensions (excluding space inversions), because rotations are the only proper orthochronous Lorentz transformations that leave at rest a particle with zero momentum, while $SO(2,1)$ and $SO(3,1)$ are the Lorentz groups in $(2+1)$- and $(3+1)$-dimensions, respectively. The group $ISO(2)$ is the group of Euclidean geometry, consisting of rotations and translations in two dimensions. Its appearance as the little group for $p^2 = 0$ is explained below.

	Standard k^μ	Little Group
(a) $p^2 = -M^2 < 0,\ p^0 > 0$	$(M,0,0,0)$	$SO(3)$
(b) $p^2 = -M^2 < 0,\ p^0 < 0$	$(-M,0,0,0)$	$SO(3)$
(c) $p^2 = 0,\ p^0 > 0$	$(\kappa,0,0,\kappa)$	$ISO(2)$
(d) $p^2 = 0,\ p^0 < 0$	$(-\kappa,0,0,\kappa)$	$ISO(2)$
(e) $p^2 = N^2 > 0$	$(0,0,0,N)$	$SO(2,1)$
(f) $p^\mu = 0$	$(0,0,0,0)$	$SO(3,1)$

Now, what about scalar products for arbitrary momenta? Using the unitarity of the operator $U(\Lambda)$ in Eqs. (2.5.5) and (2.5.11), we find for the scalar product:

$$\begin{aligned}\langle p',\sigma'|p,\sigma\rangle &= N(p)\langle U^{-1}(L(p))p',\sigma'|k,\sigma\rangle \\ &= N(p)N^*(p')D_{\sigma\sigma'}(W(L^{-1}(p),p'))^*\delta^3(\mathbf{k}'-\mathbf{k})\,,\end{aligned}$$

where $k' \equiv L^{-1}(p)p'$. Since also $k = L^{-1}(p)p$, the delta function $\delta^3(\mathbf{k}-\mathbf{k}')$ is proportional to $\delta^3(\mathbf{p}-\mathbf{p}')$. For $p' = p$, the little-group transformation here is trivial, $W(L^{-1}(p),p) = \mathbf{1}$, and so the scalar product is

$$\langle p',\sigma'|p,\sigma\rangle = |N(p)|^2\delta_{\sigma'\sigma}\delta^3(\mathbf{k}'-\mathbf{k})\,. \tag{2.5.14}$$

It remains to work out the proportionality factor relating $\delta^3(\mathbf{k}-\mathbf{k}')$ and $\delta^3(\mathbf{p}-\mathbf{p}')$. Note that the Lorentz-invariant integral of an arbitrary scalar function $f(p)$ over four-momenta with $-p^2 = M^2 \geq 0$ and $p^0 > 0$ (i.e., cases (a) or (c)) may be written

$$\begin{aligned}\int d^4p\,\delta(p^2+M^2)\theta(p^0)f(p) &= \int d^3\mathbf{p}\,dp^0\,\delta((p^0)^2-\mathbf{p}^2-M^2)\theta(p^0)f(\mathbf{p},p^0) \\ &= \int d^3\mathbf{p}\,\frac{f(\mathbf{p},\sqrt{\mathbf{p}^2+M^2})}{2\sqrt{\mathbf{p}^2+M^2}}\,.\end{aligned}$$

($\theta(p^0)$ is the step function: $\theta(x) = 1$ for $x \geq 0$, $\theta(x) = 0$ for $x < 0$.) We see that when integrating on the 'mass shell' $p^2 + M^2 = 0$, the invariant volume element is

$$d^3\mathbf{p}/\sqrt{\mathbf{p}^2+M^2}\,. \tag{2.5.15}$$

The delta function is defined by

$$
\begin{aligned}
F(\mathbf{p}) &= \int F(\mathbf{p}')\delta^3(\mathbf{p}-\mathbf{p}')d^3\mathbf{p}' \\
&= \int F(\mathbf{p}')\left[\sqrt{\mathbf{p}'^2+M^2}\,\delta^3(\mathbf{p}'-\mathbf{p})\right]\frac{d^3\mathbf{p}'}{\sqrt{\mathbf{p}'^2+M^2}}\,,
\end{aligned}
$$

so we see that the invariant delta function is

$$\sqrt{\mathbf{p}'^2+M^2}\,\delta^3(\mathbf{p}'-\mathbf{p}) = p^0\delta^3(\mathbf{p}'-\mathbf{p})\,. \tag{2.5.16}$$

Since p' and p are related to k' and k respectively by a Lorentz transformation, $L(p)$, we have then

$$p^0\delta^3(\mathbf{p}'-\mathbf{p}) = k^0\delta^3(\mathbf{k}'-\mathbf{k})$$

and therefore

$$\langle p',\sigma'|p,\sigma\rangle = |N(p)|^2\delta_{\sigma'\sigma}\left(\frac{p^0}{k^0}\right)\delta^3(\mathbf{p}'-\mathbf{p})\,. \tag{2.5.17}$$

The normalization factor $N(p)$ is sometimes chosen to be just $N(p)=1$, but then we would need to keep track of the p^0/k^0 factor in scalar products. Instead, I will here adopt the more usual convention that

$$N(p) = \sqrt{k^0/p^0} \tag{2.5.18}$$

for which

$$\langle p',\sigma'|p,\sigma\rangle = \delta_{\sigma'\sigma}\delta^3(\mathbf{p}'-\mathbf{p})\,. \tag{2.5.19}$$

We now consider the two cases of physical interest: particles of mass $M>0$, and particles of zero mass.

Mass Positive-Definite

The little group here is the three-dimensional rotation group. Its unitary representations can be broken up into a direct sum of irreducible unitary representations [7] $D^{(j)}_{\sigma'\sigma}(R)$ of dimensionality $2j+1$, with $j=0,\frac{1}{2},1,\ldots$. These can be built up from the standard matrices for infinitesimal rotations $R_{ik}=\delta_{ik}+\Theta_{ik}$, with $\Theta_{ik}=-\Theta_{ki}$ infinitesimal:

$$D^{(j)}_{\sigma'\sigma}(\mathbf{1}+\Theta) = \delta_{\sigma'\sigma}+\frac{i}{2}\Theta_{ik}(J^{(j)}_{ik})_{\sigma'\sigma}\,, \tag{2.5.20}$$

$$
\begin{aligned}
(J^{(j)}_{23}\pm iJ^{(j)}_{31})_{\sigma'\sigma} &= (J^{(j)}_1\pm iJ^{(j)}_2)_{\sigma'\sigma} \\
&= \delta_{\sigma',\sigma\pm 1}\sqrt{(j\mp\sigma)(j\pm\sigma+1)}\,,
\end{aligned} \tag{2.5.21}
$$

$$(J^{(j)}_{12})_{\sigma'\sigma} = (J^{(j)}_3)_{\sigma'\sigma} = \sigma\delta_{\sigma'\sigma}\,, \tag{2.5.22}$$

with σ running over the values $j,j-1,\ldots,-j$. For a particle of mass $M>0$ and spin j, Eq. (2.5.11) now becomes

$$U(\Lambda)|p,\sigma\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}}\sum_{\sigma'}D^{(j)}_{\sigma'\sigma}(W(\Lambda,p))|\Lambda p,\sigma'\rangle\,, \tag{2.5.23}$$

with the little-group element $W(\Lambda, p)$ (the Wigner rotation [5]) given by Eq. (2.5.10):

$$W(\Lambda, p) = L^{-1}(\Lambda p)\Lambda L(p)\,.$$

To calculate this rotation, we need to choose a 'standard boost' $L(p)$ which carries the four-momentum from $k^\mu = (M, \mathbf{0})$ to p^μ. This is conveniently chosen as

$$\begin{aligned} L^i{}_k(p) &= \delta_{ik} + (\gamma - 1)\hat{p}_i\hat{p}_k\,, \\ L^i{}_0(p) &= L^0{}_i(p) = \hat{p}_i\sqrt{\gamma^2 - 1}\,, \\ L^0{}_0(p) &= \gamma\,, \end{aligned} \tag{2.5.24}$$

where

$$\hat{p}_i \equiv p_i/|\mathbf{p}|\,, \qquad \gamma \equiv \sqrt{\mathbf{p}^2 + M^2}/M\,.$$

It is very important that when $\Lambda^\mu{}_\nu$ is an arbitrary three-dimensional rotation $\mathcal{R}$, the Wigner rotation $W(\Lambda, p)$ is the same as $\mathcal{R}$ for all p. To see this, note that the boost (2.5.24) may be expressed as

$$L(p) = R(\hat{\mathbf{p}})B(|\mathbf{p}|)R^{-1}(\hat{\mathbf{p}})\,,$$

where $R(\hat{\mathbf{p}})$ is a rotation (to be defined in a standard way below, in Eq. (2.5.47)) that takes the three-axis into the direction of $\mathbf{p}$, and

$$B(|\mathbf{p}|) = \begin{pmatrix} \gamma & 0 & 0 & \sqrt{\gamma^2 - 1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sqrt{\gamma^2 - 1} & 0 & 0 & \gamma \end{pmatrix}.$$

Then for an arbitrary rotation $\mathcal{R}$

$$W(\mathcal{R}, p) = R(\mathcal{R}\hat{\mathbf{p}})B^{-1}(|\mathbf{p}|)R^{-1}(\mathcal{R}\hat{\mathbf{p}})\mathcal{R}R(\hat{\mathbf{p}})B(|\mathbf{p}|)R^{-1}(\hat{\mathbf{p}})\,.$$

But the rotation $R^{-1}(\mathcal{R}\hat{\mathbf{p}})\mathcal{R}R(\hat{\mathbf{p}})$ takes the three-axis into the direction $\hat{\mathbf{p}}$, and then into the direction $\mathcal{R}\hat{\mathbf{p}}$, and then back to the three-axis, so it must be just a rotation by some angle θ around the three-axis

$$R^{-1}(\mathcal{R}\hat{\mathbf{p}})\mathcal{R}R(\hat{\mathbf{p}}) = R(\theta) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Since $R(\theta)$ commutes with $B(|\mathbf{p}|)$, this now gives

$$\begin{aligned} W(\mathcal{R}, p) &= R(\mathcal{R}\hat{\mathbf{p}})B^{-1}(|\mathbf{p}|)R(\theta)B(|\mathbf{p}|)R^{-1}(\hat{\mathbf{p}}) \\ &= R(\mathcal{R}\hat{\mathbf{p}})R(\theta)R^{-1}(\hat{\mathbf{p}}) \end{aligned}$$

and hence

$$W(\mathcal{R}, p) = \mathcal{R}$$

as was to be shown. Thus states of a moving massive particle (and, by extension, multi-particle states) have the same transformation under rotations as in non-relativistic quantum mechanics. This is another piece of good news—the whole apparatus of spherical harmonics, Clebsch–Gordan coefficients, etc. can be carried over wholesale from non-relativistic to relativistic quantum mechanics.

Mass Zero

First, we have to work out the structure of the little group. Consider an arbitrary little-group element $W^\mu{}_\nu$, with $W^\mu{}_\nu k^\nu = k^\mu$, where k^μ is the standard four-momentum for this case, $k^\mu = (1, 0, 0, 1)$. Acting on a time-like four-vector $t^\mu = (1, 0, 0, 0)$, such a Lorentz transformation must yield a four-vector Wt whose length and scalar product with $Wk = k$ are the same as those of t:

$$(Wt)^\mu (Wt)_\mu = t^\mu t_\mu = -1 \,, \qquad (Wt)^\mu k_\mu = t^\mu k_\mu = -1 \,.$$

Any four-vector that satisfies the second condition may be written

$$(Wt)^\mu = (1 + \zeta, \alpha, \beta, \zeta)$$

and the first condition then yields the relation

$$\zeta = (\alpha^2 + \beta^2)/2 \,. \tag{2.5.25}$$

It follows that the effect of $W^\mu{}_\nu$ on t^ν is the same as that of the Lorentz transformation

$$S^\mu{}_\nu(\alpha, \beta) = \begin{pmatrix} 1+\zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1-\zeta \end{pmatrix} . \tag{2.5.26}$$

This does not mean that W equals $S(\alpha, \beta)$, but it does mean that $S^{-1}(\alpha, \beta)W$ is a Lorentz transformation that leaves the time-like four-vector $(1, 0, 0, 0)$ invariant, and is therefore a pure rotation. Also, $S^\mu{}_\nu$ like $W^\mu{}_\nu$ leaves the light-like four-vector $(1, 0, 0, 1)$ invariant, so $S^{-1}(\alpha, \beta)W$ must be a rotation by some angle θ around the three-axis

$$S^{-1}(\alpha, \beta)W = R(\theta) \,, \tag{2.5.27}$$

where

$$R^\mu{}_\nu(\theta) \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} .$$

The most general element of the little group is therefore of the form

$$W(\theta, \alpha, \beta) = S(\alpha, \beta) R(\theta) \,. \tag{2.5.28}$$

What group is this? We note that the transformations with $\theta = 0$ or with $\alpha = \beta = 0$ form subgroups:

$$S(\bar\alpha, \bar\beta) S(\alpha, \beta) = S(\bar\alpha + \alpha, \bar\beta + \beta) \tag{2.5.29}$$

$$R(\bar\theta) R(\theta) = R(\bar\theta + \theta) \,. \tag{2.5.30}$$

These subgroups are *Abelian*—that is, their elements all commute with each other. Furthermore, the subgroup with $\theta = 0$ is *invariant*, in the sense that its elements are transformed into other elements of the same subgroup by any member of the group

$$R(\theta)S(\alpha,\beta)R^{-1}(\theta) = S(\alpha\cos\theta + \beta\sin\theta, -\alpha\sin\theta + \beta\cos\theta)\,. \tag{2.5.31}$$

From Eqs. (2.5.29)–(2.5.31) we can work out the product of any group elements. The reader will recognize these multiplication rules as those of the group $ISO(2)$, consisting of translations (by a vector (α, β)) and rotations (by an angle θ) in two dimensions.

Groups that do *not* have invariant Abelian subgroups have certain simple properties, and for this reason are called *semi-simple*. As we have seen, the little group $ISO(2)$ like the inhomogeneous Lorentz group is *not* semi-simple, and this leads to interesting complications. First, let's take a look at the Lie algebra of $ISO(2)$. For θ, α, β infinitesimal, the general group element is

$$W(\theta,\alpha,\beta)^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu\,, \qquad \omega_{\mu\nu} = \begin{pmatrix} 0 & -\alpha & -\beta & 0 \\ \alpha & 0 & \theta & -\alpha \\ \beta & -\theta & 0 & -\beta \\ 0 & \alpha & \beta & 0 \end{pmatrix}.$$

From (2.4.3), we see then that the corresponding Hilbert space operator is

$$U(W(\theta,\alpha,\beta)) = \mathbf{1} + i\alpha A + i\beta B + i\theta J_3\,, \tag{2.5.32}$$

where A and B are the Hermitian operators

$$A = -J^{13} + J^{10} = J_2 + K_1\,, \tag{2.5.33}$$

$$B = -J^{23} + J^{20} = -J_1 + K_2\,, \tag{2.5.34}$$

and, as before, $J_3 = J_{12}$. Either from (2.4.18)–(2.4.20), or directly from Eqs. (2.5.29)–(2.5.31), we see that these generators have the commutators

$$[J_3, A] = +iB\,, \tag{2.5.35}$$

$$[J_3, B] = -iA\,, \tag{2.5.36}$$

$$[A, B] = 0\,. \tag{2.5.37}$$

Since A and B are commuting Hermitian operators they (like the momentum generators of the inhomogeneous Lorentz group) can be simultaneously diagonalized by states $|k, a, b\rangle$

$$A|k,a,b\rangle = a|k,a,b\rangle\,, \qquad B|k,a,b\rangle = b|k,a,b\rangle\,.$$

The problem is that if we find one such set of non-zero eigenvalues of A, B, then we find a whole continuum. From Eq. (2.5.31), we have

$$U[R(\theta)]A\,U^{-1}[R(\theta)] = A\cos\theta - B\sin\theta\,,$$

$$U[R(\theta)]B\,U^{-1}[R(\theta)] = A\sin\theta + B\cos\theta\,,$$

and so, for arbitrary θ,

$$A|k,a,b\rangle^\theta = (a\cos\theta - b\sin\theta)|k,a,b\rangle^\theta ,$$

$$B|k,a,b\rangle^\theta = (a\sin\theta + b\cos\theta)|k,a,b\rangle^\theta ,$$

where

$$|k,a,b\rangle^\theta \equiv U^{-1}(R(\theta))|k,a,b\rangle .$$

Massless particles are not observed to have any continuous degree of freedom like θ; to avoid such a continuum of states, we must require that physical states (now called $|k,\sigma\rangle$) are eigenvectors of A and B with $a = b = 0$:

$$A|k,\sigma\rangle = B|k,\sigma\rangle = 0 . \tag{2.5.38}$$

These states are then distinguished by the eigenvalue of the remaining generator

$$J_3|k,\sigma\rangle = \sigma|k,\sigma\rangle . \tag{2.5.39}$$

Since the momentum $\mathbf{k}$ is in the three-direction, σ gives the component of angular momentum in the direction of motion, or *helicity*.

We are now in a position to calculate the Lorentz transformation properties of general massless particle states. First note that by use of the general arguments of Section 2.2, Eq. (2.5.32) generalizes for finite α and β to

$$U(S(\alpha,\beta)) = \exp(i\alpha A + i\beta B) \tag{2.5.40}$$

and for finite θ to

$$U(R(\theta)) = \exp(iJ_3\theta) . \tag{2.5.41}$$

An arbitrary element W of the little group can be put in the form (2.5.28), so that

$$U(W)|k,\sigma\rangle = \exp(i\alpha A + i\beta B)\exp(i\theta J_3)|k,\sigma\rangle = \exp(i\theta\sigma)|k,\sigma\rangle$$

and therefore Eq. (2.5.8) gives

$$D_{\sigma'\sigma}(W) = \exp(i\theta\sigma)\delta_{\sigma'\sigma} ,$$

where θ is the angle defined by expressing W as in Eq. (2.5.28). The Lorentz transformation rule for a massless particle of arbitrary helicity is now given by Eqs. (2.5.11) and (2.5.18) as

$$U(\Lambda)|p,\sigma\rangle = \sqrt{\frac{(\Lambda p)^0}{p^0}}\exp(i\sigma\theta(\Lambda,p))|\Lambda p,\sigma\rangle \tag{2.5.42}$$

with $\theta(\Lambda,p)$ defined by

$$W(\Lambda,p) \equiv L^{-1}(\Lambda p)\Lambda L(p) \equiv S(\alpha(\Lambda,p),\beta(\Lambda,p))R(\theta(\Lambda,p)) . \tag{2.5.43}$$

We shall see in Section 5.9 that electromagnetic gauge invariance arises from the part of the little group parameterized by α and β.

At this point we have not yet encountered any reason that would forbid the helicity σ of a massless particle from being an arbitrary real number. As we shall see in Section 2.7, there are topological considerations that restrict the allowed values of σ to integers and half-integers, just as for massive particles.

To calculate the little-group element (2.5.43) for a given Λ and p, (and also to enable us to calculate the effect of space or time inversion on these states in the next section) we need to fix a convention for the standard Lorentz transformation that takes us from $k^\mu = (\kappa, 0, 0, \kappa)$ to p^μ. This may conveniently be chosen to have the form

$$L(p) = R(\hat{\mathbf{p}})B(|\mathbf{p}|/\kappa) \tag{2.5.44}$$

where $B(u)$ is a pure boost along the three-direction:

$$B(u) \equiv \begin{pmatrix} (u^2+1)/2u & 0 & 0 & (u^2-1)/2u \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (u^2-1)/2u & 0 & 0 & (u^2+1)/2u \end{pmatrix} \tag{2.5.45}$$

and $R(\hat{\mathbf{p}})$ is a pure rotation that carries the three-axis into the direction of the unit vector $\hat{\mathbf{p}}$. For instance, suppose we take $\hat{\mathbf{p}}$ to have polar and azimuthal angles θ and ϕ:

$$\hat{\mathbf{p}} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)\,. \tag{2.5.46}$$

Then we can take $R(\hat{\mathbf{p}})$ as a rotation by angle θ around the two-axis, which takes $(0, 0, 1)$ into $(\sin\theta, 0, \cos\theta)$, followed by a rotation by angle ϕ around the three-axis:

$$U(R(\hat{\mathbf{p}})) = \exp(i\phi J_3)\exp(i\theta J_2)\,, \tag{2.5.47}$$

where $0 \leq \theta \leq \pi$, $0 \leq \phi < 2\pi$. (We give $U(R(\hat{\mathbf{p}}))$ rather than $R(\hat{\mathbf{p}})$, together with a specification of the range of ϕ and θ, because shifting θ or ϕ by 2π would give the same rotation $R(\hat{\mathbf{p}})$, but a different sign for $U(R(\hat{\mathbf{p}}))$ when acting on half-integer spin states.) Since (2.5.47) is a rotation, and does take the three-axis into the direction (2.5.46), any other choice of such an $R(\hat{\mathbf{p}})$ would differ from this one by at most an initial rotation around the three-axis, corresponding to a mere redefinition of the phase of the one-particle states.

Note that the helicity is Lorentz-invariant; a massless particle of a given helicity σ looks the same (aside from its momentum) in all inertial frames. Indeed, we would be justified in thinking of massless particles of each different helicity as different species of particles. However, as we shall see in the next section, particles of opposite helicity are related by the symmetry of space inversion. Thus, because electromagnetic and gravitational forces obey space inversion symmetry, the massless particles of helicity ± 1 associated with electromagnetic phenomena are both called *photons*, and the massless particles of helicity ± 2 that are believed to be associated with gravitation are both called *gravitons*. On the other hand, the supposedly massless particles of helicity $\pm 1/2$ that are emitted in nuclear beta decay have no interactions (apart from gravitation) that respect the symmetry of space inversion, so these particles are given different names: *neutrinos* for helicity $+1/2$, and *antineutrinos* for helicity $-1/2$.

Even though the helicity of a massless particle is Lorentz-invariant, the state itself is not. In particular, because of the helicity-dependent phase factor $\exp(i\sigma\theta)$ in Eq. (2.5.42), a state formed as a linear superposition of one-particle states with opposite helicities will be changed by a Lorentz transformation into a different superposition. For instance, a general one-photon state of four-momenta may be written

$$|p;\alpha\rangle = \alpha_+|p,+1\rangle + \alpha_-|p,-1\rangle\,,$$

where

$$|\alpha_+|^2 + |\alpha_-|^2 = 1\,.$$

The generic case is one of *elliptic polarization*, with $|\alpha_\pm|$ both non-zero and unequal. *Circular polarization* is the limiting case where either α_+ or α_- vanishes, and *linear polarization* is the opposite extreme, with $|\alpha_+| = |\alpha_-|$. The overall phase of α_+ and α_- has no physical significance, and for linear polarization may be adjusted so that $\alpha_- = \alpha_+^*$, but the relative phase is still important. Indeed, for linear polarizations with $\alpha_- = \alpha_+^*$, the phase of α_+ may be identified as the angle between the plane of polarization and some fixed reference direction perpendicular to $\mathbf{p}$. Eq. (2.5.42) shows that under a Lorentz transformation $\Lambda^\mu{}_\nu$, this angle rotates by an amount $\theta(\Lambda, p)$. Plane polarized gravitons can be defined in a similar way, and here Eq. (2.5.42) has the consequence that a Lorentz transformation Λ rotates the plane of polarization by an angle $2\theta(\Lambda, p)$.

2.6 Space Inversion and Time-Reversal

We saw in Section 2.3 that any homogeneous Lorentz transformation is either proper and orthochronous (i.e., $\mathrm{Det}\,\Lambda = +1$ and $\Lambda^0{}_0 \geq +1$) or else equal to a proper orthochronous transformation times either $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$, where $\mathcal{P}$ and $\mathcal{T}$ are the space inversion and time-reversal transformations

$$\mathcal{P}^\mu{}_\nu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \qquad \mathcal{T}^\mu{}_\nu = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

It used to be thought self-evident that the fundamental multiplication rule of the Poincaré group

$$U(\bar\Lambda, \bar a)U(\Lambda, a) = U(\bar\Lambda\Lambda, \bar\Lambda a + \bar a)$$

would be valid even if Λ and/or $\bar\Lambda$ involved factors of $\mathcal{P}$ or $\mathcal{T}$ or $\mathcal{PT}$. In particular, it was believed that there are operators corresponding to $\mathcal{P}$ and $\mathcal{T}$ themselves:

$$P \equiv U(\mathcal{P}, 0), \qquad T \equiv U(\mathcal{T}, 0)$$

such that

$$PU(\Lambda, a)P^{-1} = U(\mathcal{P}\Lambda\mathcal{P}^{-1}, \mathcal{P}a), \tag{2.6.1}$$

$$TU(\Lambda, a)T^{-1} = U(\mathcal{T}\Lambda\mathcal{T}^{-1}, \mathcal{T}a) \tag{2.6.2}$$

for any proper orthochronous Lorentz transformation $\Lambda^\mu{}_\nu$, and translation a^μ. These transformation rules incorporate most of what is meant when we say that P or T are 'conserved'.

In 1956–57 it became understood [8] that this is true for P only in the approximation in which one ignores the effects of weak interactions, such as those that produce nuclear beta decay. Time-reversal survived for a while, but in 1964 there appeared indirect evidence [9] that these properties of T are also only approximately satisfied. (See Section 3.3.) In what follows, we will make believe that operators P and T satisfying Eqs. (2.6.1) and (2.6.2) actually exist, but it should be kept in mind that this is only an approximation.

Let us apply Eqs. (2.6.1) and (2.6.2) in the case of an infinitesimal transformation, i.e.,

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \qquad a^\mu = \epsilon^\mu$$

with $\omega_{\mu\nu} = -\omega_{\nu\mu}$ and ϵ_μ both infinitesimal. Using (2.4.3), and equating coefficients of $\omega_{\rho\sigma}$ and ϵ_ρ in Eqs. (2.6.1) and (2.6.2), we obtain the P and T transformation properties of the Poincaré generators

$$PiJ^{\rho\sigma}P^{-1} = i\mathcal{P}_\mu{}^\rho\mathcal{P}_\nu{}^\sigma J^{\mu\nu}, \tag{2.6.3}$$

$$PiP^\rho P^{-1} = i\mathcal{P}_\mu{}^\rho P^\mu, \tag{2.6.4}$$

$$TiJ^{\rho\sigma}T^{-1} = i\mathcal{T}_\mu{}^\rho\mathcal{T}_\nu{}^\sigma J^{\mu\nu}, \tag{2.6.5}$$

$$TiP^\rho T^{-1} = i\mathcal{T}_\mu{}^\rho P^\mu. \tag{2.6.6}$$

This is much like Eqs. (2.4.8) and (2.4.9), except that we have not cancelled factors of i on both sides of these equations, because at this point we have not yet decided whether P and T are linear and unitary or antilinear and antiunitary.

The decision is an easy one. Setting $\rho = 0$ in Eq. (2.6.4) gives

$$PiHP^{-1} = iH,$$

where $H \equiv P^0$ is the energy operator. If P were antiunitary and antilinear then it would anticommute with i, so $PHP^{-1} = -H$. But then for any state $|\Psi\rangle$ of energy $E > 0$, there would have to be another state $P^{-1}|\Psi\rangle$ of energy $-E < 0$. There are no states of negative energy (energy less than that of the vacuum), so we are forced to choose the other alternative: P *is linear and unitary, and commutes rather than anticommutes with* H.

On the other hand, setting $\rho = 0$ in Eq. (2.6.6) yields

$$TiHT^{-1} = -iH.$$

If we supposed that T is linear and unitary then we could simply cancel the is, and find $THT^{-1} = -H$, with the again disastrous conclusion that for any state $|\Psi\rangle$ of energy E there is another state $T^{-1}|\Psi\rangle$ of energy $-E$. To avoid this, we are forced here to conclude that T *is antilinear and antiunitary.*

Now that we have decided that P is linear and T is antilinear, we can conveniently rewrite Eqs. (2.6.3)–(2.6.6) in terms of the generators (2.4.15)–(2.4.17) in a three-dimensional notation

$$
\begin{aligned}
P\mathbf{J}P^{-1} &= +\mathbf{J}, &(2.6.7)\\
P\mathbf{K}P^{-1} &= -\mathbf{K}, &(2.6.8)\\
P\mathbf{P}P^{-1} &= -\mathbf{P}, &(2.6.9)\\
T\mathbf{J}T^{-1} &= -\mathbf{J}, &(2.6.10)\\
T\mathbf{K}T^{-1} &= +\mathbf{K}, &(2.6.11)\\
T\mathbf{P}T^{-1} &= -\mathbf{P}, &(2.6.12)
\end{aligned}
$$

and, as shown before,

$$PHP^{-1} = THT^{-1} = H. \qquad (2.6.13)$$

It is physically sensible that P should preserve the sign of $\mathbf{J}$, because at least the orbital part is a vector product $\mathbf{r} \times \mathbf{p}$ of two vectors, both of which change sign under an inversion of the spatial coordinate system. On the other hand, T reverses $\mathbf{J}$, because after time-reversal an observer will see all bodies spinning in the opposite direction. Note by the way that Eq. (2.6.10) is consistent with the angular-momentum commutation relations $\mathbf{J} \times \mathbf{J} = i\mathbf{J}$, because T reverses not only $\mathbf{J}$, but also i. The reader can easily check that Eqs. (2.6.7)–(2.6.13) are consistent with all the commutation relations (2.4.18)–(2.4.24).

Let us now consider what P and T do to one-particle states:

$P : M > 0$

The one-particle states $|k, \sigma\rangle$ are defined as eigenvectors of $\mathbf{P}$, H, and J_3 with eigenvalues 0, M, and σ, respectively. From Eqs. (2.6.7), (2.6.9), and (2.6.13), we see that the same must be true of the state $P|k, \sigma\rangle$, and therefore (barring degeneracies) these states can only differ by a phase

$$P|k, \sigma\rangle = \eta_\sigma |k, \sigma\rangle$$

with a phase factor ($|\eta_\sigma| = 1$) that may or may not depend on the spin σ. To see that η_σ is σ-independent, we note from (2.5.8), (2.5.20), and (2.5.21) that

$$(J_1 \pm iJ_2)|k, \sigma\rangle = \sqrt{(j \mp \sigma)(j \pm \sigma + 1)}\,|k, \sigma \pm 1\rangle, \qquad (2.6.14)$$

where j is the particle's spin. Operating on both sides with P, we find

$$\eta_\sigma = \eta_{\sigma \pm 1}$$

and so η_σ is actually independent of σ. We therefore write

$$P|k, \sigma\rangle = \eta |k, \sigma\rangle \qquad (2.6.15)$$

with η a phase, known as the *intrinsic parity*, that depends only on the species of particle on which P acts.

To get to finite momentum states, we must apply the unitary operator $U(L(p))$ corresponding to the 'boost' (2.5.24):

$$|p, \sigma\rangle = \sqrt{M/p^0}\, U(L(p))|k, \sigma\rangle.$$

We note that

$$\mathcal{P}L(p)\mathcal{P}^{-1} = L(\mathcal{P}p), \qquad \mathcal{P}p = (p^0, -\mathbf{p}),$$

so using Eqs. (2.6.1) and (2.6.15), we have

$$P|p,\sigma\rangle = \sqrt{M/p^0}\, U(L(\mathcal{P}p))\eta|k,\sigma\rangle$$

or in other words

$$P|p,\sigma\rangle = \eta|\mathcal{P}p,\sigma\rangle. \tag{2.6.16}$$

$T : M > 0$

From Eqs. (2.6.10), (2.6.12), and (2.6.13), we see that the effect of T on the zero-momentum one-particle state $|k,\sigma\rangle$ is to yield a state with

$$\begin{aligned}
\mathbf{P}(T|k,\sigma\rangle) &= 0, \\
H(T|k,\sigma\rangle) &= M(T|k,\sigma\rangle), \\
J_3(T|k,\sigma\rangle) &= -\sigma(T|k,\sigma\rangle),
\end{aligned}$$

and so

$$T|k,\sigma\rangle = \zeta_\sigma|k,-\sigma\rangle,$$

where ζ_σ is a phase factor. Applying the operator T to (2.6.14), and recalling that T anticommutes with $\mathbf{J}$ *and* i, we find

$$(-J_1 \pm iJ_2)\zeta_\sigma|k,-\sigma\rangle = \sqrt{(j\mp\sigma)(j\pm\sigma+1)}\,\zeta_{\sigma\pm1}|k,-\sigma\mp1\rangle.$$

Using Eq. (2.6.14) again on the left, we see that the square-root factors cancel, and so

$$-\zeta_\sigma = \zeta_{\sigma\pm1}.$$

We write the solution as $\zeta_\sigma = \zeta(-1)^{j-\sigma}$, with ζ another phase that depends only on the species of particle:

$$T|k,\sigma\rangle = \zeta(-1)^{j-\sigma}|k,-\sigma\rangle. \tag{2.6.17}$$

However, unlike the 'intrinsic parity' η, the time-reversal phase ζ has no physical significance. This is because we can redefine the one-particle states by a change of phase

$$|k,\sigma\rangle \longrightarrow |k,\sigma\rangle' = \zeta^{1/2}|k,\sigma\rangle$$

in such a way that the phase ζ is eliminated from the transformation rule

$$T|k,\sigma\rangle' = \zeta^{*1/2}T|k,\sigma\rangle = \zeta^{*1/2}\zeta(-1)^{j-\sigma}|k,-\sigma\rangle = (-1)^{j-\sigma}|k,-\sigma\rangle'.$$

In what follows we will keep the arbitrary phase ζ in Eq. (2.6.17), just to keep open our options in choosing the phase of the one-particle states, but it should be kept in mind that this phase is of no real importance.

To deal with states of finite momentum, we again apply the 'boost' (2.5.24). Note that

$$\mathcal{T}L(p)\mathcal{T}^{-1} = L(\mathcal{P}p), \qquad \mathcal{P}p = (p^0, -\mathbf{p}).$$

(That is, changing the sign of each element of $L^\mu{}_\nu$ with an odd number of *time*-indices is the same as changing the signs of elements with an odd number of *space*-indices.) Using Eqs. (2.6.2) and (2.5.5), we have then

$$T|p,\sigma\rangle = \zeta(-1)^{j-\sigma}|\mathcal{P}p,-\sigma\rangle. \tag{2.6.18}$$

$P : M = 0$

Acting on a state $|k,\sigma\rangle$ that is defined as an eigenvector of P^μ with eigenvalue $k^\mu = (\kappa,0,0,\kappa)$ and an eigenvector of J_3 with eigenvalue σ, the parity operator P yields a state with four-momentum $(\mathcal{P}k)^\mu = (\kappa,0,0,-\kappa)$ and J_3 equal to σ. Thus it takes a state of helicity (the component of spin along the direction of motion) σ into one of helicity $-\sigma$. As mentioned earlier, this shows that the existence of a space-inversion symmetry requires that any species of massless particle with non-zero helicity must be accompanied with another of opposite helicity. Because P does not leave the standard momentum invariant, it is convenient to consider instead the operator $U(R_2^{-1})P$, where R_2 is a rotation that also takes k to $\mathcal{P}k$, conveniently chosen as a rotation by $-180°$ around the two-axis

$$U(R_2) = \exp(-i\pi J_2). \tag{2.6.19}$$

Since $U(R_2^{-1})$ reverses the sign of J_3, we have

$$U(R_2^{-1})P|k,\sigma\rangle = \eta_\sigma|k,-\sigma\rangle \tag{2.6.20}$$

with η_σ a phase factor. Now, $R_2^{-1}\mathcal{P}$ commutes with the Lorentz 'boost' (2.5.45), and $\mathcal{P}$ commutes with the rotation $R(\hat{\mathbf{p}})$ which takes the three-direction into the direction of $\mathbf{p}$, so by operating on (2.5.5) with P, we find for a general four-momentum p^μ

$$\begin{aligned} P|p,\sigma\rangle &= \sqrt{\frac{\kappa}{p^0}}\,U\left(R(\hat{\mathbf{p}})R_2B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right)U(R_2^{-1})P|k,\sigma\rangle \\ &= \sqrt{\frac{\kappa}{p^0}}\,\eta_\sigma U\left(R(\hat{\mathbf{p}})R_2B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right)|k,-\sigma\rangle. \end{aligned}$$

Note that $R(\hat{\mathbf{p}})R_2$ is a rotation that takes the three-axis into the direction of $-\hat{\mathbf{p}}$, but $U(R(\hat{\mathbf{p}})R_2)$ is not quite equal to $U(R(-\hat{\mathbf{p}}))$. According to (2.5.47),

$$U(R(-\hat{\mathbf{p}})) = \exp\big(i(\phi\pm\pi)J_3\big)\exp\big(i(\pi-\theta)J_2\big)$$

with azimuthal angle chosen as $\phi+\pi$ or $\phi-\pi$ according to whether $0\le\phi<\pi$ or $\pi\le\phi<2\pi$, so that it remains in the range of 0 to 2π. Then

$$\begin{aligned} U^{-1}(R(-\hat{\mathbf{p}}))U(R(\hat{\mathbf{p}})R_2) &= \exp\big(-i(\pi-\theta)J_2\big) \\ &\quad\times\exp\big(-i(\phi\pm\pi)J_3\big)\exp(i\phi J_3)\exp(i\theta J_2)\exp(-i\pi J_2) \\ &= \exp\big(-i(\pi-\theta)J_2\big)\exp(\mp i\pi J_3)\exp\big(-i(\pi-\theta)J_2\big). \end{aligned}$$

But a rotation of $\pm180°$ around the three-axis reverses the sign of J_2, so

$$U(R(\hat{\mathbf{p}})R_2) = U(R(-\hat{\mathbf{p}}))\exp(\pm i\pi J_3). \tag{2.6.21}$$

Also, $R(-\hat{\mathbf{p}})B(|\mathbf{p}|/\kappa)$ is just the standard boost $L(\mathcal{P}p)$ in the direction $\mathcal{P}p = (p^0, -\mathbf{p})$. We have then finally

$$P|p,\sigma\rangle = \eta_\sigma \exp(\mp i\pi\sigma)|\mathcal{P}p, -\sigma\rangle \tag{2.6.22}$$

with the phase $-\pi\sigma$ or $+\pi\sigma$ according to whether the two-component of $\mathbf{p}$ is positive or negative, respectively. This peculiar change of sign in the operation of parity for massless particles of half-integer spin is due to the convention adopted in Eq. (2.5.47) for the rotation used to define massless particle states of arbitrary momentum. Because the rotation group is not simply connected, some discontinuity of this sort is unavoidable.

$T : M = 0$

Acting on the state $|k,\sigma\rangle$, which has values $k^\mu = (\kappa, 0, 0, \kappa)$ and σ for P^μ and J_3, the time-reversal operator T yields a state which has values $(\mathcal{P}k)^\mu = (\kappa, 0, 0, -\kappa)$ and $-\sigma$ for P^μ and J_3. Thus T does not change the helicity $\mathbf{J}\cdot\hat{\mathbf{k}}$, and by itself has nothing to say about whether massless particles of one helicity σ are accompanied with others of helicity $-\sigma$. Because T like P does not leave the standard four-momentum k invariant, it is convenient to consider the generator $U(R_2^{-1})T$, where R_2 is the rotation (2.6.19), which also takes k into $\mathcal{P}k$. This commutes with J_3, so

$$U(R_2^{-1})T|k,\sigma\rangle = \zeta_\sigma|k,\sigma\rangle \tag{2.6.23}$$

with ζ_σ another phase. Since $R_2^{-1}\mathcal{T}$ commutes with the boost (2.5.45), and $\mathcal{T}$ commutes with the rotation $R(\hat{\mathbf{p}})$, operating with T on the state (2.5.5) gives

$$T|p,\sigma\rangle = \sqrt{\frac{\kappa}{p^0}}\, U\left[R(\hat{\mathbf{p}})R_2 B\left(\frac{|\mathbf{p}|}{\kappa}\right)\right]\zeta_\sigma|k,\sigma\rangle. \tag{2.6.24}$$

Using Eq. (2.6.21), this yields finally

$$T|p,\sigma\rangle = \zeta_\sigma \exp(\pm i\pi\sigma)|\mathcal{P}p,\sigma\rangle. \tag{2.6.25}$$

Again, the top or bottom sign applies according to whether the two-component of $\mathbf{p}$ is positive or negative, respectively.

* * *

It is interesting that the square T^2 of the time-reversal operator has a very simple action on both massive and massless one-particle states. Using Eq. (2.6.18), and recalling that T is antiunitary, we see that for massive one-particle states:

$$\begin{aligned} T^2|p,\sigma\rangle &= T\zeta(-1)^{j-\sigma}|\mathcal{P}p,-\sigma\rangle \\ &= \zeta^*(-1)^{j-\sigma}\zeta(-1)^{j+\sigma}|p,\sigma\rangle \end{aligned}$$

or in other words

$$T^2|p,\sigma\rangle = (-1)^{2j}|p,\sigma\rangle. \tag{2.6.26}$$

We get the same result for massless particles. If the two-component of $\mathbf{p}$ is positive then the two-component of $\mathcal{P}\mathbf{p}$ is negative, and vice-versa, so Eq. (2.6.25) gives

$$\begin{aligned} T^2|p,\sigma\rangle &= T\zeta_\sigma \exp(\pm i\pi\sigma)|\mathcal{P}p,\sigma\rangle \\ &= \zeta_\sigma^* \exp(\mp i\pi\sigma)\zeta_\sigma \exp(\mp i\pi\sigma)|p,\sigma\rangle \\ &= \exp(\mp 2\pi i\sigma)|p,\sigma\rangle. \end{aligned}$$

As long as σ is an integer or half-integer, this can be written

$$T^2|p,\sigma\rangle = (-1)^{2|\sigma|}|p,\sigma\rangle. \tag{2.6.27}$$

By the 'spin' of a massless particle, we usually mean the absolute value of the helicity, so Eq. (2.6.27) is the same as Eq. (2.6.26).

This result has an interesting consequence. When T^2 acts on any state $|\Psi\rangle$ of a system of non-interacting particles, either massive or massless, it yields a factor $(-1)^{2j}$ or $(-1)^{2|\sigma|}$ for each particle. Hence if the state contains an odd number of particles of half-integer spin or helicity (plus any number of particles of integer spin or helicity), we get an overall change of sign

$$T^2|\Psi\rangle = -|\Psi\rangle. \tag{2.6.28}$$

If we now 'turn on' various interactions, this result will be preserved, provided these interactions respect invariance under time-reversal, even if they do not respect rotational invariance. (For instance, these arguments will apply even if our system is subjected to arbitrary static gravitational and electric fields.) Now, suppose that $|\Psi\rangle$ is an eigenstate of the Hamiltonian. Since T commutes with the Hamiltonian, $T|\Psi\rangle$ will also be an eigenstate of the Hamiltonian. Is it the same state? If so, then $T|\Psi\rangle$ can differ from $|\Psi\rangle$ only by a phase

$$T|\Psi\rangle = \zeta|\Psi\rangle,$$

but then

$$T^2|\Psi\rangle = T(\zeta|\Psi\rangle) = \zeta^* T|\Psi\rangle = |\zeta|^2|\Psi\rangle = |\Psi\rangle,$$

in contradiction with Eq. (2.6.28). We see that any energy eigenstate $|\Psi\rangle$ satisfying Eq. (2.6.28) must be degenerate, with another eigenstate of the same energy. This is known as a 'Kramers degeneracy' [10]. Of course, this conclusion is trivial if the system is in a rotationally invariant environment, because the total angular-momentum j of any state of this system would have to be a half-integer, and there would therefore be $2j+1 = 2, 4, \ldots$ degenerate states. The surprising result is that at least a two-fold degeneracy persists even if rotational invariance is perturbed by external fields, such as electrostatic fields, as long as these fields are invariant under T. In particular, if any particle had an electric or gravitational dipole moment then the degeneracy among its $2j+1$ spin states would be entirely removed in a static electric or gravitational field, so such dipole moments are forbidden by time-reversal invariance.

For the sake of completeness, it should be mentioned that P and T can have more complicated effects on multiplets of particles with the same mass. This possibility will be considered in Appendix C of this chapter. No physically relevant examples are known.

2.7 Projective Representations[9]

We now return to the possibility mentioned in Section 2.2, that a group of symmetries may be represented projectively on physical states; that is, the elements T, $\bar{T}$, etc. of the symmetry group may be represented on the physical Hilbert space by unitary operators $U(T)$, $U(\bar{T})$, etc., which satisfy the composition rule

$$U(T)U(\bar{T}) = \exp\bigl(i\phi(T,\bar{T})\bigr)U(T\bar{T}) \tag{2.7.1}$$

with ϕ a real phase. (A bar is used here just to distinguish one symmetry operator from another.) The basic requirement that any phase ϕ in Eq. (2.7.1) would have to satisfy is the associativity condition

$$U(T_3)\bigl(U(T_2)U(T_1)\bigr) = \bigl(U(T_3)U(T_2)\bigr)U(T_1),$$

which imposes on ϕ the corresponding condition

$$\phi(T_2,T_1) + \phi(T_3,T_2T_1) = \phi(T_3,T_2) + \phi(T_3T_2,T_1). \tag{2.7.2}$$

Of course, any phase of the form

$$\phi(T,\bar{T}) = \alpha(T\bar{T}) - \alpha(T) - \alpha(\bar{T}) \tag{2.7.3}$$

will automatically satisfy Eq. (2.7.2), but a projective representation with such a phase can be replaced with an ordinary representation by replacing $U(T)$ with

$$\widetilde{U}(T) \equiv U(T)\exp\bigl(i\alpha(T)\bigr)$$

for which

$$\widetilde{U}(T)\widetilde{U}(\bar{T}) = \widetilde{U}(T\bar{T}).$$

Any set of functions $\phi(T,\bar{T})$ that satisfy Eq. (2.7.2), and that differ only by functions $\Delta\phi(T,\bar{T})$ of the form (2.7.3), is called a 'two-cocycle'. A trivial cocycle is one that contains the function $\phi = 0$, and hence consists of functions of the form (2.7.3), which can be eliminated by a redefinition of $U(T)$. We are interested here in whether a symmetry group allows any non-trivial two-cocyles; that is, whether it may have a representation on the physical Hilbert space that is *intrinsically* projective, in the sense that the phase $\phi(T,\bar{T})$ *cannot* be eliminated in this way.

In order to answer this question, it is useful first to consider the effect of a phase ϕ in Eq. (2.7.1) on the commutation relations of the generators of infinitesimal transformations. When either $\bar{T}$ or T is the identity, the phase ϕ must clearly vanish

$$\phi(T,\mathbf{1}) = \phi(\mathbf{1},\bar{T}) = 0. \tag{2.7.4}$$

[9]This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

When both T and $\bar{T}$ are *near* the identity the phase must be small. Using coordinates θ^a to parameterize group elements (as in Section 2.2), with $T(0) = \mathbf{1}$, Eq. (2.7.4) tells us that the expansion of $\phi[T(\theta), T(\bar{\theta})]$ around $\theta = \bar{\theta} = 0$ must start with terms of order $\theta\bar{\theta}$:

$$\phi\big(T(\theta), T(\bar{\theta})\big) = f_{ab}\theta^a\bar{\theta}^b + \cdots\,, \tag{2.7.5}$$

where f_{ab} are real numerical constants. Inserting this expansion in the power series expansion of Eq. (2.7.1), and repeating the steps that led to (2.2.22), we now have

$$[t_b, t_c] = iC^a{}_{bc}t_a + iC_{bc}\mathbf{1}, \tag{2.7.6}$$

where C_{bc} is the antisymmetric coefficient

$$C_{bc} = -f_{bc} + f_{cb}. \tag{2.7.7}$$

The appearance of terms on the right-hand side of the commutation relation proportional to the unit element (so-called *central charges*) is the counterpart for the Lie algebra of the presence of phases in a projective representation of a group.

The constants C_{bc} as well as $C^a{}_{bc}$ are subject to an important constraint, which follows from the Jacobi identity. Taking the commutator of (2.7.6) with t_d, and adding the same expressions with b, c, d replaced with c, d, b and d, b, c, the sum of the three double-commutators on the left-hand side vanishes identically, and so

$$C^a{}_{bc}C^e{}_{ad} + C^a{}_{cd}C^e{}_{ab} + C^a{}_{db}C^e{}_{ac} = 0 \tag{2.7.8}$$

and also

$$C^a{}_{bc}C_{ad} + C^a{}_{cd}C_{ab} + C^a{}_{db}C_{ac} = 0. \tag{2.7.9}$$

Eq. (2.7.9) always has one obvious class of non-zero solutions for C_{ab}:

$$C_{ab} = C^e{}_{ab}\phi_e, \tag{2.7.10}$$

where ϕ_e is an arbitrary set of real constants. For these solutions, we can eliminate the central charges from Eq. (2.7.6) by a redefinition of the generators

$$t_a \longrightarrow \widetilde{t}_a \equiv t_a + \phi_a. \tag{2.7.11}$$

The new generators then satisfy the commutation relations without central charges

$$[\widetilde{t}_b, \widetilde{t}_c] = iC^a{}_{bc}\widetilde{t}_a. \tag{2.7.12}$$

A given Lie algebra may or may not allow solutions of Eq. (2.7.9) other than Eq. (2.7.10).

We can now state the key theorem that governs the occurrence of intrinsically projective representations. The phase of any representation $U(T)$ of a given group can be chosen so that $\phi = 0$ in Eq. (2.7.1), if two conditions are met:

(a) The generators of the group in this representation can be redefined (as in Eq. (2.7.11)), so as to eliminate all central charges from the Lie algebra.

(b) The group is simply connected, i.e., any two group elements may be connected by a path lying within the group, and any two such paths may be continuously transformed into one another. (An equivalent statement is that any loop that starts and ends at the same group element may be shrunk continuously to a point.)

This theorem is proved in Appendix B of this chapter, which also offers comments about the case of groups that are not simply connected. It shows that there are just two (not exclusive) ways that intrinsically projective representations may arise: either algebraically, because the group is represented projectively even near the identity, or topologically, because the group is not simply connected, and hence a path from $\mathbf{1}$ to T and then from T to $\bar{T}$ may not be continuously deformable into some other path from $\mathbf{1}$ to $T\bar{T}$. In the latter case, the phase ϕ in Eq. (2.7.1) depends on the particular choice of standard paths, leading from the origin to the various group elements, that are used to define the corresponding U-operators.

Let's now consider each of these possibilities in turn for the special case of the inhomogeneous Lorentz group.

(A) Algebra

With central charges, the commutation relations of the generators of the inhomogeneous Lorentz group would read

$$i[J^{\mu\nu}, J^{\rho\sigma}] = \eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\sigma\mu} J^{\rho\nu} + \eta^{\sigma\nu} J^{\rho\mu} + C^{\rho\sigma,\mu\nu}, \tag{2.7.13}$$
$$i[P^{\mu}, J^{\rho\sigma}] = \eta^{\mu\rho} P^{\sigma} - \eta^{\mu\sigma} P^{\rho} + C^{\rho\sigma,\mu}, \tag{2.7.14}$$
$$i[J^{\mu\nu}, P^{\rho}] = \eta^{\nu\rho} P^{\mu} - \eta^{\mu\rho} P^{\nu} + C^{\rho,\mu\nu}, \tag{2.7.15}$$
$$i[P^{\mu}, P^{\rho}] = C^{\rho,\mu} \tag{2.7.16}$$

in place of Eqs. (2.4.12)–(2.4.14). We see that the Cs also satisfy the antisymmetry conditions

$$C^{\rho\sigma,\mu\nu} = -C^{\mu\nu,\rho\sigma}, \tag{2.7.17}$$
$$C^{\rho\sigma,\mu} = -C^{\mu,\rho\sigma}, \tag{2.7.18}$$
$$C^{\rho,\mu} = -C^{\mu,\rho}. \tag{2.7.19}$$

We will now show that all these constants have additional algebraic properties that allow them to be eliminated by shifting the definitions of $J^{\mu\nu}$ and P^{μ} by constant terms. (This corresponds to redefining the phase of the operators $U(\Lambda, a)$.) To derive these properties, we apply the Jacobi identities

$$[J^{\mu\nu}, [P^{\rho}, P^{\sigma}]] + [P^{\sigma}, [J^{\mu\nu}, P^{\rho}]] + [P^{\rho}, [P^{\sigma}, J^{\mu\nu}]] = 0, \tag{2.7.20}$$
$$[J^{\lambda\eta}, [J^{\mu\nu}, P^{\rho}]] + [P^{\rho}, [J^{\lambda\eta}, J^{\mu\nu}]] + [J^{\mu\nu}, [P^{\rho}, J^{\lambda\eta}]] = 0, \tag{2.7.21}$$
$$[J^{\lambda\eta}, [J^{\mu\nu}, J^{\rho\sigma}]] + [J^{\rho\sigma}, [J^{\lambda\eta}, J^{\mu\nu}]] + [J^{\mu\nu}, [J^{\rho\sigma}, J^{\lambda\eta}]] = 0. \tag{2.7.22}$$

(The Jacobi identity involving three Ps is automatically satisfied, and hence yields no further information.) Using Eqs. (2.7.13)–(2.7.16) in Eqs. (2.7.20)–(2.7.22), we obtain algebraic conditions on the Cs

$$0 = \eta^{\nu\rho} C^{\mu,\sigma} - \eta^{\mu\rho} C^{\nu,\sigma} - \eta^{\nu\sigma} C^{\mu,\rho} + \eta^{\mu\sigma} C^{\nu,\rho}, \tag{2.7.23}$$

$$\begin{aligned} 0 = {} & \eta^{\nu\rho} C^{\mu,\lambda\eta} - \eta^{\mu\rho} C^{\nu,\lambda\eta} - \eta^{\mu\lambda} C^{\rho,\eta\nu} + \eta^{\mu\eta} C^{\rho,\lambda\nu} \\ & + \eta^{\lambda\nu} C^{\rho,\mu\eta} - \eta^{\eta\nu} C^{\rho,\mu\lambda} + \eta^{\rho\lambda} C^{\eta,\mu\nu} - \eta^{\rho\eta} C^{\lambda,\mu\nu}, \end{aligned} \tag{2.7.24}$$

$$\begin{aligned} 0 = {} & \eta^{\nu\rho} C^{\mu\sigma,\lambda\eta} - \eta^{\mu\rho} C^{\nu\sigma,\lambda\eta} - \eta^{\sigma\mu} C^{\rho\nu,\lambda\eta} + \eta^{\sigma\nu} C^{\rho\mu,\lambda\eta} \\ & + \eta^{\eta\mu} C^{\lambda\nu,\rho\sigma} - \eta^{\lambda\mu} C^{\eta\nu,\rho\sigma} - \eta^{\nu\lambda} C^{\mu\eta,\rho\sigma} + \eta^{\nu\eta} C^{\mu\lambda,\rho\sigma} \\ & + \eta^{\sigma\lambda} C^{\rho\eta,\mu\nu} - \eta^{\rho\lambda} C^{\sigma\eta,\mu\nu} - \eta^{\sigma\eta} C^{\rho\lambda,\mu\nu} + \eta^{\rho\eta} C^{\sigma\lambda,\mu\nu}. \end{aligned} \tag{2.7.25}$$

Contracting Eq. (2.7.23) with $\eta_{\nu\rho}$ gives

$$C^{\mu,\sigma} = 0. \tag{2.7.26}$$

On the other hand, the constants $C^{\mu,\lambda\eta}$ and $C^{\rho\sigma,\mu\nu}$ are not necessarily zero, but their algebraic structure is simple enough so that they can be eliminated by shifting the definitions of P^μ and $J^{\mu\nu}$, respectively. Contracting Eq. (2.7.24) with $\eta_{\nu\rho}$ gives

$$C^{\mu,\lambda\eta} = \eta^{\eta\mu} C^\lambda - \eta^{\mu\lambda} C^\eta, \tag{2.7.27}$$

$$C^\lambda \equiv \frac{1}{3} \eta_{\rho\nu} C^{\rho,\lambda\nu}. \tag{2.7.28}$$

Also, contracting (2.7.25) with $\eta_{\nu\rho}$ gives

$$C^{\mu\sigma,\lambda\eta} = \eta^{\eta\mu} C^{\lambda\sigma} - \eta^{\lambda\mu} C^{\eta\sigma} + \eta^{\sigma\lambda} C^{\eta\mu} - \eta^{\eta\sigma} C^{\lambda\mu}, \tag{2.7.29}$$

$$C^{\lambda\sigma} \equiv \frac{1}{2} \eta_{\nu\rho} C^{\lambda\nu,\sigma\rho}. \tag{2.7.30}$$

(These expressions automatically satisfy Eqs. (2.7.24) and (2.7.25), so there is no further information to be gained from the Jacobi identities.) We now see that if the Cs are not zero, they can be eliminated by defining new generators

$$\widetilde{P}^\mu \equiv P^\mu + C^\mu, \tag{2.7.31}$$

$$\widetilde{J}^{\mu\sigma} \equiv J^{\mu\sigma} + C^{\mu\sigma}, \tag{2.7.32}$$

and the commutation relations are then what they would be for an ordinary representation

$$i[\widetilde{J}^{\mu\nu}, \widetilde{J}^{\rho\sigma}] = \eta^{\nu\rho} \widetilde{J}^{\mu\sigma} - \eta^{\mu\rho} \widetilde{J}^{\nu\sigma} - \eta^{\sigma\mu} \widetilde{J}^{\rho\nu} + \eta^{\sigma\nu} \widetilde{J}^{\rho\mu}, \tag{2.7.33}$$

$$i[\widetilde{J}^{\mu\nu}, \widetilde{P}^\rho] = \eta^{\nu\rho} \widetilde{P}^\mu - \eta^{\mu\rho} \widetilde{P}^\nu, \tag{2.7.34}$$

$$i[\widetilde{P}^\mu, \widetilde{P}^\rho] = 0. \tag{2.7.35}$$

The commutation relations will always be taken in the form Eqs. (2.7.33)–(2.7.35), but with tildas dropped.

Incidentally, the fact that there is no central charge in the algebra of the $J^{\mu\nu}$ could have been immediately inferred from the fact that this algebra is of the type known as 'semi-simple'. (Semi-simple Lie algebras are those that have no 'invariant Abelian' subalgebra, consisting of generators that commute with each other and whose commutators with any other generators also belong to the subalgebra.) There is a general theorem [11] that any central charges in semi-simple Lie algebras may always be removed by a redefinition of generators, as in Eq. (2.7.32). On the other hand, the full Poincaré algebra spanned by $J^{\mu\nu}$ and P^μ is not semi-simple (the P^μ form an invariant Abelian subalgebra), and we needed a special argument to show that its central charges can also be eliminated in this way. Indeed, the non-semi-simple Galilean algebra discussed in Section 2.4 does allow a central charge, the mass M.

We see that the inhomogeneous Lorentz group satisfies the first of the two conditions needed to rule out intrinsically projective representations. How about the second?

(B) Topology

To explore the topology of the inhomogeneous Lorentz group, it is very convenient to represent homogeneous Lorentz transformations by 2×2 complex matrices. Any real four-vector V^μ can be used to construct an Hermitian 2×2 matrix

$$v \equiv V^\mu \sigma_\mu = \begin{pmatrix} V^0 + V^3 & V^1 - iV^2 \\ V^1 + iV^2 & V^0 - V^3 \end{pmatrix}, \tag{2.7.36}$$

where σ_μ are the usual Pauli matrices with $\sigma_0 = \mathbf{1}$. Conversely any 2×2 Hermitian matrix can be put in this form, and therefore defines a real four-vector V^μ.

The property of Hermiticity will be preserved under the transformation

$$v \longrightarrow \lambda v \lambda^\dagger \tag{2.7.37}$$

with λ an arbitrary complex 2×2 matrix. Furthermore, the covariant square of the four-vector is

$$V_\mu V^\mu = (V^1)^2 + (V^2)^2 + (V^3)^2 - (V^0)^2 = -\operatorname{Det} v \tag{2.7.38}$$

and this determinant is preserved by the transformation (2.7.37) provided that

$$|\operatorname{Det} \lambda| = 1. \tag{2.7.39}$$

Each complex 2×2 matrix λ satisfying Eq. (2.7.39) thus defines a real linear transformation of V^μ that leaves Eq. (2.7.38) invariant, i.e., a homogeneous Lorentz transformation $\Lambda(\lambda)$:

$$\lambda V^\mu \sigma_\mu \lambda^\dagger = \big(\Lambda^\mu{}_\nu(\lambda) V^\nu\big)\sigma_\mu. \tag{2.7.40}$$

Furthermore, for two such matrices λ and $\bar\lambda$, we have

$$\begin{aligned}(\lambda\bar\lambda)V^\mu\sigma_\mu(\lambda\bar\lambda)^\dagger &= \lambda\big(\bar\lambda V^\mu \sigma_\mu \bar\lambda^\dagger\big)\lambda^\dagger \\ &= \lambda\Lambda^\mu{}_\nu(\bar\lambda)V^\nu\sigma_\mu\lambda^\dagger \\ &= \Lambda^\mu{}_\rho(\lambda)\Lambda^\rho{}_\nu(\bar\lambda)V^\nu\sigma_\mu\end{aligned}$$

and so

$$\Lambda(\lambda\bar{\lambda}) = \Lambda(\lambda)\Lambda(\bar{\lambda}). \tag{2.7.41}$$

However, two λs that differ only by an overall phase have the same effect on v in Eq. (2.7.37), and so correspond to the same Lorentz transformation. It is therefore convenient to adjust the phase of the λs so that

$$\operatorname{Det}\lambda = 1, \tag{2.7.42}$$

which is consistent with Eq. (2.7.41). The 2×2 complex matrices with unit determinant form a group, known as $SL(2, \mathbb{C})$. (The SL stands for 'special linear', with 'special' denoting a unit determinant, while C stands for 'complex'.) The group elements depend on $4-1=3$ complex parameters, or 6 real parameters, the same number as the Lorentz group. However, $SL(2, \mathbb{C})$ is not the same as the Lorentz group; if λ is a matrix in $SL(2, \mathbb{C})$, then so is $-\lambda$, and both λ and $-\lambda$ produce the same Lorentz transformation in Eq. (2.7.37). Indeed, it is easy to see that the matrix

$$\lambda(\theta) = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}$$

produces a Lorentz transformation $\Lambda(\lambda(\theta))$ which is just a rotation by an angle θ around the three-axis, and hence $\lambda = -\mathbf{1}$ produces a rotation by an angle 2π. The Lorentz group is not the same as $SL(2, \mathbb{C})$, but rather[10] $SL(2, \mathbb{C})/\mathbb{Z}_2$, which is the group of complex 2×2 matrices with unit determinant, and with λ identified with $-\lambda$.

Now, what is the topology of the Lorentz group? By the polar decomposition theorem [12], any complex non-singular matrix λ may be written in the form

$$\lambda = ue^h,$$

where u is unitary and h is Hermitian

$$u^\dagger u = \mathbf{1}, \qquad h^\dagger = h.$$

Since $\operatorname{Det} u$ is a phase factor, and $\operatorname{Det}\exp h = \exp \operatorname{Tr} h$ is real and positive, the condition (2.7.42) requires both

$$\operatorname{Det} u = 1, \qquad \operatorname{Tr} h = 0.$$

(The factor u simply provides the rotation subgroup of the Lorentz group; if u is unitary then $\operatorname{Tr}(uvu^\dagger) = \operatorname{Tr} v$, so $V^0 = \frac{1}{2}\operatorname{Tr} v$ is left invariant by $\Lambda(u)$.) Furthermore, this decomposition is unique, so $SL(2, \mathbb{C})$ is topologically just the direct product (i.e., the set of pairs of points) of the space of all us and the space of all hs. Any Hermitian traceless 2×2 matrix h can be expressed as

$$h = \begin{pmatrix} c & a - ib \\ a + ib & -c \end{pmatrix}$$

[10] The group $\mathbb{Z}_2$ consists of just elements $+\mathbf{1}$ and $-\mathbf{1}$. In general, when we write G/H, with H an invariant subgroup of G, we mean the group G with elements g and gh identified if $g \in G$ and $h \in H$. The subgroup $\mathbb{Z}_2$ is trivially invariant because its elements commute with all elements of $SL(2, \mathbb{C})$.

with a, b, c real but otherwise unconstrained, so the space of all hs is topologically the same as ordinary three-dimensional flat space, $\mathbb{R}^3$. On the other hand, any unitary 2×2 matrix with unit determinant can be expressed as

$$u = \begin{pmatrix} d + ie & f + ig \\ -f + ig & d - ie \end{pmatrix}$$

with d, e, f, g subject to the single non-linear constraint

$$d^2 + e^2 + f^2 + g^2 = 1,$$

so the space $SU(2)$ of all us is topologically the same as S^3, the three-dimensional surface of a spherical ball in flat four-dimensional space. Thus $SL(2, \mathbb{C})$ is topologically the same as the direct product $\mathbb{R}^3 \times S^3$. This is simply connected: any curve connecting two points of $\mathbb{R}^3$ or S^3 can be deformed into any other, and the same is true of the direct product. (All spheres S_n except the circle S_1 are simply connected.) However, we are interested in $SL(2, \mathbb{C})/\mathbb{Z}_2$, not $SL(2, \mathbb{C})$. Identifying λ with $-\lambda$ is the same as identifying the unitary factors u and $-u$ (because e^h is always positive), so the Lorentz group has the topology of $\mathbb{R}^3 \times S^3/\mathbb{Z}_2$, where $S^3/\mathbb{Z}_2$ is the three-dimensional spherical surface with opposite points of the sphere identified. This is *not* simply connected; for instance, a path on S^3 from u to u' cannot be continuously deformed into a path on S^3 from u to $-u'$, even though these two paths link the same points of $S^3/\mathbb{Z}_2$. In fact, $S^3/\mathbb{Z}_2$ is *doubly* connected; the paths between any two points fall into two classes, depending on whether or not they involve an inversion $u \to -u$, and any path in one class can be deformed into another path of that class. An equivalent statement is that a double loop, that goes twice over the same path from any element back to itself, may be continuously contracted to a point. (As discussed in Appendix B, this is summarized mathematically in the statement that the fundamental group, or first homotopy group, of $S^3/\mathbb{Z}_2$ is $\mathbb{Z}_2$.) Similarly, the inhomogeneous Lorentz group has the same topology as $\mathbb{R}^4 \times \mathbb{R}^3 \times S^3/\mathbb{Z}_2$, and is therefore also doubly connected.

Because the Lorentz group (homogeneous or inhomogeneous) is not simply connected, it does have intrinsically projective representations. However, because the double loop that goes twice from $\mathbf{1}$ to Λ to $\Lambda\bar{\Lambda}$ and then back to $\mathbf{1}$ can be contracted to a point, we must have

$$\left[U(\Lambda)U(\bar{\Lambda})U^{-1}(\Lambda\bar{\Lambda})\right]^2 = \mathbf{1}$$

and hence the phase $e^{i\phi(\Lambda,\bar{\Lambda})}$ is just a sign

$$U(\Lambda)U(\bar{\Lambda}) = \pm U(\Lambda\bar{\Lambda}). \tag{2.7.43}$$

Likewise, for the inhomogeneous Lorentz group

$$U(\Lambda, a)U(\bar{\Lambda}, \bar{a}) = \pm U(\Lambda\bar{\Lambda}, \Lambda\bar{a} + a). \tag{2.7.44}$$

These 'representations up to a sign' are familiar; they are just the states of integer spin, for which the signs in Eqs. (2.7.43) and (2.7.44) are always $+1$, and the states of half-integer spin, for which these signs are $+1$ or -1 according to whether the path from $\mathbf{1}$ to

Λ to $\Lambda\bar{\Lambda}$ and then back to $\mathbf{1}$ is or is not contractible to a point. This difference arises because a rotation of 2π around the three-axis acting on a state with angular-momentum three-component σ produces a phase $e^{2\pi i\sigma}$, and thus has no effect on a state of integer spin and produces a sign change when acting on a state of half-integer spin. (These two cases correspond to the two irreducible representations of the first homotopy group, $\mathbb{Z}_2$.) Thus Eq. (2.7.43) or Eq. (2.7.44) imposes a superselection rule: we must not mix states of integer and half-integer spin.

For finite mass, the limitation to integer or half-integer spin was previously derived by purely algebraic means from the well-known representations of the generators of the little group, which here are just the angular-momentum matrices $J^{(j)}$ with j integer or half-integer. On the other hand, for zero mass the action of the little group on physical one-particle states is just a rotation around the momentum, and here there is no *algebraic* reason for a limitation to integer or half-integer helicity. There is, however, a *topological* reason: a rotation by an angle 4π around the momentum can be continuously deformed into no rotation at all, so the factor $\exp(4\pi i\sigma)$ must be unity, and hence σ must be an integer or half-integer.

Instead of working with projective representations and imposing a superselection rule, we can just as well expand the Lorentz group, taking it as $SL(2,\mathbb{C})$ itself, instead of $SL(2,\mathbb{C})/\mathbb{Z}_2$ as before. Ordinary rotation invariance forbids transitions between states of integer and half-integer total spin, so the only difference is that now the group is simply-connected, and it therefore has only ordinary representations, not projective representations, so that we cannot infer a superselection rule. This does not mean that we actually can prepare physical systems in linear combinations of states of integer and half-integer spin, but only that the observed Lorentz invariance of nature cannot be used to show that such superpositions are impossible.

Similar remarks apply to any symmetry group. If its Lie algebra involves central charges, then we can always expand the algebra to include generators that commute with anything, and whose eigenvalues are the central charges, just as we did when we added a mass operator to the Lie algebra of the Galilean group at the end of Section 2.4. The expanded Lie algebra is then, of course, free of central charges, so the part of the group near the identity has only ordinary representations, and does not require any superselection rule. Likewise, even though a Lie group G may not be simply connected, it can always be expressed as C/H, where C is a simply connected group known as the 'universal covering group' of G, and H is an invariant subgroup[11] of C. In general, we may just as well take the symmetry group as C instead of G, because there is no difference in their consequences, except that G implies a superselection rule, while C does not. In short, the issue of superselection rules is a bit of a red herring; *it may or it may not be possible to prepare physical systems in arbitrary superpositions of states, but one cannot settle the question by reference to symmetry principles, because whatever one thinks the symmetry group of nature may be,*

[11] The first homotopy group of C/H is H. We have seen that the covering group of the homogeneous Lorentz group is $SL(2,\mathbb{C})$, and the covering group of the three-dimensional rotation group is $SU(2)$. This connection with SL and SU groups is special to the case of three, four, or six dimensions; for general dimensions d the covering group of $SO(d)$ is given a special name, 'Spin(d)'.

there is always another group whose consequences are identical except for the absence of superselection rules.

Appendix A The Symmetry Representation Theorem

This appendix presents the proof of the fundamental theorem of Wigner [2] that any symmetry transformation can be represented on the Hilbert space of physical states by an operator that is either linear and unitary or antilinear and antiunitary. For our present purposes, the property of symmetry transformations on which we chiefly rely is that they are ray transformations T that preserve transition probabilities, in the sense that if $|\Psi_1\rangle$ and $|\Psi_2\rangle$ are state-vectors belonging to rays $\mathcal{R}_1$ and $\mathcal{R}_2$ then any state-vectors $|\Psi'_1\rangle$ and $|\Psi'_2\rangle$ belonging to the transformed rays $T\mathcal{R}_1$ and $T\mathcal{R}_2$ satisfy

$$\left|\langle\Psi'_1|\Psi'_2\rangle\right|^2 = \left|\langle\Psi_1|\Psi_2\rangle\right|^2. \tag{2.A.1}$$

We also require that a symmetry transformation should have an inverse that preserves transition probabilities in the same sense.

To start, consider some complete orthonormal set of state-vectors $|\Psi_k\rangle$ belonging to rays $\mathcal{R}_k$, with

$$\langle\Psi_k|\Psi_l\rangle = \delta_{kl}, \tag{2.A.2}$$

and let $|\Psi'_k\rangle$ be some arbitrary choice of state-vectors belonging to the transformed rays $T\mathcal{R}_k$. From Eq. (2.A.1), we have

$$\left|\langle\Psi'_k|\Psi'_l\rangle\right|^2 = \left|\langle\Psi_k|\Psi_l\rangle\right|^2 = \delta_{kl}.$$

But $\langle\Psi'_k|\Psi'_k\rangle$ is automatically real and positive, so this requires that it should have the value unity, and therefore

$$\langle\Psi'_k|\Psi'_l\rangle = \delta_{kl}. \tag{2.A.3}$$

It is easy to see that these transformed states $|\Psi'_k\rangle$ also form a complete set, for if there were any non-zero state-vector $|\Psi'\rangle$ that was orthogonal to all of the $|\Psi'_k\rangle$, then the inverse transform of the ray to which $|\Psi'\rangle$ belongs would consist of non-zero state-vectors $|\Psi''\rangle$ for which, for all k:

$$\left|\langle\Psi_k|\Psi''\rangle\right|^2 = \left|\langle\Psi'_k|\Psi'\rangle\right|^2 = 0,$$

which is impossible since the $|\Psi_k\rangle$ were assumed to form a complete set.

We must now establish a phase convention for the states $|\Psi'_k\rangle$. For this purpose, we single out one of the $|\Psi_k\rangle$, say $|\Psi_1\rangle$, and consider the state-vectors

$$|Y_k\rangle = \frac{1}{\sqrt{2}}\left[|\Psi_1\rangle + |\Psi_k\rangle\right] \tag{2.A.4}$$

belonging to some ray $\mathcal{S}_k$, with $k \neq 1$. Any state-vector $|Y'_k\rangle$ belonging to the transformed ray $T\mathcal{S}_k$ may be expanded in the state-vectors $|\Psi'_l\rangle$,

$$|Y'_k\rangle = \sum_l c_{kl}|\Psi'_l\rangle.$$

From Eq. (2.A.1) we have

$$|c_{kk}| = |c_{k1}| = \frac{1}{\sqrt{2}}$$

and for $l \neq k$ and $l \neq 1$:

$$c_{kl} = 0.$$

For any given k, by an appropriate choice of phase of the two state-vectors $|Y'_k\rangle$ and $|\Psi'_k\rangle$, we can clearly adjust the phases of the two non-zero coefficients c_{kk} and c_{k1} so that both coefficients are just $1/\sqrt{2}$. From now on, the state-vectors $|Y'_k\rangle$ and $|\Psi'_k\rangle$ chosen in this way will be denoted $U|Y_k\rangle$ and $U|\Psi_k\rangle$. As we have seen,

$$U\frac{1}{\sqrt{2}}\big[|\Psi_k\rangle + |\Psi_1\rangle\big] = U|Y_k\rangle = \frac{1}{\sqrt{2}}\big[U|\Psi_k\rangle + U|\Psi_1\rangle\big]. \tag{2.A.5}$$

However, it still remains to define $U|\Psi\rangle$ for general state-vectors $|\Psi\rangle$.

Now consider an arbitrary state-vector $|\Psi\rangle$ belonging to an arbitrary ray $\mathcal{R}$, and expand it in the $|\Psi_k\rangle$:

$$|\Psi\rangle = \sum_k C_k|\Psi_k\rangle. \tag{2.A.6}$$

Any state $|\Psi'\rangle$ that belongs to the transformed ray $T\mathcal{R}$ may similarly be expanded in the complete orthonormal set $U|\Psi_k\rangle$:

$$|\Psi'\rangle = \sum_k C'_k U|\Psi_k\rangle. \tag{2.A.7}$$

The equality of $|\langle\Psi_k|\Psi\rangle|^2$ and $|\langle U\Psi_k|\Psi'\rangle|^2$ tells us that for all k (including $k = 1$):

$$|C_k|^2 = |C'_k|^2, \tag{2.A.8}$$

while the equality of $|\langle Y_k|\Psi\rangle|^2$ and $|\langle UY_k|\Psi'\rangle|^2$ tells us that for all $k \neq 1$:

$$|C_k + C_1|^2 = |C'_k + C'_1|^2. \tag{2.A.9}$$

The ratio of Eqs. (2.A.9) and (2.A.8) yields the formula

$$\mathrm{Re}(C_k/C_1) = \mathrm{Re}(C'_k/C'_1) \tag{2.A.10}$$

which with Eq. (2.A.8) also requires

$$\mathrm{Im}(C_k/C_1) = \pm\,\mathrm{Im}(C'_k/C'_1) \tag{2.A.11}$$

and therefore either

$$C_k/C_1 = C'_k/C'_1, \tag{2.A.12}$$

or else

$$C_k/C_1 = (C'_k/C'_1)^*. \tag{2.A.13}$$

Furthermore, we can show that the same choice must be made for each k. (This step in the proof was omitted by Wigner.) To see this, suppose that for some k, we have

$C_k/C_1 = C'_k/C'_1$, while for some $l \neq k$, we have instead $C_l/C_1 = (C'_l/C'_1)^*$. Suppose also that both ratios are complex, so that these are really different cases. (This incidentally requires that $k \neq 1$ and $l \neq 1$, as well as $k \neq l$.) We will show that this is impossible.

Define a state-vector $|\Phi\rangle = \frac{1}{\sqrt{3}}[|\Psi_1\rangle + |\Psi_k\rangle + |\Psi_l\rangle]$. Since all the ratios of the coefficients in this state-vector are real, we must get the same ratios in any state-vector $|\Phi'\rangle$ belonging to the transformed ray:

$$|\Phi'\rangle = \frac{\alpha}{\sqrt{3}}\big[U|\Psi_1\rangle + U|\Psi_k\rangle + U|\Psi_l\rangle\big],$$

where α is a phase factor with $|\alpha| = 1$. But then the equality of the transition probabilities $|\langle\Phi|\Psi\rangle|$ and $|\langle\Phi'|\Psi'\rangle|$ requires that

$$\left|1 + \frac{C'_k}{C'_1} + \frac{C'_l}{C'_1}\right|^2 = \left|1 + \frac{C_k}{C_1} + \frac{C_l}{C_1}\right|^2$$

and hence

$$\left|1 + \frac{C_k}{C_1} + \frac{C^*_l}{C^*_1}\right|^2 = \left|1 + \frac{C_k}{C_1} + \frac{C_l}{C_1}\right|^2 .$$

This is only possible if

$$\mathrm{Re}\left(\frac{C_k C^*_l}{C_1 C^*_1}\right) = \mathrm{Re}\left(\frac{C_k C_l}{C_1 C_1}\right)$$

or, in other words, if

$$\mathrm{Im}\left(\frac{C_k}{C_1}\right)\mathrm{Im}\left(\frac{C_l}{C_1}\right) = 0.$$

Hence either C_k/C_1 or C_l/C_1 must be real for any pair k, l, in contradiction with our assumptions. We see then that for a given symmetry transformation T applied to a given state-vector $\sum_k C_k|\Psi_k\rangle$, we must have either Eq. (2.A.12) for all k, or else Eq. (2.A.13) for all k.

Wigner ruled out the second possibility, Eq. (2.A.13), because as he showed any symmetry transformation for which this possibility is realized would have to involve a reversal in the time coordinate, and in the proof he presented he was considering only symmetries like rotations that do not affect the direction of time. Here we are treating symmetries involving time-reversal on the same basis as all other symmetries, so we will have to consider that, for each symmetry T and state-vector $\sum_k C_k|\Psi_k\rangle$, either Eq. (2.A.12) or Eq. (2.A.13) may apply. Depending on which of these alternatives is realized, we will now define $U|\Psi\rangle$ to be the particular one of the state-vectors $|\Psi'\rangle$ belonging to the ray $T\mathcal{R}$ with phase chosen so that either $C_1 = C'_1$ or $C_1 = {C'_1}^*$, respectively. Then either

$$U\left(\sum_k C_k|\Psi_k\rangle\right) = \sum_k C_k U|\Psi_k\rangle \tag{2.A.14}$$

or else

$$U\left(\sum_k C_k|\Psi_k\rangle\right) = \sum_k C^*_k U|\Psi_k\rangle. \tag{2.A.15}$$

It remains to be proved that for a given symmetry transformation, we must make the same choice between Eqs. (2.A.14) and (2.A.15) for arbitrary values of the coefficients C_k. Suppose that Eq. (2.A.14) applies for a state-vector $\sum_k A_k|\Psi_k\rangle$ while Eq. (2.A.15) applies for a state-vector $\sum_k B_k|\Psi_k\rangle$. Then the invariance of transition probabilities requires that

$$\left|\sum_k B_k^* A_k\right|^2 = \left|\sum_k B_k A_k\right|^2$$

or equivalently

$$\sum_{kl} \mathrm{Im}(A_k^* A_l)\,\mathrm{Im}(B_k^* B_l) = 0. \tag{2.A.16}$$

We cannot rule out the possibility that Eq. (2.A.16) may be satisfied for a pair of state-vectors $\sum_k A_k|\Psi_k\rangle$ and $\sum_k B_k|\Psi_k\rangle$ belonging to different rays. However, for any pair of such state-vectors, with neither A_k nor B_k all of the same phase (so that Eqs. (2.A.14) and (2.A.15) are not the same), we can always find a third state-vector $\sum_k C_k|\Psi_k\rangle$ for which[12]

$$\sum_{kl} \mathrm{Im}(C_k^* C_l)\,\mathrm{Im}(A_k^* A_l) \neq 0 \tag{2.A.17}$$

and also

$$\sum_{kl} \mathrm{Im}(C_k^* C_l)\,\mathrm{Im}(B_k^* B_l) \neq 0. \tag{2.A.18}$$

As we have seen, it follows from Eq. (2.A.17) that the same choice between Eqs. (2.A.14) and (2.A.15) must be made for $\sum_k A_k|\Psi_k\rangle$ and $\sum_k C_k|\Psi_k\rangle$, and it follows from Eq. (2.A.18) that the same choice between Eqs. (2.A.14) and (2.A.15) must be made for $\sum_k B_k|\Psi_k\rangle$ and $\sum_k C_k|\Psi_k\rangle$, so the same choice between Eqs. (2.A.14) and (2.A.15) must also be made for the two state-vectors $\sum_k A_k|\Psi_k\rangle$ and $\sum_k B_k|\Psi_k\rangle$ with which we started. We have thus shown that for a given symmetry transformation T either all state-vectors satisfy Eq. (2.A.14) or else they all satisfy Eq. (2.A.15).

It is now easy to show that as we have defined it, the quantum mechanical operator U is either linear and unitary or antilinear and antiunitary. First, suppose that Eq. (2.A.14) is satisfied for all state-vectors $\sum_k C_k|\Psi_k\rangle$. Any two state-vectors $|\Psi\rangle$ and $|\Phi\rangle$ may be expanded as

$$|\Psi\rangle = \sum_k A_k|\Psi_k\rangle, \qquad |\Phi\rangle = \sum_k B_k|\Psi_k\rangle$$

[12]If for some pair k, l both $A_k^* A_l$ and $B_k^* B_l$ are complex, then choose all C's to vanish except for C_k and C_l, and choose these two coefficients to have different phases. If $A_k^* A_l$ is complex but $B_k^* B_l$ is real for some pair k, l, then there must be some other pair m, n (possibly with either m or n but not both equal to k or l) for which $B_m^* B_n$ is complex. If also $A_m^* A_n$ is complex, then choose all C's to vanish except for C_m and C_n, and choose these two coefficients to have different phase. If $A_m^* A_n$ is real, then choose all C's to vanish except for C_k, C_l, C_m, and C_n, and choose these four coefficients all to have different phases. The case where $B_k^* B_l$ is complex but $A_k^* A_l$ is real is handled in just the same way.

and so, using Eq. (2.A.14),

$$U\big(\alpha|\Psi\rangle+\beta|\Phi\rangle\big)=U\sum_k(\alpha A_k+\beta B_k)|\Psi_k\rangle=\sum_k(\alpha A_k+\beta B_k)U|\Psi_k\rangle$$
$$=\alpha\sum_k A_kU|\Psi_k\rangle+\beta\sum_k B_kU|\Psi_k\rangle.$$

Using Eq. (2.A.14) again, this gives

$$U\big(\alpha|\Psi\rangle+\beta|\Phi\rangle\big)=\alpha U|\Psi\rangle+\beta U|\Phi\rangle, \tag{2.A.19}$$

so U is *linear.* Also, using Eqs. (2.A.2) and (2.A.3), the scalar product of the transformed states is

$$\langle U\Psi|U\Phi\rangle=\sum_{kl}A_k^*B_l\langle U\Psi_k|U\Psi_l\rangle=\sum_k A_k^*B_k,$$

and hence

$$\langle U\Psi|U\Phi\rangle=\langle\Psi|\Phi\rangle, \tag{2.A.20}$$

so U is *unitary.*

The case of a symmetry that satisfies Eq. (2.A.15) for all state-vectors may be dealt with in much the same way. The reader can probably supply the arguments without help, but since antilinear operators may be unfamiliar, we shall give the details here anyway. Suppose that Eq. (2.A.15) is satisfied for all state-vectors $\sum_k C_k|\Psi_k\rangle$. Any two state-vectors $|\Psi\rangle$ and $|\Phi\rangle$ may be expanded as before, and so:

$$U\big(\alpha|\Psi\rangle+\beta|\Phi\rangle\big)=U\sum_k(\alpha A_k+\beta B_k)|\Psi_k\rangle$$
$$=\sum_k(\alpha^*A_k^*+\beta^*B_k^*)U|\Psi_k\rangle$$
$$=\alpha^*\sum_k A_k^*U|\Psi_k\rangle+\beta^*\sum_k B_k^*U|\Psi_k\rangle.$$

Using Eq. (2.A.15) again, this gives

$$U\big(\alpha|\Psi\rangle+\beta|\Phi\rangle\big)=\alpha^*U|\Psi\rangle+\beta^*U|\Phi\rangle, \tag{2.A.21}$$

so U is *antilinear.* Also, using Eqs. (2.A.2) and (2.A.3), the scalar product of the transformed states is

$$\langle U\Psi|U\Phi\rangle=\sum_{kl}A_kB_l^*\langle U\Psi_k|U\Psi_l\rangle=\sum_k A_kB_k^*,$$

and hence

$$\langle U\Psi|U\Phi\rangle=\langle\Psi|\Phi\rangle^*, \tag{2.A.22}$$

so U is *antiunitary.*

Appendix B Group Operators and Homotopy Classes

In this appendix we shall prove the theorem stated in Section 2.7, that the phases of the operators $U(T)$ for finite symmetry transformations T may be chosen so that these operators form a representation of the symmetry group, rather than a projective representation, provided (a) the generators of the group can be defined so that there are no central charges in the Lie algebra, and (b) the group is simply connected. We shall also comment on the projective representations encountered for groups that are not simply connected, and their relation to the homotopy classes of the group.

To prove this theorem, let us recall the method by which we construct the operators corresponding to symmetry transformations. As described in Section 2.2, we introduce a set of real variables θ^a to parameterize these transformations, in such a way that the transformations satisfy the composition rule (2.2.15):

$$T(\bar{\theta})T(\theta) = T\big(f(\bar{\theta}, \theta)\big).$$

We want to construct operators $U(T(\theta)) \equiv U[\theta]$ that satisfy the corresponding condition[13]

$$U[\bar{\theta}]U[\theta] = U\big[f(\bar{\theta}, \theta)\big]. \tag{2.B.1}$$

To do this, we lay down arbitrary 'standard' paths $\Theta^a_\theta(s)$ in group parameter space, running from the origin to each point θ, with $\Theta^a_\theta(0) = 0$ and $\Theta^a_\theta(1) = \theta^a$, and define $U_\theta(s)$ along each such path by the differential equation

$$\frac{d}{ds}U_\theta(s) = it_a U_\theta(s) h^a{}_b\big(\Theta_\theta(s)\big)\frac{d\Theta^b_\theta(s)}{ds} \tag{2.B.2}$$

with the initial condition

$$U_\theta(0) = \mathbf{1}, \tag{2.B.3}$$

where

$$[h^{-1}]^a{}_b(\theta) \equiv \left[\frac{\partial f^a(\bar{\theta}, \theta)}{\partial \bar{\theta}^b}\right]_{\bar{\theta}=0}. \tag{2.B.4}$$

We are eventually going to identify the operators $U[\theta]$ with $U_\theta(1)$, but first we must establish some of the properties of $U_\theta(s)$.

In order to check the composition rule, consider two points θ_1 and θ_2, and define a path γ that runs from 0 to θ_1 and thence to $f(\theta_2, \theta_1)$:

$$\Theta^a_\gamma(s) \equiv \begin{cases} \Theta^a_{\theta_1}(2s), & 0 \le s \le \frac{1}{2}, \\ f^a\big(\Theta_{\theta_2}(2s-1), \theta_1\big), & \frac{1}{2} \le s \le 1. \end{cases} \tag{2.B.5}$$

At the end of the first segment, we are at $U_\gamma(\frac{1}{2}) = U_{\theta_1}(1)$. To evaluate $U_\gamma(s)$ along the second segment, we need the derivative of $f^a(\Theta_{\theta_2}(2s-1), \theta_1)$. For this purpose, we use the fundamental associativity condition:

$$f^a\big(f(\theta_3, \theta_2), \theta_1\big) = f^a\big(\theta_3, f(\theta_2, \theta_1)\big). \tag{2.B.6}$$

[13] Square brackets are used here to distinguish U operators constructed as functions of the group parameters from those expressed as functions of the group transformations themselves.

Matching the coefficients of θ_3^c in the limit $\theta_3 \to 0$ yields the result:

$$\frac{\partial f^a(\theta_2, \theta_1)}{\partial \theta_2^b} h^b{}_c\big(f(\theta_2, \theta_1)\big) = h^a{}_c(\theta_2). \tag{2.B.7}$$

Along the second segment, the differential equation (2.B.2) for $U_\gamma(s)$ is thus the same as the differential equation for $U_{\theta_2}(2s-1)$. They satisfy different initial conditions, but $U_\gamma(s)U_{\theta_1}^{-1}(1)$ also satisfies the same differential equation as $U_{\theta_2}(2s-1)$, and in addition the same initial condition: at $s=\frac{1}{2}$, both are unity. We therefore conclude that for $\frac{1}{2} \le s \le 1$,

$$U_\gamma(s)U_{\theta_1}^{-1}(1) = U_{\theta_2}(2s-1)$$

and in particular

$$U_\gamma(1) = U_{\theta_2}(1)U_{\theta_1}(1). \tag{2.B.8}$$

However, this does *not* say that $U_\theta(1)$ satisfies the desired composition rule (2.B.1), because although the path $\Theta_\gamma(s)$ runs from $\theta^a = 0$ to $\theta^a = f^a(\theta_2, \theta_1)$, in general it will not be the same as whatever 'standard' path $\Theta_{f(\theta_2,\theta_1)}$ we have chosen to run directly from $\theta^a = 0$ to $\theta^a = f^a(\theta_2, \theta_1)$. We need to show that $U_\theta(1)$ is independent of the path from 0 to θ in order to be able to identify $U[\theta]$ as $U_\theta(1)$.

For this purpose, consider the variation δU of $U_\theta(s)$ produced by a variation $\delta\Theta(s)$ in the path from 0 to θ. Taking the variation of Eq. (2.B.2) gives the differential equation

$$\begin{aligned}\frac{d}{ds}\delta U = {} & it_a \delta U\, h^a{}_b(\Theta)\frac{d\Theta^b}{ds} + it_a U h^a{}_{b,c}(\Theta)\delta\Theta^c \frac{d\Theta^b}{ds} \\ & + it_a U h^a{}_b(\Theta)\frac{d\delta\Theta^b}{ds},\end{aligned}$$

where $h^a{}_{b,c} \equiv \partial h^a{}_b/\partial\theta^c$. Using the Lie commutation relations (2.2.22) (without central charges) and rearranging a bit, this gives

$$\begin{aligned}\frac{d}{ds}\big(U^{-1}\delta U\big) = {} & \frac{d}{ds}\big(iU^{-1}t_a U h^a{}_b \delta\Theta^b\big) \\ & + iU^{-1}t_a U \delta\Theta^b \frac{d\Theta^c}{ds}\left(h^a{}_{c,b} - h^a{}_{b,c} + C^a{}_{ed}h^e{}_b h^d{}_c\right).\end{aligned} \tag{2.B.9}$$

However, by taking the limit $\theta_3, \theta_2 \to 0$ in the associativity condition (2.B.6), we find for all θ:

$$h(\theta)^a{}_{b,c} = -f^a{}_{de}h(\theta)^d{}_b h(\theta)^e{}_c, \tag{2.B.10}$$

where $f^a{}_{de}$ is the coefficient defined by (2.2.19). Antisymmetrizing in b and c shows that the last term in Eq. (2.B.9) vanishes

$$h^a{}_{c,b} - h^a{}_{b,c} + C^a{}_{ed}h^e{}_b h^d{}_c = 0. \tag{2.B.11}$$

Eq. (2.B.9) thus tells us that the quantity

$$U^{-1}\delta U - iU^{-1}t_a U h^a{}_b \delta\Theta^b$$

is constant along the path $\theta(s)$. It follows that $U_\theta(1)$ is stationary under any infinitesimal variation of the path that leaves the endpoints $\Theta(0) = 0$ and $\Theta(1) = \theta$ (and $U_\theta(0) = \mathbf{1}$) fixed. But assumption (b) tells us that any path from $\Theta(0) = 0$ to $\Theta(1) = \theta$ can be continuously deformed into any other, so we may now regard $U_\theta(1)$ as a path-independent function of θ alone:

$$U_\theta(1) \equiv U[\theta]. \tag{2.B.12}$$

In particular, since the path γ leads from 0 to $f(\theta_2, \theta_1)$, we have

$$U_\gamma(1) = U\big[f(\theta_2, \theta_1)\big] \tag{2.B.13}$$

so that Eq. (2.B.8) shows that $U[\theta]$ satisfies the group multiplication law (2.B.1), as was to be proved.

Now that we have constructed a non-projective representation $U[\theta]$, it remains to prove that any projective representation $\widetilde{U}[\theta]$ of the same group with the same representation generators t_a can only differ from $U[\theta]$ by a phase:

$$\widetilde{U}[\theta] = e^{i\alpha(\theta)} U[\theta]$$

so that the phase ϕ in the multiplication law for $\widetilde{U}[\theta]$:

$$\widetilde{U}[\theta']\widetilde{U}[\theta] = e^{i\phi(\theta',\theta)}\widetilde{U}\big[f(\theta', \theta)\big]$$

can be removed by a simple change of phase of $\widetilde{U}[\theta]$. To see this, consider the operator

$$U[\theta]^{-1}U[\theta']^{-1}\widetilde{U}[\theta']\widetilde{U}[\theta] = U\big[f(\theta', \theta)\big]^{-1}\widetilde{U}\big[f(\theta', \theta)\big]e^{i\phi(\theta',\theta)}.$$

Because $U[\theta]$ and $\widetilde{U}[\theta]$ have the same generators, the derivative of the left-hand side with respect to θ'^a vanishes at $\theta' = 0$, and so

$$0 = \frac{\partial}{\partial\theta^b}\Big\{U[\theta]^{-1}\widetilde{U}[\theta]\Big\} + i\phi_b(\theta)U[\theta]^{-1}\widetilde{U}[\theta],$$

where

$$\phi_b(\theta) \equiv h^a{}_b(\theta)\left[\frac{\partial}{\partial\theta'^a}\phi(\theta', \theta)\right]_{\theta'=0}.$$

Differentiating this result with respect to θ^c and antisymmetrizing in b and c gives immediately

$$0 = \frac{\partial\phi_b(\theta)}{\partial\theta^c} - \frac{\partial\phi_c(\theta)}{\partial\theta^b}.$$

A familiar theorem [13] tells us that in a simply connected space, this requires that ϕ_b is just a gradient of some function β:

$$\phi_b(\theta) = \frac{\partial\beta(\theta)}{\partial\theta^b}.$$

Thus the quantity $U[\theta]^{-1}\widetilde{U}[\theta]e^{i\beta(\theta)}$ is actually constant in θ. Setting it equal to its value at $\theta = 0$, we see that $\widetilde{U}$ is just proportional to U:

$$\widetilde{U}[\theta] = U[\theta]\exp\big(-i\beta(\theta) + i\beta(0)\big)$$

as claimed above.

* * *

The above analysis provides some information about the nature of the phase factors that can appear in the group multiplication law when the Lie algebra is free of central charges but the group is not simply connected. Suppose that the path γ from zero to θ to $f(\bar{\theta}, \theta)$ cannot be deformed into the standard path we have chosen to go from zero to $f(\bar{\theta}, \theta)$, or in other words, that the loop from zero to θ to $f(\bar{\theta}, \theta)$ and then back to zero is not continuously deformable to a point. Then $U[f(\theta_2, \theta_1)]^{-1}U[\theta_2]U[\theta_1]$ can be a phase factor $\exp(i\phi(\theta_2, \theta_1)) \neq \mathbf{1}$, but ϕ will be the same for all other loops into which this can be continuously deformed. The set consisting of all loops that start and end at the origin and that can be continuously deformed into a given loop is known as the *homotopy class* [14] of that loop; we have thus seen that $\phi(\theta_2, \theta_1)$ depends only on the homotopy class of the loop from zero to θ to $f(\bar{\theta}, \theta)$ and then back to zero. The set of homotopy classes forms a group; the 'product' of the homotopy class for loops $\mathcal{L}_1$ and $\mathcal{L}_2$ is the homotopy class of the loop formed by going around $\mathcal{L}_1$ and then $\mathcal{L}_2$; the 'inverse' of the homotopy class of the loop $\mathcal{L}$ is the homotopy class of the loop obtained by going around $\mathcal{L}$ in the opposite direction; and the 'identity' is the homotopy class of loops that can be deformed into a point at the origin. This group is known as the *first homotopy group* or *fundamental group* of the space in question. It is easy to see that the phase factors form a representation of this group: if going around loop $\mathcal{L}$ gives a phase factor $e^{i\phi}$, and going around loop $\bar{\mathcal{L}}$ gives a phase factor $e^{i\bar{\phi}}$, then going around both loops gives a phase factor $e^{i\phi}e^{i\bar{\phi}}$. Hence we can catalog all the possible types of projective representations of a given group $\mathcal{G}$ (with no central charges) if we know the one-dimensional representations of the first homotopy group of the parameter space of $\mathcal{G}$. Homotopy groups will be discussed in greater detail in Volume II.

Appendix C Inversions and Degenerate Multiplets

It is usually assumed that the inversions T and P take one-particle states into other one-particle states of the same species, perhaps with phase factors that depend on the particle species. In Section 2.6 we noted in passing that inversions might act in a more complicated way than this on degenerate multiplets of one-particle states, a possibility that seems to have been first suggested by Wigner [15] in 1964. This appendix will explore generalized versions of the inversion operators, in which finite matrices appear in place of the inversion phases, but without making some of Wigner's limiting assumptions.

Let us start with time-reversal. Wigner limited the possible action of the inversion operators by assuming that their squares are proportional to the unit operator. Because T is antiunitary, it is easy to see that the corresponding proportionality factor for T^2 can only be ± 1, perhaps with different signs for subspaces separated by superselection rules. When the sign for T^2 on the space of states with even or odd values of $2j$ is opposite to the sign $(-1)^{2j}$ found in Section 2.6, the physical states involved must furnish representations of the operator T that are more complicated than that assumed so far. But if we are willing to admit this possibility, there does not seem to be any good reason to impose Wigner's condition that T^2 is proportional to unity. It is not convincing to appeal to the structure of the extended Poincaré group; the only useful definition of any of the inversion operators is one that makes the operator exactly or approximately conserved, and this may not be the definition that makes T^2 proportional to the unit operator.

To explore more general possibilities for time-reversal, let us assume that on a massive one-particle state it has the action

$$T|\mathbf{p}, \sigma, n\rangle = (-1)^{j-\sigma} \sum_m \Theta_{mn} |-\mathbf{p}, -\sigma, m\rangle, \tag{2.C.1}$$

where $\mathbf{p}$, j, and σ are the particle's momentum, spin, and spin z-component, and n, m are indices labelling members of a degenerate multiplet of particle species. (The appearance of the factor $(-1)^{j-\sigma}$ and the reversal of $\mathbf{p}$ and σ are deduced in the same way as in Section 2.6.) The matrix Θ_{mn} is unknown, except that because T is antiunitary, Θ must be unitary.

Now let us see how we can simplify this transformation by an appropriate choice of basis for the one-particle states. Defining new states by the unitary transformation $|\mathbf{p}, \sigma, n\rangle' = \sum_m \mathcal{U}_{mn} |\mathbf{p}, \sigma, m\rangle$, we find the same transformation (2.C.1), with the matrix Θ_{mn} changed to

$$\Theta' = \mathcal{U}^{-1} \Theta \mathcal{U}^*. \tag{2.C.2}$$

We cannot in general make Θ' diagonal by such a choice of basis of the one-particle states, as we could if T were unitary. But we can instead make it block-diagonal, with the blocks either 1×1 phases, or 2×2 matrices of the form

$$\begin{pmatrix} 0 & e^{i\phi/2} \\ e^{-i\phi/2} & 0 \end{pmatrix}, \tag{2.C.3}$$

where the ϕ are various real phases.

(Here is the proof. First, note that Eq. (2.C.2) gives

$$\Theta' \Theta'^* = \mathcal{U}^{-1} \Theta \Theta^* \mathcal{U}.$$

This is a unitary transformation, so it can be chosen to diagonalize the unitary matrix $\Theta\Theta^*$. Assuming this to have been done, and dropping primes, we have

$$\Theta = D\Theta^T, \tag{2.C.4}$$

where D is a unitary diagonal matrix, say with phases $e^{i\phi_n}$ along the main diagonal. One immediate consequence is that the diagonal component Θ_{nn} vanishes unless $e^{i\phi_n} = 1$. Furthermore, if $e^{i\phi_n} = 1$ but $e^{i\phi_m} \neq 1$, then Eq. (2.C.4) tells us that $\Theta_{nm} = \Theta_{mn} = 0$. By listing first all rows and columns for which $e^{i\phi_n} = 1$, the matrix Θ is put in the form

$$\Theta = \begin{pmatrix} \mathcal{A} & 0 \\ 0 & \mathcal{B} \end{pmatrix}, \tag{2.C.5}$$

where $\mathcal{A}$ is symmetric as well as unitary, and the diagonal elements of $\mathcal{B}$ all vanish. Because $\mathcal{A}$ is symmetric, it can be expressed as the exponential of a symmetric anti-Hermitian matrix, so it can be diagonalized by a transformation (2.C.2) acting only on $\mathcal{A}$, with the corresponding submatrix of $\mathcal{U}$ real and hence orthogonal. It is therefore only necessary to consider the submatrix $\mathcal{B}$ that connects the rows and columns for which $e^{i\phi_n} \neq 0$. For $n \neq m$, Eq. (2.C.4) gives $\Theta_{nm} = e^{i\phi_n}\Theta_{mn}$ and $\Theta_{mn} = e^{i\phi_m}\Theta_{nm}$, so $\Theta_{nm} = e^{i\phi_n}e^{i\phi_m}\Theta_{nm}$ and also $\Theta_{mn} = e^{i\phi_n}e^{i\phi_m}\Theta_{mn}$. Hence $\Theta_{nm} = \Theta_{mn} = 0$ unless $e^{i\phi_n}e^{i\phi_m} = 1$. If we list first all rows and columns of $\mathcal{B}$ with a given phase $e^{i\phi_1} \neq 1$, and then all rows and columns with the opposite phase, and then all rows and columns with some other phase $e^{i\phi_2} \neq 1$ not equal to $e^{\pm i\phi_1}$, and then all rows and columns with opposite phase, and so on, the matrix $\mathcal{B}$ becomes of block diagonal form

$$\mathcal{B} = \begin{pmatrix} \mathcal{B}_1 & 0 & \cdots \\ 0 & \mathcal{B}_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}, \tag{2.C.6}$$

where

$$\mathcal{B}_i = \begin{pmatrix} 0 & e^{i\phi_i/2}\mathcal{C}_i \\ e^{-i\phi_i/2}\mathcal{C}_i^T & 0 \end{pmatrix}. \tag{2.C.7}$$

Furthermore, the unitarity of Θ and hence of $\mathcal{B}$ requires that $\mathcal{C}_i\mathcal{C}_i^\dagger = \mathcal{C}_i^\dagger\mathcal{C}_i = \mathbf{1}$, and hence $\mathcal{C}_i$ is square and unitary. By applying a transformation (2.C.2) with $\mathcal{U}$ block-diagonal in the same sense as Θ, and with the matrix in the ith block of form

$$\begin{pmatrix} V_i & 0 \\ 0 & W_i \end{pmatrix}$$

with V_i and W_i unitary, the submatrices $\mathcal{C}_i$ are subjected to the transformations $\mathcal{C}_i \longrightarrow V_i^{-1}\mathcal{C}_iW_i^*$, so we can clearly choose this transformation to make $\mathcal{C}_i = \mathbf{1}$. This establishes a correspondence between pairs of individual rows and columns within each block with phases $e^{i\phi_i}$ and $e^{-i\phi_i}$. To put the matrix $\mathcal{B}$ into block-diagonal form with 2×2 blocks of form (2.C.3), it is now only necessary to rearrange the rows and columns so that within the ith block we list rows and columns with phase $e^{i\phi_i}$ alternating with the corresponding rows and columns with phase $e^{-i\phi_i}$.)

It is important to note that where $e^{i\phi} \neq 1$, it is not possible to choose states to diagonalize the time-reversal transformation. If we have a pair of states $|\mathbf{p}, \sigma, \pm\rangle$ on which T acts with a matrix (2.C.3), then

$$T|\mathbf{p}, \sigma, \pm\rangle = e^{\pm i\phi/2}(-1)^{j-\sigma}|-\mathbf{p}, -\sigma, \mp\rangle. \tag{2.C.8}$$

Then on an arbitrary linear combination of these states, time-reversal gives

$$T\big(c_+|\mathbf{p}, \sigma, +\rangle + c_-|\mathbf{p}, \sigma, -\rangle\big) = (-1)^{j-\sigma}\big(e^{i\phi/2}c_+^*|-\mathbf{p}, -\sigma, -\rangle + e^{-i\phi/2}c_-^*|-\mathbf{p}, -\sigma, +\rangle\big).$$

For $c_+|\mathbf{p}, \sigma, +\rangle + c_-|\mathbf{p}, \sigma, -\rangle$ to be transformed under T by a phase λ, it is necessary that

$$e^{i\phi/2}c_+^* = \lambda c_-, \qquad e^{-i\phi/2}c_-^* = \lambda c_+.$$

But combining these equations gives $e^{\pm i\phi/2}c_\pm^* = |\lambda|^2 c_\pm^* e^{\mp i\phi/2}$, which is impossible unless either $c_+ = c_- = 0$ or $e^{i\phi}$ is unity. Thus for $e^{i\phi} \neq 1$, time-reversal invariance imposes a two-fold degeneracy on these states, beyond that associated with their spin.

Of course, if there is an additional 'internal' symmetry operator S which subjects these states to the transformation

$$S|\mathbf{p},\sigma,\pm\rangle = e^{\pm i\phi/2}|-\mathbf{p},\sigma,\mp\rangle,$$

then we can redefine the time-reversal operator as $T' \equiv S^{-1}T$, and this operator would not mix the states $|\mathbf{p},\sigma,\pm\rangle$ with one another. It is only in the case where no such internal symmetry exists that we can attribute the doubling of particle states to time-reversal itself.

Let's come back now to the question of the square of T. Repeating the transformation (2.C.8) gives

$$T^2|\mathbf{p},\sigma,\pm\rangle = (-1)^{2j}e^{\mp i\phi}|\mathbf{p},\sigma,\pm\rangle. \tag{2.C.9}$$

If we were to assume with Wigner that T^2 is proportional to the unit operator, then we would have to have $e^{i\phi} = e^{-i\phi}$, and since the phase is then real it would have to be $+1$ or -1. The choice $e^{i\phi} = -1$ would still require a two-fold degeneracy of one-particle states beyond that associated with their spin, and under Wigner's assumptions *all* particles would show this doubling. But there is no reason not to take a general phase ϕ in Eq. (2.C.8), one that may vanish for some particles and not for others. Thus the fact that observed particles do not show the extra two-fold degeneracy does not rule out the possibility that others might.

We may also consider the possibility of more complicated representations of the parity operator P, with

$$P|\mathbf{p},\sigma,n\rangle = \sum_m \Pi_{mn}|-\mathbf{p},\sigma,m\rangle \tag{2.C.10}$$

with a unitary but otherwise unconstrained matrix Π. Unlike the case of time-reversal, here we may always diagonalize this matrix by a choice of basis for the states. But this choice of basis may not be the one in which time-reversal acts simply, so, in principle, P and T together can impose additional degeneracies that would not be required by P or T alone.

As discussed in Chapter 5, any quantum field theory is expected to respect a symmetry known as CPT, which acts on one-particle states as

$$CPT|\mathbf{p},\sigma,n\rangle = (-1)^{j-\sigma}|\mathbf{p},-\sigma,n^c\rangle, \tag{2.C.11}$$

where n^c denotes the antiparticle (or 'charge-conjugate') of particle n. No phases or matrices are allowed in this transformation (though of course we could always introduce such phases or matrices by combining CPT with good internal symmetries.) It follows that

$$(CPT)^2|\mathbf{p},\sigma,n\rangle = (-1)^{2j}|\mathbf{p},-\sigma,n\rangle, \tag{2.C.12}$$

so the possibility suggested by Wigner of a sign $-(-1)^{2j}$ in the action of $(CPT)^2$ does not arise in quantum field theory.

To the extent that T is a good symmetry of some class of phenomena, so is the inversion $CP \equiv (CPT)T^{-1}$. For the states that transform under T in the conventional way

$$T|\mathbf{p},\sigma,n\rangle \propto |-\mathbf{p},-\sigma,n\rangle, \tag{2.C.13}$$

the CP operator also acts conventionally

$$CP|\mathbf{p},\sigma,n\rangle \propto |{-\mathbf{p}},\sigma,n^c\rangle. \tag{2.C.14}$$

The operator $C \equiv CPP^{-1}$ then just interchanges particles and antiparticles

$$C|\mathbf{p},\sigma,n\rangle \propto |\mathbf{p},\sigma,n^c\rangle. \tag{2.C.15}$$

On the other hand, where T has the unconventional representation (2.C.8), Eq. (2.C.11) gives

$$CP|\mathbf{p},\sigma,\pm\rangle = e^{\mp i\phi/2}|{-\mathbf{p}},\sigma,\mp^c\rangle \tag{2.C.16}$$

In particular, it is possible that the degeneracy indicated by the label $\pm$ may be the same as the particle–antiparticle degeneracy, so that the antiparticle (as defined by CPT) of the state $|\Psi_\pm\rangle$ is $|\Psi_\mp\rangle$. In this case, CP would have the unconventional property of *not* interchanging particles and antiparticles. As far as these particles are concerned, CP and T would be what are usually called P and CT. But this is not merely a matter of definition; on other particles CP and T would still have their usual effect.

No examples are known of particles that furnish unconventional representations of inversions, so these possibilities will not be pursued further here. From now on, the inversions will be assumed to have the conventional action assumed in Section 2.6.

Problems

1. Suppose that observer $\mathcal{O}$ sees a W-boson (spin one and mass $m \neq 0$) with momentum $\mathbf{p}$ in the y-direction and spin z-component σ. A second observer $\mathcal{O}'$ moves relative to the first with velocity $\mathbf{v}$ in the z-direction. How does $\mathcal{O}'$ describe the W state?

2. Suppose that observer $\mathcal{O}$ sees a photon with momentum $\mathbf{p}$ in the y-direction and polarization vector in the z-direction. A second observer $\mathcal{O}'$ moves relative to the first with velocity $\mathbf{v}$ in the z-direction. How does $\mathcal{O}'$ describe the same photon?

3. Derive the commutation relations for the generators of the Galilean group directly from the group multiplication law (without using our results for the Lorentz group). Include the most general set of central charges that cannot be eliminated by redefinition of the group generators.

4. Show that the operators $P_\mu P^\mu$ and $W_\mu W^\mu$ commute with all Lorentz transformation operators $U(\Lambda, a)$, where $W_\mu \equiv \epsilon_{\mu\nu\rho\lambda}J^{\nu\rho}P^\lambda$.

5. Consider physics in two space and one time dimensions, assuming invariance under a 'Lorentz' group $SO(2,1)$. How would you describe the spin states of a single massive particle? How do they behave under Lorentz transformations? What about the inversions P and T?

6. As in Problem 5, consider physics in two space and one time dimensions, assuming invariance under a 'Lorentz' group $O(2,1)$. How would you describe the spin states of a single massless particle? How do they behave under Lorentz transformations? What about the inversions P and T?

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3 Scattering Theory

The general principles of relativistic quantum mechanics described in the previous chapter have so far been applied here only to states of a single stable particle. Such one-particle states by themselves are not very exciting — it is only when two or more particles interact with each other that anything interesting can happen. But experiments do not generally follow the detailed course of events in particle interactions. Rather, the paradigmatic experiment (at least in nuclear or elementary particle physics) is one in which several particles approach each other from a macroscopically large distance, and interact in a microscopically small region, after which the products of the interaction travel out again to a macroscopically large distance. The physical states before and after the collision consist of particles that are so far apart that they are effectively non-interacting, so they can be described as direct products of the one-particle states discussed in the previous chapter. In such an experiment, all that is measured is the probability distribution, or 'cross-sections', for transitions between the initial and final states of distant and effectively non-interacting particles. This chapter will outline the formalism [1] used for calculating these probabilities and cross-sections.

3.1 'In' and 'Out' States

A state consisting of several non-interacting particles may be regarded as one that transforms under the Poincaré (inhomogeneous Lorentz) group as a direct product of one-particle states. To label the one-particle states we use their four-momenta p^μ, spin z-component (or, for massless particles, helicity) σ, and, since we now may be dealing with more than one species of particle, an additional discrete label n for the particle type, which includes a specification of its mass, spin, charge, etc. The general transformation rule is

$$\begin{aligned} U(\Lambda, a)|p_1, \sigma_1, n_1;\, p_2, \sigma_2, n_2;\, \ldots\rangle = {}& \exp\bigl[-ia_\mu\bigl(p_1^\mu + p_2^\mu + \cdots\bigr)\bigr] \sqrt{\frac{(\Lambda p_1)^0 (\Lambda p_2)^0 \cdots}{p_1^0 p_2^0 \cdots}} \\ & \times \sum_{\sigma_1' \sigma_2' \cdots} D^{(j_1)}_{\sigma_1' \sigma_1}\bigl(W(\Lambda, p_1)\bigr) D^{(j_2)}_{\sigma_2' \sigma_2}\bigl(W(\Lambda, p_2)\bigr) \cdots \\ & \times |\Lambda p_1, \sigma_1', n_1;\, \Lambda p_2, \sigma_2', n_2;\, \ldots\rangle. \end{aligned} \tag{3.1.1}$$

where $W(\Lambda, p)$ is the Wigner rotation (2.5.10), and $D^{(j)}_{\sigma'\sigma}(W)$ are the conventional $(2j+1)$-dimensional unitary matrices representing the three-dimensional rotation group. (This is for massive particles; for any massless particle, the matrix $D^{(j)}_{\sigma'\sigma}(W(\Lambda, p))$ is replaced with $\delta_{\sigma'\sigma} \exp(i\sigma\theta(\Lambda, p))$, where θ is the angle defined by Eq. (2.5.43).) The states are normalized as in Eq. (2.5.19):

$$\begin{aligned} & \bigl\langle p_1', \sigma_1', n_1';\, p_2', \sigma_2', n_2';\, \ldots \bigm| p_1, \sigma_1, n_1;\, p_2, \sigma_2, n_2;\, \ldots\bigr\rangle \\ & \quad = \delta^3(\mathbf{p}_1' - \mathbf{p}_1)\delta_{\sigma_1'\sigma_1}\delta_{n_1' n_1}\delta^3(\mathbf{p}_2' - \mathbf{p}_2)\delta_{\sigma_2'\sigma_2}\delta_{n_2' n_2}\cdots \\ & \qquad \pm \text{permutations}. \end{aligned} \tag{3.1.2}$$

with the term '$\pm$ permutations' included to take account of the possibility that it is some permutation of the particle types $n_1', n_2', \ldots$ that are of the same species of the particle types $n_1, n_2, \ldots$. (As discussed more fully in Chapter 4, its sign is -1 if this permutation includes an odd permutation of half-integer spin particles, and otherwise $+1$. This will not be important in the work of the present chapter.)

We often use an abbreviated notation, letting one Greek letter, say α, stand for the whole collection $p_1, \sigma_1, n_1;\ p_2, \sigma_2, n_2;\ \ldots$. In this notation, Eq. (3.1.2) is written simply

$$\langle \alpha' | \alpha \rangle = \delta(\alpha' - \alpha), \tag{3.1.3}$$

with $\delta(\alpha' - \alpha)$ standing for the sum of products of delta functions and Kronecker deltas appearing on the right-hand side of Eq. (3.1.2). Also, in summing over states, we write

$$\int d\alpha \cdots \equiv \sum_{n_1 \sigma_1 n_2 \sigma_2 \cdots} \int d^3p_1\, d^3p_2 \cdots . \tag{3.1.4}$$

In particular, the completeness relation for states normalized as in Eq. (3.1.3) reads

$$|\Psi\rangle = \int d\alpha\, |\alpha\rangle\langle\alpha|\Psi\rangle. \tag{3.1.5}$$

The transformation rule (3.1.1) is only possible for particles that for one reason or another are not interacting. Setting $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu$ and $a^\mu = (\tau, 0, 0, 0)$, for which $U(\Lambda, a) = \exp(iH\tau)$, Eq. (3.1.1) requires among other things that $|\alpha\rangle$ be an energy eigenstate,

$$H|\alpha\rangle = E_\alpha|\alpha\rangle, \tag{3.1.6}$$

with an energy equal to the sum of the one-particle energies,

$$E_\alpha = p_1^0 + p_2^0 + \cdots , \tag{3.1.7}$$

and with no interaction terms, terms that would involve more than one particle at a time.

On the other hand, the transformation rule (3.1.1) does apply in scattering processes at times $t \to \mp\infty$. As explained at the beginning of this chapter, in the typical scattering experiment we start with particles at time $t \to -\infty$ so far apart that they are not yet interacting, and end with particles at $t \to +\infty$ so far apart that they have ceased interacting. We therefore have not one but two sets of states that transform as in Eq. (3.1.1): the 'in' and 'out' states[14] $|\alpha;\ \text{in}\rangle$ and $|\alpha;\ \text{out}\rangle$ will be found to contain the particles described by the label α if observations are made at $t \to -\infty$ or $t \to +\infty$, respectively.

Note how this definition is framed. To maintain manifest Lorentz invariance, in the formalism we are using here, state-vectors do not change with time — a state-vector $|\Psi\rangle$ describes the whole spacetime history of a system of particles. (This is known as the *Heisenberg picture*, in distinction with the Schrödinger picture, where the operators are constant and the states change with time.) Thus we do *not* say that $|\alpha;\ \text{in}\rangle$ and $|\alpha;\ \text{out}\rangle$ are the limits at $t \to -\infty$ and $t \to +\infty$ of a time-dependent state-vector $|\Psi(t)\rangle$.

However, implicit in the definition of the states is a choice of the inertial frame from which the observer views the system; different observers see *equivalent* state-vectors, but

[14] The labels '+' and '−' for 'in' and 'out' states may seem backward, but they seem to have become traditional. They arise from the signs in Eq. (3.1.16).

not the *same* state-vector. In particular, suppose that a standard observer O sets his or her clock so that $t = 0$ is at some time during the collision process, while some other observer O' at rest with respect to the first uses a clock set so that $t' = 0$ is at a time $t = \tau$; that is, the two observers' time coordinates are related by $t' = t - \tau$. Then if O sees the system to be in a state $|\Psi\rangle$, O' will see the system in a state $U(\mathbf{1}, -\tau)|\Psi\rangle = \exp(-iH\tau)|\Psi\rangle$. Thus the appearance of the state long before or long after the collision (in whatever basis is used by O) is found by applying a time-translation operator $\exp(-iH\tau)$ with $\tau \to -\infty$ or $\tau \to +\infty$, respectively. Of course, if the state is really an energy eigenstate, then it cannot be localized in time — the operator $\exp(-iH\tau)$ yields an inconsequential phase factor $\exp(-iE_\alpha\tau)$. Therefore, we must consider wave-packets, superpositions $\int d\alpha\, g(\alpha)|\alpha\rangle$ of states, with an amplitude $g(\alpha)$ that is non-zero and smoothly varying over some finite range ΔE of energies. The 'in' and 'out' states are defined so that the superposition

$$\exp(-iH\tau)\int d\alpha\, g(\alpha)|\alpha;\,\text{in/out}\rangle = \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha;\,\text{in/out}\rangle$$

has the appearance of a corresponding superposition of free-particle states for $\tau \ll -1/\Delta E$ or $\tau \gg +1/\Delta E$, respectively.

To make this concrete, suppose we can divide the time-translation generator H into two terms, a free-particle Hamiltonian H_0 and an interaction H_{int},

$$H = H_0 + H_{\text{int}}, \tag{3.1.8}$$

in such a way that H_0 has eigenstates $|\alpha\rangle$ that have the same appearance as the eigenstates $|\alpha;\,\text{in}\rangle$ and $|\alpha;\,\text{out}\rangle$ of the complete Hamiltonian,

$$H_0|\alpha\rangle = E_\alpha|\alpha\rangle, \tag{3.1.9}$$

$$\langle\alpha'|\alpha\rangle = \delta(\alpha' - \alpha). \tag{3.1.10}$$

Note that H_0 is assumed here to have the same spectrum as the full Hamiltonian H. This requires that the masses appearing in H_0 be the physical masses that are actually measured, which are not necessarily the same as the 'bare' mass terms appearing in H; the difference if there is any must be included in the interaction H_{int}, not H_0. Also, any relevant bound states in the spectrum of H should be introduced into H_0 as if they were elementary particles.[15]

The 'in' and 'out' states can now be defined as eigenstates of H, not H_0,

$$H|\alpha;\,\text{in/out}\rangle = E_\alpha|\alpha;\,\text{in/out}\rangle, \tag{3.1.11}$$

which satisfy the condition

$$\begin{aligned}\int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha;\,\text{in}\rangle &\longrightarrow \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha\rangle \qquad (\tau \to -\infty),\\ \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha;\,\text{out}\rangle &\longrightarrow \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha\rangle \qquad (\tau \to +\infty).\end{aligned} \tag{3.1.12}$$

[15] Alternatively, in non-relativistic problems we can include the binding potential in H_0. In the application of this method to 'rearrangement collisions,' where some bound states appear in the initial state but not the final state, or vice-versa, one must use a different split of H into H_0 and H_{int} in the initial and final states.

Eq. (3.1.12) can be rewritten as the requirement that:

$$\exp(-iH\tau)\int d\alpha\, g(\alpha)|\alpha;\,\text{in/out}\rangle \longrightarrow \exp(-iH_0\tau)\int d\alpha\, g(\alpha)|\alpha\rangle$$

for $\tau \to -\infty$ or $\tau \to +\infty$, respectively. This is sometimes rewritten as a formula for the 'in' and 'out' states:

$$\begin{aligned} |\alpha;\,\text{in}\rangle &= \Omega(-\infty)|\alpha\rangle, \\ |\alpha;\,\text{out}\rangle &= \Omega(+\infty)|\alpha\rangle, \end{aligned} \tag{3.1.13}$$

where

$$\Omega(\tau) \equiv \exp(+iH\tau)\exp(-iH_0\tau). \tag{3.1.14}$$

However, it should be kept in mind that $\Omega(\mp\infty)$ in Eq. (3.1.13) gives meaningful results only when acting on a smooth superposition of energy eigenstates.

One immediate consequence of the definition (3.1.12) is that the 'in' and 'out' states are normalized just like the free-particle states. To see this, note that since the left-hand side of Eq. (3.1.12) is obtained by letting the unitary operator $\exp(-iH\tau)$ act on a time-independent state, its norm is independent of time, and therefore equals the norm of its limit for $\tau \to \mp\infty$, i.e., the norm of the right-hand side of Eq. (3.1.12):

$$\begin{aligned} &\int d\alpha\, d\beta\, \exp\bigl(-i(E_\alpha - E_\beta)\tau\bigr) g(\alpha) g^*(\beta) \bigl\langle \beta;\,\text{in/out} \bigm| \alpha;\,\text{in/out}\bigr\rangle \\ &\qquad = \int d\alpha\, d\beta\, \exp\bigl(-i(E_\alpha - E_\beta)\tau\bigr) g(\alpha) g^*(\beta) \langle\beta|\alpha\rangle. \end{aligned}$$

Since this is supposed to be true for all smooth functions $g(\alpha)$, the scalar products must be equal:

$$\bigl\langle \beta;\,\text{in/out} \bigm| \alpha;\,\text{in/out}\bigr\rangle = \langle\beta|\alpha\rangle = \delta(\beta - \alpha). \tag{3.1.15}$$

It is useful for some purposes to have an explicit though formal solution of the energy eigenvalue equation (3.1.11) satisfying the conditions (3.1.12). For this purpose, write Eq. (3.1.11) as

$$(E_\alpha - H_0)|\alpha;\,\text{in/out}\rangle = H_{\text{int}}|\alpha;\,\text{in/out}\rangle.$$

The operator $E_\alpha - H_0$ is not invertible; it annihilates not only the free-particle state $|\alpha\rangle$, but also the continuum of other free-particle states $|\beta\rangle$ of the same energy. Since the 'in' and 'out' states become just $|\alpha\rangle$ for $H_{\text{int}} \to 0$, we tentatively write the formal solutions as $|\alpha\rangle$ plus a term proportional to H_{int}:

$$\begin{aligned} |\alpha;\,\text{in}\rangle &= |\alpha\rangle + (E_\alpha - H_0 + i\epsilon)^{-1} H_{\text{int}}|\alpha;\,\text{in}\rangle, \\ |\alpha;\,\text{out}\rangle &= |\alpha\rangle + (E_\alpha - H_0 - i\epsilon)^{-1} H_{\text{int}}|\alpha;\,\text{out}\rangle, \end{aligned} \tag{3.1.16}$$

or, expanding in a complete set of free-particle states,

$$\begin{aligned} |\alpha;\,\text{in}\rangle &= |\alpha\rangle + \int d\beta\, \frac{T^{\text{in}}_{\beta\alpha}|\beta\rangle}{E_\alpha - E_\beta + i\epsilon}, \\ |\alpha;\,\text{out}\rangle &= |\alpha\rangle + \int d\beta\, \frac{T^{\text{out}}_{\beta\alpha}|\beta\rangle}{E_\alpha - E_\beta - i\epsilon}, \end{aligned} \tag{3.1.17}$$

$$T^{\text{in/out}}_{\beta\alpha} \equiv \langle\beta|H_{\text{int}}|\alpha;\ \text{in/out}\rangle, \tag{3.1.18}$$

with ϵ a positive infinitesimal quantity, inserted to give meaning to the reciprocal of $E_\alpha - H_0$. These are known as the *Lippmann–Schwinger equations* [1a]. We shall use Eq. (3.1.17) at the end of the next section to give a slightly less unrigorous proof of the orthonormality of the 'in' and 'out' states.

It remains to be shown that Eq. (3.1.17), with a $+i\epsilon$ or $-i\epsilon$ in the denominator, satisfies the condition (3.1.12) for an 'in' or an 'out' state, respectively. For this purpose, consider the superpositions

$$|\Psi^{\text{in/out}}_g(t)\rangle \equiv \int d\alpha\, e^{-iE_\alpha t} g(\alpha)|\alpha;\ \text{in/out}\rangle, \tag{3.1.19}$$

$$|\Phi_g(t)\rangle \equiv \int d\alpha\, e^{-iE_\alpha t} g(\alpha)|\alpha\rangle. \tag{3.1.20}$$

We want to show that $|\Psi^{\text{in}}_g(t)\rangle$ and $|\Psi^{\text{out}}_g(t)\rangle$ approach $|\Phi_g(t)\rangle$ for $t \to -\infty$ and $t \to +\infty$, respectively. Using Eq. (3.1.17) in Eq. (3.1.19) gives

$$|\Psi^{\text{in/out}}_g(t)\rangle = |\Phi_g(t)\rangle + \int d\alpha \int d\beta\, \frac{e^{-iE_\alpha t} g(\alpha) T^{\text{in/out}}_{\beta\alpha}|\beta\rangle}{E_\alpha - E_\beta \pm i\epsilon}, \tag{3.1.21}$$

where the upper and lower signs refer to 'in' and 'out', respectively. Let us recklessly interchange the order of integration, and consider first the integrals

$$\mathcal{I}^{\text{in/out}}_\beta \equiv \int d\alpha\, \frac{e^{-iE_\alpha t} g(\alpha) T^{\text{in/out}}_{\beta\alpha}}{E_\alpha - E_\beta \pm i\epsilon}.$$

For $t \to -\infty$, we can close the contour of integration for the energy variable E_α in the upper half-plane with a large semi-circle, with the contribution from this semi-circle killed by the factor $\exp(-iE_\alpha t)$, which is exponentially small for $t \to -\infty$ and $\operatorname{Im} E_\alpha > 0$. The integral is then given by a sum over the singularities of the integrand in the upper half-plane. The functions $g(\alpha)$ and $T^{\text{in/out}}_{\beta\alpha}$ may, in general, be expected to have some singularities at values of E_α with finite positive imaginary parts, but just as for the large semi-circle, their contribution is exponentially damped for $t \to -\infty$. (Specifically, $-t$ must be much greater than both the time-uncertainty in the wave-packet $g(\alpha)$ and the duration of the collision, which respectively govern the location of the singularities of $g(\alpha)$ and $T^{\text{in/out}}_{\beta\alpha}$ in the complex E_α plane.) This leaves the singularity in $(E_\alpha - E_\beta \pm i\epsilon)^{-1}$, which is in the upper half-plane for $\mathcal{I}^{\text{out}}_\beta$ but not $\mathcal{I}^{\text{in}}_\beta$. We conclude then that $\mathcal{I}^{\text{in}}_\beta$ vanishes for $t \to -\infty$. In the same way, for $t \to +\infty$ we must close the contour of integration in the lower half-plane, and so $\mathcal{I}^{\text{out}}_\beta$ vanishes in this limit. We conclude that $|\Psi^{\text{in/out}}_g(t)\rangle$ approaches $|\Phi_g(t)\rangle$ for $t \to \mp\infty$, in agreement with the defining condition (3.1.12).

* * *

For future use, we note a convenient representation of the factor $(E_\alpha - E_\beta \pm i\epsilon)^{-1}$ in Eq. (3.1.17). In general, we can write

$$(E \pm i\epsilon)^{-1} = \frac{\mathcal{P}_\epsilon}{E} \mp i\pi\delta_\epsilon(E), \tag{3.1.22}$$

where

$$\frac{\mathcal{P}_\epsilon}{E} \equiv \frac{E}{E^2+\epsilon^2}, \tag{3.1.23}$$

$$\delta_\epsilon(E) \equiv \frac{\epsilon}{\pi(E^2+\epsilon^2)}. \tag{3.1.24}$$

The function (3.1.23) is just $1/E$ for $|E| \gg \epsilon$, and vanishes for $E \to 0$, so for $\epsilon \to 0$ it behaves just like the 'principal value function' $\mathcal{P}/E$, which allows us to give meaning to integrals of $1/E$ times any smooth function of E, by excluding an infinitesimal interval around $E = 0$. The function (3.1.24) is of order ϵ for $|E| \gg \epsilon$, and gives unity when integrated over all E, so in the limit $\epsilon \to 0$ it behaves just like the familiar delta function $\delta(E)$. With this understanding, we can drop the label ϵ in Eq. (3.1.22), and write simply

$$(E \pm i\epsilon)^{-1} = \frac{\mathcal{P}}{E} \mp i\pi\delta(E). \tag{3.1.25}$$

3.2 The S-Matrix

An experimentalist generally prepares a state to have a definite particle content at $t \to -\infty$, and then measures what this state looks like at $t \to +\infty$. If the state is prepared to have a particle content α for $t \to -\infty$, then it is the 'in' state $|\alpha;\,\text{in}\rangle$, and if it is found to have the particle content β at $t \to +\infty$, then it is the 'out' state $|\beta;\,\text{out}\rangle$. The probability amplitude for the transition $\alpha \to \beta$ is thus the scalar product

$$S_{\beta\alpha} = \big\langle \beta;\,\text{out}\;\big|\;\alpha;\,\text{in}\big\rangle. \tag{3.2.1}$$

This array of complex amplitudes is known as the S-matrix [2]. If there were no interactions then 'in' and 'out' states would be the same, and then $S_{\beta\alpha}$ would be just $\delta(\alpha-\beta)$. The rate for a reaction $\alpha \to \beta$ is thus proportional to $|S_{\beta\alpha} - \delta(\alpha-\beta)|^2$. We shall see in detail in Section 3.4 what $S_{\beta\alpha}$ has to do with measured rates and cross-sections.

Perhaps it should be stressed that 'in' and 'out' states do not inhabit two different Hilbert spaces. They differ only in how they are labelled: by their appearance either at $t \to -\infty$ or $t \to +\infty$. Any 'in' state can be expanded as a sum of 'out' states, with expansion coefficients given by the S-matrix (3.2.1).

Since $S_{\beta\alpha}$ is the matrix connecting two sets of orthonormal states, it must be unitary. To see this in greater detail, apply the completeness relation (3.1.5) to the 'out' states, and write

$$\int d\beta\, S^*_{\beta\gamma} S_{\beta\alpha} = \int d\beta\, \big\langle \gamma;\,\text{in}\;\big|\;\beta;\,\text{out}\big\rangle\big\langle \beta;\,\text{out}\;\big|\;\alpha;\,\text{in}\big\rangle = \big\langle \gamma;\,\text{in}\;\big|\;\alpha;\,\text{in}\big\rangle.$$

Using (3.1.15), this gives

$$\int d\beta\, S^*_{\beta\gamma} S_{\beta\alpha} = \delta(\gamma-\alpha), \tag{3.2.2}$$

or in brief, $S^\dagger S = \mathbf{1}$. In the same way, completeness for the 'in' states gives[16]

$$\int d\beta\, S_{\gamma\beta} S^*_{\alpha\beta} = \delta(\gamma-\alpha), \tag{3.2.3}$$

[16] An alternative proof is given at the end of this section. Note that for infinite 'matrices,' the unitarity conditions $S^\dagger S = \mathbf{1}$ and $SS^\dagger = \mathbf{1}$ are not equivalent.

or, in other words, $SS^\dagger = \mathbf{1}$.

It is often convenient instead of dealing with the S-matrix to work with an operator S, defined to have matrix elements between *free-particle* states equal to the corresponding elements of the S-matrix:

$$\langle\beta|S|\alpha\rangle \equiv S_{\beta\alpha}. \tag{3.2.4}$$

The explicit though highly formal expression (3.1.13) for the 'in' and 'out' states yields a formula for the S-operator:

$$S = \Omega(+\infty)^\dagger\Omega(-\infty) = U(+\infty, -\infty), \tag{3.2.5}$$

where

$$U(\tau, \tau_0) \equiv \Omega(\tau)^\dagger\Omega(\tau_0) = \exp(iH_0\tau)\exp\big(-iH(\tau-\tau_0)\big)\exp(-iH_0\tau_0). \tag{3.2.6}$$

This will be used in the next section to examine the Lorentz invariance of the S-matrix, and in Sec. 3.5 to derive a formula for the S-matrix in time-dependent perturbation theory.

The methods of the previous section can be used to derive a useful alternative formula for the S-matrix. Let's return to Eq. (3.1.21) for the 'in' state $|\Psi_g^{\rm in}\rangle$, but this time take $t \to +\infty$. We must now close the contour of integration for E_α in the *lower* half-E_α-plane, and although as before the singularities in $T_{\beta\alpha}^{\rm in}$ and $g(\alpha)$ make no contribution for $t \to +\infty$, we now do pick up a contribution from the singular factor $(E_\alpha - E_\beta + i\epsilon)^{-1}$. The contour runs from $E_\alpha = -\infty$ to $E_\alpha = +\infty$, and then back to $E_\alpha = -\infty$ on a large semi-circle in the lower half-plane, so it circles the singularity in a *clockwise* direction. By the method of residues, this contribution to the integral over E_α is given by the value of the integrand at $E_\alpha = E_\beta - i\epsilon$, times a factor $-2i\pi$. That is, in the limit $\epsilon \to 0+$, for $t \to +\infty$ the integral over α in (3.1.21) has the asymptotic behavior

$$\mathcal{I}_\beta^{\rm in} \longrightarrow -2i\pi e^{-iE_\beta t}\int d\alpha\,\delta(E_\alpha - E_\beta)g(\alpha)T_{\beta\alpha}^{\rm in},$$

and hence, for $t \to +\infty$,

$$|\Psi_g^{\rm in}(t)\rangle \longrightarrow \int d\beta\, e^{-iE_\beta t}|\beta\rangle\left[g(\beta) - 2i\pi\int d\alpha\,\delta(E_\alpha - E_\beta)g(\alpha)T_{\beta\alpha}^{\rm in}\right].$$

But expanding (3.1.19) for $|\Psi_g^{\rm in}\rangle$ in a complete set of 'out' states gives

$$|\Psi_g^{\rm in}(t)\rangle = \int d\alpha\, e^{-iE_\alpha t}g(\alpha)\int d\beta\,|\beta;\,{\rm out}\rangle S_{\beta\alpha}.$$

Since $S_{\beta\alpha}$ contains a factor $\delta(E_\beta - E_\alpha)$, this may be rewritten

$$|\Psi_g^{\rm in}(t)\rangle = \int d\beta\,|\beta;\,{\rm out}\rangle e^{-iE_\beta t}\int d\alpha\, g(\alpha)S_{\beta\alpha},$$

and, using the defining property (3.1.12) for 'out' states, this has the asymptotic behavior for $t \to +\infty$

$$|\Psi_g^{\rm in}(t)\rangle \longrightarrow \int d\beta\,|\beta\rangle e^{-iE_\beta t}\int d\alpha\, g(\alpha)S_{\beta\alpha}.$$

Comparing this with our previous result, we find

$$\int d\alpha\, g(\alpha)S_{\beta\alpha} = g(\beta) - 2i\pi \int d\alpha\, \delta(E_\alpha - E_\beta)g(\alpha)T^{\rm in}_{\beta\alpha},$$

or in other words

$$S_{\beta\alpha} = \delta(\beta - \alpha) - 2i\pi\delta(E_\alpha - E_\beta)T^{\rm in}_{\beta\alpha}. \tag{3.2.7}$$

This suggests a simple approximation for the S-matrix: for a weak interaction $H_{\rm int}$, we can neglect the difference between 'in' and free-particle states in (3.1.18), in which case Eq. (3.2.7) gives

$$S_{\beta\alpha} \simeq \delta(\beta - \alpha) - 2i\pi\delta(E_\alpha - E_\beta)\langle\beta|H_{\rm int}|\alpha\rangle. \tag{3.2.8}$$

This is known as the *Born approximation* [3]. Higher-order terms are discussed in Section 3.5.

* * *

We can use the Lippmann–Schwinger equations (3.1.16) for the 'in' and 'out' states to give a proof [4] of the orthonormality of these states and the unitarity of the S-matrix, as well as Eq. (3.2.7), without having to deal with limits at $t \to \mp\infty$. First, by using (3.1.16) on either the left- or right-hand side of the matrix element $\langle\beta;\text{ in/out}|H_{\rm int}|\alpha;\text{ in/out}\rangle$ and equating the results, we find that

$$\begin{aligned}&\big\langle\beta;\text{ in/out}\,\big|\,H_{\rm int}\,\big|\,\alpha\big\rangle + \big\langle\beta;\text{ in/out}\,\big|\,H_{\rm int}(E_\alpha - H_0 \pm i\epsilon)^{-1}H_{\rm int}\,\big|\,\alpha;\text{ in/out}\big\rangle\\ &\quad= \big\langle\beta\,\big|\,H_{\rm int}\,\big|\,\alpha;\text{ in/out}\big\rangle + \big\langle\beta;\text{ in/out}\,\big|\,H_{\rm int}(E_\beta - H_0 \mp i\epsilon)^{-1}H_{\rm int}\,\big|\,\alpha;\text{ in/out}\big\rangle.\end{aligned}$$

Summing over a complete set $|\gamma\rangle$ of intermediate states, this gives the equation:

$$\begin{aligned}T^{\text{in/out}\,*}_{\alpha\beta} - T^{\text{in/out}}_{\beta\alpha} &= -\int d\gamma\, T^{\text{in/out}\,*}_{\gamma\beta}T^{\text{in/out}}_{\gamma\alpha}\\ &\qquad\times\left([E_\alpha - E_\gamma \pm i\epsilon]^{-1} - [E_\beta - E_\gamma \mp i\epsilon]^{-1}\right).\end{aligned} \tag{3.2.9}$$

To prove the orthonormality of the 'in' and 'out' states, divide Eq. (3.2.9) by $E_\alpha - E_\beta \pm 2i\epsilon$. This gives

$$\left(\frac{T^{\rm in/out}_{\alpha\beta}}{E_\beta - E_\alpha \mp 2i\epsilon}\right)^* + \frac{T^{\rm in/out}_{\beta\alpha}}{E_\alpha - E_\beta \pm 2i\epsilon} = -\int d\gamma \left(\frac{T^{\rm in/out}_{\gamma\beta}}{E_\beta - E_\alpha \mp i\epsilon}\right)^* \frac{T^{\rm in/out}_{\gamma\alpha}}{E_\alpha - E_\beta \pm i\epsilon}.$$

The 2ϵs in the denominators on the left-hand side can be replaced with ϵs, since the only important thing is that these are positive infinitesimals. We see then that $\delta(\beta - \alpha) + T^{\rm in/out}_{\beta\alpha}/(E_\beta - E_\alpha \pm i\epsilon)$ is unitary. With (3.1.17), this is just the statement that the in and out states form two orthonormal sets of state-vectors. The unitarity of the S-matrix can be proved in a similar fashion by multiplying (3.2.9) with $\delta(E_\beta - E_\alpha)$ instead of $(E_\alpha - E_\beta \pm 2i\epsilon)^{-1}$.

3.3 Symmetries of the S-Matrix

In this section we will consider both what is meant by the invariance of the S-matrix under various symmetries, and what are the conditions on the Hamiltonian that will ensure such invariance properties.

Lorentz Invariance

For any proper orthochronous Lorentz transformation $x \to \Lambda x + a$, we may define a unitary operator $U(\Lambda, a)$ by specifying that it acts as in Eq. (3.1.1) on either the 'in' or the 'out' states. When we say that a theory is Lorentz-invariant, we mean that the same operator $U(\Lambda, a)$ acts as in (3.1.1) on both 'in' and 'out' states. Since the operator $U(\Lambda, a)$ is unitary, we may write

$$S_{\beta\alpha} = \langle \beta; \mathrm{out} | \alpha; \mathrm{in} \rangle = \big\langle U(\Lambda, a)\beta; \mathrm{out} \;\big|\; U(\Lambda, a)\alpha; \mathrm{in} \big\rangle,$$

so using (3.1.1), we obtain the Lorentz invariance (actually, covariance) property of the S-matrix: for arbitrary Lorentz transformations $\Lambda^\mu{}_\nu$ and translations a^μ,

$$\begin{aligned}
&S_{p_1'\sigma_1'n_1';p_2'\sigma_2'n_2';\ldots,\,p_1\sigma_1 n_1;p_2\sigma_2 n_2;\ldots} \\
&= \exp\big(ia_\mu(p_1^\mu + p_2^\mu + \cdots - p_1'^\mu - p_2'^\mu - \cdots)\big) \\
&\quad \times \sqrt{\frac{(\Lambda p_1)^0 (\Lambda p_2)^0 \cdots (\Lambda p_1')^0 (\Lambda p_2')^0 \cdots}{p_1^0 p_2^0 \cdots p_1'^0 p_2'^0 \cdots}} \\
&\quad \times \sum_{\bar\sigma_1 \bar\sigma_2 \ldots} D^{(j_1)}_{\bar\sigma_1 \sigma_1}(W(\Lambda, p_1))\, D^{(j_2)}_{\bar\sigma_2 \sigma_2}(W(\Lambda, p_2)) \cdots \\
&\quad \times \sum_{\bar\sigma_1' \bar\sigma_2' \ldots} D^{(j_1')*}_{\bar\sigma_1' \sigma_1'}\big(W(\Lambda, p_1')\big)\, D^{(j_2')*}_{\bar\sigma_2' \sigma_2'}\big(W(\Lambda, p_2')\big) \cdots \\
&\quad \times S_{\Lambda p_1'\bar\sigma_1'n_1';\Lambda p_2'\bar\sigma_2'n_2';\ldots,\,\Lambda p_1\bar\sigma_1 n_1;\Lambda p_2\bar\sigma_2 n_2;\ldots}\,.
\end{aligned} \tag{3.3.1}$$

(Primes are used to distinguish final from initial particles; bars are used to distinguish summation variables.) In particular, since the left-hand side is independent of a^μ, so must be the right-hand side, and so the S-matrix vanishes unless the four-momentum is conserved. We can therefore write the part of the S-matrix that represents actual interactions among the particles in the form:

$$S_{\beta\alpha} - \delta(\beta - \alpha) = -2\pi i M_{\beta\alpha} \delta^4(p_\beta - p_\alpha). \tag{3.3.2}$$

(However, as we will see in the next chapter, the amplitude $M_{\beta\alpha}$ itself contains terms that involve further delta function factors.)

Eq. (3.3.1) should be regarded as a definition of what we mean by the Lorentz invariance of the S-matrix, rather than a theorem, because it is only for certain special choices of Hamiltonian that there exists a unitary operator that acts as in (3.1.1) on both 'in' and 'out' states. We need to formulate conditions on the Hamiltonian that would ensure the Lorentz invariance of the S-matrix. For this purpose, it will be convenient to work with the operator S defined by Eq. (3.2.4):

$$S_{\beta\alpha} = \langle \beta | S | \alpha \rangle_0.$$

As we have defined the free-particle states in Chapter 2, they furnish a representation of the inhomogeneous Lorentz group, so we can always define a unitary operator $U_0(\Lambda, a)$ that induces the transformation (3.1.1) on these states:

$$
\begin{aligned}
&U_0(\Lambda, a)|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_0 \\
&= \exp(-ia_\mu(p_1^\mu + p_2^\mu + \cdots)) \sqrt{\frac{(\Lambda p_1)^0 (\Lambda p_2)^0 \cdots}{p_1^0 p_2^0 \cdots}} \\
&\quad \times \sum_{\bar\sigma_1 \bar\sigma_2 \ldots} D^{(j_1)}_{\bar\sigma_1 \sigma_1}(W(\Lambda, p_1))\, D^{(j_2)}_{\bar\sigma_2 \sigma_2}(W(\Lambda, p_2)) \cdots |\Lambda p_1, \bar\sigma_1, n_1; \Lambda p_2, \bar\sigma_2, n_2; \ldots\rangle_0 .
\end{aligned}
$$

Eq. (3.3.1) will thus hold if this unitary operator commutes with the S-operator:

$$U_0(\Lambda, a)^{-1} S\, U_0(\Lambda, a) = S.$$

This condition can also be expressed in terms of infinitesimal Lorentz transformations. Just as in Section 2.4, there will exist a set of Hermitian operators, a momentum $\mathbf{P}_0$, an angular momentum $\mathbf{J}_0$, and a boost generator $\mathbf{K}_0$, that together with H_0 generate the infinitesimal version of inhomogeneous Lorentz transformations when acting on free-particle states. Eq. (3.3.1) is equivalent to the statement that the S-matrix is unaffected by such transformations, or in other words, that the S-operator commutes with these generators:

$$[H_0, S] = [\mathbf{P}_0, S] = [\mathbf{J}_0, S] = [\mathbf{K}_0, S] = 0. \tag{3.3.3}$$

Because the operators H_0, $\mathbf{P}_0$, $\mathbf{J}_0$, and $\mathbf{K}_0$ generate infinitesimal inhomogeneous Lorentz transformations on the free states, they automatically satisfy the commutation relations (2.4.18)–(2.4.24):

$$[J_0^i, J_0^j] = i\epsilon_{ijk} J_0^k, \tag{3.3.4}$$

$$[J_0^i, K_0^j] = i\epsilon_{ijk} K_0^k, \tag{3.3.5}$$

$$[K_0^i, K_0^j] = -i\epsilon_{ijk} J_0^k, \tag{3.3.6}$$

$$[J_0^i, P_0^j] = i\epsilon_{ijk} P_0^k, \tag{3.3.7}$$

$$[K_0^i, P_0^j] = iH_0 \delta_{ij}, \tag{3.3.8}$$

$$[J_0^i, H_0] = [P_0^i, H_0] = [P_0^i, P_0^j] = 0, \tag{3.3.9}$$

$$[K_0^i, H_0] = iP_0^i. \tag{3.3.10}$$

where i, j, k, etc. run over the values 1, 2, and 3, and ϵ_{ijk} is the totally antisymmetric quantity with $\epsilon_{123} = +1$.

In the same way, we may define a set of 'exact generators,' operators $\mathbf{P}$, $\mathbf{J}$, $\mathbf{K}$ that together with the full Hamiltonian H generate the transformations (3.1.1) on, say, the 'in' states. (As already mentioned, what is not obvious is that the same operators generate the same transformations on the 'out' states.) The group structure tells us that these exact generators satisfy the same commutation relations:

$$[J^i, J^j] = i\epsilon_{ijk}J^k, \tag{3.3.11}$$
$$[J^i, K^j] = i\epsilon_{ijk}K^k, \tag{3.3.12}$$
$$[K^i, K^j] = -i\epsilon_{ijk}J^k, \tag{3.3.13}$$
$$[J^i, P^j] = i\epsilon_{ijk}P^k, \tag{3.3.14}$$
$$[K^i, P^j] = iH\delta_{ij}, \tag{3.3.15}$$
$$[J^i, H] = [P^i, H] = [P^i, P^j] = 0, \tag{3.3.16}$$
$$[K^i, H] = iP^i. \tag{3.3.17}$$

In virtually all known field theories, the effect of interactions is to add an interaction term H_{int} to the Hamiltonian, while leaving the momentum and angular momentum unchanged:

$$H = H_0 + H_{\text{int}}, \qquad \mathbf{P} = \mathbf{P}_0, \qquad \mathbf{J} = \mathbf{J}_0. \tag{3.3.18}$$

(The only known exceptions are theories with topologically twisted fields, such as those with magnetic monopoles, where the angular momentum of states depends on the interactions.) Eq. (3.3.18) implies that the commutation relations (3.3.11), (3.3.14), and (3.3.16) are satisfied provided that the interaction commutes with the free-particle momentum and angular-momentum operators

$$[H_{\text{int}}, \mathbf{P}_0] = [H_{\text{int}}, \mathbf{J}_0] = 0. \tag{3.3.19}$$

It is easy to see from the Lippmann–Schwinger equation (3.1.16) or equivalently from (3.1.13) that the operators that generate translations and rotations when acting on the 'in' (and 'out') states are indeed simply $\mathbf{P}_0$ and $\mathbf{J}_0$. Also we easily see that $\mathbf{P}_0$ and $\mathbf{J}_0$ commute with the operator $U(t, t_0)$ defined by Eq. (3.2.6), and hence with the S-operator $U(\infty, -\infty)$. Further, we already know that the S-operator commutes with H_0, because there are energy-conservation delta functions in both terms in (3.2.7). This leaves just the boost generator $\mathbf{K}_0$ which we need to show commutes with the S-operator.

On the other hand, it is *not* possible to set the boost generator $\mathbf{K}$ equal to its free-particle counterpart $\mathbf{K}_0$, because then Eqs. (3.3.15) and (3.3.8) would give $H = H_0$, which is certainly not true in the presence of interactions. Thus when we add an interaction H_{int} to H_0, we must also add a correction $\mathbf{W}$ to the boost generator:

$$\mathbf{K} = \mathbf{K}_0 + \mathbf{W}. \tag{3.3.20}$$

Of the remaining commutation relations, let us concentrate on Eq. (3.3.17), which may now be put in the form

$$[\mathbf{K}_0, H_{\text{int}}] = -[\mathbf{W}, H]. \tag{3.3.21}$$

By itself, the condition (3.3.21) is empty, because for any H_{int} we could always define $\mathbf{W}$ by giving its matrix elements between H-eigenstates $|\alpha\rangle$ and $|\beta\rangle$ as $-\langle\beta|[\mathbf{K}_0, H_{\text{int}}]|\alpha\rangle/(E_\beta - E_\alpha)$. Recall that the crucial point in the Lorentz invariance of a theory is not that there should exist a set of exact generators satisfying Eqs. (3.3.11)–(3.3.17), but rather that these operators should act the same way on 'in' and 'out' states; merely finding an operator $\mathbf{K}$ that satisfies Eq. (3.3.21) is not enough. Eq. (3.3.21) does become significant if we add the requirement that matrix elements of $\mathbf{W}$ should be smooth functions of the energies, and in particular should not have singularities of the form $(E_\beta - E_\alpha)^{-1}$. We shall now show that Eq. (3.3.21), together with an appropriate smoothness condition on $\mathbf{W}$, does imply the remaining Lorentz invariance condition $[\mathbf{K}_0, S] = 0$.

To prove this, let us consider the commutator of $\mathbf{K}_0$ with the operator $U(t, t_0)$ for finite t and t_0. Using Eq. (3.3.10) and the fact that $\mathbf{P}_0$ commutes with H_0 yields:

$$[\mathbf{K}_0, \exp(iH_0t)] = -t\mathbf{P}_0 \exp(iH_0t)$$

while Eq. (3.3.21) (which is equivalent to Eq. (3.3.17)) yields

$$[\mathbf{K}, \exp(iHt)] = -t\mathbf{P}\exp(iHt) = -t\mathbf{P}_0 \exp(iHt).$$

The momentum operators then cancel in the commutator of $\mathbf{K}_0$ with U, and we find:

$$[\mathbf{K}_0, U(\tau, \tau_0)] = -\mathbf{W}(\tau)U(\tau, \tau_0) + U(\tau, \tau_0)\mathbf{W}(\tau_0), \tag{3.3.22}$$

where

$$\mathbf{W}(t) = \exp(iH_0t)\mathbf{W}\exp(-iH_0t). \tag{3.3.23}$$

If the matrix elements of $\mathbf{W}$ between H_0-eigenstates are sufficiently smooth functions of energy, then matrix elements of $\mathbf{W}(t)$ between smooth superpositions of energy eigenstates vanish for $t \to \pm\infty$, so Eq. (3.3.22) gives in effect:

$$0 = [\mathbf{K}_0, U(\infty, -\infty)] = [\mathbf{K}_0, S], \tag{3.3.24}$$

as was to be shown. This is the essential result: Eq. (3.3.21) together with the smoothness condition on matrix elements of $\mathbf{W}$ that ensures that $\mathbf{W}(t)$ vanishes for $t \to \pm\infty$ provides a sufficient condition for the Lorentz invariance of the S-matrix. This smoothness condition is a natural one, because it is much like the condition on matrix elements of H_{int} that is needed to make $H_I(t)$ vanish for $t \to \pm\infty$, as required in order to justify the very idea of an S-matrix.

We can also use Eq. (3.3.22) with $\tau = 0$ and $\tau_0 = \mp\infty$ to show that

$$\mathbf{K}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{K}_0, \tag{3.3.25}$$

where $\Omega(\mp\infty)$ is according to (3.1.13) the operator that converts a free-particle state $|\alpha\rangle_0$ into the corresponding 'in' or 'out' state. Also, it follows trivially from Eqs. (3.3.18) and (3.3.19) that the same is true for the momentum and angular momentum:

$$\mathbf{P}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{P}_0, \tag{3.3.26}$$

$$\mathbf{J}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{J}_0. \tag{3.3.27}$$

Finally, since all free and asymptotic states are eigenstates of H_0 and H respectively with the same eigenvalue E_α, we have

$$H\Omega(\mp\infty) = \Omega(\mp\infty)H_0. \tag{3.3.28}$$

Eqs. (3.3.25)–(3.3.28) show that with our assumptions, 'in' and 'out' states do transform under inhomogeneous Lorentz transformations just like the free-particle states. Also, since these are similarity transformations, we now see that the exact generators $\mathbf{K}$, $\mathbf{P}$, $\mathbf{J}$, and H satisfy the same commutation relations as $\mathbf{K}_0$, $\mathbf{P}_0$, $\mathbf{J}_0$, and H_0. This is why it turned out to be unnecessary in proving the Lorentz invariance of the S-matrix to use the other commutation relations (3.3.12), (3.3.13), and (3.3.15) that involve $\mathbf{K}$.

Internal Symmetries

There are various symmetries, like the symmetry in nuclear physics under interchange of neutrons and protons, or the 'charge-conjugation' symmetry between particles and antiparticles, that have nothing directly to do with Lorentz invariance, and further appear the same in all inertial frames. Such a symmetry transformation T acts on the Hilbert space of physical states as a unitary operator $U(T)$, that induces linear transformations on the indices labelling particle species

$$\begin{aligned}&U(T)|p_1,\sigma_1,n_1;p_2,\sigma_2,n_2;\ldots\rangle_{\text{in/out}}\\&\quad=\sum_{\bar{n}_1\bar{n}_2\ldots}\mathcal{D}_{\bar{n}_1n_1}(T)\mathcal{D}_{\bar{n}_2n_2}(T)\cdots|p_1,\sigma_1,\bar{n}_1;p_2,\sigma_2,\bar{n}_2;\ldots\rangle_{\text{in/out}}.\end{aligned}\tag{3.3.29}$$

In accordance with the general discussion in Chapter 2, the $U(T)$ must satisfy the group multiplication rule

$$U(\bar{T})U(T)=U(\bar{T}T),\tag{3.3.30}$$

where $\bar{T}T$ is the transformation obtained by first performing the transformation T, then some other transformation $\bar{T}$. Acting on Eq. (3.3.29) with $U(\bar{T})$, we see that the matrices $\mathcal{D}$ satisfy the same rule

$$\mathcal{D}(\bar{T})\mathcal{D}(T)=\mathcal{D}(\bar{T}T).\tag{3.3.31}$$

Also, taking the scalar product of the states obtained by acting with $U(T)$ on two different 'in' states or two different 'out' states, and using the normalization condition (3.1.2), we see that $\mathcal{D}(T)$ must be unitary

$$\mathcal{D}^\dagger(T)=\mathcal{D}^{-1}(T).\tag{3.3.32}$$

Finally, taking the scalar product of the states obtained by acting with $U(T)$ on one 'out' state and one 'in' state shows that $\mathcal{D}$ commutes with the S-matrix, in the sense that

$$\begin{aligned}&\sum_{\bar{N}_1\bar{N}_2\ldots}\sum_{\bar{N}'_1\bar{N}'_2\ldots}\mathcal{D}^*_{\bar{N}'_1n'_1}(T)\mathcal{D}^*_{\bar{N}'_2n'_2}(T)\cdots\mathcal{D}_{\bar{N}_1n_1}(T)\mathcal{D}_{\bar{N}_2n_2}(T)\cdots\\&\qquad\times S_{p'_1\sigma'_1\bar{N}'_1;p'_2\sigma'_2\bar{N}'_2;\ldots,\,p_1\sigma_1\bar{N}_1;p_2\sigma_2\bar{N}_2;\ldots}\\&\quad=S_{p'_1\sigma'_1n'_1;p'_2\sigma'_2n'_2;\ldots,\,p_1\sigma_1n_1;p_2\sigma_2n_2;\ldots}.\end{aligned}\tag{3.3.33}$$

Again, this is a definition of what we mean by a theory being invariant under the internal symmetry T, because to derive Eq. (3.3.33) we still need to show that the *same* unitary operator $U(T)$ will induce the transformation (3.3.29) on both 'in' and 'out' states.

This will be the case if there is an 'unperturbed' transformation operator $U_0(T)$ that induces these transformations on free-particle states,

$$U_0(T)|p_1,\sigma_1,n_1;p_2,\sigma_2,n_2;\ldots\rangle_0 = \sum_{\bar{N}_1\bar{N}_2\ldots} \mathcal{D}_{\bar{N}_1 n_1}(T)\mathcal{D}_{\bar{N}_2 n_2}(T)\cdots|p_1,\sigma_1,\bar{N}_1;p_2,\sigma_2,\bar{N}_2;\ldots\rangle_0 \tag{3.3.34}$$

and that commutes with both the free-particle and interaction parts of the Hamiltonian

$$U_0^{-1}(T)H_0U_0(T) = H_0, \tag{3.3.35}$$

$$U_0^{-1}(T)H_{\rm int}U_0(T) = H_{\rm int}. \tag{3.3.36}$$

From either the Lippmann–Schwinger equation (3.1.17) or from (3.1.13), we see that the operator $U_0(T)$ will induce the transformations (3.3.29) on 'in' and 'out' states as well as free-particle states, so that we can derive Eq. (3.3.29) taking $U(T)$ as $U_0(T)$.

A special case of great physical importance is that of a one-parameter Lie group, where T is a function of a single parameter θ, with

$$T(\bar{\theta})T(\theta) = T(\bar{\theta}+\theta). \tag{3.3.37}$$

As shown in Section 2.2, in this case the corresponding Hilbert-space operators must take the form

$$U(T(\theta)) = \exp(iQ\theta) \tag{3.3.38}$$

with Q a Hermitian operator. Likewise the matrices $\mathcal{D}(T)$ take the form

$$\mathcal{D}_{n'n}(T(\theta)) = \delta_{n'n}\exp(iq_n\theta), \tag{3.3.39}$$

where q_n are a set of real species-dependent numbers. Here Eq. (3.3.33) simply tells us that the qs are conserved: $S_{\beta\alpha}$ vanishes unless

$$q_{n_1'} + q_{n_2'} + \cdots = q_{n_1} + q_{n_2} + \cdots\,. \tag{3.3.40}$$

The classic example of such a conservation law is that of conservation of electric charge. Also, all known processes conserve baryon number (the number of baryons, such as protons, neutrons, and hyperons, minus the number of their antiparticles) and lepton number (the number of leptons, such as electrons, muons, τ particles, and neutrinos, minus the number of their antiparticles) but as we shall see in Volume II, these conservation laws are believed to be only very good approximations. There are other conservation laws of this type that are definitely only approximate, such as the conservation of the quantity known as strangeness, which was introduced to explain the relatively long life of a class of particles discovered by Rochester and Butler [5] in cosmic rays in 1947. For instance, the mesons now called K^+ and K^0[17] are assigned strangeness +1 and the hyperons Λ^0, Σ^+, Σ^0, Σ^- are assigned strangeness −1, while the more familiar protons, neutrons, and π mesons (or pions) are

[17] Superscripts denote electric charges in units of the absolute value of the electronic charge. A 'hyperon' is any particle carrying non-zero strangeness and unit baryon number.

taken to have strangeness zero. The conservation of strangeness in strong interactions explains why strange particles are always produced in association with one another, as in reactions like $\pi^+ + n \to K^+ + \Lambda^0$, while the relatively slow decays of strange particles into non-strange ones such as $\Lambda^0 \to p + \pi^-$ and $K^+ \to \pi^+ + \pi^0$ show that the interactions that do not conserve strangeness are very weak.

The classic example of a 'non-Abelian' symmetry whose generators do not commute with one another is isotopic spin symmetry, which was suggested [6] in 1937 on the basis of an experiment [7] that showed the existence of a strong proton–proton force similar to that between protons and neutrons. Mathematically, the group is $SU(2)$, like the covering group of the group of three-dimensional rotations; its generators are denoted t_i, with $i = 1, 2, 3$, and satisfy a commutation relation like (2.4.18):

$$[t_i, t_j] = i\epsilon_{ijk} t_k.$$

To the extent that isotopic spin symmetry is respected, it requires particles to form degenerate multiplets labelled with an integer or half-integer T and with $2T + 1$ components distinguished by their t_3 values, just like the degenerate spin multiplets required by rotational invariance. These include the nucleons p and n with $T = \frac{1}{2}$ and $t_3 = \frac{1}{2}, -\frac{1}{2}$; the pions π^+, π^0, and π^- with $T = 1$ and $t_3 = +1, 0, -1$; and the Λ^0 hyperon with $T = 0$ and $t_3 = 0$. These examples illustrate the relation between electric charge Q, the third component of isotopic spin t_3, the baryon number B, and the strangeness S:

$$Q = t_3 + (B + S)/2.$$

This relation was originally inferred from observed selection rules, but it was interpreted [8] by Gell-Mann and Ne'eman in 1960 to be a consequence of the embedding of both the isospin $\mathbf{T}$ and the 'hypercharge' $Y \equiv B + S$ in the Lie algebra of a larger but more badly broken non-Abelian internal symmetry, based on the non-Abelian group $SU(3)$. As we will see in Volume II, today both isospin and $SU(3)$ symmetry are understood as incidental consequences of the small masses of the two or three lightest quarks in the modern theory of strong interactions, quantum chromodynamics.

The implications of isotopic spin symmetry for reactions among strongly interacting particles can be worked out by the same familiar methods that were invented for deriving the implications of rotational invariance. In particular, for a two-body reaction $A+B \to C+D$, Eq. (3.3.33) requires that the S-matrix may be put in the form (suppressing all but isospin labels):

$$S_{t_{C3}t_{D3},t_{A3}t_{B3}} = \sum_{T,t_3} C_{T_C T_D}(Tt_3; t_{C3}t_{D3}) C_{T_A T_B}(Tt_3; t_{A3}t_{B3}) S_T,$$

where $C_{j_1 j_2}(j\sigma; \sigma_1\sigma_2)$ is the usual Clebsch–Gordan coefficient [9] for forming a spin j with three-component σ from spins j_1 and j_2 with three-components σ_1 and σ_2, respectively; and S_T is a 'reduced' S-matrix depending on T and on all the suppressed momentum and spin variables, but not on the isospin three-components $t_{A3}, t_{B3}, t_{C3}, t_{D3}$. Of course, this, like all of the consequences of isotopic spin invariance, is only approximate, because this symmetry is not respected by electromagnetic (and other) interactions, as shown for instance by the fact that different members of the same isospin multiplet like p and n have different electric charges and slightly different masses.

Parity

To the extent that the symmetry under the transformation $\mathbf{x} \to -\mathbf{x}$ is really valid, there must exist a unitary operator P under which both 'in' and 'out' states transform as a direct product of single-particle states:

$$P|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}} = \eta_{n_1}\eta_{n_2}\cdots|\mathcal{P}p_1, \sigma_1, n_1; \mathcal{P}p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}}. \quad (3.3.41)$$

where η_n is the intrinsic parity of particles of species n, and $\mathcal{P}$ reverses the space components of p^μ. (This is for massive particles; the modification for massless particles is obvious.) The parity conservation condition for the S-matrix is then:

$$\begin{aligned} &S_{p_1'\sigma_1'n_1';p_2'\sigma_2'n_2';\ldots,\, p_1\sigma_1n_1;p_2\sigma_2n_2;\ldots} \\ &\quad = \eta^*_{n_1'}\eta^*_{n_2'}\cdots\eta_{n_1}\eta_{n_2}\cdots S_{\mathcal{P}p_1'\sigma_1'n_1';\mathcal{P}p_2'\sigma_2'n_2';\ldots,\, \mathcal{P}p_1\sigma_1n_1;\mathcal{P}p_2\sigma_2n_2;\ldots}. \end{aligned} \quad (3.3.42)$$

Just as for internal symmetries, an operator P satisfying Eq. (3.3.41) will actually exist if the operator P_0 which is defined to act this way on free-particle states commutes with H_{int} as well as H_0.

The phases η_n may be inferred either from dynamical models or from experiment, but neither can provide a unique determination of the ηs. This is because we are always free to redefine P by combining it with any conserved internal symmetry operator. For instance, if P is conserved, then so is

$$P' \equiv P\exp(i\alpha B + i\beta L + i\gamma Q),$$

where B, L, and Q are respectively baryon number, lepton number, and electric charge, and α, β, and γ are arbitrary real phases; hence either P or P' could be called the parity operator. The neutron, proton, and electron have different combinations of values for B, L, and Q, so by judicious choice of the phases α, β, and γ we can define the intrinsic parities of all three particles to be $+1$. However, once we have done this the intrinsic parities of other particles like the charged pion (which can be emitted in a transition $n \to p + \pi^-$) are no longer arbitrary. Also, the intrinsic parity of any particle like the neutral pion π^0 which carries no conserved quantum numbers is always meaningful.

The foregoing remarks help to clarify the question of whether intrinsic parities must always have the values ± 1. It is easy to say that space inversion $\mathcal{P}$ has the group multiplication law $\mathcal{P}^2 = \mathbf{1}$; however, the parity operator that is conserved may not be this one, but rather may differ from it by a phase transformation of some sort. In any case, whether or not $P^2 = \mathbf{1}$, the operator P^2 behaves just like an internal symmetry transformation:

$$P^2|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}} = \eta^2_{n_1}\eta^2_{n_2}\cdots|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}}.$$

If this internal symmetry is part of a continuous symmetry group of phase transformations, such as the group of multiplication by the phases $\exp(i\alpha B + i\beta L + i\gamma Q)$ with arbitrary values of α, β, and γ, then its inverse square root must also be a member of this group, say I_P, with $I_P^2 P^2 = \mathbf{1}$ and $[I_P, P] = 0$. (For instance, if $P^2 = \exp(i\alpha B + \cdots)$, then take $I_P = \exp(-\frac{1}{2}i\alpha B + \cdots)$.) We can then define a new parity operator $P' \equiv PI_P$ with $P'^2 = \mathbf{1}$. This is conserved to the same extent as P, so there is no reason why we should not call this *the* parity operator, in which case the intrinsic parities can only take the values ± 1.

The only sort of theory in which it is not necessarily possible to define parity so that all intrinsic parities have the values ± 1 is one in which there is some discrete internal symmetry which is not a member of any continuous symmetry group of phase transformations [10]. For instance, it is a consequence of angular-momentum conservation that the total number F of all particles of half-integer spin can only change by even numbers, so the internal symmetry operator $(-1)^F$ is conserved. All known particles of half-integer spin have odd values of the sum $B + L$ of baryon number and lepton number, so as far as we know, $(-1)^F = (-1)^{B+L}$. If this is true, then $(-1)^F$ is part of a continuous symmetry group, consisting of the operators $\exp(i\alpha(B + L))$ with arbitrary real α, and has an inverse square root $\exp(-i\pi(B + L)/2)$. In this case, if $P^2 = (-1)^F$ then P can be redefined so that all intrinsic parities are ± 1. However, if there were to be discovered a particle of half-integer spin and an even value of $B + L$ (such as a so-called Majorana neutrino, with $j = \frac{1}{2}$ and $B + L = 0$), then it would be possible to have $P^2 = (-1)^F$ without our being able to redefine the parity operator itself to have eigenvalues ± 1. In this case, of course, we would have $P^4 = \mathbf{1}$, so all particles would have intrinsic parities either ± 1 or (like the Majorana neutrino) $\pm i$.

It follows from Eq. (3.3.42) that if the product of intrinsic parities in the final state is equal to the product of intrinsic parities in the initial state, or equal to minus this product, then the S-matrix must be respectively even or odd overall in the three-momenta. For instance, it was observed [11] in 1951 that a pion can be absorbed by a deuteron from the $\ell = 0$ ground state of the $\pi^- d$ atom, in the reaction $\pi^- + d \to n + n$. (As discussed in Section 3.7, the orbital angular-momentum quantum number ℓ can be used in relativistic physics in the same way as in non-relativistic wave mechanics.) The initial state has total angular-momentum $j = 1$ (the pion and deuteron having spins zero and one, respectively), so the final state must have orbital angular-momentum $\ell = 1$ and total neutron spin $s = 1$. (The other possibilities $\ell = 1$, $s = 0$; $\ell = 0$, $s = 1$; and $\ell = 2$, $s = 1$, which are allowed by angular-momentum conservation, are forbidden by the requirement that the final state be antisymmetric in the two neutrons.) Because the final state has $\ell = 1$, the matrix element is odd under reversal of the direction of all three-momenta, so we can conclude that the intrinsic parities of the particles in this reaction must be related by:

$$\eta_d \eta_{\pi^-} = -\eta_n^2 .$$

The deuteron is known to be a bound state of a proton and neutron with even orbital angular-momentum (chiefly $\ell = 0$), and as we have seen we can take the neutron and proton to have the same intrinsic parity, so $\eta_d = \eta_n^2$, and we can conclude that $\eta_{\pi^-} = -1$; that is, the negative pion is a *pseudoscalar* particle. The π^+ and π^0 have also been found to have negative parity, as would be expected from the symmetry (isospin invariance) among these three particles.

The negative parity of the pion has some striking consequences. A spin zero particle that decays into three pions must have intrinsic parity $\eta_\pi^3 = -1$, because in the Lorentz frame in which the decaying particle is at rest, rotational invariance only allows the matrix element to depend on scalar products of the pion momenta with each other, all of which are even under reversal of all momenta. (The triple scalar product $\mathbf{p}_1 \cdot (\mathbf{p}_2 \times \mathbf{p}_3)$ formed from the three pion momenta vanishes because $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 = 0$.) For the same reason, a spin zero particle that decays into two pions must have intrinsic parity $\eta_\pi^2 = +1$. In particular, among the strange particles discovered in the late 1940s there seemed to be two different particles of zero spin (inferred from the angular distribution of their decay products): one, the τ, was identified by its decay into three pions, and hence was assigned a parity -1, while the other, the θ, was identified by its decay into two pions and was assigned a parity $+1$. The trouble with all this was that as the τ and θ were studied in greater detail, they seemed increasingly to have identical masses and lifetimes. After many suggested solutions of this puzzle, Lee and Yang in 1956 finally cut the Gordian knot, and proposed that the τ and θ are the same particle, (now known as the $K^\pm$) and that parity is simply not conserved in the weak interactions that lead to its decay [12].

As we shall see in detail in the next section of this chapter, the rate for a physical process $\alpha \to \beta$ (with $\alpha \neq \beta$) is proportional to $|S_{\beta\alpha}|^2$, with proportionality factors that are invariant under reversal of all three-momenta. As long as the states α and β contain definite numbers of particles of each type, the phase factors in Eq. (3.3.42) have no effect on $|S_{\beta\alpha}|^2$, so Eq. (3.3.42) would imply that the rate for $\alpha \to \beta$ is invariant under the reversal of direction of all three-momenta. As we have seen, this is a trivial consequence of rotational invariance for the decays of a K meson into two or three pions, but it is a non-trivial restriction on rates in more complicated processes. For example, following theoretical suggestions by Lee and Yang, Wu together with a group at the National Bureau of Standards measured the angular distribution of the electron in the final state of the beta decay $\mathrm{Co}^{60} \to \mathrm{Ni}^{60} + e^- + \bar{\nu}$ with a polarized cobalt source [13]. (No attempt was made in this experiment to measure the momentum of the antineutrino or nickel nucleus.) The electrons were found to be preferentially emitted in a direction opposite to that of the spin of the decaying nucleus, which would, of course, be impossible if the decay rate were invariant under a reversal of all three-momenta. A similar result was found in the decay of a positive muon (polarized in its production in the process $\pi^+ \to \mu^+ + \nu$) into a positron, neutrino, and antineutrino [14]. In this way, it became clear that parity is indeed not conserved in the weak interactions responsible for these decays. Nevertheless, for reasons discussed in Section 12.5, parity *is* conserved in the strong and electromagnetic interactions, and therefore continues to play an important part in theoretical physics.

Time-Reversal

We saw in Section 2.6 that the time-reversal operator T acting on a one-particle state $|p, \sigma, n\rangle$ gives a state $|\mathcal{P}p, -\sigma, n\rangle$ with reversed spin and momentum, times a phase $\zeta_n(-1)^{j-\sigma}$. A multi-particle state transforms as usual as a direct product of one-particle states, except that since this is a time-reversal transformation, we expect 'in' and 'out' states to be interchanged:

$$\begin{aligned} T&|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}} \\ &= \zeta_{n_1}(-1)^{j_1-\sigma_1}\zeta_{n_2}(-1)^{j_2-\sigma_2}\cdots|\mathcal{P}p_1, -\sigma_1, n_1; \mathcal{P}p_2, -\sigma_2, n_2; \ldots\rangle_{\text{out/in}}. \end{aligned} \tag{3.3.43}$$

(Again, this is for massive particles, with obvious modifications being required for massless particles.) It will be convenient to abbreviate this assumption as

$$T|\alpha\rangle_{\text{in/out}} = |\Theta\alpha\rangle_{\text{out/in}}, \tag{3.3.44}$$

where Θ indicates a reversal of sign of three-momenta and spins as well as multiplication by the phase factors shown in Eq. (3.3.43). Because T is antiunitary, we have

$$\langle\beta; \text{out}|\alpha; \text{in}\rangle = \langle\Theta\alpha; \text{out}|\Theta\beta; \text{in}\rangle, \tag{3.3.45}$$

so the time-reversal invariance condition for the S-matrix is

$$S_{\beta\alpha} = S_{\Theta\alpha,\Theta\beta} \tag{3.3.46}$$

or in more detail

$$\begin{aligned} &S_{p'_1\sigma'_1n'_1;p'_2\sigma'_2n'_2;\ldots,\, p_1\sigma_1n_1;p_2\sigma_2n_2;\ldots} \\ &= \zeta_{n'_1}(-1)^{j'_1-\sigma'_1}\zeta_{n'_2}(-1)^{j'_2-\sigma'_2}\cdots\zeta^*_{n_1}(-1)^{j_1-\sigma_1}\zeta^*_{n_2}(-1)^{j_2-\sigma_2}\cdots \\ &\quad\times S_{\mathcal{P}p_1,-\sigma_1,n_1;\mathcal{P}p_2,-\sigma_2,n_2;\ldots,\,\mathcal{P}p'_1,-\sigma'_1,n'_1;\mathcal{P}p'_2,-\sigma'_2,n'_2;\ldots}. \end{aligned} \tag{3.3.47}$$

Note that in addition to the reversal of momenta and spins, the role of initial and final states is interchanged, as would be expected for a symmetry involving the reversal of time.

The S-matrix will satisfy this transformation rule if the operator T_0 that induces time-reversal transformations on free-particle states

$$T_0|\alpha\rangle_0 = |\Theta\alpha\rangle_0 \tag{3.3.48}$$

commutes not only with the free-particle Hamiltonian (which is automatic) but also with the interaction:

$$T_0^{-1}H_0T_0 = H_0, \tag{3.3.49}$$

$$T_0^{-1}H_{\text{int}}T_0 = H_{\text{int}}. \tag{3.3.50}$$

In this case we can take $T = T_0$, and use either (3.1.13) or (3.1.16) to show that time-reversal transformations do act as stated in Eq. (3.3.44). For instance, operating on the Lippmann–Schwinger equation (3.1.16) with T and using Eqs. (3.3.48)–(3.3.50), we have

$$T|\alpha\rangle_{\text{in/out}} = |\Theta\alpha\rangle_0 + \int d\gamma\,[E_\alpha - H_0 \mp i\epsilon]^{-1}H_{\text{int}}T|\alpha\rangle_{\text{in/out}}$$

with the sign of $\pm i\epsilon$ reversed because T is antiunitary. This is just the Lippmann–Schwinger equation for $|\Theta\alpha\rangle_{\text{out/in}}$, thus justifying Eq. (3.3.44). Similarly, because T is antiunitary it changes the sign of the i in the exponent of $\Omega(t)$, so that

$$T\Omega(-\infty)|\alpha\rangle_0 = \Omega(\infty)|\Theta\alpha\rangle_0,$$

again leading to Eq. (3.3.44).

In contrast with the case of parity conservation, the time-reversal invariance condition (3.3.46) does *not* in general tell us that the rate for the process $\alpha \to \beta$ is the same as for the process $\Theta\alpha \to \Theta\beta$. However, something like this is true in cases where the S-matrix takes the form

$$S_{\beta\alpha} = S^{(0)}_{\beta\alpha} + S^{(1)}_{\beta\alpha}, \tag{3.3.51}$$

where $S^{(1)}$ is small, while $S^{(0)}$ happens to have matrix element zero for some particular process of interest, though it generally has much larger matrix elements than $S^{(1)}$. (For instance, the process might be nuclear beta decay, $N \to N' + e^- + \bar{\nu}$, with $S^{(0)}$ the S-matrix produced by the strong nuclear and electromagnetic interactions alone, and $S^{(1)}$ the correction to the S-matrix produced by the weak interactions. Section 3.5 shows how the use of the 'distorted-wave Born approximation' leads to an S-matrix of the form (3.3.51) in cases of this type. In some cases $S^{(0)}$ is simply the unit operator.) To first order in $S^{(1)}$, the unitarity condition for the S-operator reads

$$\mathbf{1} = S^\dagger S = S^{(0)\dagger}S^{(0)} + S^{(0)\dagger}S^{(1)} + S^{(1)\dagger}S^{(0)}.$$

Using the zeroth-order relation $S^{(0)\dagger}S^{(0)} = \mathbf{1}$, this gives a reality condition for $S^{(1)}$:

$$S^{(1)} = -S^{(0)}S^{(1)\dagger}S^{(0)}. \tag{3.3.52}$$

If $S^{(1)}$ as well as $S^{(0)}$ satisfies the time-reversal condition (3.3.46), then this can be put in the form

$$S^{(1)}_{\beta\alpha} = -\int d\gamma \int d\gamma'\, S^{(0)}_{\beta\gamma'} S^{(1)*}_{\Theta\gamma',\Theta\gamma} S^{(0)}_{\gamma\alpha}. \tag{3.3.53}$$

Since $S^{(0)}$ is unitary, the rates for the processes $\alpha \to \beta$ and $\Theta\alpha \to \Theta\beta$ are thus the same if summed over sets $\mathcal{J}$ and $\mathcal{F}$ of final and initial states that are complete with respect to $S^{(0)}$. (By being 'complete' here is meant that if $S^{(0)}_{\alpha'\alpha}$ is non-zero, and either α or α' are in $\mathcal{J}$, then both states are in $\mathcal{J}$; and similarly for $\mathcal{F}$.) In the simplest case we have 'complete' sets $\mathcal{J}$ and $\mathcal{F}$ consisting of just one state each; that is, both the initial and the final states are eigenvectors of $S^{(0)}$ with eigenvalues $e^{2i\delta_\alpha}$ and $e^{2i\delta_\beta}$, respectively. (The δ_α and δ_β are called 'phase shifts'; they are real because $S^{(0)}$ is unitary.) In this case, Eq. (3.3.53) becomes simply:

$$S^{(1)}_{\beta\alpha} = -e^{2i(\delta_\alpha+\delta_\beta)} S^{(1)*}_{\Theta\beta,\Theta\alpha}, \tag{3.3.54}$$

and it is clear that the absolute value of the S-matrix for the process $\alpha \to \beta$ is the same as for the process $\Theta\alpha \to \Theta\beta$. This is the case for instance in nuclear beta decay (in the approximation in which we ignore the relatively weak Coulomb interaction between the electron and nucleus in the final state), because both the initial and final states are eigenstates of the strong interaction S-matrix (with $\delta_\alpha = \delta_\beta = 0$). Thus if time-reversal invariance is respected, the differential rate for a beta decay process should be unchanged if we reverse both the momenta and the spin z-components σ of all particles. This prediction was *not* contradicted in the 1956 experiments [13,14] that discovered the non-conservation of parity; for instance, time-reversal invariance is consistent with the observation that electrons from the decay $\mathrm{Co}^{60} \to \mathrm{Ni}^{60} + e^- + \bar{\nu}$ are preferentially emitted in a direction opposite to that of the Co^{60} spin. As described below, indirect evidence against time-reversal invariance did emerge in 1964, but it remains a useful approximate symmetry in weak as well as strong and electromagnetic interactions.

In some cases we can use a basis of states for which $\Theta\alpha = \alpha$ and $\Theta\beta = \beta$, for which Eq. (3.3.54) reads

$$S^{(1)}_{\beta\alpha} = -e^{2i(\delta_\alpha+\delta_\beta)} S^{(1)*}_{\beta\alpha}, \tag{3.3.55}$$

which just says that $iS^{(1)}_{\beta\alpha}$ has the phase $\delta_\alpha + \delta_\beta \bmod \pi$. This is known as *Watson's theorem* [15]. The phases in Eqs. (3.3.54) or (3.3.55) may be measured in processes where there is interference between different final states. For instance, in the decay of the spin 1/2 hyperon Λ into a nucleon and a pion, the final state can only have orbital angular-momentum $\ell = 0$ or $\ell = 1$; the angular distribution of the pion relative to the Λ spin involves the interference between these states, and hence according to Watson's theorem depends on the difference $\delta_s - \delta_p$ of their phase shifts.

PT

Although the 1957 experiments on parity violation did not rule out time-reversal invariance, they did show immediately that the product PT is not conserved. If conserved this operator would have to be antiunitary for the same reasons as for T, so in processes like nuclear beta decay its consequences would take the form of relations like Eq. (3.3.54):

$$S^{(1)}_{\beta\alpha} = -e^{i(\delta_\alpha+\delta_\beta)} S^{(1)*}_{\mathcal{P}\Theta\beta,\mathcal{P}\Theta\alpha},$$

where $\mathcal{P}\Theta$ reverses the signs of all spin z-components but *not* any momenta. Neglecting the final-state Coulomb interaction, it would then follow that there could be no preference for the electron in the decay $\mathrm{Co}^{60} \to \mathrm{Ni}^{60} + e^- + \bar{\nu}$ to be emitted in the same or the opposite direction to the Co^{60} spin, in contradiction with what was observed.

C, *CP*, and *CPT*

As already mentioned, there is an internal symmetry transformation, known as charge-conjugation, which interchanges particles and antiparticles. Formally, this entails the existence of a unitary operator C, whose effect on multi-particle states is:

$$C|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}} = \xi_{n_1}\xi_{n_2}\cdots |p_1, \sigma_1, n_1^c; p_2, \sigma_2, n_2^c; \ldots\rangle_{\text{in/out}}, \tag{3.3.56}$$

where n^c is the antiparticle of particle type n, and ξ_n is yet another phase. If this is true for both 'in' and 'out' states then the S-matrix satisfies the invariance conditions

$$\begin{aligned} &S_{p_1'\sigma_1'n_1';p_2'\sigma_2'n_2';\ldots,\,p_1\sigma_1n_1;p_2\sigma_2n_2;\ldots} \\ &\quad= \xi^*_{n_1'}\xi^*_{n_2'}\cdots\xi_{n_1}\xi_{n_2}\cdots S_{p_1'\sigma_1'n_1'^c;p_2'\sigma_2'n_2'^c;\ldots,\,p_1\sigma_1n_1^c;p_2\sigma_2n_2^c;\ldots}. \end{aligned} \tag{3.3.57}$$

As with other internal symmetries, the S-matrix will satisfy this condition if the operator C_0 that is defined to act as stated in Eq. (3.3.56) on free-particle states commutes with the interaction H_{int} as well as H_0; in this case, we take $C = C_0$.

The phases ξ_n are called charge-conjugation parities. Just as for the ordinary parities η_n, the ξ_n are in general not uniquely defined, because for any operator C that is defined to satisfy Eq. (3.3.56), we can find another such operator with different ξ_n by multiplying C with any internal symmetry phase transformation, such as $\exp(i\alpha B + i\beta L + i\gamma Q)$; the only particles whose charge-conjugation parities are individually measurable are those completely neutral particles like the photon or the neutral pion that carry no conserved quantum numbers and are their own antiparticles. In reactions involving only completely neutral particles, Eq. (3.3.57) tells us that the product of the charge-conjugation parities in the initial and final states must be equal; for instance, as we shall see the photon is required by quantum electrodynamics to have charge-conjugation parity $\eta_\gamma = -1$, so the observation of the neutral pion decay $\pi^0 \to 2\gamma$ requires that $\eta_{\pi^0} = +1$; it then follows that the process $\pi^0 \to 3\gamma$ should be forbidden, as is in fact known to be the case. For these two particles, the charge-conjugation parities are real, either $+1$ or -1. Just as for ordinary parity, this will always be the case if all internal phase transformation symmetries are members of continuous groups of phase transformations, because then we can redefine C by multiplying by the inverse square root of the internal symmetry equal to C^2, with the result that the new C satisfies $C^2 = \mathbf{1}$.

For general reactions, Eq. (3.3.57) requires that the rate for a process equals the rate for the same process with particles replaced with their corresponding antiparticles. This was not directly contradicted by the 1957 experiments on parity non-conservation (it will be a long time before anyone is able to study the beta decay of anticobalt), but these experiments showed that C is not conserved in the *theory* of weak interactions as modified by Lee and Yang [12] to take account of parity non-conservation. (As we shall see below, the observed violation of TP conservation would imply a violation of C conservation in *any* field theory of weak interactions, not just in the particular theory considered by Lee and Yang.) It is understood today that C as well as P is not conserved in the weak interactions responsible for processes like beta decay and the decay of the pion and muon, though both C and P are conserved in the strong and electromagnetic interactions.

Although the early experiments on parity non-conservation indicated that neither C nor P are conserved in the weak interactions, they left open the possibility that their product CP is universally conserved. For some years it was expected (though not with complete confidence) that CP would be found to be generally conserved. This had particularly important consequences for the properties of the neutral K mesons. In 1954 Gell-Mann and Pais [16] had pointed out that because the K^0 meson is not its own antiparticle (the K^0 carries a non-zero value for the approximately conserved quantity known as strangeness) the particles with definite decay rates would be not K^0 or $\bar{K}^0$, but the linear combinations $K^0 \pm \bar{K}^0$. This was originally explained in terms of C conservation, but with C not conserved in the weak interactions, the argument may be equally well be based on CP conservation.

If we arbitrarily define the phases in the CP operator and in the K^0 and $\bar{K}^0$ states so that

$$CP|K^0\rangle = |\bar{K}^0\rangle$$

and

$$CP|\bar{K}^0\rangle = |K^0\rangle$$

then we can define self-charge-conjugate one-particle states

$$|K_1^0\rangle \equiv \frac{1}{\sqrt{2}}\left[|K^0\rangle + |\bar{K}^0\rangle\right]$$

and

$$|K_2^0\rangle \equiv \frac{1}{\sqrt{2}}\left[|K^0\rangle - |\bar{K}^0\rangle\right],$$

which have CP eigenvalues $+1$ and -1, respectively. The fastest available decay mode of these particles is into two-pion states, but CP conservation would allow this only[18] for the K_1, not the K_2. The K_2^0 would thus be expected to decay only by slower modes, into three pions or into a pion, muon or electron, and neutrino. Nevertheless, it was found by Fitch and Cronin in 1964 that the long-lived neutral K-meson does have a small probability for decaying into two pions [17]. The conclusion was that CP is not exactly conserved in the weak interactions, although it seems more nearly conserved than C or P individually.

As we shall see in Chapter 5, there are good reasons to believe that although neither C nor CP is strictly conserved, the product CPT is exactly conserved in all interactions, at least in any quantum field theory. It is CPT that provides a precise correspondence between particles and antiparticles, and in particular it is the fact that CPT commutes with the Hamiltonian that tells us that stable particles and antiparticles have exactly the same mass. Because CPT is antiunitary, it relates the S-matrix for an arbitrary process to the S-matrix for the *inverse* process with all spin three-components reversed and particles replaced with antiparticles. However, in cases where the S-matrix can be divided into a weak term $S^{(1)}$ that produces a given reaction and a strong term $S^{(0)}$ that acts in the initial and final states, we can use the same arguments that were used above in studying the implications of T conservation to show that the rate for any process is equal to the rate for the same process with particles replaced with antiparticles and spin three-components reversed, provided that we sum over sets of initial and final states that are complete with respect to $S^{(0)}$. In particular, although the partial rates for decay of the particle into a pair of final states β_1, β_2 with $S^{(0)}_{\beta_1\beta_2} \neq 0$ may differ from the partial rates for the decay of the antiparticle into the corresponding final states $CPT\beta_1$ and $CPT\beta_2$, we shall see in Section 3.5 that (without any approximations) the total decay rate of any particle is equal to that of its antiparticle.

We can now understand why the 1957 experiments on parity violation could be interpreted in the context of the existing theory of weak interactions as evidence that C conservation as well as P conservation are badly violated but CP is not. These theories

[18] The neutral K-mesons have spin zero, so the two-pion final state has $\ell = 0$, and hence $P = +1$. Further, $C = +1$ for two π^0s because the π^0 has $C = +1$, and also for an $\ell = 0$ $\pi^+\pi^-$ state because C interchanges the two pions.

were field theories, and therefore automatically conserved CPT. Since the experiments showed that PT conservation but not T conservation is badly violated in nuclear beta decay, *any* theory that was consistent with these experiments and in which CPT is conserved would have to also incorporate C but not CP non-conservation. Similarly, the observation in 1964 of small violations of CP conservation in the weak interactions together with the assumed invariance of all interactions under CPT allowed the immediate inference that the weak interactions also do not exactly conserve T. This has since been verified [18] by more detailed studies of the K^0–$\bar{K}^0$ system, but it has so far been impossible to find other direct evidence of the failure of invariance under time-reversal.

3.4 Rates and Cross-Sections

The S-matrix $S_{\beta\alpha}$ is the probability amplitude for the transition $\alpha \to \beta$, but what does this have to do with the transition rates and cross-sections measured by experimentalists? In particular, (3.3.2) shows that $S_{\beta\alpha}$ has a factor $\delta^4(p_\beta - p_\alpha)$, which ensures the conservation of the total energy and momentum, so what are we to make of the factor $[\delta^4(p_\beta - p_\alpha)]^2$ in the transition probability $|S_{\beta\alpha}|^2$? The proper way to approach these problems is by studying the way that experiments are actually done, using wave packets to represent particles localized far from each other before a collision, and then following the time-history of these superpositions of multi-particle states. In what follows we will instead give a quick and easy derivation of the main results, actually more a mnemonic than a derivation, with the excuse that (as far as I know) no interesting open problems in physics hinge on getting the fine points right regarding these matters.

We consider our whole system of physical particles to be enclosed in a large box with a macroscopic volume V. For instance, we can take this box as a cube, but with points on opposite sides identified, so that the single-valuedness of the spatial wave function requires the momenta to be quantized

$$\mathbf{p} = \frac{2\pi}{L}(n_1, n_2, n_3), \tag{3.4.1}$$

where the n_i are integers, and $L^3 = V$. Then all three-dimensional delta-functions become

$$\delta^3_V(\mathbf{p}' - \mathbf{p}) \equiv \frac{1}{(2\pi)^3}\int_V d^3x\, e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}} = \frac{V}{(2\pi)^3}\delta_{\mathbf{p}',\mathbf{p}}, \tag{3.4.2}$$

where $\delta_{\mathbf{p}',\mathbf{p}}$ is an ordinary Kronecker delta symbol, equal to one if the subscripts are equal and zero otherwise. The normalization condition (3.1.2) thus implies that the states we have been using have scalar products in a box which are not just sums of products of Kronecker deltas, but also contain a factor $[V/(2\pi)^3]^N$, where N is the number of particles in the state. To calculate transition probabilities we should use states of unit norm, so let us introduce states normalized approximately for our box

$$|\alpha\rangle^{\text{Box}} \equiv \left[(2\pi)^3/V\right]^{N_\alpha/2} |\alpha\rangle, \tag{3.4.3}$$

with norm

$${}^{\text{Box}}\langle\beta|\alpha\rangle^{\text{Box}} = \delta_{\beta\alpha}, \tag{3.4.4}$$

where $\delta_{\beta\alpha}$ is a product of Kronecker deltas, one for each three-momentum, spin, and species label, plus terms with particles permuted. Correspondingly, the S-matrix may be written

$$S_{\beta\alpha} = \left[V/(2\pi)^3\right]^{(N_\beta+N_\alpha)/2} S^{\mathrm{Box}}_{\beta\alpha}, \tag{3.4.5}$$

where $S^{\mathrm{Box}}_{\beta\alpha}$ is calculated using the states (3.4.3).

Of course, if we just leave our particles in the box forever, then every possible transition will occur again and again. To calculate a meaningful transition probability we also have to put our system in a 'time box'. We suppose that the interaction is turned on for only a time T. One immediate consequence is that the energy-conservation delta function is replaced with

$$\delta_T(E_\alpha - E_\beta) = \frac{1}{2\pi}\int_{-T/2}^{T/2} \exp\bigl(i(E_\alpha - E_\beta)t\bigr)\, dt. \tag{3.4.6}$$

The probability that a multi-particle system, which is in a state α before the interaction is turned on, is found in a state β after the interaction is turned off, is

$$P(\alpha\to\beta) = |S^{\mathrm{Box}}_{\beta\alpha}|^2 = \left[(2\pi)^3/V\right]^{N_\beta+N_\alpha} |S_{\beta\alpha}|^2. \tag{3.4.7}$$

This is the probability for a transition into one specific box state β. The number of one-particle box states in a momentum-space volume d^3p is $V d^3p/(2\pi)^3$, because this is the number of triplets of integers n_1, n_2, n_3 for which the momentum (3.4.1) lies in the momentum-space volume d^3p around $\mathbf{p}$. We shall define the final-state interval $d\beta$ as a product of d^3p for each final particle, so the total number of states in this range is

$$d\mathcal{N}_\beta = \left[V/(2\pi)^3\right]^{N_\beta} d\beta. \tag{3.4.8}$$

Hence the total probability for the system to wind up in a range $d\beta$ of final states is

$$dP(\alpha\to\beta) = P(\alpha\to\beta)d\mathcal{N}_\beta = \left[(2\pi)^3/V\right]^{N_\alpha} |S_{\beta\alpha}|^2 d\beta. \tag{3.4.9}$$

We will restrict our attention throughout this section to final states β that are not only different (however slightly) from the initial state α, but that also satisfy the more stringent condition, that no subset of the particles in the state β (other than the whole state itself) has precisely the same four-momentum as some corresponding subset of the particles in the state α. (In the language to be introduced in the next chapter, this means that we are considering only the connected part of the S-matrix.) For such states, we may define a delta function-free matrix element $M_{\beta\alpha}$:

$$S_{\beta\alpha} = -2\pi i\, \delta^3_V(\mathbf{p}_\beta - \mathbf{p}_\alpha)\delta_T(E_\beta - E_\alpha)M_{\beta\alpha}. \tag{3.4.10}$$

Our introduction of the box allows us to interpret the squares of delta functions in $|S_{\beta\alpha}|^2$ for $\beta \neq \alpha$ as

$$\begin{aligned}\left[\delta^3_V(\mathbf{p}_\beta - \mathbf{p}_\alpha)\right]^2 &= \delta^3_V(\mathbf{p}_\beta - \mathbf{p}_\alpha)\delta^3_V(0) = \delta^3_V(\mathbf{p}_\beta - \mathbf{p}_\alpha)V/(2\pi)^3,\\ \left[\delta_T(E_\beta - E_\alpha)\right]^2 &= \delta_T(E_\beta - E_\alpha)\delta_T(0) = \delta_T(E_\beta - E_\alpha)T/2\pi,\end{aligned}$$

so Eq. (3.4.9) gives a differential transition probability

$$\begin{aligned} dP(\alpha \to \beta) = (2\pi)^2 \left[(2\pi)^3/V\right]^{N_\alpha - 1} (T/2\pi) |M_{\beta\alpha}|^2 \\ \times \delta_V^3(\mathbf{p}_\beta - \mathbf{p}_\alpha)\delta_T(E_\beta - E_\alpha) d\beta. \end{aligned}$$

If we let V and T be very large, the delta function product here may be interpreted as an ordinary four-dimensional delta function $\delta^4(p_\beta - p_\alpha)$. In this limit, the transition probability is simply proportional to the time T during which the interaction is acting, with a coefficient that may be interpreted as a differential transition *rate*:

$$\begin{aligned} d\Gamma(\alpha \to \beta) &\equiv dP(\alpha \to \beta)/T \\ &= (2\pi)^{3N_\alpha - 2} V^{1-N_\alpha} |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta, \end{aligned} \tag{3.4.11}$$

where now

$$S_{\beta\alpha} = -2\pi i \delta^4(p_\beta - p_\alpha) M_{\beta\alpha}. \tag{3.4.12}$$

This is the master formula which is used to interpret calculations of S-matrix elements in terms of predictions for actual experiments. We will come back to the interpretation of the factor $\delta^4(p_\alpha - p_\beta)d\beta$ later in this section.

There are two cases of special importance:

$\boldsymbol{N_\alpha = 1}$ **:**

Here the volume V cancels in Eq. (3.4.11), which gives the transition rate for a single-particle state α to decay into a general multi-particle state β as

$$d\Gamma(\alpha \to \beta) = 2\pi |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta. \tag{3.4.13}$$

Of course, this makes sense only if the time T during which the interaction acts is much less than the mean lifetime τ_α of the particle α, so we cannot pass to the limit $T \to \infty$ in $\delta_T(E_\alpha - E_\beta)$. There is an unremovable width $\Delta E \simeq 1/T \gtrsim 1/\tau_\alpha$ in this delta function, so Eq. (3.4.13) is only useful if the total decay rate $1/\tau_\alpha$ is much less than any of the characteristic energies of the process.

$\boldsymbol{N_\alpha = 2}$ **:**

Here the rate (3.4.11) is proportional to $1/V$, or in other words, to the density of either particle at the position of the other one. Experimentalists generally report not the transition rate per density, but the *rate per flux*, also known as the *cross-section*. The flux of either particle at the position of the other particle is defined as the product of the density $1/V$ and the relative velocity u_α:

$$\Phi_\alpha = u_\alpha / V. \tag{3.4.14}$$

(A general definition of u_α is given below; for the moment we will content ourselves with specifying that if either particle is at rest then u_α is defined as the velocity of the other.) Thus the differential cross-section is

$$d\sigma(\alpha \to \beta) \equiv d\Gamma(\alpha \to \beta)/\Phi_\alpha = (2\pi)^4 u_\alpha^{-1} |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta. \tag{3.4.15}$$

Even though the cases $N_\alpha = 1$ and $N_\alpha = 2$ are the most important, transition rates for $N_\alpha \geq 3$ are all measurable in principle, and some of them are very important in chemistry, astrophysics, etc. (For instance, in one of the main reactions that release energy in the sun, two protons and an electron turn into a deuteron and a neutrino.) Section 3.6 presents an application of the master transition rate formula (3.4.11) for general numbers N_α of initial particles.

We next take up the question of the Lorentz transformation properties of rates and cross-sections, which will help us to give a more general definition of the relative velocity u_α in Eq. (3.4.15). The Lorentz transformation rule (3.3.1) for the S-matrix is complicated by the momentum-dependent matrices associated with each particle's spin. To avoid this complication, consider the absolute-value squared of (3.3.1) (after factoring out the Lorentz-invariant delta function in Eq. (3.4.12)), and sum over all spins. The unitarity of the matrices $D^{(j)}(W)$ (or their analogs for zero mass) then shows that, apart from the energy factors in (3.3.1), the sum is Lorentz-invariant. That is, the quantity

$$\sum_{\text{spins}} |M_{\beta\alpha}|^2 \prod_\beta E \prod_\alpha E \equiv R_{\beta\alpha} \tag{3.4.16}$$

is a scalar function of the four-momenta of the particles in states α and β. (By $\prod_\alpha E$ and $\prod_\beta E$ is meant the product of all the single-particle energies $p^0 = \sqrt{\mathbf{p}^2 + m^2}$ for the particles in the states α and β.)

We can now write the spin-summed single-particle decay rate (3.4.13) as

$$\sum_{\text{spins}} d\Gamma(\alpha \to \beta) = 2\pi E_\alpha^{-1} R_{\beta\alpha} \delta^4(p_\beta - p_\alpha) d\beta / \prod_\beta E.$$

The factor $d\beta/\prod_\beta E$ may be recognized as the product of the Lorentz-invariant momentum-space volume elements (2.5.15), so it is Lorentz-invariant. So also are $R_{\beta\alpha}$ and $\delta^4(p_\beta - p_\alpha)$, leaving just the non-invariant factor $1/E_\alpha$, where E_α is the energy of the single initial particle. Our conclusion then is that the decay rate has the same Lorentz transformation property as $1/E_\alpha$. This is, of course, just the usual special-relativistic time dilation—the faster the particle, the slower it decays.

Similarly, our result (3.4.15) for the spin-summed cross-section may be written as

$$\sum_{\text{spins}} d\sigma(\alpha \to \beta) = (2\pi)^4 u_\alpha^{-1} E_1^{-1} E_2^{-1} R_{\beta\alpha} \delta^4(p_\alpha - p_\beta) d\beta / \prod_\beta E,$$

where E_1 and E_2 are the energies of the two particles in the initial state α. It is conventional to define the cross-section to be (when summed over spins) a Lorentz-invariant function of four-momenta. The factors $R_{\beta\alpha}$, $\delta^4(p_\beta - p_\alpha)$, and $d\beta/\prod_\beta E$ are already Lorentz-invariant, so this means that we must define the relative velocity u_α in arbitrary inertial frames so that $u_\alpha E_1 E_2$ is a scalar. We also mentioned earlier that in the Lorentz frame in which one particle (say, particle 1) is at rest, u_α is the velocity of the other particle. This uniquely determines u_α to have the value in general Lorentz frames

$$u_\alpha = \sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} / E_1 E_2, \tag{3.4.17}$$

where p_1, p_2 and m_1, m_2 are the four-momenta and mass of the two particles in the initial state α.

As a bonus, we note that in the 'center-of-mass' frame, where the total three-momentum vanishes, we have

$$p_1 = (E_1, \mathbf{p}), \qquad p_2 = (E_2, -\mathbf{p}),$$

and here Eq. (3.4.17) gives

$$u_\alpha = \frac{|\mathbf{p}|(E_1 + E_2)}{E_1 E_2} = \left| \frac{\mathbf{p}_1}{E_1} - \frac{\mathbf{p}_2}{E_2} \right| \tag{3.4.18}$$

as might have been expected for a relative velocity. However, in this frame u_α is not really a physical velocity; in particular, Eq. (3.4.18) shows that for extremely relativistic particles, it can take values as large as 2. [19]

We now turn to the interpretation of the so-called phase-space factor $\delta^4(p_\beta - p_\alpha)d\beta$, which appears in the general formula (3.4.11) for transition rates, and also in Eqs. (3.4.13) and (3.4.15) for decay rates and cross-sections. We here specialize to the case of the 'center-of-mass' Lorentz frame, where the total three-momentum of the initial state vanishes

$$\mathbf{p}_\alpha = 0. \tag{3.4.19}$$

(For $N_\alpha = 1$, this is just the case of a particle decaying at rest.) If the final state consists of particles with momenta $\mathbf{p}'_1, \mathbf{p}'_2, \ldots$, then

$$\delta^4(p_\beta - p_\alpha)d\beta = \delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 + \cdots)\delta(E'_1 + E'_2 + \cdots - E)d^3p'_1 d^3p'_2 \cdots , \tag{3.4.20}$$

where $E \equiv E_\alpha$ is the total energy of the initial state. Any one of the $\mathbf{p}'_k$ integrals, say over $\mathbf{p}'_1$, can be done trivially by just dropping the momentum delta function

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta(E'_1 + E'_2 + \cdots - E)d^3p'_2 \cdots \tag{3.4.21}$$

with the understanding that wherever $\mathbf{p}'_1$ appears (as in E'_1) it must be replaced with

$$\mathbf{p}'_1 = -\mathbf{p}'_2 - \mathbf{p}'_3 - \cdots . \tag{3.4.22}$$

We can similarly use the remaining delta function to eliminate any one of the remaining integrals.

In the simplest case, there are just two particles in the final state. Here (3.4.21) gives

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta(E'_1 + E'_2 - E)d^3p'_2.$$

In more detail, this is

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta\left(\sqrt{|\mathbf{p}'_1|^2 + m'^2_1} + \sqrt{|\mathbf{p}'_1|^2 + m'^2_2} - E\right)|\mathbf{p}'_1|^2 d|\mathbf{p}'_1| d\Omega, \tag{3.4.23}$$

where $\mathbf{p}'_2 = -\mathbf{p}'_1$ and $d\Omega \equiv \sin\theta\, d\theta\, d\phi$ is the solid-angle differential for $\mathbf{p}'_1$. This can be simplified by using the standard formula

$$\delta(f(x)) = \delta(x - x_0)/|f'(x_0)|,$$

[19] Eq. (3.4.17) makes it obvious that $E_1 E_2 u_\alpha$ is a scalar. Also, when particle 1 is at rest, we have $\mathbf{p}_1 = 0$, $E_1 = m_1$, so $p_1 \cdot p_2 = -m_1 E_2$, and so Eq. (3.4.17) gives $u_\alpha = \sqrt{E_2^2 - m_2^2}/E_2 = |\mathbf{p}_2|/E_2$, which is just the velocity of particle 2.

where $f(x)$ is an arbitrary real function with a single simple zero at $x = x_0$. In our case, the argument $E_1' + E_2' - E$ of the delta function in Eq. (3.4.23) has a unique zero at $|\mathbf{p}_1'| = k'$, where

$$k' = \frac{\sqrt{(E^2 - m_1'^2 - m_2'^2)^2 - 4m_1'^2 m_2'^2}}{2E}, \tag{3.4.24}$$

$$E_1' = \sqrt{k'^2 + m_1'^2} = \frac{E^2 - m_2'^2 + m_1'^2}{2E}, \tag{3.4.25}$$

$$E_2' = \sqrt{k'^2 + m_2'^2} = \frac{E^2 - m_1'^2 + m_2'^2}{2E}, \tag{3.4.26}$$

with derivative

$$\left[\frac{d}{d|\mathbf{p}_1'|}\left(\sqrt{|\mathbf{p}_1'|^2 + m_1'^2} + \sqrt{|\mathbf{p}_1'|^2 + m_2'^2} - E\right)\right]_{|\mathbf{p}_1'| = k'} = \frac{k'}{E_1'} + \frac{k'}{E_2'} = \frac{k'E}{E_1'E_2'}. \tag{3.4.27}$$

We can thus drop the delta function and the differential $d|\mathbf{p}_1'|$ in Eq. (3.4.23), by dividing by (3.4.27),

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \frac{k'E_1'E_2'}{E}\, d\Omega \tag{3.4.28}$$

with the understanding that k', E_1', and E_2' are given everywhere by Eqs. (3.4.24)–(3.4.26). In particular, the differential rate (3.4.13) for decay of a one-particle state of zero momentum and energy E into two particles is

$$\frac{d\Gamma(\alpha \to \beta)}{d\Omega} = \frac{2\pi k'E_1'E_2'}{E}|M_{\beta\alpha}|^2 \tag{3.4.29}$$

and the differential cross-section for the two-body scattering process $12 \to 1'2'$ is given by Eq. (3.4.15) as

$$\frac{d\sigma(\alpha \to \beta)}{d\Omega} = \frac{(2\pi)^4 k'E_1'E_2'}{Eu_\alpha}|M_{\beta\alpha}|^2 = \frac{(2\pi)^4 k'E_1'E_2'E_1E_2}{E^2 k}|M_{\beta\alpha}|^2, \tag{3.4.30}$$

where $k \equiv |\mathbf{p}_1| = |\mathbf{p}_2|$.

The above case $N_\beta = 2$ is particularly simple, but there is one nice result for $N_\beta = 3$ that is also worth recording. For $N_\beta = 3$, Eq. (3.4.21) gives

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow d^3p_2' d^3p_3' \times \delta\left(\sqrt{|\mathbf{p}_2' + \mathbf{p}_3'|^2 + m_1'^2} + \sqrt{|\mathbf{p}_2'|^2 + m_2'^2} + \sqrt{|\mathbf{p}_3'|^2 + m_3'^2} - E\right).$$

We write the momentum-space volume as

$$d^3p_2' d^3p_3' = |\mathbf{p}_2'|^2 d|\mathbf{p}_2'||\mathbf{p}_3'|^2 d|\mathbf{p}_3'| d\Omega_3 d\phi_{23} d\cos\theta_{23},$$

where $d\Omega_3$ is the differential element of solid angle for $\mathbf{p}_3'$, and θ_{23} and ϕ_{23} are the polar and azimuthal angles of $\mathbf{p}_2'$ relative to the $\mathbf{p}_3'$ direction. The orientation of the plane spanned by $\mathbf{p}_2'$ and $\mathbf{p}_3'$ is specified by ϕ_{23} and the direction of $\mathbf{p}_3'$, with the remaining angle θ_{23} fixed by the energy-conservation condition

$$\sqrt{|\mathbf{p}_2'|^2 + 2|\mathbf{p}_2'||\mathbf{p}_3'|\cos\theta_{23} + |\mathbf{p}_3'|^2 + m_1'^2} + \sqrt{|\mathbf{p}_2'|^2 + m_2'^2} + \sqrt{|\mathbf{p}_3'|^2 + m_3'^2} = E.$$

The derivative of the argument of the delta function with respect to $\cos\theta_{23}$ is

$$\frac{\partial E_1'}{\partial \cos\theta_{23}} = \frac{|\mathbf{p}_2'||\mathbf{p}_3'|}{E_1'},$$

so we can do the integral over $\cos\theta_{23}$ by just dropping the delta function and dividing by this derivative

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow |\mathbf{p}_2'|d|\mathbf{p}_2'||\mathbf{p}_3'|d|\mathbf{p}_3'|E_1' d\Omega_3 d\phi_{23}.$$

Replacing momenta with energies, this is finally

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow E_1'E_2'E_3'\, dE_2'dE_3'd\Omega_3 d\phi_{23}. \tag{3.4.31}$$

But recall that the quantity (3.4.16), obtained by summing $|M_{\beta\alpha}|^2$ over spins and multiplying with the product of energies, is a scalar function of four-momenta. If we approximate this scalar as a constant, then Eq. (3.4.31) tells us that for a fixed initial state, the distribution of events plotted in the E_2', E_3' plane is uniform. Any departure from a uniform distribution of events in this plot thus provides a useful clue to the dynamics of the decay process, including possible centrifugal barriers or resonant intermediate states. This is known as a Dalitz plot [19], because of its use by Dalitz in 1953 to analyze the decay $K^+ \to \pi^+ + \pi^+ + \pi^-$.

3.5 Perturbation Theory

The technique that has historically been most useful in calculating the S-matrix is perturbation theory, an expansion in powers of the interaction term H_{int} in the Hamiltonian $H = H_0 + H_{\text{int}}$. Eqs. (3.2.7) and (3.1.18) give the S-matrix as

$$S_{\beta\alpha} = \delta(\beta - \alpha) - 2\pi i\delta(E_\beta - E_\alpha)T^+_{\beta\alpha}, \qquad T^+_{\beta\alpha} = \langle\beta\|H_{\text{int}}\|\alpha;\text{in}\rangle,$$

where $|\alpha;\text{in}\rangle$ satisfies the Lippmann–Schwinger equation (3.1.17):

$$|\alpha;\text{in}\rangle = |\alpha\rangle + \int d\gamma\, \frac{T^+_{\gamma\alpha}|\gamma\rangle}{E_\alpha - E_\gamma + i\epsilon}.$$

Operating on this equation with H_{int} and taking the scalar product with $|\beta\rangle$ yields an integral equation for T^+

$$T^+_{\beta\alpha} = (H_{\text{int}})_{\beta\alpha} + \int d\gamma\, \frac{(H_{\text{int}})_{\beta\gamma}T^+_{\gamma\alpha}}{E_\alpha - E_\gamma + i\epsilon}, \tag{3.5.1}$$

where

$$(H_{\text{int}})_{\beta\alpha} \equiv \langle\beta\|H_{\text{int}}\|\alpha\rangle. \tag{3.5.2}$$

The perturbation series for $T^+_{\beta\alpha}$ is obtained by iteration from Eq. (3.5.1)

$$\begin{aligned} T^+_{\beta\alpha} = (H_{\text{int}})_{\beta\alpha} &+ \int d\gamma\, \frac{(H_{\text{int}})_{\beta\gamma}(H_{\text{int}})_{\gamma\alpha}}{E_\alpha - E_\gamma + i\epsilon} \\ &+ \int d\gamma\, d\gamma'\, \frac{(H_{\text{int}})_{\beta\gamma}(H_{\text{int}})_{\gamma\gamma'}(H_{\text{int}})_{\gamma'\alpha}}{(E_\alpha - E_\gamma + i\epsilon)(E_\alpha - E_{\gamma'} + i\epsilon)} + \cdots . \end{aligned} \tag{3.5.3}$$

The method of calculation based on Eq. (3.5.3), which dominated calculations of the S-matrix in the 1930s, is today known as *old-fashioned perturbation theory*. Its obvious drawback is that the energy denominators obscure the underlying Lorentz invariance of the S-matrix. It still has some uses, however, in clarifying the way that singularities of the S-matrix arise from various intermediate states. For the most part in this book, we will rely on a rewritten version of Eq. (3.5.3), known as *time-dependent perturbation theory*, which has the virtue of making Lorentz invariance much more transparent, while somewhat obscuring the contribution of individual intermediate states.

The easiest way to derive the time-ordered perturbation expansion is to use Eq. (3.2.5), which gives the S-operator as

$$S = U(\infty, -\infty),$$

where

$$U(\tau, \tau_0) \equiv \exp(iH_0\tau)\exp\big(-iH(\tau-\tau_0)\big)\exp(-iH_0\tau_0).$$

Differentiating this formula for $U(\tau,\tau_0)$ with respect to τ gives the differential equation

$$i\frac{d}{d\tau}U(\tau,\tau_0) = H_I(\tau)U(\tau,\tau_0), \tag{3.5.4}$$

where

$$H_I(t) \equiv \exp(iH_0t)H_{\rm int}\exp(-iH_0t). \tag{3.5.5}$$

(Operators with this sort of time-dependence are said to be defined in the *interaction picture*, to distinguish their time-dependence from the time-dependence

$$O_H(t) = \exp(iHt)O_H\exp(-iHt)$$

required in the Heisenberg picture of quantum mechanics.) Eq. (3.5.4) as well as the initial condition $U(\tau_0,\tau_0) = \mathbf{1}$ is obviously satisfied by the solution of the integral equation

$$U(\tau,\tau_0) = \mathbf{1} - i\int_{\tau_0}^{\tau} dt\, H_I(t)U(t,\tau_0). \tag{3.5.6}$$

By iteration of this integral equation, we obtain an expansion for $U(\tau,\tau_0)$ in powers of $H_{\rm int}$

$$\begin{aligned} U(\tau,\tau_0) = \mathbf{1} &- i\int_{\tau_0}^{\tau} dt_1\, H_I(t_1) + (-i)^2\int_{\tau_0}^{\tau} dt_1\int_{\tau_0}^{t_1} dt_2\, H_I(t_1)H_I(t_2) \\ &+ (-i)^3\int_{\tau_0}^{\tau} dt_1\int_{\tau_0}^{t_1} dt_2\int_{\tau_0}^{t_2} dt_3\, H_I(t_1)H_I(t_2)H_I(t_3) + \cdots . \end{aligned} \tag{3.5.7}$$

Setting $\tau = \infty$ and $\tau_0 = -\infty$ then gives the perturbation expansion for the S-operator:

$$\begin{aligned} S = \mathbf{1} &- i\int_{-\infty}^{\infty} dt_1\, H_I(t_1) + (-i)^2\int_{-\infty}^{\infty} dt_1\int_{-\infty}^{t_1} dt_2\, H_I(t_1)H_I(t_2) \\ &+ (-i)^3\int_{-\infty}^{\infty} dt_1\int_{-\infty}^{t_1} dt_2\int_{-\infty}^{t_2} dt_3\, H_I(t_1)H_I(t_2)H_I(t_3) + \cdots . \end{aligned} \tag{3.5.8}$$

This can also be derived directly from the old-fashioned perturbation expansion (3.5.3), by using the Fourier representation of the energy factors in Eq. (3.5.3):

$$(E_\alpha - E_\gamma + i\epsilon)^{-1} = -i \int_0^\infty d\tau \, \exp\big(i(E_\alpha - E_\gamma)\tau\big) \tag{3.5.9}$$

with the understanding that such integrals are to be evaluated by inserting a convergence factor $e^{-\epsilon\tau}$ in the integrand, with $\epsilon \to 0+$.

There is a way of rewriting Eq. (3.5.8) that proves very useful in carrying out manifestly Lorentz-invariant calculations. Define the *time-ordered product* of any time-dependent operators as the product with factors arranged so that the one with the latest time-argument is placed leftmost, the next-latest next to the leftmost, and so on. For instance,

$$T\{H_I(t)\} = H_I(t),$$

$$T\{H_I(t_1)H_I(t_2)\} = \theta(t_1 - t_2)H_I(t_1)H_I(t_2) + \theta(t_2 - t_1)H_I(t_2)H_I(t_1),$$

and so on, where $\theta(\tau)$ is the step function, equal to $+1$ for $\tau > 0$ and to zero for $\tau < 0$. The time-ordered product of n H_Is is a sum over all $n!$ permutations of the H_Is, each of which gives the same integral over all $t_1 \cdots t_n$, so Eq. (3.5.8) may be written

$$S = \mathbf{1} + \sum_{n=1}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 dt_2 \cdots dt_n \, T\{H_I(t_1) \cdots H_I(t_n)\}. \tag{3.5.10}$$

This is sometimes known as the Dyson series [20]. This series can be summed if the $H_I(t)$ at different times all commute; the sum is then

$$S = \exp\left(-i \int_{-\infty}^{\infty} dt \, H_I(t)\right).$$

Of course, this is not usually the case; in general (3.5.10) does not even converge, and is at best an asymptotic expansion in whatever coupling constant factors appear in H_{int}. However Eq. (3.5.10) is sometimes written in the general case as

$$S = T \exp\left(-i \int_{-\infty}^{\infty} dt \, H_I(t)\right)$$

with T indicating here that the expression is to be evaluated by time-ordering each term in the series expansion for the exponential.

We can now readily find one large class of theories for which the S-matrix is manifestly Lorentz-invariant. Since the elements of the S-matrix are the matrix elements of the S-operator between free-particle states $|\alpha\rangle$, $|\beta\rangle$, etc., what we want is that the S-operator should commute with the operator $U_0(\Lambda, a)$ that produces Lorentz transformations on these free-particle states. Equivalently, the S-operator must commute with the generators of $U_0(\Lambda, a)$: H_0, $\mathbf{P}_0$, $\mathbf{J}_0$, and $\mathbf{K}_0$. To satisfy this requirement, let's try the hypothesis that $H_I(t)$ is an integral over three-space

$$H_I(t) = \int d^3x \, \mathcal{H}_I(\mathbf{x}, t) \tag{3.5.11}$$

with $\mathcal{H}_I(x)$ a scalar in the sense that

$$U_0(\Lambda, a)\mathcal{H}_I(x)U_0^{-1}(\Lambda, a) = \mathcal{H}_I(\Lambda x + a). \tag{3.5.12}$$

(By equating the coefficients of a^0 for infinitesimal transformations it can be checked that $\mathcal{H}_I(x)$ has a time-dependence consistent with Eq. (3.5.5).) Then S may be written as a sum of four-dimensional integrals

$$S = \mathbf{1} + \sum_{n=1}^{\infty} \frac{(-i)^n}{n!} \int d^4x_1 \cdots d^4x_n \, T\{\mathcal{H}_I(x_1) \cdots \mathcal{H}_I(x_n)\}. \tag{3.5.13}$$

Everything is now manifestly Lorentz invariant, except for the time-ordering of the operator product.

Now, the time-ordering of two spacetime points x_1, x_2 is Lorentz-invariant unless $x_1 - x_2$ is space-like, i.e., unless $(x_1 - x_2)^2 > 0$, so the time-ordering in Eq. (3.5.13) introduces no special Lorentz frame if (though not only if) the $\mathcal{H}_I(x)$ all commute at space-like or light-like separations:

$$[\mathcal{H}_I(x), \mathcal{H}_I(x')] = 0 \qquad \text{for} \qquad (x - x')^2 \geq 0. \tag{3.5.14}$$

We can use the results of Section 3.3 to give a formal non-perturbative proof that an interaction (3.5.11) satisfying Eqs. (3.5.12) and (3.5.14) does lead to an S-matrix with the correct Lorentz transformation properties. For an infinitesimal boost, Eq. (3.5.12) gives

$$i[\mathbf{K}_0, \mathcal{H}_I(\mathbf{x}, t)] = t\boldsymbol{\nabla}\mathcal{H}_I(\mathbf{x}, t) + \mathbf{x}\frac{\partial}{\partial t}\mathcal{H}_I(\mathbf{x}, t), \tag{3.5.15}$$

so integrating over $\mathbf{x}$ and setting $t = 0$,

$$[\mathbf{K}_0, H_{\text{int}}] = \left[\mathbf{K}_0, \int d^3x \, \mathcal{H}_I(\mathbf{x}, 0)\right] = [H_0, \mathbf{W}], \tag{3.5.16}$$

where

$$\mathbf{W} \equiv \int d^3x \, \mathbf{x}\mathcal{H}_I(\mathbf{x}, 0). \tag{3.5.17}$$

If (as is usually the case) the matrix elements of $\mathcal{H}_I(\mathbf{x}, 0)$ between eigenstates of H_0 are smooth functions of the energy eigenvalues, then the same is true of H_{int}, as is necessary for the validity of scattering theory, and also true of $\mathbf{W}$, which is necessary in the proof of Lorentz invariance. The other condition for Lorentz invariance, the commutation relation (3.3.21), is also valid if and only if

$$0 = [\mathbf{W}, H_{\text{int}}] = \int d^3x \int d^3y \, \mathbf{x}[\mathcal{H}_I(\mathbf{x}, 0), \mathcal{H}_I(\mathbf{y}, 0)]. \tag{3.5.18}$$

This condition would follow from the 'causality' condition (3.5.14), but provides a somewhat less restrictive sufficient condition for Lorentz invariance of the S-matrix.

Theories of this class are not the only ones that are Lorentz invariant, but the most general Lorentz invariant theories are not very different. In particular, there is always a commutation condition something like (3.5.14) that needs to be satisfied. This condition has no counterpart for non-relativistic systems, for which time-ordering is always Galilean-invariant. *It is this condition that makes the combination of Lorentz invariance and quantum mechanics so restrictive.* [20]

* * *

The methods described so far in this section are useful when the interaction operator $H_{\rm int}$ is sufficiently small. There is also a modified version of this approximation, known as the *distorted-wave Born approximation*, that is useful when the interaction contains two terms

$$H_{\rm int} = H_s + H_w \tag{3.5.19}$$

with H_w weak but H_s strong. We can define $|\alpha; s, {\rm in/out}\rangle$ as what the 'in' and 'out' states would be if H_s were the whole interaction

$$|\alpha; s, {\rm in/out}\rangle = |\alpha\rangle + (E_\alpha - H_0 \pm i\epsilon)^{-1} H_s |\alpha; s, {\rm in/out}\rangle. \tag{3.5.20}$$

We can then write (3.1.16) as

$$\begin{aligned}
T^+_{\beta\alpha} &= \langle\beta\| H_{\rm int} \|\alpha; {\rm in}\rangle \\
&= \left(\langle\beta; s, {\rm out}| - \langle\beta; s, {\rm out}| H_s (E_\beta - H_0 + i\epsilon)^{-1}\right)(H_s + H_w)|\alpha; {\rm in}\rangle \\
&= \langle\beta; s, {\rm out}\| H_w \|\alpha; {\rm in}\rangle \\
&\quad + \langle\beta; s, {\rm out}\| [H_s - H_s (E_\beta - H_0 + i\epsilon)^{-1}(H_s + H_w)] \|\alpha; {\rm in}\rangle,
\end{aligned}$$

and so

$$T^+_{\beta\alpha} = \langle\beta; s, {\rm out}\| H_w \|\alpha; {\rm in}\rangle + \langle\beta; s, {\rm out}\| H_s \|\alpha\rangle. \tag{3.5.21}$$

The second term on the right-hand side is just what $T^+_{s,\beta\alpha}$ would be in the presence of strong interactions alone

$$T^+_{s,\beta\alpha} = \langle\beta\| H_s \|\alpha; s, {\rm in}\rangle = \langle\beta; s, {\rm out}\| H_s \|\alpha\rangle. \tag{3.5.22}$$

(For a proof of Eq. (3.5.22), just drop H_w everywhere in the derivation of Eq. (3.5.21).) Eq. (3.5.21) is most useful when this second term vanishes: that is, when the process $\alpha \to \beta$ cannot be produced by the strong interactions alone. (For example, in nuclear beta decay we need a weak nuclear force to turn neutrons into protons, even though we cannot ignore the presence of the strong nuclear force acting in the initial and final nuclear states.) For such processes, the matrix element (3.5.22) vanishes, so Eq. (3.5.21) reads

$$T^+_{\beta\alpha} = \langle\beta; s, {\rm out}\| H_w \|\alpha; {\rm in}\rangle. \tag{3.5.23}$$

So far, this is all exact. However, this way of rewriting the T-matrix becomes worthwhile when H_w is so weak that we may neglect its effect on the state $|\alpha; {\rm in}\rangle$ in Eq. (3.5.23), and hence replace $|\alpha; {\rm in}\rangle$ with the state $|\alpha; s, {\rm in}\rangle$, which takes account only of the strong interaction H_s. In this approximation, Eq. (3.5.23) becomes

$$T^+_{\beta\alpha} \simeq \langle\beta; s, {\rm out}\| H_w \|\alpha; s, {\rm in}\rangle. \tag{3.5.24}$$

[20] We write the condition on x and x' here as $(x - x')^2 \geq 0$ instead of $(x - x')^2 > 0$, because as we shall see in Chapter 6, Lorentz invariance can be disturbed by troublesome singularities at $x = x'$.

This is valid to first order in H_w, but to all orders in H_s. This approximation is ubiquitous in physics; for instance, the S-matrix element for nuclear beta or gamma decay is calculated using Eq. (3.5.24) with H_s the strong nuclear interaction and H_w respectively either the weak nuclear interaction or the electromagnetic interaction, and with $|\beta; s, \mathrm{out}\rangle$ and $|\alpha; s, \mathrm{in}\rangle$ the final and initial nuclear states.

3.6 Implications of Unitarity

The unitarity of the S-matrix imposes an interesting and useful condition relating the amplitude $M_{\alpha\alpha}$ for forward scattering in an arbitrary multi-particle state α to the total rate for all reactions in that state. Recall that in the general case, where the state β may or may not be the same as the state α, the S-matrix may be written as in (3.3.2):

$$S_{\beta\alpha} = \delta(\beta - \alpha) - 2\pi i \delta^4(p_\beta - p_\alpha) M_{\beta\alpha}.$$

The unitarity condition then gives

$$\begin{aligned}\delta(\gamma - \alpha) &= \int d\beta\, S^*_{\beta\gamma} S_{\beta\alpha} \\ &= \delta(\gamma - \alpha) - 2\pi i \delta^4(p_\gamma - p_\alpha) M_{\gamma\alpha} \\ &\quad + 2\pi i \delta^4(p_\gamma - p_\alpha) M^*_{\alpha\gamma} + 4\pi^2 \int d\beta\, \delta^4(p_\beta - p_\gamma)\delta^4(p_\beta - p_\alpha) M^*_{\beta\gamma} M_{\beta\alpha}.\end{aligned}$$

Cancelling the term $\delta(\gamma - \alpha)$ and a factor $2\pi\delta^4(p_\gamma - p_\alpha)$, we find that for $p_\gamma = p_\alpha$

$$0 = -iM_{\gamma\alpha} + iM^*_{\alpha\gamma} + 2\pi \int d\beta\, \delta^4(p_\beta - p_\alpha) M^*_{\beta\gamma} M_{\beta\alpha}. \tag{3.6.1}$$

This is most useful in the special case $\alpha = \gamma$, where it reads

$$\mathrm{Im}\, M_{\alpha\alpha} = -\pi \int d\beta\, \delta^4(p_\beta - p_\alpha)|M_{\beta\alpha}|^2. \tag{3.6.2}$$

Using Eq. (3.4.11), this can be expressed as a formula for the total rate for all reactions produced by an initial state α in a volume V

$$\begin{aligned}\Gamma_\alpha &\equiv \int d\beta\, \frac{d\Gamma(\alpha \to \beta)}{d\beta} \\ &= (2\pi)^{3N_\alpha - 2} V^{1 - N_\alpha} \int d\beta\, \delta^4(p_\beta - p_\alpha)|M_{\beta\alpha}|^2 \\ &= -\frac{1}{\pi}(2\pi)^{3N_\alpha - 2} V^{1 - N_\alpha}\, \mathrm{Im}\, M_{\alpha\alpha}.\end{aligned} \tag{3.6.3}$$

In particular, where α is a two-particle state, this can be written

$$\mathrm{Im}\, M_{\alpha\alpha} = -u_\alpha \sigma_\alpha / 16\pi^3, \tag{3.6.4}$$

where u_α is the relative velocity (3.4.17) in state α, and σ_α is the *total* cross-section in this state, given by (3.4.15) as

$$\sigma_\alpha \equiv \int d\beta\, d\sigma(\alpha \to \beta)/d\beta = (2\pi)^4 u_\alpha^{-1} \int d\beta\, |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha). \tag{3.6.5}$$

This is usually expressed in a slightly different way, in terms of a *scattering amplitude* $f(\alpha \to \beta)$. Eq. (3.4.30) shows that the differential cross-section for *two-body* scattering in the center-of-mass frame is

$$\frac{d\sigma(\alpha \to \beta)}{d\Omega} = \frac{(2\pi)^4 k' E_1' E_2' E_1 E_2}{kE^2} |M_{\beta\alpha}|^2, \tag{3.6.6}$$

where k' and k are the magnitudes of the momenta in the initial and final states. We therefore define the scattering amplitude as[21]

$$f(\alpha \to \beta) \equiv -\frac{4\pi^2}{E} \sqrt{\frac{k' E_1' E_2' E_1 E_2}{k}}\, M_{\beta\alpha}, \tag{3.6.7}$$

so that the differential cross-section is simply

$$\frac{d\sigma(\alpha \to \beta)}{d\Omega} = |f(\alpha \to \beta)|^2. \tag{3.6.8}$$

In particular, for *elastic* two-body scattering, we have

$$f(\alpha \to \beta) \equiv -\frac{4\pi^2 E_1 E_2}{E} M_{\beta\alpha}. \tag{3.6.9}$$

Using (3.4.18) for the relative velocity u_α, the unitarity prediction (3.6.3) now reads

$$\operatorname{Im} f(\alpha \to \alpha) = \frac{k}{4\pi} \sigma_\alpha. \tag{3.6.10}$$

This form of the unitarity condition (3.6.3) is known as the *optical theorem* [22].

There is a pretty consequence of the optical theorem that tells much about the pattern of scattering at high energy. The scattering amplitude f may be expected to be a smooth function of angle, so there must be some solid angle $\Delta\Omega$ within which $|f|^2$ has nearly the same value (say, within a factor 2) as in the forward direction. The total cross-section is then bounded by

$$\sigma_\alpha \geq \int |f|^2 d\Omega \geq \frac{1}{2} |f(\alpha \to \alpha)|^2 \Delta\Omega \geq \frac{1}{2} |\operatorname{Im} f(\alpha \to \alpha)|^2 \Delta\Omega.$$

Using Eq. (3.6.10) then yields an upper bound on $\Delta\Omega$

$$\Delta\Omega \leq \frac{32\pi^2}{k^2 \sigma_\alpha}. \tag{3.6.11}$$

As we shall see in the next section, total cross-sections are usually expected to approach constants or grow slowly at high energy, so Eq. (3.6.11) shows that the solid angle around the forward direction within which the differential cross-section is roughly constant shrinks at least as fast as $1/k^2$ for $k \to \infty$. This increasingly narrow peak in the forward direction at high energies is known as a *diffraction peak*.

[21] The phase of f is conventional, and is motivated by the wave mechanical interpretation [21] of f as the coefficient of the outgoing wave in the solution of the time-independent Schrödinger equation. The normalization of f used here is slightly unconventional for inelastic scattering; usually f is defined so that a ratio of final and initial velocities appears in the formula for the differential cross-section.

Returning now to the general case of reactions involving arbitrary numbers of particles, we can use Eq. (3.6.2) together with CPT invariance to say something about the relations for total interaction rates of particles and antiparticles. Because CPT is antiunitary, its conservation does not in general imply any simple relation between a process $\alpha \to \beta$ and the process with particles replaced with their antiparticles. Instead, it provides a relation between a process and the *inverse* process involving antiparticles: we can use the same arguments that allowed us to deduce (3.3.46) from time-reversal invariance to show that CPT invariance requires the S-matrix to satisfy the condition

$$S_{\beta,\alpha} = S_{\mathcal{CPT}\alpha,\mathcal{CPT}\beta}, \tag{3.6.12}$$

where $\mathcal{CPT}$ indicates that we must reverse all spin z-components, change all particles into their corresponding antiparticles, and multiply the matrix element by various phase factors for the particles in the initial state and by their complex conjugates for the particles in the final state. Since CPT invariance also requires particles to have the same masses as their corresponding antiparticles, the same relation holds for the coefficient of $\delta^4(p_\alpha - p_\beta)$ in $S_{\beta\alpha}$:

$$M_{\beta,\alpha} = M_{\mathcal{CPT}\alpha,\mathcal{CPT}\beta}. \tag{3.6.13}$$

In particular, when the initial and final states are the same the phase factors all cancel, and Eq. (3.6.13) says that

$$\begin{aligned} &M_{p_1\sigma_1 n_1;\,p_2\sigma_2 n_2;\,\dots,\,p_1\sigma_1 n_1;\,p_2\sigma_2 n_2;\,\dots} \\ &\quad = M_{p_1,-\sigma_1,n_1^c;\,p_2,-\sigma_2,n_2^c;\,\dots,\,p_1,-\sigma_1,n_1^c;\,p_2,-\sigma_2,n_2^c;\,\dots}, \end{aligned} \tag{3.6.14}$$

where a superscript c on n indicates the antiparticle of n. The generalized optical theorem (3.6.2) then tells us that *the total reaction rate from an initial state consisting of some set of particles is the same as for an initial state consisting of the corresponding antiparticles with spins reversed*:

$$\Gamma_{p_1\sigma_1 n_1;\,p_2\sigma_2 n_2;\,\dots} = \Gamma_{p_1,-\sigma_1,n_1^c;\,p_2,-\sigma_2,n_2^c;\,\dots}. \tag{3.6.15}$$

In particular, applying this to one-particle states, we see that the decay rate of any particle equals the decay rate of the antiparticle with reversed spin. Rotational invariance does not allow particle decay rates to depend on the spin z-component of the decaying particle, so a special case of the general result (3.6.15) is that unstable particles and their corresponding antiparticles have precisely the same lifetimes.

* * *

The same argument that led from the unitarity condition $S^\dagger S = \mathbf{1}$ to our result (3.6.2) also allows us to use the other unitarity relation $SS^\dagger = \mathbf{1}$ to derive the result

$$\operatorname{Im} M_{\alpha\alpha} = -\pi \int d\beta\, \delta^4(p_\beta - p_\alpha)|M_{\alpha\beta}|^2. \tag{3.6.16}$$

Putting this together with Eq. (3.6.2) then yields the reciprocity relation

$$\int d\beta\, \delta^4(p_\beta - p_\alpha)|M_{\beta\alpha}|^2 = \int d\beta\, \delta^4(p_\beta - p_\alpha)|M_{\alpha\beta}|^2 \tag{3.6.17}$$

or in other words

$$\int d\beta\,\frac{d\Gamma(\alpha\to\beta)}{d\beta}=\int d\beta\,\frac{d\Gamma(\beta\to\alpha)}{d\alpha}. \tag{3.6.18}$$

This result can be used in deriving some of the most important results of kinetic theory [23]. If $P_\alpha d\alpha$ is the probability of finding the system in a volume $d\alpha$ of the space of multi-particle states $|\alpha\rangle$, then the rate of decrease in P_α due to transitions to all other states is $P_\alpha\int d\beta\,d\Gamma(\alpha\to\beta)/d\beta$, while the rate of increase of P_α due to transitions from all other states is $\int d\beta\,P_\beta\,d\Gamma(\beta\to\alpha)/d\alpha$; the rate of change of P_α is then

$$\frac{dP_\alpha}{dt}=\int d\beta\,P_\beta\frac{d\Gamma(\beta\to\alpha)}{d\alpha}-P_\alpha\int d\beta\,\frac{d\Gamma(\alpha\to\beta)}{d\beta}. \tag{3.6.19}$$

It follows immediately that $\int P_\alpha d\alpha$ is time-independent. (Just interchange the labelling of the integration variables in the integral in the second term in Eq. (3.6.19).) On the other hand, the rate of change of the entropy $-\int d\alpha\,P_\alpha\ln P_\alpha$ is

$$-\frac{d}{dt}\int d\alpha\,P_\alpha\ln P_\alpha=-\int d\alpha\int d\beta\,(\ln P_\alpha+1)\left[P_\beta\frac{d\Gamma(\beta\to\alpha)}{d\alpha}-P_\alpha\frac{d\Gamma(\alpha\to\beta)}{d\beta}\right].$$

Interchanging the labelling of the integration variables in the second term, this may be written

$$-\frac{d}{dt}\int d\alpha\,P_\alpha\ln P_\alpha=\int d\alpha\int d\beta\,P_\beta\ln\left(\frac{P_\beta}{P_\alpha}\right)\frac{d\Gamma(\beta\to\alpha)}{d\alpha}.$$

Now, for any positive probabilities P_α and P_β, the function $P_\beta\ln(P_\beta/P_\alpha)$ satisfies the inequality[22]

$$P_\beta\ln\left(\frac{P_\beta}{P_\alpha}\right)\geq P_\beta-P_\alpha.$$

The rate of change of the entropy is thus bounded by

$$-\frac{d}{dt}\int d\alpha\,P_\alpha\ln P_\alpha\geq\int d\alpha\int d\beta\,[P_\beta-P_\alpha]\frac{d\Gamma(\beta\to\alpha)}{d\alpha}$$

or interchanging variables of integration in the second term

$$-\frac{d}{dt}\int d\alpha\,P_\alpha\ln P_\alpha\geq\int d\alpha\int d\beta\,P_\beta\left[\frac{d\Gamma(\beta\to\alpha)}{d\alpha}-\frac{d\Gamma(\alpha\to\beta)}{d\beta}\right].$$

But the unitarity relation (3.6.18) tells us that the integral over α on the right-hand side vanishes, so we may conclude that the entropy always increases:

$$-\frac{d}{dt}\int d\alpha\,P_\alpha\ln P_\alpha\geq 0. \tag{3.6.20}$$

This is the 'Boltzmann H-theorem.' This theorem is often derived in statistical mechanics textbooks either by using the Born approximation, for which $|M_{\beta\alpha}|^2$ is symmetric in α and β so that $d\Gamma(\beta\to\alpha)/d\alpha=d\Gamma(\alpha\to\beta)/d\beta$, or by assuming time-reversal invariance,

[22] The difference between the left- and right-hand sides approaches the positive quantity $(P_\alpha-P_\beta)^2/2P_\beta$ for $P_\alpha\to P_\beta$, and has a derivative with respect to P_α that is positive- or negative-definite for all $P_\alpha>P_\beta$ or $P_\alpha<P_\beta$, respectively.

which would tell us that $|M_{\beta\alpha}|^2$ is unchanged if we interchange α and β and also reverse all momenta and spins. Of course, neither the Born approximation nor time-reversal invariance are exact, so it is a good thing that the unitarity result (3.6.18) is all we need in order to derive the H-theorem.

The increase of the entropy stops when the probability P_α becomes a function only of conserved quantities such as the total energy and charge. In this case the conservation laws require $d\Gamma(\beta \to \alpha)/d\alpha$ to vanish unless $P_\alpha = P_\beta$, so we can replace P_β with P_α in the first term of Eq. (3.6.19). Using Eq. (3.6.18) again then shows that in this case, P_α is time-independent. Here again, we need only the unitarity relation (3.6.18), not the Born approximation or time-reversal invariance.

3.7 Partial-Wave Expansions[23]

It is often convenient to work with the S-matrix in a basis of free-particle states in which all variables are discrete, except for the total momentum and energy. This is possible because the components of the momenta $\mathbf{p}_1, \ldots, \mathbf{p}_n$ in an n-particle state of definite total momentum $\mathbf{p}$ and total energy E form a $(3n-4)$-dimensional compact space; for instance, for $n = 2$ particles in the center-of-mass frame with $\mathbf{p} = 0$, this space is a two-dimensional spherical surface. Any function on such a compact space may be expanded in a series of generalized 'partial waves', such as the spherical harmonics that are commonly used in representing functions on the two-sphere. We may thus define a basis for these n-particle states that apart from the continuous variables $\mathbf{p}$ and E is discrete: we label the free-particle states in such a basis as $|E, \mathbf{p}, N\rangle$, with the index N incorporating all spin and species labels as well as whatever indices are used to label the generalized partial waves. These states may conveniently be chosen to be normalized so that their scalar products are:

$$\langle E', \mathbf{p}', N'|E, \mathbf{p}, N\rangle = \delta(E'-E)\delta^3(\mathbf{p}'-\mathbf{p})\delta_{N'N}. \tag{3.7.1}$$

The S-operator then has matrix elements in this basis of the form

$$\langle E', \mathbf{p}', N'|S|E, \mathbf{p}, N\rangle = \delta(E'-E)\delta^3(\mathbf{p}'-\mathbf{p})S_{N'N}(E, \mathbf{p}), \tag{3.7.2}$$

where $S_{N'N}$ is a unitary matrix. Similarly, the T-operator, whose free-particle matrix elements $\langle\beta|T|\alpha\rangle$ are defined to be the quantities $T^+_{\beta\alpha}$ defined by Eq. (3.1.18), may be expressed in our new basis (in accordance with Eq. (3.4.12)) as

$$\langle E', \mathbf{p}', N'|T|E, \mathbf{p}, N\rangle = \delta^3(\mathbf{p}'-\mathbf{p})M_{N'N}(E, \mathbf{p}) \tag{3.7.3}$$

and the relation (3.2.7) is now an ordinary matrix equation:

$$S_{N'N}(E, \mathbf{p}) = \delta_{N'N} - 2\pi i M_{N'N}(E, \mathbf{p}). \tag{3.7.4}$$

We shall use this general formalism in the following section; for the present we will concentrate on reactions in which the initial state involves just two particles.

[23] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

For example, consider a state consisting of two non-identical particles of species n_1, n_2 with non-zero masses M_1, M_2 and arbitrary spins s_1, s_2. In this case, the states may be labelled by their total momentum $\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2$, the energy E, the species labels n_1, n_2, the spin z-components σ_1, σ_2, and a pair of integers ℓ, m (with $|m| \leq \ell$) that specify the dependence of the state on the directions of, say, $\mathbf{p}_1$. Alternatively, we can form a convenient discrete basis by using Clebsch–Gordan coefficients [9] to combine the two spins to give a total spin s with z-component μ, and then using Clebsch–Gordan coefficients again to combine this with the orbital angular momentum ℓ with three-component m to form a total angular momentum j with three-component σ. This gives a basis of states $|E, \mathbf{p}, j, \sigma, \ell, s, n\rangle$ (with n a 'channel index' labelling the two particle species n_1, n_2), defined by their scalar products with the states of definite individual momenta and spin three-components:

$$\langle \mathbf{p}_1, \sigma_1; \mathbf{p}_2, \sigma_2; n' | E, \mathbf{p}, j, \sigma, \ell, s, n\rangle = \left(\frac{|\mathbf{p}_1| E_1 E_2}{E}\right)^{-1/2} \delta^3(\mathbf{p} - \mathbf{p}_1 - \mathbf{p}_2)$$
$$\times \delta\left(E - \sqrt{\mathbf{p}_1^2 + M_1^2} - \sqrt{\mathbf{p}_2^2 + M_2^2}\right) \delta_{n'n} \sum_{m,\mu} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{\ell s}(j, \sigma; m, \mu) Y_\ell^m(\widehat{\mathbf{p}}_1). \tag{3.7.5}$$

where Y_ℓ^m are the usual spherical harmonics [24]. The factor $(|\mathbf{p}_1| E_1 E_2 / E)^{-1/2}$ is inserted so that in the center-of-mass frame these states will be properly normalized:

$$\begin{aligned} &\langle E', \mathbf{p}', j', \sigma', \ell', s', n' | E, \mathbf{0}, j, \sigma, \ell, s, n\rangle \\ &\quad = \delta^3(\mathbf{p}') \delta(E' - E) \delta_{j'j} \delta_{\sigma'\sigma} \delta_{\ell'\ell} \delta_{s's} \delta_{n'n}. \end{aligned} \tag{3.7.6}$$

For identical particles to avoid double counting we must integrate only over half of the two-particle momentum space, so an extra factor $\sqrt{2}$ should appear in the scalar product (3.7.6).

In the center-of-mass frame the matrix elements of any momentum-conserving and rotationally-invariant operator O must take the form:

$$\langle E, \mathbf{p}', j', \sigma', \ell', s', n' | O | E, \mathbf{0}, j, \sigma, \ell, s, n\rangle = \delta^3(\mathbf{p}') O^j_{\ell' s' n', \ell s n}(E) \delta_{j'j} \delta_{\sigma'\sigma}. \tag{3.7.7}$$

(The fact that this is diagonal in j and σ follows from the commutation of O with $\mathbf{J}^2$ and J_3, and the further fact that the coefficient of $\delta_{\sigma\sigma'}$ is independent of σ follows from the commutation of O with $J_1 \pm i J_2$. This is a special case of a general result known as the Wigner–Eckart theorem [25].) Applying this to the operator M whose matrix elements are the quantities $M_{\beta\alpha}$, it follows that the scattering amplitude (3.6.7) in the center-of-mass system takes the form

$$\begin{aligned} &f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \to \mathbf{k}'\sigma_1', -\mathbf{k}'\sigma_2', n') \\ &\quad \equiv -4\pi^2 \sqrt{\frac{k' E_1' E_2' E_1 E_2}{E^2 k}} M_{\mathbf{k}'\sigma_1', -\mathbf{k}'\sigma_2', n'; \mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n} \\ &\quad = -\frac{4\pi^2}{k} \sum_{j\sigma\ell m\ell' m' s\mu s'\mu'} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{\ell s}(j, \sigma; m, \mu) \\ &\qquad \times C_{s_1' s_2'}(s', \mu'; \sigma_1', \sigma_2') C_{\ell' s'}(j, \sigma; m', \mu') Y_{\ell'}^{m'*}(\widehat{\mathbf{k}}') Y_\ell^m(\widehat{\mathbf{k}}) M^j_{\ell' s' n', \ell s n}(E). \end{aligned} \tag{3.7.8}$$

The differential scattering cross-section is $|f|^2$. We will take the direction of the initial momentum $\mathbf{k}$ to be along the three-direction, in which case

$$Y_\ell^m(\widehat{\mathbf{k}}) = \delta_{m0}\sqrt{\frac{2\ell+1}{4\pi}}. \tag{3.7.9}$$

Integrating $|f|^2$ over the direction of the final momentum $\mathbf{k}'$ and respectively summing and averaging over final and initial spin three-components gives the total cross-section[24] for transitions from channel n to n':

$$\begin{aligned}\sigma(n \to n'; E) = \frac{\pi}{k^2(2s_1+1)(2s_2+1)} &\sum_{j\ell s\ell' s'}(2j+1) \\ &\times \left|\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell' s' n', \ell s n}(E)\right|^2.\end{aligned} \tag{3.7.10}$$

Summing (3.7.10) over all two-body channels gives the total cross-section for all elastic or inelastic two-body reactions:

$$\begin{aligned}\sum_{n'}\sigma(n \to n'; E) = \frac{\pi}{k^2(2s_1+1)(2s_2+1)} &\sum_{j\ell s}(2j+1) \\ &\times \left[(\mathbf{1} - S^j(E))^\dagger(\mathbf{1} - S^j(E))\right]_{\ell sn, \ell sn}.\end{aligned} \tag{3.7.11}$$

For comparison, Eqs. (3.7.8), (3.7.9), (3.7.4), and the Clebsch–Gordan sum rules give the spin-averaged forward scattering amplitude as

$$f(n; E) = \frac{i}{2k(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)[\mathbf{1} - S^j(E)]_{\ell sn, \ell sn}.$$

The optical theorem (3.6.10) then gives the total cross-section as

$$\sigma_{\text{total}}(n; E) = \frac{2\pi}{k^2(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)\,\text{Re}[\mathbf{1} - S^j(E)]_{\ell sn, \ell sn}. \tag{3.7.12}$$

If only two-body channels can be reached from the channel n at energy E, then the matrix $S^j(E)$ (or at least some submatrix that includes channel n) is unitary, and thus

$$\left[(\mathbf{1} - S^j(E))^\dagger(\mathbf{1} - S^j(E))\right]_{\ell sn, \ell sn} = 2\,\text{Re}[\mathbf{1} - S^j(E)]_{\ell sn, \ell sn}, \tag{3.7.13}$$

[24] In deriving this result, we use standard sum rules [9] for the Clebsch–Gordan coefficients: first

$$\sum_{\sigma_1\sigma_2} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{s_1 s_2}(\bar{s}, \bar{\mu}; \sigma_1, \sigma_2) = \delta_{s\bar{s}}\delta_{\mu\bar{\mu}},$$

and the same with primes; then

$$\sum_{m\sigma} C_{\ell s}(j, \sigma; m, \bar{\sigma}) C_{\ell' s}(\bar{j}, \bar{\sigma}; m, \bar{\sigma}) = \delta_{j\bar{j}}\delta_{\sigma\bar{\sigma}},$$

and finally

$$\sum_{\sigma\mu} C_{\ell s}(j, \sigma; 0, \mu) C_{\ell' s}(j, \sigma; 0, \mu) = \frac{2j+1}{2\ell+1}\delta_{\ell\ell'}.$$

so (3.7.12) and (3.7.11) are equal. On the other hand, if channels involving three or more particles are open, then the difference of (3.7.12) and (3.7.11) gives the total cross-section for producing extra particles:

$$\sigma_{\text{production}}(n;E) = \frac{\pi}{k^2(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)[\mathbf{1} - S^j(E)^\dagger S^j(E)]_{\ell sn,\ell sn}, \tag{3.7.14}$$

and this must be positive.

The partial wave expansion is particularly useful when applied to processes where the relevant part of the S-matrix is diagonal. This is the case, for instance, if the initial channel n contains just two spinless particles, and no other channels are open at this energy, as in $\pi^+ - \pi^+$ or $\pi^+ - \pi^0$ scattering at energies below the threshold for producing extra pions (provided one ignores weak and electromagnetic interactions). For a pair of spinless particles we have $j = \ell$, and angular momentum conservation keeps the S-matrix diagonal. It is also possible for the S-matrix to be diagonal in certain processes involving particles with spin; for instance in pion–nucleon scattering we can have $j = \ell + \frac{1}{2}$ or $j = \ell - \frac{1}{2}$, but for a given j these two states have opposite parity, so they cannot be connected by non-zero S-matrix elements. In any case, if for some n and E the S-matrix elements $S^j_{N',\ell sn}(E,\mathbf{0})$ all vanish unless N' is the two-body state j,ℓ,s,n, then unitarity requires that

$$S^j_{\ell' s' n',\ell sn}(E) = \exp[2i\delta_{j\ell sn}(E)]\delta_{\ell'\ell}\delta_{s's}\delta_{n'n}, \tag{3.7.15}$$

where $\delta_{j\ell sn}(E)$ is a real phase, commonly known as the *phase shift*. This formula is also often used where the two-body part of the S-matrix is diagonal but channels containing three or more particles are also open; in such cases the phase shift must have a positive imaginary part, to keep (3.7.14) positive. For real phase shifts, the elastic and total cross-sections are then given by Eq. (3.7.10) or Eq. (3.7.12) as:

$$\sigma(n \to n;E) = \sigma_{\text{total}}(n;E) = \frac{4\pi}{k^2(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)\sin^2\delta_{j\ell sn}(E). \tag{3.7.16}$$

This familiar result is usually derived in non-relativistic quantum mechanics by studying the coordinate space wave function for a particle in a potential. The derivation given here is offered both to show that the partial wave expansion applies for elastic scattering even at relativistic velocities, and also to emphasize that it depends on no particular dynamical assumptions, only on unitarity and invariance principles.

It is also often useful to introduce phase shifts in dealing with problems where several channels are open, forming a few irreducible representations of some internal symmetry group. The classic example of such an internal symmetry is isotopic spin symmetry, for which the channel index n includes a specification of the isospins T_1, T_2 of the two particles together with their three-components t_1, t_2; the states in channel n may be expressed as linear combinations of the tth components of irreducible representations T, with coefficients given by the familiar Clebsch–Gordan coefficients $C_{T_1T_2}(T,t;t_1,t_2)$. Suppose that for the channels and energy of interest the S-matrix is diagonal in ℓ and s as well as j, T, and t. Unitarity and isospin symmetry then allow us to write the S-matrix as

$$S^j_{\ell' s' T' t',\ell sTt} = \exp[2i\delta_{j\ell sT}(E)]\delta_{\ell'\ell}\delta_{s's}\delta_{T'T}\delta_{t't}, \tag{3.7.17}$$

with $\delta_{j\ell sT}(E)$ a real phase shift, t-independent according to the Wigner–Eckart theorem. The partial cross-sections can again be calculated from Eq. (3.7.10), and the total cross-section is given by Eq. (3.7.12) as

$$\sigma_{\text{total}}(t_1, t_2; E) = \frac{4\pi}{k^2(2s_1+1)(2s_2+1)} \times \sum_{j\ell sTt} (2j+1) C_{T_1 T_2}(T, t; t_1, t_2)^2 \sin^2 \delta_{j\ell sT}(E). \tag{3.7.18}$$

For instance, in pion–pion scattering we have phase shifts $\delta_{\ell\ell 0T}(E)$ with $T=0$ or $T=2$ for each even ℓ and $T=1$ for each odd ℓ, while for pion–nucleon scattering we have phase shifts $\delta_{j,j\pm\frac{1}{2},\frac{1}{2},T}$ with $T=\frac{1}{2}$ or $T=\frac{3}{2}$.

We can gain some useful insight about the threshold behavior of the scattering amplitudes and phase shifts from considerations of analyticity that are nearly independent of any dynamical assumptions. Unless there are special circumstances that would produce singularities in momentum space, we would expect the matrix element $M_{\mathbf{k}'\sigma'_1,-\mathbf{k}'\sigma'_2,n';\mathbf{k}\sigma_1,-\mathbf{k}\sigma_2,n}$ to be an analytic function[25] of the three-momenta $\mathbf{k}$ and $\mathbf{k}'$ near $k=0$ or $k'=0$ or (for elastic scattering) $k=k'=0$. Turning to the partial wave expansion (3.7.8) for M, we note that $k^\ell Y_\ell^m(\widehat{\mathbf{k}})$ is a simple polynomial function of the three-vector $\mathbf{k}$, so in order for the matrix element to be an analytic function of the three-momenta $\mathbf{k}$ and $\mathbf{k}'$ near $k=0$ or $k'=0$, the coefficients $M^j_{\ell's'n',\ell sn}$ or equivalently $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}$ must go as $k^{\ell+1/2}k'^{\ell'+1/2}$ when k and/or k' go to zero. Hence for small k and/or k', it is only the lowest partial waves in the initial and/or final state that contribute appreciably to the scattering amplitude. We have three possible cases:

Exothermic reactions

Here k' approaches a finite value as $k \to 0$, and in this limit $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}$ goes as $k^{\ell+1/2}$. The cross-section (3.7.11) in this case goes as $k^{2\ell-1}$, where ℓ is here the *lowest* orbital angular momentum that can lead to the reaction. In the most usual case $\ell=0$, so the reaction cross-section goes as $1/k$. (This is the case, for instance, in the absorption of slow neutrons by complex nuclei, or for the annihilation of electron–positron pairs into photons at low energy, aside from the higher-order effects of Coulomb forces.) The reaction rate is the cross-section times the flux, which goes like k, so the rate for an exothermic reaction behaves like a constant for $k \to 0$. However, it is the cross-section rather than the reaction rate that determines the probability of absorption when a beam crosses a given thickness of target material, and the factor $1/k$ makes this probability very high for slow neutrons in an absorbing material like boron.

Endothermic Reactions

[25] For instance, in the Born approximation (3.2.8), M is proportional to the Fourier transform of the coordinate space matrix elements of the interaction, and hence is analytic at zero momenta as long as these matrix elements fall off sufficiently rapidly at large separations. The chief exception is for scattering involving long-distance forces, such as the Coulomb force.

Here the reaction is forbidden until k reaches a finite threshold value, where $k' = 0$. Just above this threshold $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}$ goes as $(k')^{\ell'+1/2}$. The cross-section (3.7.11) in this case goes as $(k')^{2\ell'+1}$, where ℓ' is here the *lowest* orbital angular momentum that can be produced at threshold. In the most usual case $\ell' = 0$, so the reaction cross-section rises above threshold like k', and hence like $\sqrt{E - E_{\text{threshold}}}$. (This is the case, for instance, in the associated production of strange particles, or the production of electron–positron pairs in the scattering of photons.)

Elastic Reactions

Here $k = k'$, so k and k' go to zero together. (This is the case where $n' = n$, or where n' consists of particles in the same isotopic spin multiplets as are those in n.) In elastic scattering the partial waves with $\ell = \ell' = 0$ are always present, so in the limit $k \to 0$ the scattering amplitude (3.7.8) becomes a constant:

$$f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \to \mathbf{k}'\sigma'_1, -\mathbf{k}'\sigma'_2, n') \\ \to \sum_{s\sigma} C_{s_1 s_2}(s,\sigma;\sigma_1,\sigma_2) C_{s'_1 s'_2}(s,\sigma;\sigma'_1,\sigma'_2) a_s(n \to n'), \tag{3.7.19}$$

where a is a constant, known as the *scattering length*, defined by the limit

$$S^s_{0sn',0sn} \longrightarrow \delta_{n'n} + 2ika_s(n \to n') \qquad (k = k' \to 0). \tag{3.7.20}$$

Summing $4\pi|f|^2$ over final spins and averaging over initial spins gives the total cross-section for the transition $n \to n'$ at $k = k' = 0$:

$$\sigma(n \to n'; k = 0) = \frac{4\pi}{(2s_1+1)(2s_2+1)} \sum_s (2s+1)|a_s(n \to n')|^2. \tag{3.7.21}$$

The classic instance of the use of this formula is in neutron–proton scattering, where there are two scattering lengths, with the spin singlet length a_0 considerably larger than the spin triplet length a_1.

The partial wave expansion can also be used to make a crude guess about the behavior of cross-sections at high energy. With decreasing wavelengths, we may expect scattering to be described more or less classically: a particle of momentum k and orbital angular momentum ℓ would have an impact parameter ℓ/k, and will therefore strike a disk of radius R if $\ell \leq kR$. This can be interpreted as a statement about S-matrix elements:

$$S^j_{\ell's'n',\ell sn} \longrightarrow \begin{cases} 0, & \ell \ll kR_n, \\ 1, & \ell \gg kR_n, \end{cases} \tag{3.7.22}$$

where R_n is some sort of interaction radius for channel n. For a given $\ell \gg s$, there are $2s+1$ values of j, all close enough to ℓ to approximate $2j+1 \simeq 2\ell+1$, so the sum over j and s in Eq. (3.7.12) merely gives a factor of order

$$\sum_{js}(2j+1) = (2\ell+1)\sum_s(2s+1) = (2\ell+1)(2s_1+1)(2s_2+1).$$

The total cross-section is then given for $k \gg 1/R_n$ by Eq. (3.7.12) as

$$\sigma_{\text{total}}(n; E) \longrightarrow \frac{2\pi}{k^2} \sum_{\ell \lesssim kR_n} (2\ell + 1) \longrightarrow 2\pi R_n^2. \tag{3.7.23}$$

In exactly the same way, Eq. (3.7.10) gives the elastic scattering cross-section

$$\sigma(n \to n; E) \longrightarrow \pi R_n^2. \tag{3.7.24}$$

The difference between Eqs. (3.7.23) and (3.7.24) gives an inelastic cross-section πR_n^2, which is what we would expect for collisions with an opaque disk of radius R_n. (The somewhat surprising elastic scattering cross-section πR_n^2 may be attributed to diffraction by the disk.) On the other hand, if we assume along with Eq. (3.7.22) that $S^j_{\ell' s' n', \ell s n}$ is complex only for impact parameters ℓ/k within a small range of width $\Delta_n \ll R_n$ around $\ell/k = R_n$, then using the inequality $|\operatorname{Im}(1 - S^j_{\ell' s' n', \ell s n})| \leq 2$, the same analysis gives a bound on the real part of the forward scattering amplitude

$$|\operatorname{Re} f(n; E)| \leq 2kR_n\Delta_n \ll |\operatorname{Im} f(n; E)|. \tag{3.7.25}$$

The smallness of the real part of the forward scattering amplitude at high energy is confirmed by experiment.

So far we have not said anything about whether the interaction radius R_n itself may depend on energy. As a very crude guess, we may take R_n as the distance at which the factor $\exp(-\mu r)$ in the Yukawa potential (1.2.74) takes a value proportional to some unknown power of E, in which case R_n goes as $\log E$ for $E \to \infty$, and the cross-sections go as $(\log E)^2$. As it happens, it has been rigorously shown [26] on the basis of very general assumptions that the total cross-section can grow no faster than $(\log E)^2$ for $E \to \infty$, and in fact the observed proton–proton total cross-section rises something like $(\log E)^2$ at high energies, so this rough picture of high-energy scattering does seem to have some correspondence with reality.

3.8 Resonances[26]

It often happens that the particles participating in a multi-particle collision can form an intermediate state consisting of a *single* unstable particle R, that eventually decays into the particles observed as the final state. If the total decay rate of R is small the cross-section exhibits a rapid variation (usually a peak), known as a *resonance*, at the energy of the intermediate state R.

We shall see that the behavior of the cross-sections near a resonance is pretty much prescribed by the unitarity condition alone, which is a good thing since there are a number of very different mechanisms that can produce a nearly stable state:

(a) The simplest possibility is that the Hamiltonian can be decomposed into two terms, a 'strong' Hamiltonian $H_0 + H_s$, which has the particle R as an eigenstate, plus a

[26] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

weak perturbation H_w, which allows R to decay into various states, including the initial and final states α, β of our collision process. For instance there is a neutral particle, the Z^0, with $j = 1$ and mass 91 GeV, that would be stable in the absence of the electroweak interactions. These interactions allow the Z^0 to decay into electron–positron pairs, muon–antimuon pairs, etc., but with a total decay rate that is much less than the Z^0 mass. In 1989 the Z^0 particle was seen as a resonance[27] in electron–positron collisions at CERN and Stanford, in the reactions $e^+ + e^- \to Z^0 \to e^+ + e^-$, $e^+ + e^- \to Z^0 \to \mu^+ + \mu^-$, etc.

(b) In some cases a particle is long-lived because there is a potential barrier that nearly prevents its constituents from escaping. The classic example is nuclear alpha decay: it may be energetically possible for a nucleus to emit an alpha particle (a He^4 nucleus) but the strong electrostatic repulsion between the alpha particle and the nucleus creates a barrier region around the daughter nucleus, which the alpha particle is classically forbidden to enter. The decay then can proceed only by quantum mechanical barrier penetration, and is exponentially slow. Such an unstable state shows up as a resonance in the scattering of the alpha particle on the daughter nucleus. For instance, the lowest-energy state of the Be^8 nucleus is unstable against decay into two alpha particles, and is seen as a resonance in He^4–He^4 scattering. (In addition to Coulomb barriers, there are also centrifugal barriers that help lengthen the life of alpha-, beta-, and gamma-unstable nuclei of high spin.)

(c) It is possible for complicated systems to be nearly unstable for statistical reasons, without the presence of any potential barriers or weak interactions. For instance, an excited state of a heavy nucleus may be able to decay only if, through a statistical fluctuation, a large part of its energy is concentrated on a single neutron. This state will then show up as a resonance in the scattering of a neutron on the daughter nucleus.

These mechanisms for producing long-lived states are so different that it is truly fortunate that most of the properties of resonances follow from unitarity alone, without regard to the dynamical mechanism that produces the resonance.

First, let's consider the energy dependence of the matrix element for a reaction near a resonance. A wave packet $\int d\alpha\, g(\alpha)|\alpha;\mathrm{in}\rangle \exp(-iE_\alpha t)$ of 'in' states has a time-dependence given by (3.1.19)

$$\int d\alpha\, g(\alpha)|\alpha;\mathrm{in}\rangle e^{-iE_\alpha t} = \int d\alpha\, g(\alpha)|\alpha\rangle_0 e^{-iE_\alpha t} + \int d\beta\, |\beta\rangle_0 \int d\alpha\, \frac{e^{-iE_\alpha t} g(\alpha) T^+_{\beta\alpha}}{E_\alpha - E_\beta + i\epsilon}.$$

As mentioned in Section 3.1, a pole in the function $T^+_{\beta\alpha}$ in the lower-half complex E_α plane would make a contribution to the second term that decays exponentially as $t \to \infty$. Specifically, a pole at $E_\alpha = E_R - i\Gamma/2$ yields a term in the *amplitude* that behaves like

[27] Incidentally, this example shows that a resonant state only needs to decay *relatively* slowly; the Z^0 lifetime is 2.6×10^{-25} seconds, which is not long enough for a Z^0 travelling near the speed of light to cross an atomic nucleus. What is important is that the decay rate is 36 times smaller than the rate $\hbar/M_Z$ of oscillation of the Z^0 wave function in its rest frame.

$\exp(-iE_Rt - \Gamma t/2)$, so it corresponds to a state whose *probability* decays like $\exp(-\Gamma t)$. We conclude then that a long-lived state of energy E_R with a slow decay rate Γ produces a term in the scattering amplitude that varies as

$$T^+_{\beta\alpha} \sim (E_\alpha - E_R + i\Gamma/2)^{-1} + \text{constant}. \tag{3.8.1}$$

To go further, it will be convenient to adopt as a basis the orthonormal discrete multi-particle states $|\mathbf{p}, E, N\rangle$ discussed in the previous section; $\mathbf{p}$ and E are the total momentum and energy, and N is an index that takes only discrete (though infinitely many) values. In this basis, the S-matrix may be written

$$S_{\mathbf{p}'E'N',\mathbf{p}EN} = \delta^3(\mathbf{p}' - \mathbf{p})\delta(E' - E)S_{N'N}(\mathbf{p}, E). \tag{3.8.2}$$

Near a resonance, we expect the center-of-mass frame amplitude $S(\mathbf{0}, E) \equiv \mathcal{S}(E)$ to have the form

$$\mathcal{S}_{N'N}(E) \equiv S_{N'N}(\mathbf{0}, E) = \mathcal{S}_{0N'N} + \frac{\mathcal{R}_{N'N}}{E - E_R + i\Gamma/2}, \tag{3.8.3}$$

where $\mathcal{S}_0$ and $\mathcal{R}$ are approximately constant at least over the relatively small range of energies $|E - E_R| \lesssim \Gamma$.

In this basis, the unitarity of the S-matrix is an ordinary matrix equation

$$\mathcal{S}(E)^\dagger \mathcal{S}(E) = \mathbf{1}. \tag{3.8.4}$$

Applied to Eq. (3.8.3), this tells us that the non-resonant background S-matrix is unitary

$$\mathcal{S}_0^\dagger \mathcal{S}_0 = \mathbf{1}, \tag{3.8.5}$$

and also that the residue matrix $\mathcal{R}$ satisfies the two conditions

$$\mathcal{S}_0^\dagger \mathcal{R} - \mathcal{R}^\dagger \mathcal{S}_0 = 0, \tag{3.8.6}$$

$$-\frac{i}{2}\Gamma \mathcal{S}_0^\dagger \mathcal{R} + \frac{i}{2}\Gamma \mathcal{R}^\dagger \mathcal{S}_0 + \mathcal{R}^\dagger \mathcal{R} = 0. \tag{3.8.7}$$

These conditions can be put in a more transparent form by setting

$$\mathcal{R} \equiv -i\Gamma \mathcal{A} \mathcal{S}_0. \tag{3.8.8}$$

The unitarity conditions on the matrix $\mathcal{A}$ are then simply

$$\mathcal{A}^\dagger = \mathcal{A}, \qquad \mathcal{A}^2 = \mathcal{A}. \tag{3.8.9}$$

Any such Hermitian idempotent matrix is called a *projection matrix.* Such matrices can always be expressed as a sum of dyads of orthonormal vectors $u^{(r)}$:

$$\mathcal{A}_{N'N} = \sum_r u^{(r)}_{N'} u^{(r)*}_N, \qquad \sum_N u^{(r)*}_N u^{(s)}_N = \delta_{rs}. \tag{3.8.10}$$

The discrete part of the S-matrix is then

$$\mathcal{S}_{N'N}(E) = \sum_{N''} \left[\delta_{N'N''} - i\frac{\Gamma}{E - E_R + i\Gamma/2} \sum_r u^{(r)}_{N'} u^{(r)*}_{N''}\right] \mathcal{S}_{0N''N}. \tag{3.8.11}$$

Each term in the sum over r can be thought of as arising from a different resonant state, all these states having the same values for E_R and Γ.

What has this to do with rates and cross-sections? For simplicity let's now ignore the non-resonant background scattering, setting $\mathcal{S}_{0N'N}$ equal to $\delta_{N'N}$; we will come back to the more general case a little later. Then for the two-body discrete center-of-mass states described in the previous section, Eq. (3.8.11) reads:

$$\mathcal{S}_{j'\sigma'\ell's'n',j\sigma\ell sn}(E) = \delta_{j'j}\delta_{\sigma'\sigma}\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - i\frac{\Gamma}{E - E_R + i\Gamma/2}\sum_r u^{(r)}_{j'\sigma'\ell's'n'}u^{(r)*}_{j\sigma\ell sn}. \tag{3.8.12}$$

In all cases the label r will include an index σ_R giving the z-component of the total angular momentum of the resonant state; for a resonant state of total angular momentum j_R, σ_R takes $2j_R+1$ values. If there is no other degeneracy, then r just labels the value of σ_R, and

$$u^{(\sigma_R)}_{j\sigma\ell sn} = \delta_{jj_R}\delta_{\sigma\sigma_R}u_{\ell sn}, \tag{3.8.13}$$

where $u_{\ell sn}$ are a set of complex amplitudes that (because of the Wigner–Eckart theorem) are independent of σ. Now Eq. (3.8.12) gives the amplitude S^j defined by Eq. (3.7.7) as

$$S^j_{\ell's'n',\ell sn}(E) = \delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - i\delta_{jj_R}\frac{\Gamma}{E - E_R + i\Gamma/2}u_{\ell's'n'}u^*_{\ell sn}. \tag{3.8.14}$$

Also, Eq. (3.8.10) now reads

$$\sum_{\ell sn}|u_{\ell sn}|^2 + \cdots = 1 \tag{3.8.15}$$

with the dots representing the positive contribution of any states containing three or more particles. As we shall see, the quantities $|u_{\ell sn}|^2$ have the interpretation of branching ratios for the decay of the resonant state into the various accessible two-body states.

Eq. (3.7.12) now gives the total cross-section for all reactions in channel n:

$$\sigma_{\text{total}}(n;E) = \frac{\pi(2j_R+1)}{k^2(2s_1+1)(2s_2+1)}\frac{\Gamma\Gamma_n}{(E-E_R)^2+\Gamma^2/4}, \tag{3.8.16}$$

where

$$\Gamma_n \equiv \Gamma\sum_{\ell s}|u_{\ell sn}|^2. \tag{3.8.17}$$

This is a version of the celebrated Breit–Wigner single-level formula [27]. We can also use these results to calculate the cross-section for resonant scattering from an initial two-body channel n to a final two-body channel n'. Using Eq. (3.8.14) in Eq. (3.7.10) gives

$$\sigma(n \to n';E) = \frac{\pi(2j_R+1)}{k^2(2s_1+1)(2s_2+1)}\frac{\Gamma_n\Gamma_{n'}}{(E-E_R)^2+\Gamma^2/4}. \tag{3.8.18}$$

This shows that the probabilities that the resonant state will decay into any one of the final two-body channels n' are proportional to the $\Gamma_{n'}$. According to Eq. (3.8.15), the sum of the Γ_n (including contributions from final states containing three or more particles) is just equal to the total decay rate Γ, so we can conclude that Γ_n is just the rate for the decay of the resonant state into channel n.

We see in Eqs. (3.8.16) and (3.8.18) the characteristic resonant peak at energy E_R, with a width (the full width at half maximum) equal to the decay rate Γ. (The individual Γ_n are often called partial widths.) Since $\Gamma_n \leq \Gamma$, the total cross-section at the peak of the resonance is roughly bounded by one square wavelength, $(2\pi/k)^2$. This rule, that cross-sections at a single resonance are roughly bounded by a square wavelength, is universally applicable even in classical physics (where energy conservation plays the role played here by unitarity), as for instance in the resonant interaction of sound waves with bubbles in the sea, or gravitational waves with gravitational wave antennae. (In the latter case, the branching ratio for oscillations in any laboratory mass to lose their energy through gravitational radiation is tiny, so the cross-section even at a resonance peak is vastly less then a square wavelength [28].)

Incidentally, it often happens that a resonance is detected, but energy measurements are insufficiently precise to resolve its width. In this case, what is measured experimentally is the integral of the cross-section over the resonant peak. For the total cross-section (3.8.14), this is

$$\int \sigma_{\text{total}}(n; E)\, dE = \frac{2\pi^2(2j_R+1)\Gamma_n}{k_R^2(2s_1+1)(2s_2+1)}. \tag{3.8.19}$$

Such experiments can reveal only the partial width for decay of the resonant state into the initial particles, not the total width or branching ratios.

This formalism can also be applied when the resonant states with a given spin z-component form a multiplet related by some symmetry group. For instance, to the extent that isospin symmetry is respected, for a resonance of total isospin T_R the index r labelling resonant states includes a specification not only of the angular-momentum z-component σ_R, but also of an isospin three-component t_R, taking values $-T_R, -T_R+1, \ldots, T_R$. In this case there is no change in the above results for the total and partial cross-sections, because each two-body channel n has definite values t_1, t_2 of the isospin z-components of the two particles, and hence can only couple to the resonant state with the single value t_1+t_2 for t_R. The partial widths Γ_n here depend on t_1 and t_2 only through factors $C_{T_1T_2}(T_R, t_R; t_1, t_2)^2$.

The presence of a resonance shows itself in a characteristic behavior of phase shifts near the resonance. Returning to the general formula (3.8.11) (but still taking $\mathcal{S}_0 = \mathbf{1}$), we see from Eq. (3.8.10) that for each individual resonant state r, there is an eigenvector $u_N^{(r)}$ of $\mathcal{S}_{N'N}(E)$ with eigenvalue

$$\exp[2i\delta^{(r)}(E)] = 1 - i\frac{\Gamma}{E - E_R + i\Gamma/2}$$

or, in other words,

$$\tan\delta^{(r)}(E) = -\frac{\Gamma/2}{E - E_R}. \tag{3.8.20}$$

We see that over an energy range of order Γ centered around the resonant energy, the the 'eigenphase' $\delta^{(r)}(E)$ jumps from a value $\nu\pi$ (with ν a positive or negative integer) below the resonance to $(\nu+1)\pi$ above it. However, in order to use this result to say something about reaction rates, we need to know the eigenvectors $u_N^{(r)}$, which, in general, have components with arbitrary numbers of particles with various momenta, spin, and species.

These results are much more useful in those special cases when the particles in a particular channel N are forbidden (usually by conservation laws) from making a transition to any other channel. With this assumption, it is not difficult to include the effects of a non-resonant background scattering matrix $\mathcal{S}_0$ in the general result (3.8.11). In order for $\mathcal{S}_{N'N}$ to vanish for some particular N and all $N' \neq N$, it is necessary that the same is true of $\mathcal{S}_{0N'N}$, and also true of $u_{N'}^{(r)}$ for any r for which $u_N^{(r)} \neq 0$. The unitarity requirement (3.8.5) then requires that for this N

$$\mathcal{S}_{0N'N} = \exp(2i\delta_{0N})\delta_{N'N}$$

and Eq. (3.8.10) requires that

$$u_N^{(r)*} u_N^{(s)} = \delta_{rs},$$

so that there can be only one term r in Eq. (3.8.11) for which $u_N^{(r)} \neq 0$. In this case, Eq. (3.8.11) gives

$$\mathcal{S}_{N'N}(E) = \delta_{N'N}\left[1 - \frac{i\Gamma}{E - E_R + i\Gamma/2}\right]\exp(2i\delta_{0N}) \equiv \delta_{N'N}\exp[2i\delta_N(E)]$$

with total phase shift

$$\delta_N(E) = \delta_{0N} - \arctan\left(\frac{\Gamma/2}{E - E_R}\right). \tag{3.8.21}$$

We see that over an energy range of order Γ centered around the resonant energy E_R, the phase shift $\delta_N(E)$ jumps from a value δ_{0N} below the

resonance to $\delta_{0N} + \pi$ above it. For instance, as we saw in the previous section, these assumptions are satisfied in various two-body reactions such as pion–pion and pion–nucleon scattering at energies below the threshold for producing extra pions, with N incorporating the total and orbital angular-momenta j, ℓ (with $j = \ell$ for pion–pion scattering) and the total angular-momentum z-component σ, as well as the total isospin T and its three-component t. The Wigner–Eckart theorem allows the phase shifts to depend only on j, ℓ, and T, not on t or σ. There are famous resonances in these channels: in pion–pion scattering there is a resonance at 770 MeV called ρ with $j = \ell = 1$, $T = 1$, and $\Gamma = 150$ MeV; in pion–nucleon scattering there is a resonance at 1232 MeV called Δ with $j = \frac{3}{2}$, $\ell = 1$, $T = \frac{3}{2}$ and $\Gamma = 110$ to 120 MeV.

Inspection of Eq. (3.7.12) or Eq. (3.7.18) shows that the total cross-section reaches a peak when the resonant phase shift passes through $\pi/2$ (or odd-integer multiples of $\pi/2$.) The non-resonant phase shifts are typically rather small, so as we saw earlier, σ_{total} will exhibit a sharp peak when the phase shift δ_ℓ goes through $\pi/2$, at an energy close to E_R. However, it sometimes happens that the non-resonant background phase shift δ_{0N} is near $\pi/2$, in which case the cross-section will exhibit a sharp *dip* as the phase shift rises through π near E_R, due to destructive interference between the resonance and the non-resonant background amplitude. Such dips were first observed by Ramsauer and Townsend [29] in 1922, in the scattering of electrons by noble gas atoms.

Problems

1. Consider a theory with a separable interaction; that is,

$$\langle\beta|H_{\rm int}|\alpha\rangle = g\, u_\beta u_\alpha^*,$$

where g is a real coupling constant, and u_α is a set of complex quantities with

$$\sum_\alpha |u_\alpha|^2 = 1.$$

Use the Lippmann–Schwinger equation (3.1.16) to find explicit solutions for the 'in' and 'out' states and the S-matrix.

2. Suppose that a resonance of spin one is discovered in e^+e^- scattering at a total energy of 150 GeV and with a cross-section (in the center-of-mass frame, averaged over initial spins, and summed over final spins) for elastic e^+e^- scattering at the peak of the resonance equal to $10^{-34}\,\text{cm}^2$. What is the branching ratio for the decay of the resonant state R by the mode $R \to e^- + e^+$? What is the total cross-section for e^+e^- scattering at the peak of the resonance? (In answering both questions, ignore the non-resonant background scattering.)

3. Express the differential cross-section for two-body scattering in the *laboratory* frame, in which one of the two particles is initially at rest, in terms of kinematic variables and the matrix element $M_{\beta\alpha}$. (Derive the result directly, without using the results derived in this chapter for the differential cross-section in the center-of-mass frame.)

4. Derive the perturbation expansion (3.5.8) directly from the expansion (3.5.3) of old-fashioned perturbation theory.

5. We can define 'standing wave' states $|\alpha\rangle^0$ by a modified version of the Lippmann–Schwinger equation

$$|\alpha\rangle^0 = |\alpha\rangle_0 + \text{P. V.}\,\frac{1}{E_\alpha - H_0} H_{\rm int}|\alpha\rangle^0.$$

Show that the matrix $K_{\beta\alpha} = \pi\delta(E_\beta - E_\alpha)\langle\beta|H_{\rm int}|\alpha\rangle^0$ is Hermitian. Show how to express the S-matrix in terms of the K-matrix.

6. Express the differential cross-section for elastic π^+–proton and π^-–proton scattering in terms of the phase shifts for states of definite total angular momentum, parity, and isospin.

7. Show that the states $|E, \mathbf{p}, j, \sigma, s, n\rangle$ defined by Eq. (3.7.5) are correctly normalized to have the scalar products (3.7.6).

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4 The Cluster Decomposition Principle

Up to this point we have not had much to say about the detailed structure of the Hamiltonian operator H. This operator can be defined by giving all its matrix elements between states with arbitrary numbers of particles. Equivalently, as we shall show here, any such operator may be expressed as a function of certain operators that create and destroy single particles. We saw in Chapter 1 that such creation and annihilation operators were first encountered in the canonical quantization of the electromagnetic field and other fields in the early days of quantum mechanics. They provided a natural formalism for theories in which massive particles as well as photons can be produced and destroyed, beginning in the early 1930s with Fermi's theory of beta decay.

However, there is a deeper reason for constructing the Hamiltonian out of creation and annihilation operators, which goes beyond the need to quantize any pre-existing field theory like electrodynamics, and has nothing to do with whether particles can actually be produced or destroyed. The great advantage of this formalism is that if we express the Hamiltonian as a sum of products of creation and annihilation operators, with suitable non-singular coefficients, then the S-matrix will automatically satisfy a crucial physical requirement, the cluster decomposition principle [1], which says in effect that distant experiments yield uncorrelated results. Indeed, it is for this reason that the formalism of creation and annihilation operators is widely used in non-relativistic quantum statistical mechanics, where the number of particles is typically fixed. In relativistic quantum theories, the cluster decomposition principle plays a crucial part in making field theory inevitable. There have been many attempts to formulate a relativistically invariant theory that would not be a local field theory, and it is indeed possible to construct theories that are not field theories and yet yield a Lorentz-invariant S-matrix for two-particle scattering [2], but such efforts have always run into trouble in sectors with more than two particles: either the three-particle S-matrix is not Lorentz-invariant, or else it violates the cluster decomposition principle.

In this chapter we will first discuss the basis of states containing arbitrary numbers of bosons and fermions, then define the creation and annihilation operators, and finally show how their use facilitates the construction of Hamiltonians that yield S-matrices satisfying the cluster decomposition condition.

4.1 Bosons and Fermions

The Hilbert space of physical states is spanned by states containing $0, 1, 2, \cdots$ free particles. These can be free-particle states, or 'in' states, or 'out' states; for definiteness we shall deal here with the free-particle states $|\mathbf{p}_1\sigma_1 n_1;\, \mathbf{p}_2\sigma_2 n_2;\, \ldots\rangle$, but all our results will apply equally to 'in' or 'out' states. As usual, σ labels spin z-components (or helicities, for massless particles) and n labels particle species.

We must now go into a matter that has been passed over in Chapter 3; the symmetry properties of these states. As far as we know, all particles are either *bosons* or *fermions*, the difference being that a state is unchanged by the interchange of two identical bosons, and changes sign under the interchange of two identical fermions. That is

$$|\ldots\, \mathbf{p}\sigma n\, \ldots\, \mathbf{p}'\sigma' n\, \ldots\rangle = \pm|\ldots\, \mathbf{p}'\sigma' n\, \ldots\, \mathbf{p}\sigma n\, \ldots\rangle \tag{4.1.1}$$

with an upper or lower sign if n is a boson or a fermion, respectively, and dots representing other particles that may be present in the state. (Equivalently, this could be stated as a condition on the 'wave functions,' the coefficients of these multi-particle basis vectors in physically allowable state-vectors.) These two cases are often referred to as Bose or Fermi 'statistics.' We will see in the next chapter that Bose and Fermi statistics are only possible for particles that have integer or half-integer spins, respectively, but we shall not need this information in the present chapter. In this section we shall offer a non-rigorous argument that all particles must be either bosons or fermions, and then set up normalization conditions for multi-boson or multi-fermion states.

First note that if two particles with spins and momenta $\mathbf{p}, \sigma$ and $\mathbf{p}', \sigma'$ belong to identical species n, then the state-vectors $|\ldots\, \mathbf{p}\sigma n\, \ldots\, \mathbf{p}'\sigma' n\, \ldots\rangle$ and $|\ldots\, \mathbf{p}'\sigma' n\, \ldots\, \mathbf{p}\sigma n\, \ldots\rangle$ represent the same physical state; if this were not the case then the particles would be distinguished by their order in the labelling of the state-vector, and the first listed would not be identical with the second. Since the two state-vectors are physically indistinguishable, they must belong to the same ray, and so

$$|\ldots\, \mathbf{p}\sigma n\, \ldots\, \mathbf{p}'\sigma' n\, \ldots\rangle = \alpha_n |\ldots\, \mathbf{p}'\sigma' n\, \ldots\, \mathbf{p}\sigma n\, \ldots\rangle, \tag{4.1.2}$$

where α_n is a complex number of unit absolute value. We may regard this as part of the definition of what we mean by identical particles.

The crux of the matter is to decide on what the phase factor α_n may depend. *If* it depends only on the species index n, then we are nearly done. Interchanging the two particles in Eq. (4.1.2) again, we find

$$|\ldots\, \mathbf{p}\sigma n\, \ldots\, \mathbf{p}'\sigma' n\, \ldots\rangle = \alpha_n^2 |\ldots\, \mathbf{p}\sigma n\, \ldots\, \mathbf{p}'\sigma' n\, \ldots\rangle$$

so that $\alpha_n^2 = 1$, yielding Eq. (4.1.1) as the only two possibilities.

On what else could α_n depend? It might depend on the numbers and species of the other particles in the state (indicated by dots in Eqs. (4.1.1) and (4.1.2)), but this would lead to the uncomfortable result that the symmetry of state-vectors under interchange of particles here on earth may depend on the presence of particles elsewhere in the universe. This is the sort of thing that is ruled out by the cluster decomposition principle, to be discussed later in this chapter. The phase α_n cannot have any non-trivial dependence on the spins of the two particles that are interchanged, because then these spin-dependent phase factors would have to furnish a representation of the rotation group, and there are no non-trivial representations of the three-dimensional rotation group that are one-dimensional—that is, by phase factors. The phase α_n might conceivably depend on the momenta of the two particles that are interchanged, but Lorentz invariance would require α_n to depend only on the scalar $p_1^\mu p_{2\mu}$; this is symmetric under interchange of particles 1 and 2, and therefore such dependence would not change the argument leading to the conclusion that $\alpha_n^2 = 1$.

The logical gap in the above argument is that (although our notation hides the fact) the states $|\mathbf{p}_1\sigma_1 n;\, \mathbf{p}_2\sigma_2 n;\, \ldots\rangle$ may carry a phase factor that depends on the path through momentum space by which the momenta of the particles are brought to the values p_1, p_2, etc. In this case the interchange of two particles twice might change the state by a phase factor, so that $\alpha_n^2 \neq 1$. We will see in Section 9.7 that this is a real possibility in two-dimensional space, but not for three or more spatial dimensions.

What about interchanges of particles belonging to different species? If we like, we can avoid this question by simply agreeing from the beginning to label the state-vector by listing all photon momenta and helicities first, then all electron momenta and spin z-components, and so on through the table of elementary particle types. Alternatively, we can allow the particle labels to appear in any order, and *define* the state-vectors with particle labels in an arbitrary order as equal to the state-vector with particle labels in some standard order times phase factors, whose dependence on the interchange of particles of different species can be anything we like. In order to deal with symmetries like isospin invariance that relate particles of different species, it is convenient to adopt a convention that generalizes Eq. (4.1.1): the state-vector will be taken to be symmetric under interchange of any bosons with each other, or any bosons with any fermions, and antisymmetric with respect to interchange of any two fermions with each other, in all cases, whether the particles are of the same species or not.[28]

The normalization of these states must be defined in consistency with these symmetry conditions. To save writing, we will use a label q to denote all the quantum numbers of a single particle: its momentum, $\mathbf{p}$, spin z-component (or, for massless particles, helicity) σ, and species n. The N-particle states are thus labelled $|q_1 \cdots q_N\rangle$ (with $N = 0$ for the vacuum state $|\Omega\rangle$.) For $N = 0$ and $N = 1$ the question of symmetry does not arise: here we have

$$\langle\Omega|\Omega\rangle = 1 \tag{4.1.3}$$

and

$$\langle q'|q\rangle = \delta(q' - q), \tag{4.1.4}$$

where $\delta(q' - q)$ is a product of all the delta functions and Kronecker deltas for the particle's quantum numbers,

$$\delta(q' - q) = \delta^3(\mathbf{p}' - \mathbf{p})\delta_{\sigma'\sigma}\delta_{n'n}. \tag{4.1.5}$$

On the other hand, for $N = 2$ the states $|q_1' q_2'\rangle$ and $|q_2' q_1'\rangle$ are physically the same, so here we must take

$$\langle q_1' q_2'|q_1 q_2\rangle = \delta(q_1' - q_1)\delta(q_2' - q_2) \pm \delta(q_2' - q_1)\delta(q_1' - q_2) \tag{4.1.6}$$

the sign $\pm$ being $-$ if both particles are fermions and $+$ otherwise. This obviously is consistent with the above stated symmetry properties of the states. More generally,

$$\langle q_1' q_2' \cdots q_M'|q_1 q_2 \cdots q_N\rangle = \delta_{NM} \sum_{\mathcal{P}} \delta_{\mathcal{P}} \prod_i \delta(q_i - q'_{\mathcal{P}i}). \tag{4.1.7}$$

[28] In fact, by the same reasoning, the symmetry or antisymmetry of the state-vector under interchange of particles of the same species but different helicities or spin z-components is purely conventional, because we could have agreed from the beginning to list first the momenta of photons of helicity $+1$, then the momenta of all photons of helicity -1, then the momenta of all electrons of spin z-component $+\frac{1}{2}$, and so on. We adopt the *convention* that the state-vector is symmetric or antisymmetric under interchange of identical bosons or fermions of different helicities or spin z-components in order to facilitate the use of rotational invariance.

The sum here is over all permutations $\mathcal{P}$ of the integers $1, 2, \ldots, N$. (For instance, in the first term in Eq. (4.1.6), $\mathcal{P}$ is the identity, $\mathcal{P}1 = 1$, $\mathcal{P}2 = 2$, while in the second term $\mathcal{P}1 = 2$, $\mathcal{P}2 = 1$.) Also, $\delta_{\mathcal{P}}$ is a sign factor equal to -1 if $\mathcal{P}$ involves an odd permutation of fermions (an odd number of fermion interchanges) and $+1$ otherwise. It is easy to see that Eq. (4.1.7) has the desired symmetry or antisymmetry properties under interchange of the q_i, and also under interchange of the q'_j.

4.2 Creation and Annihilation Operators

Creation and annihilation operators may be defined in terms of their effect on the normalized multi-particle states discussed in the previous section. The *creation operator* $a^\dagger(q)$ (or in more detail, $a^\dagger(\mathbf{p}, \sigma, n)$) is defined as the operator that simply adds a particle with quantum numbers q at the front of the list of particles in the state

$$a^\dagger(q)|q_1 q_2 \cdots q_N\rangle = |q q_1 q_2 \cdots q_N\rangle. \tag{4.2.1}$$

In particular, the N-particle state can be obtained by acting on the vacuum with N creation operators

$$a^\dagger(q_1) a^\dagger(q_2) \cdots a^\dagger(q_N)|\Omega\rangle = |q_1 \cdots q_N\rangle. \tag{4.2.2}$$

It is conventional for this operator to be called $a^\dagger(q)$; its adjoint, which is then called $a(q)$, may be calculated from Eq. (4.1.7). As we shall now show, $a(q)$ removes a particle from any state on which it acts, and is therefore known as an *annihilation operator*. In particular, when the particles $q, q_1, \ldots, q_N$ are either all bosons or all fermions, we have

$$a(q)|q_1 q_2 \cdots q_N\rangle = \sum_{r=1}^{N} (\pm)^{r+1} \delta(q - q_r)|q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle, \tag{4.2.3}$$

with a $+1$ or -1 sign for bosons or fermions, respectively. (Here is the proof. We want to calculate the scalar product of $a(q)|q_1 q_2 \cdots q_N\rangle$ with an arbitrary state $|q'_1 \cdots q'_M\rangle$. Using Eq. (4.2.1), this is

$$\langle q'_1 \cdots q'_M|a(q)|q_1 \cdots q_N\rangle = \big(a^\dagger(q)|q'_1 \cdots q'_M\rangle\big)^\dagger |q_1 \cdots q_N\rangle = \langle q q'_1 \cdots q'_M|q_1 \cdots q_N\rangle.$$

We now use Eq. (4.1.7). The sum over permutations $\mathcal{P}$ of $1, 2, \ldots, N$ can be written as a sum over the integer r that is permuted into the first place, i.e. $\mathcal{P}r = 1$, and over mappings $\overline{\mathcal{P}}$ of the remaining integers $1, \ldots, r-1, r+1, \ldots, N$ into $1, \ldots, N-1$. Furthermore, the sign factor is

$$\delta_{\mathcal{P}} = (\pm)^{r-1} \delta_{\overline{\mathcal{P}}}$$

with upper and lower signs for bosons and fermions, respectively. Hence, using Eq. (4.1.7) twice,

$$\begin{aligned}
\langle q'_1 \cdots q'_M|a(q)|q_1 \cdots q_N\rangle &= \delta_{N,M+1} \sum_{r=1}^{N} \sum_{\overline{\mathcal{P}}} (\pm)^{r-1} \delta_{\overline{\mathcal{P}}} \delta(q - q_r) \prod_{i=1}^{M} \delta(q'_i - q_{\overline{\mathcal{P}}i}) \\
&= \delta_{N,M+1} \sum_{r=1}^{N} (\pm)^{r-1} \langle q'_1 \cdots q'_M|q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle.
\end{aligned}$$

Both sides of Eq. (4.2.3) thus have the same matrix element with any state $|q'_1 \cdots q'_M\rangle$, and are therefore equal, as was to be shown.) As a special case of Eq. (4.2.3), we note that for both bosons and fermions, $a(q)$ annihilates the vacuum

$$a(q)|\Omega\rangle = 0. \tag{4.2.4}$$

As defined here, the creation and annihilation operators satisfy an important commutation or anticommutation relation. Applying the operator $a(q')$ to Eq. (4.2.1) and using Eq. (4.2.3) gives

$$\begin{aligned} a(q')a^\dagger(q)|q_1 \cdots q_N\rangle = \delta(q' - q)|q_1 \cdots q_N\rangle \\ + \sum_{r=1}^{N} (\pm)^{r+2} \delta(q' - q_r)|q q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle. \end{aligned}$$

(The sign in the second term is $(\pm)^{r+2}$ because q_r is in the $(r+1)$-th place in $|qq_1 \cdots q_N\rangle$.) On the other hand, applying the operator $a^\dagger(q)$ to Eq. (4.2.3) gives

$$a^\dagger(q)a(q')|q_1 \cdots q_N\rangle = \sum_{r=1}^{N} (\pm)^{r+1} \delta(q' - q_r)|q q_1 \cdots q_{r-1} q_{r+1} \cdots q_N\rangle.$$

Subtracting or adding, we have then

$$\left[a(q')a^\dagger(q) \mp a^\dagger(q)a(q')\right]|q_1 \cdots q_N\rangle = \delta(q' - q)|q_1 \cdots q_N\rangle.$$

But this holds for all states $|q_1 \cdots q_N\rangle$ (and may easily be seen to hold also for states containing both bosons and fermions) and therefore implies the operator relation

$$a(q')a^\dagger(q) \mp a^\dagger(q)a(q') = \delta(q' - q). \tag{4.2.5}$$

In addition, Eq. (4.2.2) gives immediately

$$a^\dagger(q')a^\dagger(q) \mp a^\dagger(q)a^\dagger(q') = 0 \tag{4.2.6}$$

and so also

$$a(q')a(q) \mp a(q)a(q') = 0. \tag{4.2.7}$$

As always, the top and bottom signs apply for bosons and fermions, respectively. According to the conventions discussed in the previous section, the creation and/or annihilation operators for particles of two different species commute if either particle is a boson, and anticommute if both are fermions.

The above discussion could have been presented in reverse order (and in most textbooks usually is). That is, we could have started with the commutation or anticommutation relations Eqs. (4.2.5)–(4.2.7), derived from the canonical quantization of some given field theory. Multi-particle states would have then been defined by Eq. (4.2.2), and their scalar products Eq. (4.1.7) derived from the commutation or anticommutation relations. In fact, as discussed in Chapter 1, such a treatment would be much closer to the way that this formalism developed historically. We have followed an unhistorical approach here because we want to free ourselves from any dependence on pre-existing field theories, and rather wish to understand why field theories are the way they are.

We will now prove the fundamental theorem quoted at the beginning of this chapter: *any operator $\mathcal{O}$ may be expressed as a sum of products of creation and annihilation operators*

$$\begin{aligned}\mathcal{O} = \sum_{N=0}^{\infty}\sum_{M=0}^{\infty}\int & dq_1'\cdots dq_N'\, dq_1\cdots dq_M \\ & \times a^\dagger(q_1')\cdots a^\dagger(q_N')a(q_M)\cdots a(q_1) \\ & \times C_{NM}(q_1'\cdots q_N' q_1\cdots q_M). \end{aligned} \tag{4.2.8}$$

That is, we want to show that the C_{NM} coefficients can be chosen to give the matrix elements of this expression any desired values. We do this by mathematical induction. First, it is trivial that by choosing C_{00} properly, we can give $\langle\Omega|\mathcal{O}|\Omega\rangle$ any desired value, irrespective of the values of C_{NM} with $N > 0$ and/or $M > 0$. We need only use Eq. (4.2.4) to see that Eq. (4.2.8) has the vacuum expectation value

$$\langle\Omega|\mathcal{O}|\Omega\rangle = C_{00}.$$

Now suppose that the same is true for all matrix elements of $\mathcal{O}$ between N- and M-particle states, with $N < L, M \leq K$ or $N \leq L, M < K$; that is, that these matrix elements have been given some desired values by an appropriate choice of the corresponding coefficients C_{NM}. To see that the same is then also true of matrix elements of $\mathcal{O}$ between any L- and K-particle states, use Eq. (4.2.8) to evaluate

$$\begin{aligned}\langle q_1'\cdots q_L'|\mathcal{O}|q_1\cdots q_K\rangle = {} & L!K!C_{LK}(q_1'\cdots q_L' q_1\cdots q_K) \\ & + \text{terms involving } C_{NM} \text{ with } N < L, M \leq K \text{ or } N \leq L, M < K.\end{aligned}$$

Whatever values have already been given to C_{NM} with $N < L, M \leq K$ or $N \leq L, M < K$, there is clearly some choice of C_{LK} which gives this matrix element any desired value.

Of course, an operator need not be expressed in the form (4.2.8), with all creation operators to the left of all annihilation operators. (This is often called the 'normal' order of the operators.) However, if the formula for some operator has the creation and annihilation operators in some other order, we can always bring the creation operators to the left of the annihilation operators by repeated use of the commutation or anticommutation relations, picking up new terms from the delta function in Eq. (4.2.5).

For instance, consider any sort of additive operator F (like momentum, charge, etc.) for which

$$F|q_1\cdots q_N\rangle = \big(f(q_1)+\cdots+f(q_N)\big)|q_1\cdots q_N\rangle. \tag{4.2.9}$$

Such an operator can be written as in Eq. (4.2.8), but using only the term with $N = M = 1$:

$$F = \int dq\, a^\dagger(q)a(q)f(q). \tag{4.2.10}$$

In particular, the free-particle Hamiltonian is always

$$H_0 = \int dq\, a^\dagger(q)a(q)E(q) \tag{4.2.11}$$

where $E(q)$ is the single-particle energy

$$E(\mathbf{p}, \sigma, n) = \sqrt{\mathbf{p}^2 + m_n^2}.$$

We will need the transformation properties of the creation and annihilation operators for various symmetries. First, let's consider inhomogeneous proper orthochronous Lorentz transformations. Recall that the N-particle states have the Lorentz transformation property

$$\begin{aligned}
&U_0(\Lambda, a)|\mathbf{p}_1\sigma_1 n_1;\ \mathbf{p}_2\sigma_2 n_2;\ \ldots\rangle \\
&\quad = e^{-i(\Lambda p_1)\cdot a} e^{-i(\Lambda p_2)\cdot a} \cdots \sqrt{\frac{(\Lambda p_1)^0 (\Lambda p_2)^0 \cdots}{p_1^0 p_2^0 \cdots}} \\
&\qquad \times \sum_{\bar\sigma_1 \bar\sigma_2 \cdots} D^{(j_1)}_{\bar\sigma_1 \sigma_1}\big(W(\Lambda, p_1)\big) D^{(j_2)}_{\bar\sigma_2 \sigma_2}\big(W(\Lambda, p_2)\big) \cdots \\
&\qquad \times |\mathbf{p}_{1\Lambda}\bar\sigma_1 n_1;\ \mathbf{p}_{2\Lambda}\bar\sigma_2 n_2;\ \ldots\rangle.
\end{aligned}$$

Here $\mathbf{p}_\Lambda$ is the three-vector part of Λp, $D^{(j)}_{\bar\sigma\sigma}(R)$ is the same unitary spin-j representation of the three-dimensional rotation group as used in Section 2.5, and $W(\Lambda, p)$ is the particular rotation

$$W(\Lambda, p) = L^{-1}(\Lambda p)\Lambda L(p),$$

where $L(p)$ is the standard 'boost' that takes a particle of mass m from rest to four-momentum p^μ. (Of course, m and j depend on the species label n. This is all for $m \neq 0$; we will return to the massless particle case in the following chapter.) Now, these states can be expressed as in Eq. (4.2.2)

$$|\mathbf{p}_1\sigma_1 n_1;\ \mathbf{p}_2\sigma_2 n_2;\ \ldots\rangle = a^\dagger(\mathbf{p}_1, \sigma_1, n_1)a^\dagger(\mathbf{p}_2, \sigma_2, n_2)\cdots|\Omega\rangle,$$

where $|\Omega\rangle$ is the Lorentz-invariant vacuum state

$$U_0(\Lambda, a)|\Omega\rangle = |\Omega\rangle.$$

In order that the state (4.2.2) should transform properly, it is necessary and sufficient that the creation operator have the transformation rule

$$\begin{aligned}
&U_0(\Lambda, a)a^\dagger(\mathbf{p}, \sigma, n)U_0^{-1}(\Lambda, a) \\
&\quad = e^{-i(\Lambda p)\cdot a}\sqrt{\frac{(\Lambda p)^0}{p^0}} \sum_{\bar\sigma} D^{(j)}_{\bar\sigma\sigma}\big(W(\Lambda, p)\big) a^\dagger(\mathbf{p}_\Lambda, \bar\sigma, n).
\end{aligned} \tag{4.2.12}$$

In the same way, the operators C, P, and T, that induce charge-conjugation, space inversion, and time-reversal transformations on free particle states[29] transform the creation operators

[29]We omit the subscript '0' on these operators, because in virtually all cases where C, P, and/or T are conserved, the operators that induce these transformations on 'in' and 'out' states are the same as those defined by their action on free-particle states. This is not the case for continuous Lorentz transformations, for which it is necessary to distinguish between the operators $U(\Lambda, a)$ and $U_0(\Lambda, a)$.

as:

$$Ca^\dagger(\mathbf{p},\sigma,n)C^{-1} = \xi_n a^\dagger(\mathbf{p},\sigma,n^c), \tag{4.2.13}$$

$$Pa^\dagger(\mathbf{p},\sigma,n)P^{-1} = \eta_n a^\dagger(-\mathbf{p},\sigma,n), \tag{4.2.14}$$

$$Ta^\dagger(\mathbf{p},\sigma,n)T^{-1} = \zeta_n(-1)^{j-\sigma} a^\dagger(-\mathbf{p},-\sigma,n). \tag{4.2.15}$$

As mentioned in the previous section, although we have been dealing with operators that create and annihilate particles in free-particle states, the whole formalism can be applied to 'in' and 'out' states, in which case we would introduce operators $a_{\rm in}$ and $a_{\rm out}$ defined in the same way by their action on these states. These operators satisfy a Lorentz transformation rule just like Eq. (4.2.12), but with the true Lorentz transformation operator $U(\Lambda, a)$ instead of the free-particle operator $U_0(\Lambda, a)$.

4.3 Cluster Decomposition and Connected Amplitudes

It is one of the fundamental principles of physics (indeed, of all science) that experiments that are sufficiently separated in space have unrelated results. The probabilities for various collisions measured at Fermilab should not depend on what sort of experiments are being done at CERN at the same time. If this principle were not valid, then we could never make any predictions about any experiment without knowing everything about the universe.

In S-matrix theory, the cluster decomposition principle states that if multi-particle processes $\alpha_1 \to \beta_1$, $\alpha_2 \to \beta_2$, ..., $\alpha_{\mathcal{N}} \to \beta_{\mathcal{N}}$ are studied in $\mathcal{N}$ very distant laboratories, then the S-matrix element for the overall process factorizes. That is,[30]

$$S_{\beta_1+\beta_2+\cdots+\beta_{\mathcal{N}},\,\alpha_1+\alpha_2+\cdots+\alpha_{\mathcal{N}}} \longrightarrow S_{\beta_1\alpha_1} S_{\beta_2\alpha_2} \cdots S_{\beta_{\mathcal{N}}\alpha_{\mathcal{N}}} \tag{4.3.1}$$

if for all $i \neq j$, *all* of the particles in states α_i and β_i are at a great spatial distance from *all* of the particles in states α_j and β_j. This factorization of S-matrix elements will ensure a factorization of the corresponding transition probabilities, corresponding to uncorrelated experimental results.

There is a combinatoric trick that allows us to rewrite Eq. (4.3.1) in a more transparent way. Suppose we define the *connected* part of the S-matrix, $S^C_{\beta\alpha}$, by the formula[31]

$$S_{\beta\alpha} = \sum_{\rm PART} (\pm) S^C_{\beta_1\alpha_1} S^C_{\beta_2\alpha_2} \cdots . \tag{4.3.2}$$

[30] We are here returning to the notation used in Chapter 3; Greek letters α or β stand for a collection of particles, including for each particle a specification of its momentum, spin, and species. Also, $\alpha_1 + \alpha_2 + \cdots + \alpha_{\mathcal{N}}$ is the state formed by combining all the particles in the states $\alpha_1, \alpha_2, \ldots$, and $\alpha_{\mathcal{N}}$, and likewise for $\beta_1 + \beta_2 + \cdots + \beta_{\mathcal{N}}$.

[31] This decomposition has been used in classical statistical mechanics by Ursell, Mayer, and others, and in quantum statistical mechanics by Lee and Yang and others [3]. It has also been used to calculate many-body ground state energies by Goldstone [4] and Hugenholtz [5]. In all of these applications the purpose of isolating the connected parts of Green's functions, partition functions, resolvents, etc., is to deal with objects with a simple volume dependence. This is essentially our purpose too, because as we shall see, the crucial property of the connected parts of the S-matrix is that they are proportional to a single momentum-conservation delta function, and in a box the delta function becomes a Kronecker delta times the volume. The cluster decomposition is also the same formal device as that used in the theory of noise [6] to decompose the correlation function of several random variables into its 'cumulants'; if the random variable receives contributions from a large number N of independent fluctuations, then each cumulant is proportional to N.

Here the sum is over all different ways of partitioning the particles in the state α into clusters $\alpha_1, \alpha_2, \ldots$, and likewise a sum over all ways of partitioning the particles in the state β into clusters $\beta_1, \beta_2, \ldots$; not counting as different those that merely arrange particles within a given cluster or permute whole clusters. The sign is $+$ or $-$ according to whether the rearrangements $\alpha \to \alpha_1\alpha_2\cdots$ and $\beta \to \beta_1\beta_2\cdots$ involve altogether an even or an odd number of fermion interchanges, respectively. The term 'connected' is used because of the interpretation of $S^C_{\beta\alpha}$ in terms of diagrams representing different contributions in perturbation theory, to be discussed in the next section.

This is a recursive definition. For each α and β, the sum on the right-hand side of Eq. (4.3.2) consists of a term $S^C_{\beta\alpha}$, plus a sum $\sum'$ over products of two or more S^C-matrix elements, with a total number of particles in each of the states α_j and β_j that is *less* than the number of particles in the states α and β

$$S_{\beta\alpha} = S^C_{\beta\alpha} + \sum_{\text{PART}}{}' (\pm) S^C_{\beta_1\alpha_1} S^C_{\beta_2\alpha_2} \cdots .$$

Suppose that the S^C-matrix elements in this sum have already been chosen in such a way that Eq. (4.3.2) is satisfied for states β, α containing together fewer than, say, N particles. Then no matter what values are found in this way for the S^C-matrix elements appearing in the sum $\sum'$, we can always choose the remaining term $S^C_{\beta\alpha}$ so that Eq. (4.3.2) is also satisfied for states α, β containing a total of N particles.[32] Thus Eq. (4.3.2) contains no information in itself; it is merely a definition of S^C.

If the states α and β each consist of just a single particle, say with quantum numbers q and q' respectively, then the only term on the right-hand-side of Eq. (4.3.2) is just $S^C_{\beta\alpha}$ itself, so for one-particle states

$$S^C_{q'q} = S_{q'q} = \delta(q' - q) . \tag{4.3.3}$$

(Apart from possible degeneracies, the fact that $S_{q'q}$ is proportional to $\delta(q' - q)$ follows from conservation laws. The absence of any proportionality factor in Eq. (4.3.3) is based on a suitable choice of the relative phase of 'in' and 'out' states.) We are here assuming that single-particle states are stable, so that there are no transitions between single-particle states and any others, such as the vacuum.

For transitions between two-particle states, Eq. (4.3.2) reads

$$\begin{aligned} S_{q_1'q_2',\,q_1q_2} = S^C_{q_1'q_2',\,q_1q_2} &+ \delta(q_1' - q_1)\delta(q_2' - q_2) \\ &\pm \delta(q_1' - q_2)\delta(q_2' - q_1) . \end{aligned} \tag{4.3.4}$$

(We are here using Eq. (4.3.3).) The sign $\pm$ is $-$ if both particles are fermions, and otherwise $+$. We recognize that the two delta function terms just add up to the norm (4.1.6), so here $S^C_{\beta\alpha}$ is just $(S - \mathbf{1})_{\beta\alpha}$. But the general case is more complicated.

[32] A technicality should be mentioned here. This argument works only if we neglect the possibility that for one or more of the connected S-matrix elements in Eq. (4.3.2), the states α_j and β_j both contain no particles at all. We must therefore define the connected vacuum-vacuum element $S^C_{0,0}$ to be zero. We do not use Eq. (4.3.2) for the vacuum-vacuum S-matrix $S_{0,0}$, which in the absence of time-varying external fields is simply defined to be unity, $S_{0,0} = 1$. We will have more to say about the vacuum-vacuum amplitude in the presence of external fields in Volume II.

For transitions between three-particle or four-particle states, Eq. (4.3.2) reads

$$\begin{aligned} S_{q_1'q_2'q_3',\, q_1q_2q_3} = S^C_{q_1'q_2'q_3',\, q_1q_2q_3} \\ + \delta(q_1'-q_1)S^C_{q_2'q_3',\, q_2q_3} \pm \text{ permutations} \\ + \delta(q_1'-q_1)\delta(q_2'-q_2)\delta(q_3'-q_3) \pm \text{ permutations} \end{aligned} \tag{4.3.5}$$

and

$$\begin{aligned} S_{q_1'q_2'q_3'q_4',\, q_1q_2q_3q_4} = S^C_{q_1'q_2'q_3'q_4',\, q_1q_2q_3q_4} \\ + S^C_{q_1'q_2',\, q_1q_2}S^C_{q_3'q_4',\, q_3q_4} \pm \text{ permutations} \\ + \delta(q_1'-q_1)S^C_{q_2'q_3'q_4',\, q_2q_3q_4} \pm \text{ permutations} \\ + \delta(q_1'-q_1)\delta(q_2'-q_2)S^C_{q_3'q_4',\, q_3q_4} \pm \text{ permutations} \\ + \delta(q_1'-q_1)\delta(q_2'-q_2)\delta(q_3'-q_3)\delta(q_4'-q_4) \pm \text{ permutations}\,. \end{aligned} \tag{4.3.6}$$

(Taking account of all permutations, there are a total of $1+9+6=16$ terms in Eq. (4.3.5) and $1+18+16+72+24=131$ terms in Eq. (4.3.6). If we had not assumed that one-particle states are stable, there would be even more terms.) As explained previously, the definition of $S^C_{\beta\alpha}$ is recursive: we use Eq. (4.3.4) to define $S^C_{\beta\alpha}$ for two-particle states, then use this definition in Eq. (4.3.5) when we define $S^C_{\beta\alpha}$ for three-particle states, then use both of these definitions in Eq. (4.3.6) to obtain the definition of $S^C_{\beta\alpha}$ for four-particle states, and so on.

The point of this definition of the connected part of the S-matrix is that the cluster decomposition principle is equivalent to the requirement that $S^C_{\beta\alpha}$ must vanish when any one or more of the particles in the states β and/or α are far away in space from the others.[33] To see this, suppose that the particles in the states β and α are grouped into clusters $\beta_1, \beta_2, \ldots$ and $\alpha_1, \alpha_2, \ldots$, and that all particles in the set $\alpha_i + \beta_i$ are far from all particles in the set $\alpha_j + \beta_j$ for any $j \neq i$. Then if $S^C_{\beta'\alpha'}$ vanishes if any particles in β' or α' are far from the others, it vanishes if any particles in these states are in different clusters, so the definition (4.3.2) yields

$$\begin{aligned} S_{\beta\alpha} \longrightarrow \sum_{\text{PART}}^{(1)} (\pm) S^C_{\beta_{11}\alpha_{11}} S^C_{\beta_{12}\alpha_{12}} \cdots \\ \times \sum_{\text{PART}}^{(2)} (\pm) S^C_{\beta_{21}\alpha_{21}} S^C_{\beta_{22}\alpha_{22}} \cdots \times \cdots\,, \end{aligned} \tag{4.3.7}$$

where $\sum^{(j)}$ is a sum over all different ways of partitioning the clusters β_j and α_j into subclusters $\beta_{j1}, \beta_{j2}, \ldots$ and $\alpha_{j1}, \alpha_{j2}, \ldots$. But referring back to Eq. (4.3.2), this is just the desired factorization property (4.3.1).

For instance, suppose that in the four-particle reaction $1234 \to 1'2'3'4'$, we let particles $1, 2, 1'$, and $2'$ be very far from $3, 4, 3'$, and $4'$. Then if $S^C_{\beta\alpha}$ vanishes when any particles in β and/or α are far from the others, the only terms in Eq. (4.3.6) that survive (in an even

[33] In order to give a meaning to 'far,' we will have to Fourier transform S^C, so that each three-momentum label $\mathbf{p}$ is replaced with a spatial coordinate three-vector $\mathbf{x}$.

more abbreviated notation) are

$$\begin{aligned} S_{1'2'3'4',\,1234} \longrightarrow\ & S^C_{1'2',\,12} S^C_{3'4',\,34} \\ & + (\delta_{1'1}\delta_{2'2} \pm \delta_{1'2}\delta_{2'1}) S^C_{3'4',\,34} \\ & + (\delta_{3'3}\delta_{4'4} \pm \delta_{3'4}\delta_{4'3}) S^C_{1'2',\,12} \\ & + (\delta_{1'1}\delta_{2'2} \pm \delta_{1'2}\delta_{2'1})(\delta_{3'3}\delta_{4'4} \pm \delta_{3'4}\delta_{4'3})\,. \end{aligned}$$

Comparison with Eq. (4.3.4) shows that this is just the required factorization condition (4.3.1)

$$S_{1'2'3'4',\,1234} \longrightarrow S_{1'2',\,12} S_{3'4',\,34}\,.$$

We have formulated the cluster decomposition principle in coordinate space, as the condition that $S^C_{\beta\alpha}$ vanishes if any particles in the states β or α are far from any others. It is convenient for us to reexpress this in momentum space. The coordinate space matrix elements are defined as a Fourier transform

$$\begin{aligned} S^C_{\mathbf{x}'_1\mathbf{x}'_2\cdots,\,\mathbf{x}_1\mathbf{x}_2\cdots} \equiv \int & d^3p'_1\, d^3p'_2 \cdots d^3p_1\, d^3p_2 \cdots\ S^C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\,\mathbf{p}_1\mathbf{p}_2\cdots} \\ & \times e^{i\mathbf{p}'_1\cdot\mathbf{x}'_1} e^{i\mathbf{p}'_2\cdot\mathbf{x}'_2} \cdots e^{-i\mathbf{p}_1\cdot\mathbf{x}_1} e^{-i\mathbf{p}_2\cdot\mathbf{x}_2} \cdots\,. \end{aligned} \tag{4.3.8}$$

(We are here temporarily dropping spin and species labels, which just go along with the momentum or coordinate labels.) If $|S^C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\,\mathbf{p}_1\mathbf{p}_2\cdots}|$ were sufficiently well behaved (to be specific, if it were Lebesgue integrable) then according to the Riemann–Lebesgue theorem [7] the integral (4.3.8) would vanish when any combination of spatial coordinates goes to infinity. Now, this is certainly too strong a requirement. Translational invariance tells us that the connected part of the S-matrix, like the S-matrix itself, can only depend on differences of coordinate vectors, and therefore does not change at all if all of the x_i and x'_j vary together, with their differences held constant. This requires that the elements of S^C in a momentum basis must, like those of S, be proportional to a three-dimensional delta function that ensures momentum conservation (and makes $|S^C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\,\mathbf{p}_1\mathbf{p}_2\cdots}|$ *not* Lebesgue integrable), as well as the energy-conservation delta function required by scattering theory. That is, we can write

$$\begin{aligned} S^C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\,\mathbf{p}_1\mathbf{p}_2\cdots} = & \ \delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 + \cdots - \mathbf{p}_1 - \mathbf{p}_2 - \cdots) \\ & \times \delta(E'_1 + E'_2 + \cdots - E_1 - E_2 - \cdots) \\ & \times C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\,\mathbf{p}_1\mathbf{p}_2\cdots}\,. \end{aligned} \tag{4.3.9}$$

This is no problem: the cluster decomposition principle only requires that Eq. (4.3.8) vanish when the *differences* among some of the x_i and/or x'_i become large. However, if C itself in Eq. (4.3.9) contained additional delta functions of linear combinations of the three-momenta, then this principle would not be satisfied. For instance, suppose that there were a delta function in C that required that the sum of the $\mathbf{p}'_i$ and $-\mathbf{p}_i$ for some subset of the particles vanished. Then Eq. (4.3.8) would not vary if all of the $\mathbf{x}'_i$ and $\mathbf{x}_i$ for the particles in that subset moved together (with constant differences) away from all the other $\mathbf{x}'_k$ and $\mathbf{x}_k$, in contradiction to the cluster decomposition principle. Loosely speaking then, the cluster decomposition principle simply says that *the connected part of the S-matrix, unlike the S-matrix itself, contains just a single momentum-conservation delta function.*

In order to put this a bit more precisely, we can say that the coefficient function $C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots}$ in Eq. (4.3.9) is a smooth function of its momentum labels. But how smooth? It would be most straightforward if we could simply require that $C_{\mathbf{p}'_1\mathbf{p}'_2\cdots,\mathbf{p}_1\mathbf{p}_2\cdots}$ be analytic in all of the momenta at $\mathbf{p}'_1 = \mathbf{p}'_2 = \cdots = \mathbf{p}_1 = \mathbf{p}_2 = \cdots = 0$. This requirement would indeed guarantee that $S^C_{\mathbf{x}'_1\mathbf{x}'_2\cdots,\mathbf{x}_1\mathbf{x}_2\cdots}$ vanishes exponentially fast when any of the $\mathbf{x}$ and $\mathbf{x}'$ is very distant from any of the other $\mathbf{x}$ and $\mathbf{x}'$. However, an exponential fall-off of S^C is not an essential part of the cluster decomposition principle, and, in fact, the requirement of analyticity is not met in all theories. Most notably, in theories with massless particles, S^C can have poles at certain values of the $\mathbf{p}$ and $\mathbf{p}'$. For instance, as we will see in Chapter 10, if a massless particle can be emitted in the transition $1 \to 3$ and absorbed in the transition $2 \to 4$, then $S^C_{34,12}$ will have a term proportional to $1/(\mathbf{p}_1 - \mathbf{p}_3)^2$. After Fourier transforming, such poles yield terms in $S^C_{\mathbf{x}'_1\mathbf{x}'_2\cdots,\mathbf{x}_1\mathbf{x}_2\cdots}$ that fall off only as negative powers of coordinate differences [1]. There is no need to formulate the cluster decomposition principle so stringently that such behavior is ruled out. Thus the 'smoothness' condition on S^C should be understood to allow various poles and branch-cuts at certain values of the $\mathbf{p}$ and $\mathbf{p}'$, but not singularities as severe as delta functions.

4.4 Structure of the Interaction

We now ask, what sort of Hamiltonian will yield an S-matrix that satisfies the cluster decomposition principle? It is here that the formalism of creation and annihilation operators comes into its own. The answer is contained in the theorem that the S-matrix satisfies the cluster decomposition principle if (and as far as I know, only if) the Hamiltonian can be expressed as in Eq. (4.2.8):

$$\begin{aligned} H = \sum_{N=0}^{\infty}\sum_{M=0}^{\infty} \int & dq'_1 \cdots dq'_N \, dq_1 \cdots dq_M \\ & \times a^\dagger(q'_1)\cdots a^\dagger(q'_N) a(q_M)\cdots a(q_1) \\ & \times h_{NM}(q'_1\cdots q'_N, q_1\cdots q_M). \end{aligned} \tag{4.4.1}$$

with coefficient functions h_{NM} that contain just a *single* three-dimensional momentum-conservation delta function (returning here briefly to a more explicit notation)

$$\begin{aligned} & h_{NM}(\mathbf{p}'_1\sigma'_1 n'_1 \cdots \mathbf{p}'_N\sigma'_N n'_N, \mathbf{p}_1\sigma_1 n_1 \cdots \mathbf{p}_M\sigma_M n_M) \\ & \quad = \delta^3(\mathbf{p}'_1 + \cdots + \mathbf{p}'_N - \mathbf{p}_1 - \cdots - \mathbf{p}_N) \\ & \qquad \times \widetilde{h}_{NM}(\mathbf{p}'_1\sigma'_1 n'_1 \cdots \mathbf{p}'_N\sigma'_N n'_N, \mathbf{p}_1\sigma_1 n_1 \cdots \mathbf{p}_M\sigma_M n_M), \end{aligned} \tag{4.4.2}$$

where $\widetilde{h}_{NM}$ contains no delta function factors. Note that Eq. (4.4.1) by itself has no content—we saw in Section 4.2 that *any* operator can be put in this form. It is only Eq. (4.4.1) combined with the requirement that h_{NM} has only the single delta function shown in Eq. (4.4.2) that guarantees that the S-matrix satisfies the cluster decomposition principle.

The validity of this theorem in perturbation theory will become obvious when we develop the Feynman diagram formalism in Chapter 6. The trusting reader may prefer to skip the rest of the present chapter, and move on to consider the implications of this theorem in Chapter 5. However, the proof has some instructive features, and will help to clarify in what sense the field theory of the next chapter is inevitable.

To prove this theorem, we make use of perturbation theory in its time-dependent form. (One of the advantages of time-dependent perturbation theory is that it makes the combinatorics underlying the cluster decomposition principle much more transparent; if E is a sum of one-particle energies then e^{-iEt} is a product of functions of the individual energies, while $[E - E_\alpha + i\epsilon]^{-1}$ is not.) The S-matrix is given by Eq. (3.5.10) as[34]

$$S_{\beta\alpha} = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 \cdots dt_n \, \langle \beta | T\{H_I(t_1) \cdots H_I(t_n)\} | \alpha \rangle, \tag{4.4.3}$$

where the Hamiltonian is split into a free-particle part H_0 and an interaction H_{int}, and

$$H_I(t) \equiv \exp(iH_0 t) H_{\text{int}} \exp(-iH_0 t). \tag{4.4.4}$$

Now, the states $|\alpha\rangle$ and $|\beta\rangle$ may be expressed as in Eq. (4.2.2) as products of creation operators acting on the vacuum $|\Omega\rangle$, and $H_I(t)$ is itself a sum of products of creation and annihilation operators, so each term in the sum (4.4.3) may be written as a sum of vacuum expectation values of products of creation and annihilation operators. By using the commutation or anticommutation relations (4.2.5) we may move each annihilation operator in turn to the right past all the creation operators. For each annihilation operator moved to the right past a creation operator we have two terms, as shown by writing Eq. (4.2.5) in the form

$$a(q')a^\dagger(q) = \pm a^\dagger(q)a(q') + \delta(q' - q).$$

Moving other creation operators past the annihilation operator in the first term generates yet more terms. But Eq. (4.2.4) shows that any annihilation operator that moves all the way to the right and acts on $|\Omega\rangle$ gives zero, so in the end all we have left is the delta functions. In this way, the vacuum expectation value of a product of creation and annihilation operators is given by a sum of different terms, each term equal to a product of delta functions and $\pm$ signs from the commutators or anticommutators. It follows that each term in Eq. (4.4.3) may be expressed as a sum of terms, each term equal to a product of delta functions and $\pm$ signs from the commutators or anticommutators and whatever factors are contributed by $H_I(t)$, integrated over all the times and integrated and summed over the momenta, spins, and species in the arguments of the delta functions.

Each of the terms generated in this way may be symbolized by a diagram. (This is not yet the full Feynman diagram formalism, because we are not yet going to associate numerical quantities with the ingredients in the diagrams; we are using the diagrams here only as a way of keeping track of three-momentum delta functions.) Draw n points, called

[34] We are now adopting the convention that for $n = 0$, the time-ordered product in Eq. (4.4.3) is taken as the unit operator, so the $n = 0$ term in the sum just yields the term $\delta(\beta - \alpha)$ in $S_{\beta\alpha}$.

vertices, one for each $H_I(t)$ operator. For each delta function produced when an annihilation operator in one of these $H_I(t)$ operators moves past a creation operator in the initial state $|\alpha\rangle$, draw a line coming into the diagram from below that ends at the corresponding vertex. For each delta function produced when an annihilation operator in the adjoint of the final state $|\beta\rangle$ moves past a creation operator in one of the $H_I(t)$, draw a line from the corresponding vertex upwards out of the diagram. For each delta function produced when an annihilation operator in one $H_I(t)$ moves past a creation operator in another $H_I(t)$ draw a line between the two corresponding vertices. Finally, for each delta function produced when an annihilation operator in the adjoint of the final state moves past a creation operator in the initial state, draw a line from bottom to top, right through the diagrams. Each of the delta functions associated with one of these lines enforces the equality of the momentum arguments of the pair of creation and annihilation operators represented by the line. There is also at least one delta function contributed by each of the vertices, which enforces the conservation of the total three-momentum at the vertex.

Such a diagram may be connected (every point connected to every other by a set of lines) and if not connected, it breaks up into a number of connected pieces. The $H_I(t)$ operator associated with a vertex in one connected component effectively commutes with the $H_I(t)$ associated with any vertex in any other connected component, because for this diagram, we are not including any terms in which an annihilation operator in one vertex destroys a particle that is produced by a creation operator in the other vertex—if we did, the two vertices would be in the same connected component. Thus the matrix element in Eq. (4.4.3) can be expressed as a sum over *products* of contributions, one from each connected component:

$$\begin{aligned}
&\langle\beta|T\{H_I(t_1)\cdots H_I(t_n)\}|\alpha\rangle \\
&\qquad = \sum_{\text{clusterings}} (\pm) \prod_{j=1}^{\nu} \Big(\langle\beta_j|T\{H_I(t_{j1})\cdots H_I(t_{jn_j})\}|\alpha_j\rangle\Big)_C .
\end{aligned} \tag{4.4.5}$$

Here the sum is over all ways of splitting up the incoming and outgoing particles and $H_I(t)$ operators into ν clusters (including a sum over ν from 1 to n) with the n_j operators $H_I(t_{j1})\cdots H_I(t_{jn_j})$ and the subsets of initial particles α_j and final particles β_j all in the jth cluster. Of course, this means that

$$n = n_1 + \cdots + n_\nu$$

and also the set α is the union of all the particles in the subsets $\alpha_1, \alpha_2, \ldots, \alpha_\nu$, and likewise for the final state. Some of the clusters in Eq. (4.4.5) may contain no vertices at all, i.e., $n_j = 0$; for these factors, we must take the matrix element factor in Eq. (4.4.5) to vanish unless β_j and α_j are both one-particle states (in which case it is just a delta function $\delta(\alpha_j - \beta_j)$), because the only connected diagrams without vertices consist of a single line running through the diagram from bottom to top. Most important, the subscript C in Eq. (4.4.5) means that we exclude any contributions corresponding to disconnected diagrams, that is, any contributions in which any $H_I(t)$ operator or any initial or final particle is not connected to every other by a sequence of particle creations and annihilations.

Now let us use Eq. (4.4.5) in the sum (4.4.3). Every time variable is integrated from $-\infty$ to $+\infty$, so it makes no difference which of the $t_1, \dots, t_n$ are sorted out into each cluster. The sum over clusterings therefore yields a factor $n!/n_1!n_2!\cdots n_\nu!$, equal to the number of ways of sorting out n vertices into ν clusters, each containing $n_1, n_2, \dots$ vertices:

$$\begin{aligned}\int_{-\infty}^{\infty} & dt_1 \cdots dt_n \, \langle\beta| T\{H_I(t_1)\cdots H_I(t_n)\}|\alpha\rangle \\ &= \frac{n!}{n_1!n_2!\cdots n_\nu!} \sum_{\text{PART}} (\pm) \sum_{n_1+\cdots+n_\nu=n} \prod_{j=1}^{\nu} \int_{-\infty}^{\infty} dt_{j1}\cdots dt_{jn_j} \\ &\quad \times \Big(\langle\beta_j|T\{H_I(t_{j1})\cdots H_I(t_{jn_j})\}|\alpha_j\rangle\Big)_C .\end{aligned}$$

The first sum here is over all ways of partitioning the particles in the initial and final states into clusters $\alpha_1 \cdots \alpha_\nu$ and $\beta_1 \cdots \beta_\nu$ (including a sum over the number ν of clusters). The factor $n!$ here cancels the $1/n!$ in Eq. (4.4.3), and the factor $(-i)^n$ in the perturbation series for (4.4.5) can be written as a product $(-i)^{n_1}\cdots(-i)^{n_\nu}$, so instead of summing over n and then summing separately over $n_1, \dots, n_\nu$, constrained by $n_1 + \cdots + n_\nu = n$, we can simply sum independently over each $n_1, \dots, n_\nu$. This gives finally

$$\begin{aligned}S_{\beta\alpha} = \sum_{\text{PART}} (\pm) \prod_{j=1}^{\nu} \sum_{n_j=0}^{\infty} \frac{(-i)^{n_j}}{n_j!} \int_{-\infty}^{\infty} dt_{j1}\cdots dt_{jn_j} \\ \times \Big(\langle\beta_j|T\{H_I(t_{j1})\cdots H_I(t_{jn_j})\}|\alpha_j\rangle\Big)_C .\end{aligned}$$

Comparing this with the definition (4.2.2) of the connected matrix elements $S^C_{\beta\alpha}$, we see that these matrix elements are just given by the factors in the product here

$$S^C_{\beta\alpha} = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1\cdots dt_n \Big(\langle\beta|T\{H_I(t_1)\cdots H_I(t_n)\}|\alpha\rangle\Big)_C . \tag{4.4.6}$$

(The subscript j is dropped on all the ts and ns, as these are now mere integration and summation variables.) We see that $S^C_{\beta\alpha}$ is calculated by a very simple prescription: *$S^C_{\beta\alpha}$ is the sum of all contributions to the S-matrix that are connected, in the sense that we drop all terms in which any initial or final particle or any operator $H_I(t)$ is not connected to all the others by a sequence of particle creations and annihilations.* This justifies the adjective 'connected' for S^C.

As we have seen, momentum is conserved at each vertex and along every line, so the connected parts of the S-matrix individually conserve momentum: $S^C_{\beta\alpha}$ contains a factor $\delta^3(\mathbf{p}_\beta - \mathbf{p}_\alpha)$. What we want to prove is that $S^C_{\beta\alpha}$ contains no other delta functions.

We now make the assumption that the coefficient fractions h_{NM} in the expansion (4.4.1) of the Hamiltonian in terms of creation and annihilation operators are proportional to a *single* three-dimensional delta function, that ensures momenta conservation. This is automatically true for the free-particle Hamiltonian H_0, so it is also then true separately for the interaction H_{int}. Returning to the graphical interpretation of the matrix elements that we have been using, this means that each vertex contributes one three-dimensional delta function. (The other delta functions in matrix elements $(H_{\rm int})_{\gamma\delta}$ simply keep the momentum of any particle that is not created or annihilated at the corresponding vertex unchanged.) Now, most of these delta functions simply go to fix the momentum of intermediate particles. The only momenta that are left unfixed by such delta functions are those that circulate in loops of internal lines. (Any line which if cut leaves the diagram disconnected carries a momentum that is fixed by momentum conservation as some linear combination of the momenta of the lines coming into or going out of the diagram. If the diagram has L lines that can all be cut at the same time without the diagram becoming disconnected, then we say it has L independent loops, and there are L momenta that are not fixed by momentum conservation.) With V vertices, I internal lines, and L loops, there are V delta functions, of which $I-L$ go to fix internal momenta, leaving $V-I+L$ delta functions relating the momenta of incoming or outgoing particles. But a well-known topological identity[35] tells that for any graph consisting of C connected pieces, the numbers of vertices, internal lines, and loops are related by

$$V-I+L=C. \tag{4.4.7}$$

Hence for a connected matrix element like $S^C_{\beta\alpha}$, which arises from graphs with $C=1$, we find just a single three-dimensional delta function $\delta^3(\mathbf{p}_\beta-\mathbf{p}_\alpha)$, as was to be proved.

It was not important in the above argument that the time variables were integrated from $-\infty$ to $+\infty$. Thus exactly the same arguments can be used to show that if the coefficients $h_{N,M}$ in the Hamiltonian contain just single delta functions, then $U(t,t_0)$ can also be decomposed into connected parts, each containing a single momentum-conservation delta function factor. On the other hand, the connected part of the S-matrix also contains an energy-conservation delta function, and when we come to Feynman diagrams in Chapter 6 we shall see that $S^C_{\beta\alpha}$ contains only a single energy-conservation delta function, $\delta(E_\beta-E_\alpha)$, while $U(t,t_0)$ contains no energy-conservation delta functions at all.

It should be emphasized that the requirement that h_{NM} in Eq. (4.4.1) should have only a single three-dimensional momentum conservation delta function factor is very far from trivial, and has far-reaching implications. For instance, assume that $H_{\rm int}$ has non-vanishing matrix elements between two-particle states. Then Eq. (4.4.1) must contain a term with $N=M=2$, and coefficient

$$v_{2,2}(\mathbf{p}'_1\mathbf{p}'_2,\mathbf{p}_1\mathbf{p}_2)=(H_{\rm int})_{\mathbf{p}'_1\mathbf{p}'_2;\mathbf{p}_1\mathbf{p}_2}. \tag{4.4.8}$$

(We are here temporarily dropping spin and species labels.) But then the matrix element of the interaction between three-particle states is

$$\begin{aligned}(H_{\rm int})_{\mathbf{p}'_1\mathbf{p}'_2\mathbf{p}'_3;\mathbf{p}_1\mathbf{p}_2\mathbf{p}_3}&=v_{3,3}(\mathbf{p}'_1\mathbf{p}'_2\mathbf{p}'_3,\mathbf{p}_1\mathbf{p}_2\mathbf{p}_3)\\&\quad+v_{2,2}(\mathbf{p}'_1\mathbf{p}'_2,\mathbf{p}_1\mathbf{p}_2)\,\delta^3(\mathbf{p}'_3-\mathbf{p}_3)\pm\text{ permutations.}\end{aligned} \tag{4.4.9}$$

[35] A graph consisting of a single vertex has $V=1$, $L=0$, and $C=1$. If we add $V-1$ vertices with just enough internal lines to keep the graph connected, we have $I=V-1$, $L=0$, and $C=1$. Any additional internal lines attached (without new vertices) to the same connected graph produce an equal number of loops, so $I=V+L-1$ and $C=1$. If a disconnected graph consists of C such connected parts, the sums of I, V, and L in each connected part will then satisfy $\sum I=\sum V+\sum L-C$.

As mentioned at the beginning of this chapter, we might try to make a relativistic quantum theory that is not a field theory by choosing $v_{2,2}$ so that the two-body S-matrix is Lorentz-invariant, and adjusting the rest of the Hamiltonian so that there is no scattering in states containing three or more particles. We would then have to take $v_{3,3}$ to cancel the other terms in Eq. (4.4.9)

$$v_{3,3}(\mathbf{p}'_1\mathbf{p}'_2\mathbf{p}'_3, \mathbf{p}_1\mathbf{p}_2\mathbf{p}_3) = -\, v_{2,2}(\mathbf{p}'_1\mathbf{p}'_2, \mathbf{p}_1\mathbf{p}_2)\, \delta^3(\mathbf{p}'_3 - \mathbf{p}_3) \mp \text{ permutations.} \tag{4.4.10}$$

However, this would mean that each term in $v_{3,3}$ contains *two* delta function factors (recall that $v_{2,2}(\mathbf{p}'_1\mathbf{p}'_2, \mathbf{p}_1\mathbf{p}_2)$ has a factor $\delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 - \mathbf{p}_1 - \mathbf{p}_2)$) and this would violate the cluster decomposition principle. Thus in a theory satisfying the cluster decomposition principle, the existence of scattering processes involving two particles makes processes involving three or more particles inevitable.

* * *

When we set out to solve three-body problems in quantum theories that satisfy the cluster decomposition principle, the term $v_{3,3}$ in Eq. (4.4.9) gives no particular trouble, but the extra delta function in the other terms makes the Lippmann–Schwinger equation difficult to solve directly. The problem is that these delta functions make the kernel $[E_\alpha - E_\beta + i\epsilon]^{-1}(H_{\text{int}})_{\beta\alpha}$ of this equation not square-integrable, even after we factor out an overall momentum conservation delta function. In consequence, it cannot be approximated by a finite matrix, even one of very large rank. To solve problems involving three or more particles, it is necessary to replace the Lippmann–Schwinger equation with one that has a connected right-hand side. Such equations have been developed for the scattering of three or more particles [8,9], and in non-relativistic scattering problems they can be solved recursively, but they have not turned out to be useful in relativistic theories and so will not be described in detail here.

However, recasting the Lippmann–Schwinger equation in this manner is useful in another way. Our arguments in this section have so far relied on perturbation theory. I do not know of any non-perturbative proof of the main theorem of this section, but it has been shown [9] that these reformulated non-perturbative dynamical equations are *consistent* with the requirement that $U^C(t, t_0)$ (and hence S^C) should also contain only a single momentum-conservation delta function, as required by the cluster decomposition principle, provided that the Hamiltonian satisfies our condition that the coefficient functions $h_{N,M}$ each contain only a single momentum-conservation delta function.

Problems

1. Define generating functionals for the S-matrix and its connected part:

$$F[v] \equiv 1 + \sum_{N=1}^{\infty}\sum_{M=1}^{\infty}\frac{1}{N!M!}\int v^*(q_1')\cdots v^*(q_N')v(q_1)\cdots v(q_M) \times S_{q_1'\cdots q_N',\,q_1\cdots q_M}\,dq_1'\cdots dq_N'\,dq_1\cdots dq_M\,,$$

$$F^C[v] \equiv \sum_{N=1}^{\infty}\sum_{M=1}^{\infty}\frac{1}{N!M!}\int v^*(q_1')\cdots v^*(q_N')v(q_1)\cdots v(q_M) \times S^C_{q_1'\cdots q_N',\,q_1\cdots q_M}\,dq_1'\cdots dq_N'\,dq_1\cdots dq_M\,.$$

Derive a formula relating $F[v]$ and $F^C[v]$. (You may consider the purely bosonic case.)

2. Consider an interaction

$$H_{\text{int}} = g\int d^3\mathbf{p}_1\,d^3\mathbf{p}_2\,d^3\mathbf{p}_3\,d^3\mathbf{p}_4\,\delta^3(\mathbf{p}_1+\mathbf{p}_2-\mathbf{p}_3-\mathbf{p}_4) \times a^\dagger(\mathbf{p}_1)a^\dagger(\mathbf{p}_2)a(\mathbf{p}_3)a(\mathbf{p}_4),$$

where g is a real constant and $a(\mathbf{p})$ is the annihilation operator of a spinless boson of mass $M > 0$. Use perturbation theory to calculate the S-matrix element for scattering of these particles in the center-of-mass frame to order g^2. What is the corresponding differential cross-section?

3. A coherent state $|\lambda\rangle$ is defined to be an eigenstate of the annihilation operators $a(q)$ with eigenvalues $\lambda(q)$. Construct such a state as a superposition of the multi-particle states $|q_1, q_2, \ldots, q_N\rangle$.

References

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5 Quantum Fields and Antiparticles

6 The Feynman Rules

6.1 Derivation of the Rules

6.2 Calculation of the Propagator

6.3 Momentum Space Rules

6.4 Off the Mass Shell

Problems

References

7 The Canonical Formalism

9 Path-Integral Methods

10 Non-Perturbative Methods

10.1 Symmetries

10.2 Polology

10.3 Field and Mass Renormalization

10.4 Renormalized Charge and Ward Identities

10.5 Gauge Invariance

10.6 Electromagnetic Form Factors and Magnetic Moment

10.7 The Källén–Lehmann Representation*

10.8 Dispersion Relations*

Problems

References

11 One-Loop Radiative Corrections in Quantum Electrodynamics

13 Infrared Effects

13.1 Soft Photon Amplitudes

13.2 Virtual Soft Photons

13.3 Real Soft Photons; Cancellation of Divergences

13.4 General Infrared Divergences

13.5 Soft Photon Scattering*

13.6 The External Field Approximation*

Problems

References

14 Bound States in External Fields

14.1 The Dirac Equation

14.2 Radiative Corrections in External Fields

14.3 The Lamb Shift in Light Atoms

Problems

References